

FORM

version v5.0.0-beta.1-213-ge5c6050

Reference manual

J.A.M. Vermaseren, T. Kaneko, J. Kuipers, B. Ruijl, M. Tentyukov, T. Ueda and J. Vollinga

9 July 2025

Contents

1	Running FORM	1
2	Variables	4
2.1	Names	4
2.2	Symbols	5
2.3	Vectors	6
2.4	Indices	7
2.5	Functions	8
2.6	Sets	9
2.7	The autodeclare conventions	12
2.8	Name lists	13
2.9	Dummy indices	13
2.10	Kronecker delta's	15
2.11	Extra Symbols	16
2.12	Restrictions	18
2.13	Some common bugs	18
3	The preprocessor	20
3.1	The preprocessor variables	20
3.2	The preprocessor calculator	22
3.3	The triple dot operator	24
3.4	#add	25
3.5	#addseparator	25
3.6	#append	25
3.7	#appendpath	25
3.8	#break	25
3.9	#breakdo	26
3.10	#call	26
3.11	#case	27
3.12	#clearoptimize	28
3.13	#close	28
3.14	#closedictionary	28
3.15	#commentchar	28
3.16	#create	28
3.17	#default	29
3.18	#define	29
3.19	#do	30
3.20	#else	32

3.21	<code>#elseif</code>	32
3.22	<code>#enddo</code>	32
3.23	<code>#endif</code>	32
3.24	<code>#endinside</code>	33
3.25	<code>#endprocedure</code>	33
3.26	<code>#endswitch</code>	33
3.27	<code>#exchange</code>	33
3.28	<code>#external</code>	33
3.29	<code>#factdollar</code>	34
3.30	<code>#fromexternal</code>	34
3.31	<code>#if</code>	34
3.32	<code>#ifdef</code>	36
3.33	<code>#ifndef</code>	36
3.34	<code>#include</code>	36
3.35	<code>#inside</code>	37
3.36	<code>#message</code>	37
3.37	<code>#opendictionary</code>	38
3.38	<code>#optimize</code>	38
3.39	<code>#pipe</code>	38
3.40	<code>#preout</code>	38
3.41	<code>#prependpath</code>	39
3.42	<code>#printtimes</code>	39
3.43	<code>#procedure</code>	39
3.44	<code>#procedureextension</code>	40
3.45	<code>#prompt</code>	40
3.46	<code>#redefine</code>	40
3.47	<code>#remove</code>	41
3.48	<code>#reset</code>	41
3.49	<code>#reverseinclude</code>	41
3.50	<code>#rmexternal</code>	41
3.51	<code>#rmseparator</code>	42
3.52	<code>#setexternal</code>	42
3.53	<code>#setexternalattr</code>	42
3.54	<code>#setrandom</code>	43
3.55	<code>#show</code>	43
3.56	<code>#skipextrasymbols</code>	44
3.57	<code>#sortreallocate</code>	45
3.58	<code>#switch</code>	45
3.59	<code>#system</code>	46
3.60	<code>#terminate</code>	46
3.61	<code>#timeoutafter</code>	46
3.62	<code>#toexternal</code>	47
3.63	<code>#undefine</code>	47
3.64	<code>#usedictionary</code>	47
3.65	<code>#write</code>	47
3.66	Some remarks	50

4	Modules	51
4.1	Checkpoints	55
5	Pattern matching	57
6	The dollar variables	61
6.1	Dollar variables in a parallel environment	65
7	Statements	67
7.1	abracquets, antibracquets	67
7.2	also	67
7.3	antiputinside	68
7.4	antisymmetrize	68
7.5	apply	68
7.6	argexplode	68
7.7	argimplode	69
7.8	argtoextrasymbol	69
7.9	argument	70
7.10	auto, autodeclare	70
7.11	bracket	71
7.12	break	72
7.13	case	72
7.14	cfunctions	73
7.15	chainin	73
7.16	chainout	74
7.17	chisholm	74
7.18	cleartable	74
7.19	collect	75
7.20	commuteinset	76
7.21	commuting	76
7.22	compress	76
7.23	contract	77
7.24	copyspectator	77
7.25	createspectator	78
7.26	ctable	78
7.27	ctensors	78
7.28	cyclesymmetrize	78
7.29	deallocatetable	79
7.30	default	79
7.31	delete	79
7.32	denominators	80
7.33	dimension	80
7.34	discard	81
7.35	disorder	81
7.36	do	81
7.37	drop	82
7.38	dropcoefficient	83
7.39	dropsymbols	83

7.40	else	83
7.41	elseif	83
7.42	emptyspectator	84
7.43	endargument	84
7.44	enddo	84
7.45	endif	84
7.46	endinexpression	85
7.47	endinside	85
7.48	endrepeat	85
7.49	endswitch	85
7.50	endterm	86
7.51	endwhile	86
7.52	exit	86
7.53	extrasymbols	86
7.54	factarg	87
7.55	factdollar	88
7.56	factorize	88
7.57	fill	89
7.58	fillexpression	91
7.59	fixindex	91
7.60	format	92
7.61	frompolynomial	94
7.62	functions	94
7.63	funpowers	95
7.64	gfactorized	95
7.65	global	95
7.66	goto	96
7.67	hide	96
7.68	identify	97
7.69	idnew	99
7.70	idold	99
7.71	if	100
7.72	ifmatch	102
7.73	ifnomatch	102
7.74	index, indices	102
7.75	inexpression	103
7.76	inparallel	103
7.77	inside	104
7.78	insidefirst	104
7.79	intohide	104
7.80	keep	105
7.81	label	105
7.82	lfactorized	106
7.83	load	106
7.84	local	107
7.85	makeinteger	107
7.86	many	108
7.87	merge	108

7.88	metric	108
7.89	moduleoption	108
7.90	modulus	110
7.91	multi	111
7.92	multiply	111
7.93	ndrop	112
7.94	nfactorize	112
7.95	nfunctions	112
7.96	nhide	113
7.97	nintohide	113
7.98	normalize	113
7.99	notinparallel	114
7.100	nprint	114
7.101	nskip	114
7.102	ntable	115
7.103	ntensors	115
7.104	nunfactorize	115
7.105	nunhide	116
7.106	nwrite	116
7.107	off	117
7.108	on	119
7.109	once	123
7.110	only	123
7.111	polyfun	123
7.112	polyratfun	124
7.113	pophide	125
7.114	print	125
7.115	print[]	127
7.116	printtable	128
7.117	processbucketsize	128
7.118	propercount	129
7.119	pushhide	129
7.120	putinside	129
7.121	ratio	129
7.122	rcyclesymmetrize	130
7.123	redefine	130
7.124	removespectator	131
7.125	renumber	131
7.126	repeat	132
7.127	replaceloop	133
7.128	save	134
7.129	select	135
7.130	set	135
7.131	setexitflag	135
7.132	shuffle	136
7.133	skip	137
7.134	sort	137
7.135	splitarg	137

7.136	splitfirstarg	138
7.137	splitlastarg	138
7.138	shuffle	139
7.139	sum	140
7.140	switch	140
7.141	symbols	141
7.142	symmetrize	142
7.143	table	143
7.144	tablebase	145
7.145	tensors	145
7.146	term	145
7.147	testuse	146
7.148	threadbucketsize	146
7.149	topolynomial	146
7.150	tospectator	147
7.151	totensor	147
7.152	tovector	148
7.153	trace4	148
7.154	tracen	149
7.155	transform	149
7.156	tryreplace	152
7.157	unfactorize	152
7.158	unhide	153
7.159	unittrace	153
7.160	vectors	154
7.161	while	154
7.162	write	154

8 Functions 156

8.1	abs_	157
8.2	bernoulli_	157
8.3	binom_	157
8.4	conjg_	157
8.5	content_	157
8.6	count_	158
8.7	d_	158
8.8	dd_	158
8.9	delta_	158
8.10	deltap_	158
8.11	denom_	158
8.12	distrib_	159
8.13	div_	159
8.14	dum_	159
8.15	dummy_	160
8.16	dummyten_	160
8.17	e_	160
8.18	exp_	160
8.19	exteuclidean_	160

8.20	extrasymbol_	161
8.21	fac_	162
8.22	factorin_	162
8.23	farg_	162
8.24	firstbracket_	162
8.25	firstterm_	162
8.26	g5_	162
8.27	g6_	162
8.28	g7_	163
8.29	g_	163
8.30	gcd_	163
8.31	gi_	163
8.32	id_	163
8.33	integer_	164
8.34	inverse_	164
8.35	invfac_	164
8.36	makerational_	164
8.37	match_	165
8.38	max_	165
8.39	maxpowerof_	165
8.40	min_	165
8.41	minpowerof_	165
8.42	mod_	165
8.43	mod2_	166
8.44	moebius_	166
8.45	mul_	166
8.46	nargs_	166
8.47	node_	166
8.48	nterms_	166
8.49	numfactors_	167
8.50	partitions_	167
8.51	pattern_	168
8.52	perm_	168
8.53	poly_	168
8.54	prime_	168
8.55	putfirst_	169
8.56	random_	169
8.57	ranperm_	170
8.58	rem_	170
8.59	replace_	170
8.60	reverse_	171
8.61	root_	171
8.62	setfun_	171
8.63	sig_	171
8.64	sign_	171
8.65	sizeof_	171
8.66	sum_	171
8.67	sump_	172

8.68	table_	172
8.69	tbl_	172
8.70	term_	172
8.71	termsin_	173
8.72	termsinbracket_	173
8.73	theta_	173
8.74	thetap_	173
8.75	topologies_	173
8.76	Extra reserved names	173
9	Brackets	175
10	Output optimization	180
10.0.1	Optimization options of the Format statement	182
11	Polynomials and Factorization	187
12	The TableBase	199
12.1	addto	201
12.2	apply	202
12.3	audit	202
12.4	create	202
12.5	enter	202
12.6	load	203
12.7	off	203
12.8	on	203
12.9	open	203
12.10	testuse	204
12.11	use	204
13	Dictionaries	205
14	Dirac algebra	213
15	A few notes on the use of a metric	218
16	Sorting and statistics	223
17	The setup	228
18	The parallel version	237
18.1	TFORM	238
18.2	ParFORM	240
18.3	Some problems	241
19	External communication	244
19.1	#external	244
19.2	#toexternal	245
19.3	#fromexternal	245
19.4	#prompt	245

19.5	<code>#setexternal</code>	245
19.6	<code>#rmexternal</code>	246
19.7	<code>#setexternalattr</code>	246
19.8	An example	247
19.9	Embedding FORM in other applications	248
20	Spectators	250
21	Diagram generation	253
22	Floating point	262

Chapter 1

Running FORM

The proper way to invoke the running of FORM depends on the operating system that is being used. Here we will consider the UNIX operating system and its derivatives. The version for computers with the Windows operating system use Cygwin, which is a UNIX derivative as well and hence it functions similarly. In all cases a proper call of FORM is

```
form [options] inputfile
```

The input file should have a name that ends in the extension `.frm`. It is however not needed to specify this extension. If this extension is absent, FORM will add it. Example:

```
form myformprogram
```

and FORM will look for the file `myformprogram.frm`. It is also possible to use the standard input as the input, rather than a file, by giving `-` for the `inputfile` (which is currently not supported by PARFORM, though). The options are separated by blanks and start with a minus sign, followed by one or more alphabetic characters. They are:

- c** Error checking only. Notice that this will not work properly if there are conditionals in the preprocessor phase that depend on results obtained at earlier stages of the program.
- C** Next argument/option is a custom filename for the log file.
- d** Next argument/option is the name of a preprocessor variable that will be defined before the run starts. A specific value can be assigned with the syntax `-d VARIABLENAME=VALUE`. The default value is 1.
- D** Same as `-d`.
- f** Output goes only to log file.
- F** Output only to log file. Further like `-L` or `-ll`.
- h** Wait for some key to be touched before finishing the run. Basically only for some old window based systems.
- I** Next argument/option is the path of a directory for include, procedure and subroutine files.
- l** Make a regular log file.
- ll** Make a log file without intermediate statistics.

- L** Same as **-ll**.
- M** This option is ignored. The PID (process identifier) of the process is now always included in the name of the temporary files.
- p** Next argument/option is the path of a directory for input, include, procedure and subroutine files.
- pipe** Indicates that FORM is started up as the receiving end of a pipe. Action will be taken to set up the proper communication channels.
- q** Quiet option. Only output expressions are printed.
- R** Recover from a crash. See the checkpoint mechanism in 4.1.
- s** Next argument/option is the path of a directory for a setup file.
- si** Same as **-q**.
- S** Next argument/option is the name of a setup file.
- t** Next argument/option is the path of a directory for temporary files.
- ts** Next argument/option is the path of a directory for temporary sort files.
- T** Puts FORM in a mode in which the maximum totalsize is measured and printed at the end of the program. For more information see the "On TotalSize;" statement 7.108.
- v** Only the version will be printed. The program terminates immediately after it.
- w** This should be followed immediately by a number. The number indicates the number of worker threads for TFORM. All other versions of FORM ignore this parameter. It should be noted that TFORM is a different program. For more information, please consult chapter 18.
- W** Turn on the wall-clock time mode in the statistics. See the 'On wtimestats' statement 7.108.
- y** Run only the preprocessor and dump its output.
- z** The number following is a timelimit for the program in second.
- Z** Removes the .str file on crash, whatever its contents. Under ordinary circumstances at a crash a .str file will not be removed if it has a nonzero content.

The log file is a file in which all output is collected, even when the output appears on the screen already. This makes it possible to follow the progress of the program and have a record of everything at the same time. The default name of the log file is identical to the name of the program without the extension `.frm` but with the extra extension `.log`. Example:

```
form -t /LocalDisk/mydir -l myformprogram
```

FORM will run the program in the file `myformprogram.frm`. Its output will both be written to the screen and into the file `myformprogram.log`. The temporary files (if any) will be made in the directory `/LocalDisk/mydir`. This last feature is very useful, because writing temporary files across a network can sometimes slow things down considerably.

The second way to pass parameters to FORM during startup is by means of environment variables, assuming of course that the system supports them. The following variables are supported:

FORMPATH The directory in which FORM will look for procedures and header files, assuming it cannot find them in the current directory.

FORMTMP The directory in which FORM will make its temporary files.

FORMTMPSORT The directory in which FORM will make its temporary sort files.

FORMSETUP The full path and name of a setup file.

It should be noted that when a parameter is specified both in the command tail and in the environment the value of the command tail will be used.

The third way to pass parameters at startup is by means of a setup file. One of the first things FORM does is to locate such a startup file. The procedure that is being followed for this is:

- If the command tail specifies a setup file, FORM will use this file, ignoring all other indications with respect to the setup file. This assumes of course that this file exists. If it does not exist FORM passes on to the next option.
- If the command tail specifies a path for the setup file, FORM will try to open the file "form.set" in this directory. If this cannot be done (by lack of rights or because the file does not exist) FORM passes on to the next option.
- Next FORM tries to open the file "form.set" in the current directory.
- If all else fails, FORM will look for the environment parameter FORMSETUP and use its value as the name of a setup file.

If all the above attempts fail, FORM will not use a setup file. For more information about the setup file one should consult the corresponding chapter on page 17.

Chapter 2

Variables

The objects of symbolic manipulations are expressions. Expressions are built up from terms and terms are composed of variables. FORM knows several types of variables, each of which has special rules assigned to it. The types of variables are symbols, vectors, indices, functions, sets, and expressions. In addition there are tensors and tables which are special functions, preprocessor variables (see chapter 3), and there are dollar variables (see chapter 6). The expressions are used either in the definition of an expression or in the right hand side of an expression or a substitution. When an expression is used in the right hand side of another expression or a substitution, it will be replaced by its contents at the first opportunity. Therefore an expression will never occur as a variable in the output of other expressions and we will ignore their potential presence in the remainder of this chapter. Similarly preprocessor variables and dollar variables will be replaced immediately when they are encountered.

The right hand side of an expression can consist of symbols, vectors, indices, functions and elements of a set. All these objects have to be declared before they can be used. The rules connected to each of these types of variables are described in the sections below.

2.1 Names

There are two types of names. Regular names consist of alphabetic and numeric characters with the condition that the first character must be alphabetic. FORM is case sensitive with respect to names. In addition there are **formal names**. These names start with the character `[` and end with a matching character `]`. In between there can be any characters that are not intercepted by the preprocessor. This allows the use of variables like `[x+a]`. Using formal names can improve the readability of programs very much, while at the same time giving the user the benefits of the greater speed. The use of denominators that are composite (like $1/(x+a)$) is usually rather costly in time. Often $1/[x+a]$ is equally readable, while leading to the same results. Note however that the variable `[x+a]` will have to be declared properly. On the other hand: FORM may not have to know about `x` and `a`. These formal names can also be used for the names of expressions, but they are not valid for the names of dollar variables and the names of preprocessor variables.

Some names may contain special characters. All built in objects have for their last character an underscore (`_`). Dotproducts (the scalar product of two vectors) consist of two vectors separated either by a period or by a dollar sign. The dollar sign is used by FORM, when the output of the program has to be Fortran compatible. The user can replace the dollar sign in the output by an arbitrary character by defining the variable "DotChar" in the setup file. How this is done is explained in chapter 17. In the input the user may apply either the notation with the period

or the notation with the dollar. It is however recommended to use the period because in future versions the notation with the dollar may be dropped. The above conventions avoid the possibility of conflicts with reserved names, allowing the user full freedom when choosing names.

The dollar sign is also used as the first character in the name of dollar variables. The rest of the name should consist of alphanumeric characters of which the first should be alphabetic. The names of preprocessor variables should also consist of alphanumeric characters of which the first should be alphabetic. Also here the ones that are defined by the system have a trailing underscore (_) character.

With respect to the user defined names FORM is case sensitive. This means that the variables `a` and `A` are different objects. With respect to system defined objects FORM is case insensitive. Hence both `d_` and `D_` indicate the same Kronecker delta.

In many languages the use of the underscore (_) character is also permitted in the definition of user defined names. In FORM this is NOT the case. Even though the earlier manuals ‘forbade’ this specifically there was a bug in earlier versions that permitted it to some degree. And because people don’t read manuals, there were those who used this character and even made it into a vital part of their naming conventions. This then broke when version 3 was introduced. It should be clear though that the underscore character is reserved for a completely different type of future use and hence nothing can be done about this. Just remember: it is never a good idea to use undocumented features without consulting with the development team first.

The complex conjugate of a complex quantity is indicated by the character `#` appended to the name of the variable. In the current version of FORM not much is done with it. The latest approach is that it is seen as obsolete. If possible, please avoid using it.

The length of names is not restricted in FORM. There is one exception to this rule: names of expressions cannot be longer than 16 characters. Of course in practise there are physical limits on the size of names, posed by the size of the memory of the computer being used.

2.2 Symbols

Symbols are plain objects that behave most like normal variables in hand manipulations. Many hand manipulations concern polynomial formulae of simple algebraic variables. FORM assumes that symbols commute with all other objects and have a power connected to them. This power is limited to an installation dependent maximum and minimum. A power outside this range will lead to an error message. The user may override this built in restriction by one of private design that is more restrictive. Any power that falls outside the user defined range leads to the removal of the term that contains the variable with this power. Such a power restriction can be defined for each symbol separately.

Symbols can also have complex conjugation properties. A symbol can be declared to be real, imaginary or complex. This property is only relevant, when the complex conjugation operator is used. This operator has not been implemented and currently there are no plans to do so.

The syntax of the statement that defines symbols is given by (see also 7.141):

```
S[ymbols]    name[#{R|I|C}][ (min:max) ] ;
```

Each variable is declared by the presence of its name in a symbol-statement. If the `#` symbol is appended, it should be followed by either the character `C`, `I` or `R` to indicate whether the variable is complex, imaginary or real. The `#R` is not really necessary, as the type ‘real’ is the default. It is not relevant whether the `C`, `I`, `R` are in upper or in lower case. A power restriction is indicated with a range between regular parentheses. If one of the two numbers is not present, the default value is taken. This default value is installation dependent, but it is at least -10000 and 10000 respectively.

Each symbol-statement can define more than one variable. In that case the variables have to be separated either by comma's or by blanks. Example:

```
S      x,y,z,a#c,b#c,c#c,r(-5:5),s(:20),t#i(6:9);
```

In this statement x , y and z are normal real algebraic variables. The variables a , b and c are complex. This means that for each of these variables two entries are reserved in the property lists: one for the variable and one for its complex conjugate. The variable r has a power restriction: Any power outside the specified range will cause the term containing this power to be eliminated. This is particularly useful in power series expansions. The restrictions on s are such that there is no limitation on the minimum power of s —with the exception of the built in restrictions— but a term with a power of s that is larger than 20 is eliminated. The variable t is imaginary. This means that under complex conjugation it changes sign. Its power restrictions are somewhat uncommon. Any power outside the range 6 to 9 is eliminated. There is however one exception: a term that does not contain t to any power (t^0) is not affected.

```
s      x(:10),y;
L      F=y^7;
id     y=x+x^2;
print;
.end
```

Time =	0.01 sec	Generated terms =	4
	F	Terms in output =	4
		Bytes used =	54

```
F =
      x^7 + 7*x^8 + 21*x^9 + 35*x^10;
```

Note that all terms with a power greater than 10 do not even count as generated terms. They are intercepted immediately after the replacement, before any possible additional statements can be carried out.

There are several built in symbols. They are:

i_: it is defined by $i_-^2 = -1$ and this property is used by FORM to simplify terms. It is the only symbol that cannot be used as a dimension or a wildcard.

pi_: a reserved variable which indicates the constant π .

ee_: a reserved variable which indicates the basis of the natural logarithm e .

em_: a reserved variable which indicates the Euler-Mascheroni constant γ_E .

coeff_: this variable is automatically replaced by the coefficient of the current term.

num_: this variable is automatically replaced by the numerator of the coefficient of the current term.

den_: this variable is automatically replaced by the denominator of the coefficient of the current term.

extrasymbols_: this symbol represents the number of extra symbols (see 2.11).

2.3 Vectors

A vector is an object with a single index. This index represents a number that indicates which component of the vector is meant. Vectors have a dimension connected to them which is the

dimension of the vector space in which they are defined. In FORM this dimension is by default set to 4. If the user likes to change this default, this can be done with the ‘Dimension’-statement. The use of this command affects the dimension of all vectors and the default dimension of indices. Its syntax is (see also 7.33):

```
Dimension number;
```

or

```
Dimension symbol;
```

The number must be a number that fits inside a FORM word which is an installation dependent size, but it will be at least 32767. The number must be positive or zero. Negative values are illegal. If a symbol is specified, it must have been declared before. Any symbol may be used with the exception of `i_`.

The declaration of vectors (see 7.160) is rather straightforward:

```
V[ector] name [,MoreNames];
```

The names of the vectors may be separated either by comma’s or by blanks. Example:

```
V    p,q;
I    mu,nu;
L    F=p(mu)*q(nu);
```

2.4 Indices

Indices are objects that represent a number that is used as an integer argument for counting purposes. They are used mostly as the arguments of vectors or multidimensional arrays (or tensors). Their main property is that they have a dimension. This dimension indicates what values the index can take. A four-dimensional index can usually take the values 1 to 4. A very important property of an index is found in the convention that it is assumed that an index that is used twice in the same term is summed over. This is called the Einstein summation convention. Hence the term $p(\mu)q(\mu)$ is equivalent to the scalar product of the vectors p and q (which can also be written as $p.q$).

There are of course also indices that should not be summed over. Such indices we call zero-dimensional. This is just a convention. To declare indices we use the statement (see also 7.74):

```
Index name[={number|symbol}]
        [,othername[={number|symbol}]];
```

When the equals sign is used, this indicates the specification of a dimension. Indices that are not followed by an equals sign get the dimension that is currently the default dimension (see also 7.33)). The dimension can be either a number that is zero or positive (zero indicates that the summation convention does not apply for this index) or it can be any symbol with the exception of the symbol `i_`. The symbol must have been declared before.

The most important use of the dimension of an index is the built in rule that a Kronecker delta with twice the same index is replaced by the dimension of this index, provided this index has a non-zero dimension. Therefore when μ is 4-dimensional, $d_-(\mu, \mu)$ will be replaced by 4 and when ν is n -dimensional, $d_-(\nu, \nu)$ will be replaced by n . If ρ is zero dimensional, the expression $d_-(\rho, \rho)$ is left untouched.

In addition to the symbolic indices there is a number of fixed indices with a numeric value. The values of these indices runs from zero to an installation dependent number (usually 127). Users who like a different maximum value should consult chapter 17 about the setup parameters. The numeric indices are all assumed to have dimension zero, hence no summation is applied to them. This means that they can be used for vector components. It is therefore perfectly legal to use:

```
V    p,q,r;
L    F=p(1)*q(1)*r(1)+p(2)*q(2)*r(2);
```

When two numeric indices occur inside the same Kronecker delta, a value is substituted for this delta. Normally this value is one, when the two indices are identical and zero, when they are different. The value for the diagonal elements can be changed with the ‘FixIndex’-statement (see also 7.59):

```
Fi[xIndex] number:value [,number:value];
```

This command assigns to `d.(number,number)` the given value. This value must fit inside a single FORM word. This means that this value can at least be in the range -32768 to +32767. For more details on the size of a FORM word one should consult the installation manual.

In the case of summable indices the use of three times the same index in the same term would cause problems. FORM will execute the contraction for the first pair it encounters, after which the third index is left. In the case of four or more indices the pairing for the contractions depends on the order in which the parts of the term are processed. Hence to the user the result may seem to be quasi random. Nothing can be done about this and the user should guard against such ambiguous notation.

There is a special version of the index declarations that is used for traces of gamma matrices in n dimensions. If an index is declared with

```
Symbols n,epsilon;
Index m=n:epsilon;
```

its dimension will be n and it is assumed that `epsilon` can be used for $(n - 4)$ during the taking of the trace of a string of gamma matrices. It is also possible to use this notation in the dimension-statement. See also chapter 14 on the gamma matrices.

2.5 Functions

There are two classes of functions: **commuting functions** which commute automatically with all other objects, and **non-commuting functions** which do not necessarily commute with other non-commuting functions. An object is declared to be a commuting function with the ‘cfunction’ command. Of this command the first two characters are mandatory, the others optional. An object is declared to be a non-commuting function with the ‘function’ command. Here only the `f` is mandatory. The declaration of a function knows one option. This option concerns the complexity properties of the function. It is indicated by a `#` following the name, after which one of the characters `R`, `I`, `C` specifies whether the function is real, imaginary or complex. The declaration that a function is real is unnecessary as ‘real’ is the default property. Example:

```
CF    fa,fb,fc;
F     ga,gb,gc#c;
```

In this example the functions fa, fb, fc are commuting and the functions ga, gb and gc are not necessarily commuting. In addition the function gc is complex. More about functions and their conventions is explained in chapter 8.

Within the commutation classes there are several types of special functions. Currently these are tensors and tables. The tables are described in section 7.143 and in chapter 12.

Tensors are special functions. Their arguments can be indices and vectors only. When an argument is a vector, it is assumed that this vector has been put in this position as the result of an Einstein summation, i.e., there used to be an index in this position, but the index was contracted with the index of the vector. Hence FORM assumes that there is a linearity property with respect to such vectors. Tensors are declared with one of the following statements (see also pages 7.145, 7.103, 7.27):

```
T[ensors] t1;
CT[ensors] t2;
NT[ensors] t3;
```

The type 'ntensor' indicates a non-commuting tensor, while the other two types indicate commuting tensors. Note that the 'T' is a commuting tensor, while the 'F' indicates a non-commuting function. In addition to the above declarations one may add the same complexity properties that can be added for functions. This is currently not very useful though as there exists no complex conjugation operator yet. Internally a tensor is a function with special properties. Hence when function properties are discussed, usually these properties refer also to tensors, unless the type of the arguments would not allow the operations or arguments specified.

2.6 Sets

A set is a (non-empty) collection of variables that should all be of the same type. This type can be symbols, vectors, indices or functions. A set has a name which can be used to refer to it, and this name may not coincide with any of the other names in the program. A set is declared by giving its name, followed by a colon, after which the elements of the set are listed. The first element determines the type of all the elements of the set. All elements must have been declared as variables before the set-statement. There can be only one set per statement. Example (see also 7.130):

```
s    xa, xb, xc, xd, ya, x, y;
i    mu, nu, rho;
set exxes: xa, xb, xc, xd;
set yyy: xc, xd, xb, ya;
set indi: mu, nu, rho, 1, 2, 3;
set xandy: xa, ya;
```

We see here that a single symbol (xa) can belong to more than one set. Also the fixed indices (1, 2 and 3) can be elements of a set of indices and the numbers that can be powers can also be members of a set of symbols (usually -9999 to + 9999). If this can cause confusion, FORM will give a warning and interpret the set as a set of symbols.

In addition to the user defined sets there are some built in sets with a special meaning. These are:

int_ This is a set of symbols. It refers to all integer numbers that fit inside a FORM word.

pos_ This is a set of symbols. They are the positive integers that fit inside a FORM word.

pos0_ A set of symbols. They are all non-negative integers that fit inside a FORM word.

neg_ A set of symbols. They are all negative integers that fit inside a FORM word.

neg0_ A set of symbols. They are all non-positive integers that fit inside a FORM word.

symbol_ The set of all formal symbols. It excludes integers, numbers and whole function arguments.

fixed_ The set of all fixed indices.

index_ The set of all indices.

vector_ The set of all (auto)declared vectors.

number_ The set of all rational numbers.

even_ This is a set of symbols. It refers to all even integer numbers that fit inside a FORM word.

odd_ This is a set of symbols. It refers to all odd integer numbers that fit inside a FORM word.

dummyindices_ This is a set of indices. It refers to all indices of the type $Nm_?$ (m a positive integer) that were obtained by summing over indices with a sum statement 7.139.

Sets can be used during wildcarding. When x is a symbol, the notation $x?$ indicates ‘any symbol’. This is sometimes more than we want. In the case that we would like ‘any symbol that belongs to the set `exxes`’ we would write $x?exxes$ which is an unique notation as usually the question mark cannot be followed by a name. There should be no blank between the question mark and the name of the set. The object $x?indi$ would result in a type mismatch error, if x is a symbol and `indi` a set of indices.

This use of wildcards belonging to sets can be extended even more: The notation $x?exxes?yyy$ means that x should belong to the set `exxes`, and its replacement should be the corresponding element of set `yyy`. At first this notation looks unnecessarily complicated. The statement

```
id   x?exxes?yyy = x;
```

should have the much simpler syntax

```
id   exxes = yyy;
```

This last notation cannot be maintained, when the patterns are more complicated, hence it has been omitted altogether.

When things become really complicated, the sets can be used as kind of an array. They can be used with a fixed array index (running from 1 for the first element). When they have a symbolic argument (must be a symbol), they are either in the right hand side of an `id`-statement and the symbol must be replaced by a number by means of a wildcard substitution or in the left hand side and the symbol is automatically seen as a wildcard. The set must still follow the question mark of a wildcard. An example will clarify the above:

```

s a1,a2,a3,b1,b2,b3,x,n;
f g1,g2,g3,g;
set aa:a1,a2,a3;
set bb:b1,b2,b3;
set gg:g1,g2,g3;

id g(x?aa[n]) = gg[n](bb[n]) + bb[2]*n;

```

The `n` in the left hand side is automatically a symbol wildcard. `x` must match an element in `aa` and `n` takes its number. In the right hand side `gg[n]` becomes an array element, when the `n` is substituted. The same holds for `bb[n]`. The element `bb[2]` is immediately replaced by `b2`, so there is rarely profit by using this, unless the preprocessor had something to do with the construction of this quantity. As should be clear from the above: the array elements are indicated with straight braces.

Another use of sets is in the `select` option of the `id`-statement. This is discussed in chapter 5 on pattern matching.

Neither the array properties of the sets nor the `select` option of the `id`-statement can be used in conjunction with the built in sets. These sets are not supposed to have a finite number of indices.

Apart from the above sets that were formally declared and used by name there is a second way to use sets. These sets are called **implicitly declared sets**. They are declared at the position that they are used and their use defines their contents. The elements of the set should be enclosed by a pair of curly brackets and the set is placed at the position where otherwise the name of the set would be used:

```

Symbols a1,a2,a3,b1,b2,b3,x,n;
CFunctions g1,g2,g3,g;
Local expr =
    g(a1)+g(a2)+g(a3)+g(x);
id,g(x?{a1,a2,a3}[n]) = {g1,g2,g3}[n]({b1,b2,b3}[n]);
print;
.end

expr =
    g1(b1) + g2(b2) + g3(b3) + g(x);

```

Such a set exists internally only till the end of the module in which it is used. It can be used at all positions where named sets can be used. Hence they can also be used, when the array properties of sets are considered.

The preprocessor has to be able to distinguish these sets from strings for its calculator (see chapter 3). Usually this is no problem, because any regular name contains at least one character that is not accepted by this calculator. If the only elements in the set are numeric the comma will tell the preprocessor that it is a set and the calculator should not be used. This leaves the case of a set with a single numeric element. By placing a comma either before or after it the use of the calculator is vetoed. For the interpretation of the set this makes no difference.

When it is possible to demand an object to be inside a set, it should also be possible to demand that an object be outside a set. This is done with the `'?!'` operator instead of the `'?'` operator. The extra exclamation mark is like a `'not'` operator. It can be used only, when its use makes sense. Hence it cannot be used in conjunction with the array properties of sets and together with the `select` option of the `id`-statement. So its only use is in patterns of the type

```

x?!setname
x?!{a,b,c}

```

as is done in

```

id  x^n?!{,-1} = x^(n+1)/(n+1);

```

There is a variation of the second type that is not possible with named sets:

```

Symbols a,b,x,y,z;
CFunction f;

id  f(x?!{a,y?,z?})*f(y?!{b,x?,z?})*f(z?!{x?,y?})
    = .....

```

In this complicated pattern the z is easiest: It is not allowed to be equal to the objects that will be substituted for the wildcards x and y . The symbol x cannot be equal to the wildcards y and z , but in addition it should not be equal to a . A similar condition holds for y . One could argue that at least one of these conditions is superfluous from the strictly logical viewpoint. It depends however on the order of the declarations in how FORM runs through the pattern, so it would require some trying to see which ‘not’ specifications are superfluous. If for instance the first function is matched first, there is still no assignment for z . This means that the $z?$ in the set cannot be used yet and hence it places no restrictions on x . Therefore it is the $x?$ in the last function that causes x and z to be different. If on the other hand the last function would be matched first, we need the $z?$ in the set of the first function. From the strict logical viewpoint, FORM could go back over the pattern and still make the appropriate rejections, but this would cost too much extra time. As one can see, it is safer to specify both.

2.7 The autodeclare conventions

As we have seen above, all variables that are introduced by the user have to be declared. As such FORM is a strong typing language. This isn’t always handy. Hence it is possible to introduce some rules about the automatic declaration of classes of variables. This is done with the AutoDeclare statement (see also 7.10). If we use the statements

```

AutoDeclare Symbol x,tt;
AutoDeclare CFunction f,t;

```

any object encountered by the compiler of which the name starts with the character x will automatically be declared as a symbol. Also objects of which the name starts with the characters tt will be declared as symbols. Objects of which the name starts with the characters f or t , but not with the string tt , and that have not yet been declared will be declared automatically as commuting functions. As one can see, in the case of potential conflicts (like with t and tt) the more restrictive one takes precedence. This is independent of the order of the AutoDeclare statements. One disadvantage of the use of the AutoDeclare statement is that one loses a certain amount of control over the order of declaration of the variables, as now they will be declared in the order in which they occur in the statements. The order of the declaration determines the ordering of the objects in the output.

2.8 Name lists

Sometimes it is necessary to see how FORM has interpreted a set of declarations. It can also be that declarations were made in an unlisted include file and that the user wants to know what variables have been defined. The lists of active variables can be printed with the statement

```
On names;
```

This statement sets a flag that causes the listing of all name tables and default properties that are active at the moment that the compiler has finished compiling the current module and all modules after. The printing is just before the algebra processor takes over for the execution of the module – assuming that no error condition exists. If the ‘On names’ is specified in a module that ends with a .global-instruction, the name lists will be printed at the end of each module, as printing the name lists will then be the default option. If one likes to switch this flag off, this can be done with the statement

```
Off names;
```

which prohibits the printing of the name lists in the current module and all modules following.

2.9 Dummy indices

Sometimes indices are to be summed over but due to the evaluation procedures some terms contain the index mu and other terms contain the index nu. There is a command to sum over indices in such a way that FORM recognizes that the exact name of the index is irrelevant. This is the ‘sum’-statement (see also 7.139):

```
i  mu,nu;
f  f1,f2;
L  F=f1(mu)*f2(mu)+f1(nu)*f2(nu);
sum mu;
sum nu;
print;
.end
```

At first the expression contains two terms. After the summations FORM recognizes the terms as identical. In the output we see the term:

```
2*f1(N1_?)*f2(N1_?)
```

The N1_? are dummy indices. The dimension of these dummy indices is the current default dimension as set with the last dimension-statement. This may look like it is a restriction, but in practice it is possible to declare the default dimension to have one value in one module, take some sums, and do some more operations, and then give the default dimension another value in the next module. It should be realized however that then the dimension of the already existing dummy indices may change with it.

The scheme that is used to renumber the indices in a term is quite involved. It will catch nearly all possibilities, but in order to avoid to try all $n!$ permutations, when there are n pairs of dummy indices, FORM does not try everything. It is possible to come up with examples in which the scheme is not perfect. It is left as a challenge for the reader to find such an example. In the case that the scheme isn’t sufficient one can use the Renumber statement (see 7.125) to force a complete

renumbering. As this involves $n!$ attempts in which n is the number of different dummy indices, this can become time consuming.

These dummy indices can be used to solve a well known problem in the automatic summation of indices. This problem occurs, when summed indices are found inside a subexpression that is raised to a power:

```

Index mu,nu;
CFunctions f,g;
Vectors p,q;
Local F = (f(mu)*g(mu))^2;
sum mu;
id f(nu?) = p(nu);
id g(nu?) = q(nu);
print;
.end

```

```

F =
    p.p*q.q;

```

Clearly the answer is not what we had in mind, when we made the program. There is an easy way out:

```

Index mu,nu;
Symbol x;
CFunctions f,g;
Vectors p,q;
Local F = x^2;
repeat;
    id,once,x = f(mu)*g(mu);
    sum mu;
endrepeat;
id f(nu?) = p(nu);
id g(nu?) = q(nu);
print;
.end

```

```

F =
    p.q^2;

```

This time things went better, because each sum-statement moves an index μ to a new dummy index.

There are some extra problems connected to dummy indices. Assume that we have the expression F which contains

$$F = f(N1_?, N2_?) * f(N2_?, N1_?);$$

and next we have the module

```

Indices mu,nu,rho,si;
Vectors p1,p2,p3,v;
Tensor g;

```

```

Local G = e_(mu,nu,rho,si)*g(mu,nu,p1,v)*g(rho,si,p2,v);
sum mu,nu,rho,si;
Multiply F^3;
id v = e_(p1,p2,p3,?);
print;
.end

```

```

G =
f(N1_?,N2_?)*f(N2_?,N1_?)*f(N3_?,N4_?)*f(N4_?,N3_?)*
f(N5_?,N6_?)*f(N6_?,N5_?)*g(N7_?,N8_?,p1,N9_?)*
g(N10_?,N11_?,p2,N12_?)*e_(p1,p2,p3,N9_?)*
e_(p1,p2,p3,N12_?)*e_(N7_?,N8_?,N10_?,N11_?);

```

Here the situation with the dummy indices becomes rather messy, and all earlier versions of FORM were not prepared for this. Their answer could be:

```

G =
f(N1_?,N2_?)*f(N1_?,N2_?)*f(N1_?,N2_?)*f(N2_?,N1_?)*
f(N2_?,N1_?)*f(N2_?,N1_?)*g(N1_?,N2_?,p2,N3_?)*
g(N4_?,N5_?,p1,N6_?)*e_(p1,p2,p3,N3_?)*
e_(p1,p2,p3,N6_?)*e_(N1_?,N2_?,N4_?,N5_?);

```

which is clearly not what the program is supposed to give. In the current version we have made the tracing of the dummy indices and the renumbering of them at the proper moment a lot better. It is however not complete as a complete implementation might severely influence the speed of execution at some points. The scheme is complete for the inclusion of local and global expressions. On the other hand it doesn't work for the contents of dollar variables. Neither does it work for dummy indices introduced in user defined code as in

```
id x^n? = (f(N1_?)*g(N1_?))^n;
```

For the latter case we showed a workaround above. Anyway there is a certain ambiguity here. Just imagine we write

```
id x^n? = f(N1_?)^n*g(N1_?)^n;
```

Formally it is exactly the same, but what we mean is far from clear. For the dollar variables we considered the contracted dummy indices rare enough that it doesn't merit sacrificing speed. And then there is one more little caveat. Global expressions that were stored with older versions of FORM than version 3.2, but are read with version 3.2 or later would have a problem if the expression were to contain dummy indices. The newer version of the .sav files will contain information about the dummy indices. FORM can still read the old versions but will have to 'invent' information by assuming that there are no dummy indices. If there are expressions with such dummy indices the best is to copy the expressions to a new expression and let the copying be followed by a .sort. That should set things straight. A final remark: if an elegant solution is found with which the above cases could be made to work without the penalty in execution time, it will be built in in the future.

2.10 Kronecker delta's

The built in object `d_` represents the Kronecker delta. Even though this object looks a little bit like a tensor, internally it isn't treated as such. Actually it has its own data type. It must have

exactly two arguments and these arguments should be either indices or vectors. A $d_$ with at least one vector is immediately replaced, either by a vector with an index (if there is one vector and one index) or by a dotproduct (when there are two vectors). If a Kronecker delta contains an index that occurs also at another position in the same term, and if that index is summable, and if the index occurs as the index of a vector, inside a tensor, inside another $d_$ or as the argument of a function, and the object inside which it occurs is not inside the argument of a function itself (unless the $d_$ is inside the same argument) then the Einstein summation convention is used and the $d_$ is eliminated, while the second occurrence of the index is replaced by the other index in the $d_$. (Are you still with us?). When a Kronecker delta has two identical indices and these indices are summable, the $d_$ is replaced by the dimension of the index. If they are fixed indices, the $d_$ is replaced by one, unless this value has been altered with the `fixindex`-statement. Some examples of Kronecker delta's are given in section 8.7.

2.11 Extra Symbols

Starting with version 4.0 FORM is equipped with a mechanism to replace non-symbol objects by internally generated symbols. These are called the extra symbols. Their numbering starts at maximum number allowed for internal objects and then counts down. Hence their ordering will be opposite to what might otherwise be expected. It is possible to control their representation when they are to be printed in the output. For this there is the `ExtraSymbols` (7.53) statement. The definitions of the extra symbols can be made visible with the `%X` option in the `#write` preprocessor instruction.

Extra symbols can be introduced by the user with the `ToPolynomial` statement (7.149). This statement replaces all objects that are not numbers or symbols to positive powers by extra symbols. This may be needed for some new manipulations and can also be very handy for output that is to be treated by for instance a FORTRAN or C compiler. The `FromPolynomial` statement replaces the extra symbols again by their original meaning.

```

Vector p,q,p1,p2;
CFunction f;
CFunction Dot,InvDot;
Symbol x,x1,x2;
Set pdot:p,q;
Off Statistics;
Local F = x+x^2+1/x+1/x^2+f(x1)+f(x2)*p.q*x+f(x2)/p.q^2;
id p1?pdot.p2?pdot = Dot(p1,p2);
id 1/p1?pdot.p2?pdot = InvDot(p1,p2);
Print;
.sort

F =
    x^-2 + x^-1 + x + x^2 + f(x1) + f(x2)*Dot(p,q)*x + f(x2)*InvDot(p,q)^2;

ExtraSymbols,array,Y;
Format DOUBLEFORTRAN;
ToPolynomial;
Print;
.sort
```

```

F =
& Y(1) + Y(1)**2 + Y(2) + Y(5)**2*Y(3) + x + x*Y(4)*Y(3) + x**2

#write <sub.f> "      SUBROUTINE sub(Y)"
#write <sub.f> "*"
#write <sub.f> "*"      Compute the extra symbols. Generated on 'DATE_'"
#write <sub.f> "*"
#write <sub.f> "      REAL*8 Y('EXTRASYNBOLS_')"
#write <sub.f> "      REAL*8 Dot,InvDot"
#write <sub.f> "      Dot(p1,p2)=p1(1)*p2(1)-p1(2)*p2(2)-p1(3)*p2(3)\
                                -p1(4)*p2(4)"
#write <sub.f> "      InvDot(p1,p2)=1.D0/(Dot(p1,p2))"
#write <sub.f> "*"
#write <sub.f> "*"      We still have to add definitions here."
#write <sub.f> "*"      And we have to import all the variables."
#write <sub.f> "*"
#write <sub.f> "%X"
#write <sub.f> "*"
#write <sub.f> "      RETURN"
#write <sub.f> "      END"
ExtraSymbols,underscore,Z;
Format Normal;
Format 80;
Print;
.end

F =
Z1_ + Z1_^2 + Z2_ + Z5_^2*Z3_ + x + x*Z4_*Z3_ + x^2;

FromPolynomial;
Print;
.end

F =
x^-2 + x^-1 + x + x^2 + f(x1) + f(x2)*Dot(p,q)*x + f(x2)*InvDot(p,q)^2;

```

In the ExtraSymbols statement we say that we want the extra symbols to be presented as an array with the name Y. The alternative is a set of symbols with names ending in an underscore, but that would not make the FORTRAN compiler very happy. Then we convert the expression to symbols. As one can see, everything got converted to elements of an array Y which are treated as symbols. After we have written the file sub.f (notice that EXTRASYMBOLS_ is a built in symbol indicating the number of extra symbols) we change the representation to the (default) notation with an underscore and the character Z. The contents of the file sub.f are:

```

SUBROUTINE sub(Y)
*
*      Compute the extra symbols. Generated on Sat Apr  2 20:40:33 2011

```

```

*
    REAL*8 Y(5)
    REAL*8 Dot,InvDot
    Dot(p1,p2)=p1(1)*p2(1)-p1(2)*p2(2)-p1(3)*p2(3)-p1(4)*p2(4)
    InvDot(p1,p2)=1.D0/(Dot(p1,p2))
*
*       We still have to add definitions here.
*       And we have to import all the variables.
*
    Y(1)=x**(-1)
    Y(2)=f(x1)
    Y(3)=f(x2)
    Y(4)=Dot(p,q)
    Y(5)=InvDot(p,q)
*
    RETURN
    END

```

As one can see, with very little effort this routine can be made into a proper subroutine that computes all elements of the array Y which can then be used for computing the expression F.

2.12 Restrictions

There is a restriction on the total number of variables that FORM can handle. For the number of symbols, vectors, indices, functions and sets together the exact number depends on the type of computer. For a computer with a 32-bits processor this number is 32768. This includes the built in objects. Individual types of variables (like symbols) are usually restricted to about 8000. For a computer with a 64-bits processor the maximum has been set arbitrarily at 2000000000. In addition there are restrictions on the total amount of memory needed by FORM to maintain an administration of all these variables. These restrictions are set by the memory allocator of the computer on which FORM is running.

2.13 Some common bugs

There is a type of error by the user (including at times the author) that is so common that it deserves mentioning here. Consider the code:

```

Symbol x1,x2
Index m1,m2;

```

As a statement it is perfectly legal, but it may produce rather funny errors at a later stage when we try to use m1 or m2. Inspection with the ‘On names;’ statement shows that we have the symbols x1,x2,Index,m1,m2. This is most likely not what the user wanted. Closer inspection shows that we forgot the semicolon at the end of the symbol statement. We should have had:

```

Symbol x1,x2;
Index m1,m2;

```

This is the most common error for which FORM cannot give a direct error message (it is after all a legal statement). Hence when faced with mysterious errors or error messages, one could have a good look by using the ‘On names’ statement. Maybe it shows something, and if not, one has to look for other causes.

Chapter 3

The preprocessor

The preprocessor is a program segment that reads and edits the input, after which the processed input is offered to the compiler part of FORM. When a module instruction is encountered by the preprocessor, the compilation is halted and the module is executed. The compiler buffers are cleared and FORM will continue with the next module. The preprocessor acts almost purely on character strings. As such it does not know about the algebraic properties of the objects it processes. Additionally the preprocessor also filters out the commentary.

The commands for the preprocessor are called instructions. Preprocessor instructions start with the character `#` as the first non-blank character in a line. After this there are several possibilities.

#: Special syntax for setup parameters at the beginning of the program. See the chapter on the setup parameters.

#-, #+ Turns the listing of the input off or on.

#name Preprocessor command. The syntax of the various commands will be discussed below.

#\$name Giving a value to a dollar variable in the preprocessor. See chapter 6 on dollar variables.

3.1 The preprocessor variables

In order to help in the edit function the preprocessor is equipped with variables that can be defined or redefined by the user or by other preprocessor actions. Preprocessor variables have regular names that are composed of strings of alphanumeric characters of which the first one must be alphabetic. When they are defined one just uses this name. When they are used the name should be enclosed between a backquote and a quote as if these were some type of brackets. Hence `'a2r'` is the reference to a regular preprocessor variable. Preprocessor variables contain strings of characters. No interpretation is given to these strings. The backquote/quote pairs can be nested. Hence `'a'i'r'` will result in the preprocessor variable `'i'` to be substituted first. If this happens to be the string `"2"`, the result after the first substitution would be `'a2r'` and then FORM would look for its string value.

The use of the backquotes is different from the earlier versions of FORM. There the preprocessor variables would be enclosed in a pair of quotes and no nesting was possible. FORM still understands this old notation because it does not lead to ambiguities. The user is however strongly advised to use the new notation with the backquotes, because in future versions the old notation may not be recognized any longer.

FORM has a number of built in preprocessor variables. They are:

VERSION_ The current version as the 5 in 5.0.

SUBVERSION_ The sub-version as the 0 in 5.0.

NAME_ The name of the program file.

DATE_ The date of the current run.

CMODULE_ The number of the current module.

SHOWINPUT_ If input listing is on: 1, if off: 0.

EXTRASYNBOLS_ The current number of extra symbols (see 7.53).

OLDNUMEXTRASYNBOLS_ The number of extra symbols before the current optimization started (see chapter 10).

OPTIMMINVAR_ The number of the first extra symbol needed for the current optimization (see chapter 10).

OPTIMMAXVAR_ The number of the last extra symbol needed for the current optimization (see chapter 10).

OPTIMSCHEME_ The best Horner scheme found for the current optimization (see chapter 10).

OPTIMVALUE_ The number of arithmetic operations in the resulting expression for the current optimization (see chapter 10).

PID_ The process identifier (PID) of the running process. In PARFORM (18.2), it represents the PID of the master process in order to ensure that all the processes in a job use the same number. A recovered session from a checkpoint (4.1) keeps using the PID of the crushed session.

STOPWATCH_ Same as ‘TIMER_’.

SYSTEMERROR_ The return value of the last `#system3.59` command when invoked with the `-e` option.

TIME_ The running time till the moment of call in the string format with a decimal point and two digits after the decimal point. This is the same format as in the statistics.

TIMER_ The running time since the last reset in milliseconds. Hence, unlike ‘time_’ this value can be used in the preprocessor calculator and in numerical compares in `#if` instructions. See also the `#reset` (see 3.48) instruction.

NUMACTIVEEXPR_ The number of the currently active expressions.

ACTIVEEXPRNAMES_ The list of the currently active expression names separated by commas. This can be passed to `#do lvar={...}` instruction (3.19) like:

```
#do e = {'activeexprnames_'}
  #ifdef 'e'
    Local 'e' = 'e' + something;
  #endif
#enddo
```

UNCHANGED_ Takes the value 1 if all active expressions were unchanged in the previous module, and 0 otherwise. 'UNCHANGED_exprname' is 1 if the expression "exprname" was unchanged in the previous module, and 0 otherwise.

ZERO_ Takes the value 1 if all active expressions were zero in the previous module, and 0 otherwise. 'ZERO_exprname' is 1 if the expression "exprname" was zero in the previous module, and 0 otherwise.

If FORM cannot find a preprocessor variable, because it has neither been defined by the user, nor is it one of the built in variables, it will look in the systems environment to see whether there is an environment variable by that name. If this is the case its string value will be substituted.

Preprocessor variables can have arguments and thereby become macro's. One should consult the description of the #define 3.18 instruction about the delayed substitution feature to avoid the value of the preprocessor variables in the macro would be substituted immediately during the definition. Hence proper use is

```
#define EXCHANGE(x,y) "Multiply replace_('~x','~y','~y','~x');"
```

FORM has the following built in macro's:

TOLOWER_(string) in which the character string in the argument is converted to lower case. After this it will become input.

TOUPPER_(string) in which the character string in the argument is converted to upper case. After this it will become input.

It is anticipated that some more macro's will become available to allow for the editing of names of variables.

3.2 The preprocessor calculator

Sometimes a preprocessor variable should be interpreted as a number and some arithmetic should be done with it. For this FORM is equipped with what is called the preprocessor calculator. When the input reading device encounters a left curly bracket {, it will read till the matching right curly bracket } and then test whether the characters (after substitution of preprocessor variables) can be interpreted as a numerical expression. If it is not a valid numerical expression the whole string, including the curly brackets, will be passed on to the later stages of the program. If it is a numerical expression, it will be evaluated, and the whole string, including the curly brackets, will be replaced by a textual representation of the result. Example:

```
Local F'i' = F{'i'-1}+F{'i'-2};
```

If the preprocessor variable i has the value 11, the calculator makes this into

```
Local F11 = F10+F9;
```

Valid numerical expressions can contain the characters

```
0 1 2 3 4 5 6 7 8 9 + - * / % ( ) { } & | ^ !
```

The use of parentheses is as in regular arithmetic. The curly brackets fulfil the same role, as one can nest these brackets of course. Operators are:

+ Regular addition.

− Regular subtraction.

* Regular multiplication.

/ Regular (integer) division.

% The remainder after (integer) division as in the language C.

& And operator. This is a bitwise operator.

| Or operator. This is a bitwise or.

^ Exponent operator.

! Factorial. This is a postfix operator.

^% A postfix ²log. This means that it takes the ²log of the object to the left of it.

^/ A postfix square root. This means that it takes the square root of the object to the left of it.

Note that all arithmetic is done over the integers and that there is a finite range. On 32 bit systems this range will be $2^{31} - 1$ to -2^{31} , while on 64 bit systems this will be $2^{63} - 1$ to -2^{63} . In particular this means that {13^/} becomes 3. The preprocessor calculator is only meant for some simple counting and organization of the program flow. Hence there is no large degree of sophistication. Very important is that the comma character is not a legal character for the preprocessor calculator. This can be used to avoid some problems. Suppose one needs to make a substitution of the type:

```
id f(x?!{0}) = 1/x;
```

in which the value zero should be excluded from the pattern matching (see dynamical sets in chapter 5 on pattern matching). This would not work, because the preprocessor would make this into

```
id f(x?!0) = 1/x;
```

which is illegal syntax. Hence the proper trick is to write

```
id f(x?!{,0}) = 1/x;
```

With the comma the preprocessor will leave this untouched, and hence now the set is passed properly.

Good use of the preprocessor calculator can make life much easier for FORM. For example the following statements

```
id f('i') = 1/('i'+1);  
id f('i') = 1/{'i'+1};
```

are quite different in nature. In the first statement the compiler gets an expression with a composite denominator. The compiler never tries to simplify expressions by doing algebra on them. Sometimes this may not be optimal, but there are cases in which it would cause wrong results (in particular when noncommuting and commuting functions are mixed and wildcards are used). Hence the composite denominator has to be worked out during run time for each term separately. The second statement has the preprocessor work out the sum and hence the compiler gets a simple fraction and less time will be needed during running. Note that

```
id f('i') = {1/('i'+1)};
```

would most likely not produce the desired result, because the preprocessor calculator works only over the integers. Hence, unless *i* is equal to zero or -2, the result would be zero (excluding of course the fatal error when *i* is equal to -1).

3.3 The triple dot operator

The last stage of the actions of the preprocessor involves the triple dot operator. It indicates a repeated pattern as in `a1+...+a4` which would expand into `a1+a2+a3+a4`. This operator is used in two different ways. First the most general way:

```
<pattern1>operator1...operator2<pattern2>
```

in which the less than and greater than signs serve as boundaries for the patterns. The operators can be any pair of the following:

+ + Repetitions will be separated by plus signs.

- - Repetitions will be separated by minus signs.

+ - Repetitions will be separated by alternating signs. First will be plus.

- + Repetitions will be separated by alternating signs. First will be minus.

* * Repetitions will be separated by *.

/ / Repetitions will be separated by /.

, , Repetitions will be separated by comma's.

: : Repetitions will be separated by *single* dots.

For such a pair of operators FORM will inspect the patterns and see whether the differences between the two patterns are just numbers. If the differences are numbers and the absolute value of the difference of each matching pair is always the same (a difference of zero is allowed too; it leads to no action for the pair), then FORM will expand the pattern, running from the first to the last in increments of one. For each pair the counter can either run up or run down, depending on whether the number in the first pattern is greater or less than the number in the second pattern. Example:

```
Local F = <a1b6(c3)>-...+<a4b3(c6)>;
```

leads to

```
Local F = a1b6(c3)-a2b5(c4)+a3b4(c5)-a4b3(c6);
```

The second form is a bit simpler. It recognizes that there are special cases that can be written in a more intuitive way. If there is only a single number to be varied, and it is the end of the pattern, and the rest of the patterns consists only of alphanumeric characters of which the first is an alphabetic character, we do not need the less than/greater than combination. This is shown in

```
Symbol a1,...,a12;
```

There is one extra exception. The variables used this way may have a question mark after them to indicate that they are wildcards:

```
id f(a1?,...,a4?) = g(a1,...,a4,a1+...+a4);
```

This construction did not exist in earlier versions of FORM (version 1 and version 2). There one needed the `#do` instruction for many of the above constructions, creating code that was very hard to read. The `...` operator should improve the readability of the programs very much.

3.4 `#add`

Syntax:

```
#add object: "string"
```

See chapter 13 on dictionaries.

Adds words to an open dictionary.

3.5 `#addseparator`

Syntax:

```
#addseparator character
```

See also `#rmseparator` (3.51), `#call` (3.10), `#do` (3.19)

Adds a character to the list of permissible separator characters for arguments of `#call` or `#do` instructions. By default the two characters that are permitted are the comma and the character `|`. Blanks, tabs and double quotes are ignored. Note that the comma must be specified between double quotes as in

```
#addseparator ",",
```

3.6 `#append`

Syntax:

```
#append <filename>
```

See also `write` (3.65), `close` (3.13), `create` (3.16), `remove` (3.47)

Opens the named file for writing. The file will be positioned at the end. The next `#write` instruction will add to it.

3.7 `#appendpath`

Syntax:

```
#appendpath pathname
```

See also `prependpath` (3.41)

Appends the given path relative to the current file to the end of the FORM path.

3.8 `#break`

Syntax:

```
#break
```

See also `switch` (3.58), `endswitch` (3.26), `case` (3.11), `default` (3.17)

If the lines before were not part of the control flow (*i.e.* these lines are used for the later stages of the program), this instruction is ignored. If they are part of the control flow, the flow will continue after the matching `#endswitch` instruction. The `#break` instruction must of course be inside the range of a `#switch/#endswitch` construction.

3.9 `#breakdo`

Syntax:

```
#breakdo [<number>]
```

See also `#do` (3.19) and `#enddo` (3.22)

The `#breakdo` instruction allows one to jump out of a `#do` loop. If a (nonzero integer) number is specified it indicates the number of loops the program should terminate. Control will continue after the `#enddo` instruction of the number of loops indicated by ‘number’. The default value is one. If the value is zero the statement has no effect.

3.10 `#call`

Syntax:

```
#call procname(var1,...,varn)
```

See also `procedure` (3.43), `endprocedure` (3.25)

This instruction calls the procedure with the name `procname`. The result is that FORM looks for this procedure, first in its procedure buffers (for procedures that were defined in the regular text stream as explained under the `#procedure` instruction), then it looks for a file by the name `procname.prc` in the current directory, and if it still has not found the procedure, it looks in the directories indicated by the path variable in either the setup file or at the start of the program (see chapter 17 on the setup file). Next it looks for the `-p` option in the command that started FORM (see the chapter on running FORM). If this `-p` option has not been used FORM will see whether there is an environment variable by the name `FORMPATH`. The directories indicated there will be searched for the file `procname.prc`. If FORM cannot find the file, there will be an error message and execution will be stopped immediately.

Once the procedure has been located, FORM reads the whole file and then determines whether the number of parameters is identical in the `#call` instruction and the `#procedure` instruction. A difference is a fatal error.

The parameter field consists of strings, separated by commas. If a string contains a comma, this comma should be preceded by a backslash character (`\`). If a string should contain a linefeed, one should ‘escape’ this linefeed by putting a backslash and continue on the next line.

Before version 3 of FORM the syntax was different. The parentheses were curly brackets and the separators the symbol `|`. This was made to facilitate the use of strings that might contain commas. In practise however, this turned out to be far from handy. In addition the new preprocessor calculator is a bit more active and hence an instruction of the type

```
#call test{1}
```

will now be intercepted by the preprocessor calculator and changed into

```
#call test1
```

Because there are many advantages to the preprocessor calculator treating the parameters of the procedures before they are called (in the older versions it did not do this), the notation has been changed. FORM still understands the old notation, provided that there is no conflict with the preprocessor calculator. Hence

```
#call test{1|a}
#call test{1,a}
#call test(1|a)
#call test(1,a)
```

are all legal and give the same result, but only the last notation will work in future versions of FORM.

Nowadays also the use of the argument field wildcard (see chapter 5 on pattern matching) is allowed as in the regular functions:

```
#define a "1"
#define bc2 "x"
#define bc3 "y"
#define b "c'~a'"
#procedure hop(c,?d);
#redefine a "3"
#message This is the call: 'c','?d'
#endprocedure

#redefine a "2"
#message This is b: 'b'
~~~This is b: c2

#call hop('b'!b''!b'!b'!b','~a','b','a')
~~~This is the call: xc2c3c2c3,3,c3,2

.end
```

We also see here that the rules about delayed substitution (see also the `#define` instruction in section 3.18) apply. The use of `!b` cancels the delayed substitution that is asked for in the definition of `b`.

The default extension for procedure files is `.prc`, but it is possible to change this. There are two different ways: One is with the `#procedureExtension` instruction in section 3.44. The other is via the `setup` (see the chapter on the `setup` file, chapter 17).

3.11 `#case`

Syntax:

```
#case string
```

See also `switch` (3.58), `endswitch` (3.26), `break` (3.8), `default` (3.17)

The lines after the `#case` instruction will be used if either this is the first `#case` instruction of which the string matches the string in the `#switch` instruction, or the control flow was already using the lines before this `#case` instruction and there was no `#break` instruction (this is called fall-through). The control flow will include lines either until the next matching `#break` instruction, or until the matching `#endswitch` instruction.

3.12 `#clearoptimize`

Syntax:

`#clearoptimize`

See the chapter about optimization 10

3.13 `#close`

Syntax:

`#close <filename>`

See also `write` (3.65), `append` (3.6), `create` (3.16), `remove` (3.47)

This instruction closes the file by the given name, if such a file had been opened by the previous `#write` instruction. Normally FORM closes all such files at the end of execution. Hence the user would not have to worry about this. The use of a subsequent `#write` instruction with the same file name will remove the old contents and hence start basically a new file. There are times that this is useful.

3.14 `#closedictionary`

Syntax:

`#closedictionary`

See chapter 13 on dictionaries.

Either closes an open dictionary (3.37) or stops using the dictionary (3.64) that is currently used for output translation.

3.15 `#commentchar`

Syntax:

`#commentchar character`

The specified character should be a single non-whitespace character. There may be white space (blanks and/or tabs) before or after it. The character will take over the role of the comment character. *i.e.* any line that starts with this character in column 1 will be considered commentary. This feature was provided because output of some other algebra programs could put the multiplication sign in column 1 in longer expressions.

The default commentary character is `*`.

3.16 `#create`

Syntax:

`#append <filename>`

See also `write` (3.65), `close` (3.13), `append` (3.6), `remove` (3.47)

Opens the named file for writing. If the file existed already, its previous contents will be lost. The next `#write` instruction will add to it. In principle this instruction is not needed, because the `#write` instruction would create the file if it had not been opened yet at the moment of writing.

3.17 #default

Syntax:

```
#default
```

See also switch (3.58), endswitch (3.26), case (3.11), break (3.8)

Control flow continues after this instruction if there is no #case instruction of which the string matches the string in the #switch instruction. Control flow also continues after this instruction, if the lines before were included and there was no #break instruction to stop the control flow (fall-through). Control flow will stop either when a matching #break instruction is reached, or when a matching #endswitch is encountered. In the last case of course control flow will continue after the #endswitch instruction.

3.18 #define

Syntax:

```
#define name "string"
```

See also redefine (3.46), undefine (3.63)

in which name refers to the name of the preprocessor variable to be defined and the contents of the string will form the value of the variable. The double quotes are mandatory delimiters of the string.

The use of the #define instruction creates a new instance of the preprocessor variable with the given name. This means that the old instance remains. If for some reason the later instance becomes undefined (see for instance #undefine), the older instance will be the one that is active. If the old definition is to be overwritten, one should use the #redefine instruction.

As of version 3.2 preprocessor variables can also have arguments as in the C language. Hence

```
#define var(a,b) "('~a'+ '~b'+ '~c')"
```

is allowed. The parameters should be referred to inside a pair of " as with all preprocessor variables. A special feature is the so-called delayed substitution. With macro's like the above the question is always *when* a preprocessor variable will be substituted. Take for instance

```
#define c "3"
#define var1(a,b) "('~a'+ '~b'+ '~c')"
#define var2(a,b) "('~a'+ '~b'+ '~c')"
#define c "4"
Local F1 = 'var1(1,2)';
Local F2 = 'var2(1,2)';
Print;
.end
```

```
F1 =
    6;
```

```
F2 =
    7;
```

The parameter c will be substituted immediately when var1 is defined. In var2 it will be only substituted when var2 is used. It should be clear that a and b should also be used in the delayed fashion because they do not exist yet at the moment of the definition of var1 and var2. Notice also

that the whole macro, with its arguments should be placed between the backquote and the quote. Another example can be found with the `#call` instruction. See section 3.10

3.19 `#do`

Syntax:

```
#do lvar = i1,i2
#do lvar = i1,i2,i3
#do lvar = {string1|...|stringn}
#do lvar = {string1,...,stringn}
#do lvar = nameofexpression
```

See also `enddo` (3.22)

The `#do` instruction needs a matching `#enddo` instruction. All code in-between these two instructions will be read as many times as indicated in the parameter field of the `#do` instruction. The parameter `lvar` is a preprocessor variable of which the value is determined by the other parameters. Inside the loop it should be referred to by enclosing its name between a backquote/quote pair as is usual for preprocessor variables. The various possible parameter fields have the following meaning:

`#do lvar = i1,i2` The parameters `i1` and `i2` should be integers or names of dollar expressions that evaluate into integers. The first time in the loop `lvar` will get the value of `i1` (as a string) and each next time its value will be one greater (translated into a string again). The last time in the loop the value of `lvar` will be the greatest integer that is less or equal to `i2`. If `i2` is less than `i1`, the loop is skipped completely. If `i2` is the name of a dollar variable, each time the control reaches the end of the loop the dollar variable is evaluated and the current value is used.

`#do lvar = i1,i2,i3` The parameters `i1,i2` and `i3` should be integers or names of dollar expressions that evaluate into integers. The first time in the loop `lvar` will get the value of `i1` (as a string) and each next time its value will be incremented by adding `i3` (translated into a string again). If `i3` is positive, the last value of `lvar` will be the one for which `lvar+i3` is greater than `i2`. If `i2` is less than `i1`, the loop is skipped completely. If `i3` is negative the last value of `lvar` will be the one for which `lvar+i3` is less than `i2`. If `i3` is zero there will be an error. If `i2` or `i3` are the names of a dollar variable, each time the control reaches the end of the loop the dollar variable(s) is/are evaluated and the current value is used.

`#do lvar = {string1|...|stringn}` The first time in the loop the value of `lvar` is the string indicated by `string1`, the next time will be `string2` etc till the last time when it will be `stringn`. This is called a listed loop. The notation with the `|` is an old notation which is still accepted. The new notation uses a comma instead.

`#do lvar = {string1,...,stringn}` The first time in the loop the value of `lvar` is the string indicated by `string1`, the next time will be `string2` etc till the last time when it will be `stringn`. This is called a listed loop.

`#do lvar = expression` The loop variable will take one by one for its value all the terms of the given expression. This is protected against changing the expression inside the loop by making a copy of the expression inside the memory. Hence one should be careful with very big expressions. An expression that is zero gives a loop over zero terms, hence the loop is never executed.

The first two types of `#do` instructions are called numerical loops. In the parameters of numerical loops the preprocessor calculator is invoked automatically. One should make sure not to use a leading `{` for the first numerical parameter in such a loop. This would be interpreted as belonging to a listed loop.

After a loop has been finished, the corresponding preprocessor variable will be undefined. This means that if there is a previous preprocessor variable by the same name, the value of the `#do` instruction will be used inside the loop, and afterwards the old value will be active again.

It is allowed to overwrite the value of a preprocessor `#do` instruction variable. This can be very useful to create the equivalent of a repeat loop that contains `.sort` instructions as in

```
#do i = 1,1
  id,once,x = y+2;
  if ( count(x,1) > 0 ) redefine i "0";
  .sort
#enddo
```

A few remarks are necessary here. The `redefine` statement (see section 7.123) should be before the last `.sort` inside the loop, because the `#do` instruction is part of the preprocessor. Hence the value of `i` is considered before the module is executed. This means that if the `redefine` would be after the `.sort`, two things would go wrong: First the loop would be terminated before the `redefine` would ever make a chance of being executed. Second the statement would be compiled in the expectation that there is a variable `i`, but then the loop would be terminated. Afterwards, when the statement is being executed it would refer to a variable that does not exist any longer.

If one wants to make a loop over the externals of the brackets of an expression only, one needs to do some work. Assume we have the expression `F` and we want to loop over the brackets in `x` and `y`:

```
L   FF = F;
Bracket x,y;
.sort
CF acc,acc2;
Skip F;
Collect acc,acc2;
id  acc(x?) = 1;
id  acc2(x?)= 1;
B   x,y;
.sort
Skip F;
Collect acc;
id  acc(x?) = 1;
.sort
#do i = FF
L   G = F['i'];
.
.
#enddo
```

Notice that we have to do the collect trick twice because the first time the bracket could be too long for one term. The second time that restriction doesn't exist because besides the `x` and the `y` there are only integer coefficients.

3.20 `#else`

Syntax:

`#else`

See also `if` (3.31), `endif` (3.23), `elseif` (3.21), `ifdef` (3.32), `ifndef` (3.33)

This instruction is used inside a `#if/#endif` construction. The code that follows it until the `#endif` instruction will be read if the condition of the `#if` instruction (and of none of the corresponding `#elseif` instructions) is not true. If any of these conditions is true, this code is skipped. The reading is stopped after the matching `#endif` is encountered and continued after this matching `#endif` instruction.

3.21 `#elseif`

Syntax:

`#elseif ( condition )`

See also `if` (3.31), `endif` (3.23), `else` (3.20)

The syntax of the condition is identical to the syntax for the condition in the `#if` instruction. The `#elseif` instruction can occur between an `#if` and an `#endif` instruction, before a possible matching `#else` instruction. The code after this condition till the next `#elseif` instruction, or till a `#else` instruction or till a `#endif` instruction, whatever comes first, will be read if the condition in the `#elseif` instruction is true and none of the conditions in matching previous `#if` or `#elseif` instructions were true. The reading is stopped after the matching `#elseif/#else/#endif` is encountered and continued after the matching `#endif` instruction.

Example

```
#if ( 'i' == 2 )
    some code
#elif ( 'i' == 3 )
    more code
#elif ( 'j' >= "x2y" )
    more code
#else
    more code
#endif
```

3.22 `#enddo`

Syntax:

`#enddo`

See also `do` (3.19)

Used to terminate a preprocessor `do` loop. See the `#do` instruction.

3.23 `#endif`

Syntax:

`#endif`

See also `if` (3.31), `else` (3.20), `elseif` (3.21), `ifdef` (3.32), `ifndef` (3.33)

Used to terminate a `#if`, `#ifdef` or `#ifndef` construction. Reading will continue after it.

3.24 `#endinside`

Syntax:

`#endinside`

See also `#inside` (3.35)

Used to terminate a `#inside` construction in the preprocessor. For more details, see the `#inside` instruction.

3.25 `#endprocedure`

Syntax:

`#endprocedure`

See also `procedure` (3.43), `call` (3.10)

Each procedure must be terminated by an `#endprocedure` instruction. If the procedure resides in its own file, the `#endprocedure` will cause the closing of the file. Hence any text that is in the file after the `#endprocedure` instruction will be ignored.

When control reaches the `#endprocedure` instruction, all (local) preprocessor variables that were defined inside the procedure and all parameters of the call of the procedure will become undefined.

3.26 `#endswitch`

Syntax:

`#endswitch`

See also `switch` (3.58), `case` (3.11), `break` (3.8), `default` (3.17)

This instruction marks the end of a `#switch` construction. After none or one of the cases of the `#switch` construction has been included in the control flow, reading will continue after the matching `#endswitch` instruction. Each `#switch` needs a `#endswitch`, unless a `.end` instruction is encountered first.

3.27 `#exchange`

Syntax:

`#exchange expr1,expr2`

`#exchange $var1,$var2`

Exchanges the names of two expressions. This means that the contents of the expressions remain where they are. Hence the order in which the expressions are processed remains the same, but the name under which one has to refer to them has been changed.

In the variety with the dollar variables the contents of the variables are exchanged. This is not much work, because dollar variables reside in memory and hence only two pointers to the contents have to be exchanged (and some extra information about the contents).

This instruction can be very useful when sorting expressions or dollar variables by their contents.

3.28 `#external`

Syntax:

`#external ["prevar"] systemcommand`

Starts the command in the background, connecting to its standard input and output. By default, the `#external` command has no controlling terminal, the standard error stream is redirected to `/dev/null` and the command is run in a subshell in a new session and in a new process group (see the preprocessor instruction `#setexternalattr`).

The optional parameter “prevar” is the name of a preprocessor variable placed between double quotes. If it is present, the “descriptor” (small positive integer number) of the external command is stored into this variable and can be used for references to this external command (if there is more than one external command running simultaneously).

The external command that is started last becomes the “current” (active) external command. All further instructions `#fromexternal` and `#toexternal` deal with the current external command.

3.29 `#factdollar`

Syntax:

`#factdollar $-variable`

See also the chapters on polynomials 11 and `$`-variables 6

The `#factdollar` instruction causes the factorization of the indicated `$`-variable. After this instruction and until the `$`-variable is redefined there will be two versions of the variable: one is the original unfactorized version and the other is a list of factors. If the name of the variable is `$a` the factors can be accessed as `$a[1], ..., $a[n]`. The total number of factors is given by `$a[0]`. These factors can also be treated as preprocessor variables by putting them between quotes as in ‘`$a[2]`’.

3.30 `#fromexternal`

Syntax:

`#fromexternal[+-] ["$]varname" [maxlength]`

Appends the output of the current external command to the FORM program. The semantics differ depending on the optional arguments. After the external command sends the prompt, FORM will continue with a next line after the line containing the `#fromexternal` instruction. The prompt string is not appended. The optional `+` or `-` sign after the name has influence on the listing of the content. The varieties are:

`#fromexternal[+-]`

The semantics is similar to the `#include` instruction but folders are not supported.

`#fromexternal[+-] ["$]varname"`

is used to read the text from the running external command into the preprocessor variable `varname`, or into the dollar variable `$varname` if the name of the variable starts with the dollar sign “`$`”.

`#fromexternal[+-] ["$]varname" maxlength`

is used to read the text from the running external command into the preprocessor (or dollar) variable `varname`. Only the first `maxlength` characters are stored.

3.31 `#if`

Syntax:

`#if ( condition )`

See also `endif` (3.23), `else` (3.20), `elseif` (3.21), `ifdef` (3.32), `ifndef` (3.33)

The `#if` instruction should be accompanied by a matching `#endif` instruction. In addition there can be between the `#if` and the `#endif` some `#elseif` instructions and/or a single `#else` instruction.

The condition is a logical variable that is true if its value is not equal to zero, and false if its value is zero. Hence it is allowed to use

```
#if 'i'
    statements
#endif
```

provided that `i` has a value which can be interpreted as a number. If there is just a string that cannot be seen as a logical condition or a number it will be interpreted as false. The regular syntax of the simple condition is

```
#if 'i' == st2x
    statements
#endif
```

or

```
#if ( 'i' == st2x )
    statements
#endif
```

in which the compare is a numerical compare if both strings can be seen as numbers, while it will be a string compare if at least one of the two cannot be seen as a numerical object. One can also use more complicated conditions as in

```
#if ( ( 'i' > 5 ) && ( 'j' > 'i' ) )
```

These are referred to as composite conditions. The possible operators are

> Greater than, either in numerical or in lexicographical sense.

< Less than, either in numerical or in lexicographical sense.

>= Greater than or equal to, either in numerical or in lexicographical sense.

<= Less than or equal to, either in numerical or in lexicographical sense.

== **or** = Equal to.

!= Not equal to.

&& Logical and operator to combine conditions.

|| Logical or operator to combine conditions.

If the condition evaluates to true, the lines after the `#if` instruction will be read until the first matching `#elseif` instruction, or a `#else` instruction or a `#endif` instruction, whatever comes first. After such an instruction is encountered input reading stops and continues after the matching `#endif` instruction.

Like with the regular if-statement (see 7.71), there are some special functions that allow the asking of questions about objects. These are

<code>exists()</code>	The argument of <code>exists</code> is the name of an expression or a <code>\$</code> -variable. This function then returns one if this object exists, cq. has been defined. Otherwise it returns zero.
-----------------------	---

<code>isdefined()</code>	The argument of <code>isdefined</code> is the name of a preprocessor variable. This function then returns one if this object has been defined. Otherwise it returns zero. Technically <code>#ifdef 'VAR'</code> and <code>#if (isdefined(VAR))</code> are the same. The <code>isdefined</code> function allows for greater flexibility in composite conditions.
<code>isfactorized()</code>	The argument of <code>isfactorized</code> is the name of an expression or a <code>\$</code> -variable. This function then returns one if the object has been factorized. Otherwise it returns zero.
<code>isnumerical()</code>	The argument of <code>isnumerical</code> is the name of an expression or a <code>\$</code> -variable. This function then returns one if the object contains a single term that is purely numerical in nature. Otherwise it returns zero.
<code>maxpowerof()</code>	The argument of <code>maxpowerof</code> is the name of a symbol. This function then evaluates into the maximum power of that symbol as it has been declared. If no maximum power has been set in the declaration of the symbol, the general maximum power for symbols is returned (see 7.141).
<code>minpowerof()</code>	The argument of <code>minpowerof</code> is the name of a symbol. This function then evaluates into the minimum power of that symbol as it has been declared. If no minimum power has been set in the declaration of the symbol, the general minimum power for symbols is returned (see 7.141).
<code>sizeof()</code>	The argument of <code>termsin</code> is the name of an expression or a <code>\$</code> -variable. This function then evaluates into the number of FORM words in that expression or variable.
<code>termsin()</code>	The argument of <code>termsin</code> is the name of an expression or a <code>\$</code> -variable. This function then evaluates into the number of terms in that expression.

3.32 `#ifdef`

Syntax:

`#ifdef 'prevar'`

See also `if` (3.31), `endif` (3.23), `else` (3.20), `ifndef` (3.33)

If the named preprocessor variable has been defined the condition is true, else it is false. For the rest the instruction behaves like the `#if` instruction.

An alternative is to use the `isdefined` object inside the `#if` instruction.

3.33 `#ifndef`

Syntax:

`#ifndef 'prevar'`

See also `if` (3.31), `endif` (3.23), `else` (3.20), `ifdef` (3.32)

If the named preprocessor variable has been defined the condition is false, else it is true. For the rest the instruction behaves like the `#if` instruction.

3.34 `#include`

Syntax:

`#include[ -+ ] filename`

`#include[ -+ ] filename # foldname`

The named file is searched for and opened. Reading continues from this file until its end. Then the file will be closed and reading continues after the `#include` instruction. If a foldname is specified, FORM will only read the contents of the first fold it encounters in the given file that has the specified name.

The file is searched for in the current directory, then in the path specified in the path variable in the setup file or at the beginning of the program (see chapter 17 on the setup file). Next it will look in the path specified in the `-p` option when FORM is started (see the chapter on running FORM). If this option has not been used, FORM will look for the environment variable FORMPATH. If this variable exists it will be interpreted as a path and FORM will search the indicated directories for the given file. If none is found there will be an error message and execution will be halted.

The optional `+` or `-` sign after the name has influence on the listing of the contents of the file. A `-` sign will have the effect of a `#-` instruction during the reading of the file. A plus sign will have the effect of a `#+` instruction during the reading of the file.

A fold is defined by a starting line of the format:

```
*--#[ name :
```

and a closing line of the format

```
*--#] name :
```

in which the first character is actually the current commentary character (see the `#commentchar` instruction). All lines between two such lines are considered to be the contents of the fold. If FORM decides that it needs this fold, it will read these contents and put them in its input stream. More about folds is explained in the manual of the STedi editor which is also provided in the FORM distribution.

3.35 `#inside`

Syntax:

```
#inside $var1 [more $variables]
```

See also `#endinside` (3.24)

Used to execute a few statements on the contents of one or more dollar variables (see 6) during compilation time. Although this is a preprocessor instruction one can use the triple dot operator provided one uses the generic version with the `<>`.

The statements in the scope of the `#inside` / `#endinside` construction must be regular executable statements. They may not contain end-of-module instructions like the `.sort` instruction. It is allowed to use dollar variables, procedures and preprocessor `do` loops and `if`'s, but it is not allowed to nest the `#inside` / `#endinside` constructions.

3.36 `#message`

Syntax:

```
#message themessagestring
```

This instruction places a message in the output that is clearly marked as such. It is printed with an initial three characters in front as in

```
Symbols a,b,c;  
#message Simple example;
```

```

~~~Simple example;
  Local F = (a+b+c)^10;
  .end

```

```

Time =          0.00 sec    Generated terms =          66
          F          Terms in output =          66
                          Bytes used      =          1138

```

Note that the semicolon is not needed and if present is printed as well. If one needs messages without this clear marking, one should use the `#write` instruction.

3.37 `#opendictionary`

Syntax:

```
#opendictionary name
```

See chapter 13 on dictionaries.

Opens a dictionary and makes it ready for adding words to it. If the dictionary does not exist yet, it will be created.

3.38 `#optimize`

Syntax:

```
#optimize nameofoneexpression
```

See the chapter about optimization 10

3.39 `#pipe`

Syntax:

```
#pipe systemcommand
```

See also `system` (3.59)

This forces a system command to be executed by the operating system. The complete string (excluding initial blanks or tabs) is passed to the operating system. Next FORM will intercept the output of whatever is produced and read that as input. Hence, whenever output is produced FORM will take action, and it will wait when no output is ready. After the command has been finished, FORM will continue with the next line. This instruction has only been implemented on systems that support pipes. This is mainly UNIX and derived systems. Note that this instruction also introduces operating system dependent code. Hence it should be used with great care.

3.40 `#preout`

Syntax:

```
#preout ON
```

```
#preout OFF
```

Turns listing of the output of the preprocessor to the compiler on or off. Example:

```

#PreOut ON
S  a1,...,a4;

```

```

S,a1,a2,a3,a4
  L   F = (a1+...+a4)^2;
L,F=(a1+a2+a3+a4)^2
  id  a4 = -a1;
id,a4=-a1
  .end

```

Time =	0.00 sec	Generated terms =	10
	F	Terms in output =	3
		Bytes used =	52

3.41 #prependpath

Syntax:

```
#prependpath pathname
```

See also `appendpath` (3.7)

Prepends the given path relative to the current file to the beginning of the FORM path.

3.42 #printtimes

Syntax:

```
#printtimes
```

Prints the current execution time and real time in the same way as done at the end of the program.

Helps in monitoring the real time passed in TFORM jobs. Example:

```

#Printtimes
423.59 sec + 5815.88 sec: 6239.47 sec out of 1215.29 sec

```

3.43 #procedure

Syntax:

```
#procedure name(var1,...,varn)
```

See also `endprocedure` (3.25), `call` (3.10)

Name is the name of the procedure. It will be referred to by this name. If the procedure resides in a separate file the name of the file should be `name.prc` and the `#procedure` instruction should form the first line of the file. The `#` should be the first character of the file. The parameter field is optional. If there are no parameters, the procedure should also be called without parameters (see the `#call` instruction). The parameters (here called `var1` to `varn`) are preprocessor variables and hence they should be referred to between a backquote/quote pair as in `'var1'` to `'varn'`. If there exist already variables with such names when the procedure is called, the new definition comes on top of the old one. Hence in the procedure (and procedures called from it, unless the same problems occurs there too, as would be the case with recursions) the new definition is used, and it is released again when control returns from the procedure. After that the old definition will be in effect again.

If the procedure is included in the regular input stream, FORM will read the text of the procedure until the `#endprocedure` instruction and store it in a special buffer. When the procedure is called, FORM will read the procedure from this buffer, rather than from a file. In systems where

file transfer is slow (very busy server with a slow network) this may be faster, especially when many small procedures are called.

One way to make libraries that contain many procedures and maybe more code is to put all procedures into one header (.h) file and include this file at the beginning of the program with a `#include` instruction. This way one has all procedures load and one knows for sure that it are the proper procedures as it guards against the inadvertently picking up of procedures from other directories. It also makes for fewer files and hence makes for better housekeeping.

3.44 `#procedureextension`

Syntax:

`#procedureextension string`

See also `#call` (3.10)

The default extension of procedures is `.prc` in FORM. It is however possible that this clashes with the extensions used by other programs like the Grace system (Yuasa et al, Prog. Theor. Phys. Suppl. 138(2000)18). In that case it is possible to change the extension of the procedures in the current program. This is either done via the setup (page 17) or by the `#procedureextension` instruction of the preprocessor. The new string replaces the string `prc`, used by default. For the new string the following restrictions hold:

1. The first character must be alphabetic
2. No whitespace characters (blanks and/or tabs) are allowed

For the rest any characters can be used.

The new extension will remain valid either till the next `#procedureextension` instruction or to the next `.clear` instruction (page 4), whatever comes first.

3.45 `#prompt`

Syntax:

`#prompt [newprompt]`

Sets a new prompt for the current external command (if present) and all further (newly started) external commands.

If `newprompt` is an empty string, the default prompt (an empty line) will be used.

The prompt is a line consisting of a single prompt string. By default, this is an empty string.

3.46 `#redefine`

Syntax:

`#redefine name "string"`

See also `define` (3.18), `undefine` (3.63)

in which `name` refers to the name of the preprocessor variable to be redefined. The contents of the string will be its new value. If no variable of the given name exists yet, the instruction will be equivalent to the `#define` instruction.

3.47 **#remove**

Syntax:

```
#remove <filename>
```

See also write (3.65), append (3.6), create (3.16), close (3.13)

Deletes the named file from the system. Under UNIX this would be equivalent to the instruction

```
#system rm filename
```

and under MS-DOS oriented systems like Windows it would be equivalent to

```
#system del filename
```

The difference with the `#system` instruction is that the `#remove` instruction does not depend on the particular syntax of the operating system. Hence the `#remove` instruction can always be used.

3.48 **#reset**

Syntax:

```
#reset [<keyword>]
```

See also 'TIMER_' preprocessor variable.

Currently the only keywords that are allowed are timer and stopwatch. They have the same effect, which is to reset the timer for the 'timer_' (or 'stopwatch_') preprocessor variable (see 3.1).

3.49 **#reverseinclude**

Syntax:

```
#reverseinclude[-+] filename
```

```
#reverseinclude[-+] filename # foldname
```

This instruction is identical to the `#include` 3.34 instruction, with the exception that the statements and instructions in the file are read in reverse order. This can be useful at times when code is generated in a particular order in a file and one would like to 'undo' this code. It is somewhat related to the effects of the `debugflag` option (10.0.1) in the optimization options of the `format` statement 10.

There are a few limitations. If, for instance, linefeeds or semicolons occur inside preprocessor variables, the reading routines cannot see this. Additionally unfinished strings (unmatched double quotes) will result in a fatal error. On the other hand the fold structure remains preserved.

3.50 **#rmexternal**

Syntax:

```
#rmexternal [n]
```

Terminates an external command. The integer number `n` must be either the descriptor of a running external command, or 0.

If `n` is 0, then all external programs will be terminated.

If `n` is not specified, the current external command will be terminated.

The action of this instruction depends on the attributes of the external channel (see the `#setexternalattr` (section 19.5) instruction). By default, the instruction closes the commands' IO channels, sends a KILL signal to every process in its process group and waits for the external command to be finished.

3.51 `#rmseparator`

Syntax:

`#rmseparator character`

See also `#addseparator` (3.5), `#call` (3.10), `#do` (3.19)

Removes a character from the list of permissible separator characters for arguments of `#call` or `#do` instructions. By default the two characters that are permitted are the comma and the character `|`. Blanks, tabs and double quotes are ignored. Note that the comma must be specified between double quotes as in

```
#rmseparator ", "
```

3.52 `#setexternal`

Syntax:

`#setexternal n`

Sets the “current” external command. The instructions `#toexternal` and `#fromexternal` deal with the current external command. The integer number `n` must be the descriptor of a running external command.

3.53 `#setexternalattr`

Syntax:

`#setexternalattr list_of_attributes`

sets attributes for *newly started* external commands. Already running external commands are not affected. The list of attributes is a comma separated list of pairs `attribute=value`, e.g.:

```
#setexternalattr shell=noshell,kill=9,killall=false
```

Possible attributes are:

kill specifies the signal to be sent to the external command either before the termination of the FORM program or by the preprocessor instruction `#rmexternal`. By default this is 9 (SIGKILL). Number 0 means that no signal will be sent.

killall Indicates whether the kill signal will be sent to the whole group or only to the initial process. Possible values are “**true**” and “**false**”. By default, the kill signal will be sent to the whole group.

daemon Indicates whether the command should be “daemonized”, i.e. the initial process will be passed to the init process and will belong to the new process group in the new session. Possible values are “**true**” and “**false**”. By default, “**true**”.

shell specifies which shell is used to run a command. (Starting an external command in a subshell permits to start not only executable files but also scripts and pipelined jobs. The disadvantage is that there is no way to detect failure upon startup since usually the shell is started successfully.) By default this is “`/bin/sh -c`”. If set `shell=noshell`, the command will be started by the instruction `#external` directly but not in a subshell, so the command should be a name of the executable file rather than a system command. The instruction `#external` will duplicate the actions of the shell in searching for an executable file if the specified file name does not

contain a slash (/) character. The search path is the path specified in the environment by the PATH variable. If this variable isn't specified, the default path `"/bin:/usr/bin"` is used.

`stderr` specifies a file to redirect the standard error stream to. By default it is `"/dev/null"`. If set `stderr=terminal`, no redirection occurs.

Only attributes that are explicitly mentioned are changed, all others remain unchanged. Note, changing attributes should be done with care. For example,

```
#setexternalattr daemon=false
```

starts a command in the subshell within the current process group with default attributes `kill=9` and `killall=true`. The instruction `#rmexternal` sends the KILL signal to the wholegroup, which means that also FORM itself will be killed.

3.54 `#setrandom`

Syntax:

```
#setrandom number
```

See also `random_` (8.56) and `ranperm_` (8.57)

The `#setrandom` instruction initializes the random number generator `random_` 8.56. The number that is used as a seed can have the length of two words in FORM. This means that on a 32-bits computer it can be an (unsigned) 32-bits integer and on a 64-bits computer it can be an (unsigned) 64 bits integer. If there is no `#setrandom` instruction the random number generator is initialized in a built in standard way. The `#setrandom` instruction also initializes the random number generators of the workers when one uses TFORM or PARFORM. They are initialized with different seeds that are derived in a non-trivial way from the seed given by the user and the number of the worker.

3.55 `#show`

Syntax:

```
#show [preprocessorvariablename[s]]
```

If no names are present, the contents of all preprocessor variables will be printed to the regular output. If one or more preprocessor variables are specified (separated by comma's), only their contents will be printed. The preprocessor variables should be represented by their name only. No enclosing backquote/quote should be used, because that would force a substitution of the preprocessor variable before the instruction gets to see the name. Example:

```
#define MAX "3"
Symbols a1,...,a'MAX';
L F = (a1+...+a'MAX')^2;
#show
```

#The preprocessor variables:

```
0: VERSION_ = "3"
1: SUBVERSION_ = "2"
2: NAMEVERSION_ = ""
3: DATE_ = "Wed Feb 28 08:43:20 2007"
4: NAME_ = "testpre.frm"
5: CMODULE_ = "1"
```

```
6: MAX = "3"
   .end
```

```
Time =          0.00 sec    Generated terms =          6
                        F      Terms in output =          6
                        Bytes used    =          102
```

We see that the variable MAX has indeed the value 3. There are six additional variables which have been defined by FORM itself. Hence the trailing underscore which cannot be used in user defined names. The current version of FORM is shown in the variable VERSION_ and the name of the current program is given in the variable NAME_. For more about the system defined preprocessor variables see 3.1.

There is another preprocessor variable that does not show in the listings. Its name is SHOW-INPUT_. This variable has the value one if the listing of the input is on and the value zero if the listing of the input is off.

3.56 #skipextrasymbols

Syntax:

```
#skipextrasymbols positivenumber
```

See also ExtraSymbols (7.53) and the chapter on optimization (10).

This instructions adds a number of dummy extra symbols to the list of extra symbols (7.53). This can be used when several optimizations are done on an expression in such a way that the extra symbols of previous optimizations are still present. Normally the number space for them is erased in a #clearoptimize instruction. This can be avoided with a sequence like

```
#skipextrasymbols,{‘optimmaxvar_’-‘optimminvar_’+1}
```

In this case the numbering of the next optimization will start after the last extra symbol of the previous optimization. One should realize however that the definitions of the extra symbols are not kept once the new optimization is started or once a #clearoptimize instruction is issued. Example:

```
#-
S   a,b,c,d,e;
L   F = (a+b+c+d+3*e)^3;
B   b;
.sort
ExtraSymbols,array,w;
Format O3,stats=ON;
#optimize F
#write <> "   %40"
.sort
#SkipExtraSymbols,{‘optimmaxvar_’-‘optimminvar_’+1}
id  b = b+1;
Print +f;
B   b;
.end
```

Because the O3 format is still active, the final printing uses the optimization as well. If the #SkipExtraSymbols instruction would have been omitted, the numbering would start again from

one, while the rhs. of their definitions would contain the old extra symbols. The result would be incorrect.

3.57 `#sortreallocate`

Syntax:

```
#sortreallocate
```

See also “On sortreallocate;” (7.108).

Reallocates the small and large buffer (also on the worker threads) at the end of the current module. In some cases this can significantly reduce FORM’s memory usage as measured by “resident set size”.

3.58 `#switch`

Syntax:

```
#switch string
```

See also `endswitch` (3.26), `case` (3.11), `break` (3.8), `default` (3.17)

the string could for instance be a preprocessor variable as in

```
#switch ‘i’
```

The `#switch` instruction, together with `#case`, `#break`, `#default` and `#endswitch`, allows the user to conveniently make code for a number of cases that are distinguished by the value of a preprocessor variable. In the past this was only possible with the use of folds in the `#include` instruction and the corresponding include file (see 3.34). Because few people have an editor like STedi (see the FORM distribution site) that can handle the folds in a proper way, it was judged that the more common switch mechanism might be friendlier. The proper syntax of a complete construction would be

```
#switch ‘par’
#case 1
    some statements
#break
#case ax2
    other statements
#break
#default
    more statements
#break
#endswitch
```

The number of cases is not limited. The compare between the strings in the `#switch` instruction and in the `#case` instructions is as a text string. Hence numerical strings have no special meaning. If a `#break` instruction is omitted, control may go into another case. This is called fall-through. This is a way in which one can have the same statements for several cases. The `#default` instruction is not mandatory.

FORM will look for the first case of which the string matches the string in the `#switch` instruction. Input reading (control flow) starts after this `#case` instruction, and continues till either a `#break` instruction is encountered, or the `#endswitch` is met. After that input reading continues after the `#endswitch` instruction. If no case has a matching string, input reading starts after

the `#default` instruction. If no `#default` instruction is found, input reading continues after the matching `#endswitch` instruction.

`#switch` constructions can be nested. They can be combined with `#if` constructions, `#do` instructions, etc. but they should obey normal nesting rules (as with nesting of brackets of different types).

3.59 `#system`

Syntax:

```
#system [-e] systemcommand
```

See also pipe (3.39)

This forces a system command to be executed by the operating system. The complete string (excluding initial blanks or tabs) is passed to the operating system. FORM will then wait until control is returned. Note that this instruction introduces operating system dependent code. Hence it should be used with great care.

Without the `-e` option execution of the Form program will halt if the system command returns an error. With the `-e` option the Form program will continue execution. The value of the return code of the system command can be found in the `SYSTEMERROR_` preprocessor variable.

3.60 `#terminate`

Syntax:

```
#terminate [exitcode]
```

This forces FORM to terminate execution immediately. If an exit code is given (an integer number), this will be the return value that FORM gives to the shell program from which it was run. If no return value is specified, the value -1 will be returned.

3.61 `#timeoutafter`

Syntax:

```
#timeoutafter <Number of seconds>
```

This instruction starts a timer. When the given time expires the current program will be terminated, unless the timer is reset before this time. Resetting the timer is done with the `"#timeoutafter 0"` instruction.

The purpose of this instruction is to prevent runaway programs, because a given subpart takes much more time than it should. Example:

```
.sort
#timeoutafter 1000
#call problematicprocedure
.sort
#timeoutafter 0
```

If one runs many diagrams with a make-like facility like minos, diagrams that behave in an unexpected way can be killed this way and minos can continue with the next diagram. Later one can see which diagrams caused problems and one may study what the problem was.

3.62 `#toexternal`

Syntax:

```
#toexternal "formatstring" <,variables>
```

Sends the output to the current external command. The semantics of the "**formatstring**" and the **[,variables]** is the same as for the `#write` instruction, except for the trailing end-of-line symbol. In contrast to the `#write` instruction, the `#toexternal` instruction does not append any new line symbol to the end of its output.

3.63 `#undefine`

Syntax:

```
#undefine name
```

See also `define` (3.18), `redefine` (3.46)

Name refers to the name of the preprocessor variable to be undefined. This statement causes the given preprocessor variable to be removed from the stack of preprocessor variables. If an earlier instance of this variable existed (other variable with the same name), it will become active again. There are various other ways by which preprocessor variables can become undefined. All variables belonging to a procedure are undefined at the end of a procedure, and so are all other preprocessor variables that were defined inside this procedure. The same holds for the preprocessor variable that is used as a loop parameter in the `#do` instruction.

3.64 `#usedictionary`

Syntax:

```
#usedictionary name #usedictionary name (options)
```

See chapter 13 on dictionaries.

Starts using a dictionary for output translation.

3.65 `#write`

Syntax:

```
#write [<filename>] "formatstring" [,variables]
```

See also `append` (3.6), `create` (3.16), `remove` (3.47), `close` (3.13)

If there is no file specified, the output will be to the regular output channel. If a file is specified, FORM will look whether this file is open already. If it is open already, the specified output will be added to the file. If it is not open yet it will be opened. Any previous contents will be lost. This would be equivalent to using the `#create` instruction first. If output has to be added to an existing file, the `#append` instruction should be used first.

The format string is like a format string in the language C. This means that it is placed between double quotes. It will contain text that will be printed, and it will contain special character sequences for special actions. These sequences and the corresponding actions are:

`\n` A newline character.

`\t` A tab character.

`\"` A double quote character.

\b A backslash character.

%% The character %.

% If the last character in the string, it causes the omission of a linefeed at the end of the printing. Note that if this happens in the regular output (as opposed to a file) there may be interference with the listing of the input.

%%\$ A dollar variable. The variable should be indicated in the list of variables. Each occurrence of %%\$ will look for the next variable.

%e An active expression. The expression should be indicated in the list of variables. Each occurrence of %e will look for the next variable. Unlike the output caused by the print statement the expression will be printed without its name and there will also be no = sign unless there is one in the format string of course. If the current output format is fortran output there is an extra option. After the name of the expression one should put between parentheses the name to be used when there are too many continuation cards.

%+e Like %e, but like the +s option in the Print statement 7.114 where each term starts on a new line.

%E Like %e, but whereas the %e terminates the expression with a ;, the %E does not give this trailing semicolon.

%+E Like %E, but like the +s option in the Print statement 7.114 where each term starts on a new line.

%s A string. The string should be given in the list of variables and be enclosed between double quotes. Each occurrence of %s will look for the next variable in the list.

%f A file. The name of the file will be expected in the list of variables. The file is searched for in the current directory, then in path indicated by the path variable in the setup file or at the beginning of the file (see chapter 17 on the setup file), then in the path specified in the -p option when FORM is started (see the chapter on running FORM). If this option has not been used, FORM will look for the environment variable FORMPATH. If this variable exists it will be interpreted as a path and FORM will search the indicated directories for the given file. If none is found there will be an error message and execution will be halted.

%X Forces the printing of the list of extra symbols (2.11) and their definitions.

%O Forces the printing of the definitions of the extra symbols in the buffer with the temporary variables from the previous optimization (see the chapter on optimizations 10).

If no special variables are asked for (by means of %%\$, %e, %E or %s) the list of variables will be ignored (if present). Example:

```
Symbols a,b;
L    F = a+b;
#$a1 = a+b;
#$a2 = (a+b)^2;
#$a3 = $a1^3;
#write " One power: %$\n Two powers: %$\n Three powers: %$\n%s"\
      , $a1, $a2, $a3, " The end"
```

```

One power: b+a
Two powers: b^2+2*a*b+a^2
Three powers: b^3+3*a*b^2+3*a^2*b+a^3
The end
.end

```

```

Time =          0.00 sec      Generated terms =          2
          F          Terms in output =          2
          Bytes used      =          32

```

We see that the writing occurs immediately after the `#write` instruction, because it is done by the preprocessor. Hence the output comes before the execution of the expression `F`.

```

S   x1,...,x10;
L   MyExpression = (x1+...+x10)^4;
.sort
Format Fortran;
#write <fun.f> "      FUNCTION fun(x1,x2,x3,x4,x5,x6,x7,x8,x9,x10)"
#write <fun.f> "      REAL  x1,x2,x3,x4,x5,x6,x7,x8,x9,x10"
#write <fun.f> "      fun = %e",MyExpression(fun)
#write <fun.f> "      RETURN"
#write <fun.f> "      END"
.end

```

Some remarks are necessary here. Because the `#write` is a preprocessor instruction, the `.sort` is essential. Without it, the expression has not been worked out at the moment we want to write. The name of the expression is too long for fortran, and hence the output file will use a different name (in this case the name 'fun' was selected). The output file looks like

```

FUNCTION fun(x1,x2,x3,x4,x5,x6,x7,x8,x9,x10)
REAL x1,x2,x3,x4,x5,x6,x7,x8,x9,x10
fun = 24*x1*x2*x3*x4 + 24*x1*x2*x3*x5 + 24*x1*x2*x3*x6 + 24*x1*x2
& *x3*x7 + 24*x1*x2*x3*x8 + 24*x1*x2*x3*x9 + 24*x1*x2*x3*x10 + 12*
      . . . . .
& x8 + 4*x6**3*x9 + 4*x6**3*x10 + x6**4 + 24*x7*x8*x9*x10 + 12*x7*
& x8*x9**2
fun = fun + 12*x7*x8*x10**2 + 12*x7*x8**2*x9 + 12*x7*x8**2*x10 +
& 4*x7*x8**3 + 12*x7*x9*x10**2 + 12*x7*x9**2*x10 + 4*x7*x9**3 + 4*
& x7*x10**3 + 12*x7**2*x8*x9 + 12*x7**2*x8*x10 + 6*x7**2*x8**2 +
& 12*x7**2*x9*x10 + 6*x7**2*x9**2 + 6*x7**2*x10**2 + 4*x7**3*x8 +
& 4*x7**3*x9 + 4*x7**3*x10 + x7**4 + 12*x8*x9*x10**2 + 12*x8*x9**2
& *x10 + 4*x8*x9**3 + 4*x8*x10**3 + 12*x8**2*x9*x10 + 6*x8**2*
& x9**2 + 6*x8**2*x10**2 + 4*x8**3*x9 + 4*x8**3*x10 + x8**4 + 4*x9
& *x10**3 + 6*x9**2*x10**2 + 4*x9**3*x10 + x9**4 + x10**4

RETURN
END

```

and each time after 19 continuation lines we have to break the expression and use the `fun = fun +` trick to continue.

3.66 Some remarks

It should be noted that the various constructions like `#do/#enddo`, `#procedure/#endprocedure`, `#switch/#endswitch` and `#if/#endif` all create a certain environment. These environments cannot be interleaved. This means that one cannot make code of the type

```
#do i = 1,5
  #if ( 'MAX' > 'i' )
    id f('i') = g('i')(x);
  #enddo
  some statements
#do i = 1,5
  #endif
#enddo
```

whether this could be considered useful or not. Similarly one cannot make a construction that might be very useful:

```
#do i = 1,5
  #do j'i' = 1,3
  #enddo
  some statements
#do i = 1,5
  #enddo
#enddo
```

Currently the syntax does not allow this. This may change in the future.

Chapter 4

Modules

Modules are the basic execution blocks. Statements are always part of a module, and they will be executed only when the module is executed. This is directly opposite to preprocessor instructions which are executed when they are encountered in the input stream.

Modules are terminated by a line that starts with a period. Such a line is called the module instruction. Once the module instruction has been recognized, the compilation of the module is terminated and the module will be executed. All active expressions will be processed one by one, term by term. When each term of an expression has been through all statements of the module, the combined results of all operations on all the terms of the expression will be sorted and the resulting expression will be sent to the output. This can be an intermediate file, or it can be some memory, depending on the size of the output. If the combined output of all active expressions is less than the parameter “ScratchSize”, the results stay in memory. ScratchSize is one of the setup parameters (see chapter 17).

A module consists in general of several types of statements:

Declarations These are the declarations of variables.

Specifications These tell what to do with existing expressions as a whole.

Definitions These define new expressions.

Executable statements The operations on all active expressions.

OutputSpecifications These specify the output representation.

End-of-module specifications Extra settings that are for this module only.

Mixed statements They can occur in various classes. Most notably the print statement.

Statements must occur in such an order that no statement follows a statement of a later category. The only exception is formed by the mixed statements, which can occur anywhere. This is different from earlier versions of FORM in which the order of the statements was not fixed. This did cause a certain amount of confusion about the workings of FORM.

There are several types of modules.

.sort The general end-of-module. Causes execution of all active expressions, and prepares them for the next module.

.end Executes all active expressions and terminates the program.

.store Executes all active expressions. Then it writes all active global expressions to an intermediate storage file and removes all other non-global expressions. Removes all memory of declarations except for those that were made before a `.global` instruction.

.global No execution of expressions. It just saves declarations made thus far from being erased by a `.store` instruction.

.clear Executes all active expressions. Then it clears all buffers with the exception of the main input stream. Continues execution in the main input stream as if the program had started at this point. The only parameters that cannot be changed at this point are the setup parameters. They remain. By default also the clock is reset. If this is not desired this can be changed by means of the `ResetTimeOnClear` setup variable (see chapter 17).

Each program must be terminated by a `.end` instruction. If such an instruction is absent and FORM encounters an end-of-input it will issue a warning and generate a `.end` instruction.

Module instructions can contain a special commentary that will be printed in all statistics that are generated during the execution of the module. This special commentary is restricted to 24 characters (the statistics have a fixed format and hence there is only a limited amount of space available). This commentary is initiated by a colon and terminated by a semicolon. The characters between this colon and the semicolon are the special message, also called advertisement. Example

```
.sort:Eliminate x;
```

would give in the statistics something like

Time =	0.46 sec	Generated terms =	360
	F	Terms in output =	360
	Eliminate x	Bytes used =	4506

If the statistics are switched off, there will be no printing of this advertisement either.

For backwards compatibility there is still an obsolete mechanism to pass module options via the module instructions. This is a feature which will probably disappear in future versions of FORM. We do give the syntax to allow the user to identify the option properly and enable proper translation into the `moduleoption` statement (see 7.89).

```
.sort(PolyFun=functionname);
.sort(PolyFun=functionname):advertisement;
```

causes the given function to be treated as a polynomial function. This means that its (single) argument would be treated as the coefficient of the terms. The action of FORM on individual terms is

1. Ignore polynomial functions with more than one argument.
2. If there is no polynomial function with a single argument, generate one with the argument 1.
3. If there is more than one polynomial function with a single argument, multiply the arguments and replace these functions with a single polynomial function with the product of the arguments for a single argument.
4. Multiply the argument of the polynomial function with the coefficient of the term. Replace the coefficient itself by one.

If, after this, two terms differ only in the argument of their polynomial function FORM will add the arguments and replace the two terms by a single term which is identical to the two previous terms except for that the argument of its polynomial function is the sum of their two arguments.

It should be noted that the proper placement of .sort instructions in a FORM program is an art by itself. Too many .sort instructions cause too much sorting, which can slow execution down considerably. It can also cause the writing of intermediate expressions which are much larger than necessary, if the next statements would cause great simplifications. Not enough .sort instructions can make that cancellations are postponed unnecessarily and hence much work will be done double. This can slow down execution by a big factor. First an example of a superfluous .sort:

```
S a1,...,a7;
L F = (a1+...+a7)^16;
.sort
```

Time =	31.98 sec	Generated terms =	74613
	F	Terms in output =	74613
		Bytes used =	1904316

```
id a7 = a1+a2+a3;
.end
```

Time =	290.34 sec	Terms active =	87027
	F	Bytes used =	2253572

Time =	295.20 sec	Generated terms =	735471
	F	Terms in output =	20349
		Bytes used =	538884

Without the sort the same program gives:

```
S a1,...,a7;
L F = (a1+...+a7)^16;
id a7 = a1+a2+a3;
.end
```

Time =	262.79 sec	Terms active =	94372
	F	Bytes used =	2643640

Time =	267.81 sec	Generated terms =	735471
	F	Terms in output =	20349
		Bytes used =	538884

and we see that the sorting in the beginning is nearly completely wasted. Now a clear example of not enough .sort instructions. A common problem is the substitution of one power series into another. If one does this in one step one could have:

```
#define MAX "36"
S j,x(:'MAX'),y(:'MAX');
*
```

```

* Power series expansion of ln_(1+x)
*
L F = -sum_(j,1,'MAX',sign_(j)*x^j/j);
*
* Substitute the expansion of x = exp_(y)-1
*
id x = x*y;
#do j = 2,'MAX'+1
id x = 1+x*y/'j';
#enddo
Print;
.end

```

Time =	76.84 sec	Generated terms =	99132
	F	Terms in output =	1
		Bytes used =	18

```

F =
y;

```

With an extra .sort inside the loop one obtains for the same program (after suppressing some of the statistics:

```

#define MAX "36"
S j,x(:'MAX'),y(:'MAX');
*
* Power series expansion of ln_(1+x)
*
L F = -sum_(j,1,'MAX',sign_(j)*x^j/j);
*
* Substitute the expansion of x = exp_(y)-1
*
id x = x*y;
#do j = 2,'MAX'+1
id x = 1+x*y/'j';
.sort: step 'j';

```

Time =	0.46 sec	Generated terms =	360
	F	Terms in output =	360
	step 2	Bytes used =	4506

```

#enddo

```

```

.
.
.

```

Time =	3.07 sec	Generated terms =	3
	F	Terms in output =	1
	step 37	Bytes used =	18

```

Print;
.end

```

```

Time =          3.07 sec    Generated terms =          1
          F              Terms in output =          1
                          Bytes used      =          18

```

```

F =
  y;

```

It is very hard to give general rules that are more specific than what has been said above. The user should experiment with the placements of the `.sort` before making a very large run.

4.1 Checkpoints

If FORM programs have to run for a long time, the reliability of the hardware (computer system or network) or of the software infrastructure becomes a critical issue. Program termination due to unforeseen failures may waste days or weeks of invested execution time. The checkpoint mechanism was introduced to protect long running FORM programs as good as possible from such accidental interruptions. With activated checkpoints FORM will save its internal state and data from time to time on the hard disk. This data then allows a recovery from a crash.

The checkpoint mechanism can be activated or deactivated by `On` and `Off` statements. If the user has activated checkpoints, recovery data will be written to disk at the end of a module execution. Options allow to influence the details of the saving mechanism. If a program is terminated during execution, FORM can be restarted with the `-R` option and it will continue its execution at the last saved recovery point.

The syntax of the checkpoint activation and deactivation is

```

On checkpoint [<OPTIONS>];
Off checkpoint;

```

If no options are given, the recovery data will be saved at the end of every module. If one gives a time

```

On checkpoint <NUMBER>[<UNIT>];

```

the saving will only be done if the given time has passed after the last saving. Possible unit specifiers are `s`, `m`, `h`, `d` and the number will then be interpreted as seconds, minutes, hours, or days, respectively. The default unit is seconds.

If one needs to run a script before or after the saving, one can specify a script filename.

```

On checkpoint runbefore="<SCRIPTFILENAME>";
On checkpoint runafter="<SCRIPTFILENAME>";
On checkpoint run="<SCRIPTFILENAME>";

```

The option `run` sets both the scripts to be run before and after saving. The scripts must have the executable flag set and they must reside in the execution path of the shell (unless the filename already contains the proper path).

The scripts receive the module number as an argument (accessible as `$1` inside the script). The return value of the script running before the saving will be interpreted. If the script returns an error (non-zero return value), a message will be issued and the saving will be skipped.

The recovery data will be written to files named `FORMrecv.*` with various name extensions. If a file `FORMrecv.tmp` exists, FORM will not run unless one gives it the recovery option `-R`. This

is to prevent the unintentional loss of recovery data. If FORM terminates successfully, all the additional data files will be removed.

The additional recovery files will be created in the directory containing the scratch files. The extra files will occupy roughly as much space as the scratch files and the save and hide files combined. This extra space must be made available, of course.

If recovery data exists and FORM is started with the `-R` option, FORM will continue execution after the last module that successfully wrote the recovery data. All the command line parameters that have been given to the crashed FORM program must also be given to the recovering FORM program. The input files are not part of the recovery data and will be read in anew when recovering. Therefore it is strongly discouraged to change any of these files between saving and recovery.

Chapter 5

Pattern matching

Substitutions are made in FORM by specifying a generic object that should be replaced by an expression. This generic object is called a pattern. Patterns that the user may already be familiar with are the regular expressions in many UNIX based systems or just a statement like `ls *.frm` to list only files of which the name ends in `.frm`. In this case the `*` is called a wildcard that can take any string value. In symbolic manipulation there will be wildcards also, but their nature will be different. They are also indicated in a different way.

In FORM wildcard variables are indicated by attaching a question mark (?) to the name of a variable. The type of the variable indicates what type of object we are looking for. Assume the following id statements:

```
Functions f,g;
Symbol x;

id f(g?,x) = g(x,x);
```

In this statement `g` will match any function and hence all occurrences of `f`, in which the first argument is a function and the second argument is the symbol `x`, will match. In the right hand side the function `g` will be substituted by whatever identity `g` had to assume in the left hand side to make the match. Hence `f(f,x)` will be replaced by `f(x,x)`.

In general function wildcards can only match functions. Even though tensors are special functions, regular function wildcards cannot match tensors, and tensor wildcards cannot match functions. However commuting function wildcards can match noncommuting functions *et vice versa*.

Index wildcards can only match indices. The dimension of the indices is not relevant. Hence:

```
id f(mu?,mu?) = 4;
```

would match both `f(ka,ka)` and `f(2,2)`. We will see later how to be more selective about such matches.

When the same wildcard occurs more than once in a pattern, it should be matched by the same object in all its occurrences. Hence the above pattern would not match `f(mu,nu)`.

There is one complication concerning the above rule of index wildcards only matching indices. FORM writes contractions with vectors in a special shorthand notation called Schoonschip notation. Hence `f(mu)*p(mu)` becomes `f(p)`. This means that the substitution

```
id f(mu?)*g(nu?) = fg(mu,nu);
```

should also replace the term `f(p)*g(q)` by `fg(p,q)`. In this case it looks like the wildcard indices matched the vectors. This is however not the case, because if we take the previous pattern (with

the $f(\mu?, \mu?)$, it is not going to match the term $f(p, p)$, because this term should be read as something of the type $f(\mu, \nu) * p(\mu) * p(\nu)$ and that term does not fit the pattern $f(\mu?, \mu?)$.

Vector wildcards can match vectors, but they can also match vector-like expressions in function arguments. A vector-like expression is an expression in which all terms contain one single vector without indices, possibly multiplied by other objects like coefficients, functions or symbols. Hence

```
id f(p?) = p.p;
```

would match $f(q)$, $f(2*q-r)$ and $f(a*q+f(x)*r)$, if p , q and r are vectors, and a and x are symbols, and f is a function. It would not match $f(x)$ and neither would it match $f(q*r)$, nor $f(a*q+x)$.

Wildcard symbols are the most flexible objects. They can match symbols, numbers and expressions that do not contain loose indices or vectors without indices. These last objects are called scalar objects. Hence wildcard symbols can match all scalar objects. In

```
id x^n? = x^(n+1)/(n+1);
```

the wildcard symbol n would normally match a numerical integer power. In

```
id f(x?) = x^2;
```

there would be a match with $f(y)$, with $f(1+2*y)$ and with $f(p.p)$, but there would not be a match with $f(p)$ if p is a vector.

There is one extra type of wildcards. This type is rather special. It refers to groups of function arguments. The number of arguments is not specified. These variables are indicated by a question mark followed by a name (just the opposite of the other wildcard variables), and in the right hand side they are also written with the leading question mark:

```
id f(?name) = g(1, ?name);
```

In this statement all occurrences of f with any number of arguments (including no arguments) will match. Hence $f(\mu, \nu)$ will be replaced by $g(1, \mu, \nu)$. In the case that f is a regular function and g is a tensor, it is conceivable that the arguments in $?name$ will not fit inside a tensor. For instance $f(x)$, with x a symbol, would match and FORM would try to put the symbol inside the tensor g . This would result in a runtime error. In general FORM will only accept arguments that are indices or single vectors for a substitution into a tensor. The object $?name$ is called an **argument field wildcard**.

One should realize that the use of multiple argument field wildcards can make the pattern matching slow.

```
id f(?a, p1?, ?b, p2?, ?c, p3?, ?d) * g(?e, p3?, ?f, p1?, ?g, p2?, ?h) = ....
```

may involve considerable numbers of tries, especially when there are many occurrences of f and g in a term. One should be very careful with this.

A complication is the pattern matching in functions with symmetry properties. In principle FORM has to try all possible permutations before it can conclude that a match does not occur. This can become rather time consuming when many wildcards are involved. FORM has a number of tricks built in, in an attempt to speed this up, but it is clear that for many cases these tricks are not enough. This type of pattern matching is one of the weakest aspects of ‘artificial intelligence’ in general. It is hoped that in future versions it can be improved. For the moment the practical consequence is that argument field wildcards cannot be used in symmetric and antisymmetric functions. If one needs to make a generic replacement in a symmetric function one cannot use

```
CFunction f(symmetric),g(symmetric);
id f(?a) = ....;
```

but one could try something like

```
CFunction f(symmetric),ff,g(symmetric);
id f(x1?,...,x5?) = ff(x1,...,x5);
id ff(?a) = ...;
id ff(?a) = f(?a);
```

if f has for instance 5 arguments. If different numbers of arguments are involved, one may need more than one statement here or a statement with the `replace_` function:

```
Multiply replace_(f,ff);
```

It just shows that one should at times be a bit careful with overuse of (anti)symmetric functions. Cyclic functions do not have this restriction.

When there are various possibilities for a match, FORM will just take the first one it encounters. Because it is not fixed how FORM searches for matches (in future versions the order of trying may be changed without notice) one should try to avoid ambiguities as in

```
id f(?a,?b) = g(?a)*h(?b);
```

Of course the current search method is fully consistent (and starts with all arguments in `?a` and none in `?b` etc, but a future pattern matcher may do it in a different order.

When two argument field wildcards in the left hand side have the same name, a match will only occur, when they match the same objects. Hence

```
id f(?a,?a) = g(?a);
```

will match $f(a,b,a,b)$ or just f (in which case `?a` will have zero arguments), but it will not match $f(b,b,b)$.

Sometimes it is useful when a search can be restricted to a limited set of objects. For this FORM knows the concept of sets. If the name of a set is attached after the question mark, this is an indication for FORM to look only for matches in which the wildcard becomes one of the members of the set:

```
Symbols a,a1,a2,a3,b,c;
Set aa:a1,a2,a3;
```

```
id f(a?aa) = ...
```

would match $f(a1)$ but not $f(b)$. Sets can also be defined dynamically by enclosing the elements between curly brackets as in:

```
Symbols a,a1,a2,a3,b,c;
```

```
id f(a?{a1,a2,a3}) = ...
```

Sets of symbols can contain (small integer) numbers as well. Similarly sets of indices can contain fixed indices (positive numbers less than the value of `fixindex` (see the chapter on the setup 17). This means that some sets can be ambiguous in their nature.

Sometimes sets can be used as some type of array. In the case of

```
Symbols a,a1,a2,a3,b,c,n;
Set aa:a1,a2,a3;
```

```
id f(a?aa[n]) = ...
```

not only does 'a' have to be an element of the set aa, but if it is an element of that set, n will become the number of the element that has been matched. Hence for `f(a2)` the wildcard a would become a2 and the wildcard n would become 2. These objects can be used in the right-hand side. One can also use sets in the right-hand side with an index like the n of the previous example:

```
Symbols a,a1,a2,a3,b1,b2,b3,c,n;
Functions f,g1,g2,g3;
Set aa:a1,a2,a3;
Set bb:b1,b2,b3;
Set gg:g1,g2,g3;
```

```
id f(a?aa[n]) = gg[n](bb[n]);
```

which would replace `f(a2)` by `g2(b2)`. One cannot do arithmetic with the number of the array element. Constructions like `bb[n+1]` are not allowed.

There is one more mechanism by which the array nature of sets can be used. In the statement (declarations as before)

```
id f(a?aa?bb) = a*f(a);
```

a will have to be an element of the set aa, but after the matching it takes the identity of the corresponding element of the set bb. Hence `f(a2)` becomes after this statement `b2*f(b2)`.

Wildcards can also give their value directly to \$-variables (see chapter 6 about the \$-variables). If a \$-variable is attached to a wildcard (if there is a set restriction, it should be after the set) the \$-variable will obtain the same contents as the wildcard, provided a match occurs. If there is more than one match, the last match will be in the \$-variable.

```
id f(a?$w) = f(a);
```

will put the match of a in \$w. Hence in the case of `f(a2)` the \$-variable will have the value a2. In the case of `f(a2)*f(a3)` the eventual value of \$w depends on the order in which FORM does the matching. This is not specified and it would not be a good strategy to make programs that will depend on it. A future pattern matcher might do it differently! But one could do things like

```
while ( match(f(a?$w)) );
  id f($w) = ....
  id g($w) = ....
endwhile;
```

just to make sure with which match one is working.

Chapter 6

The dollar variables

In the older versions of FORM there were two types of variables: the preprocessor variables and the algebraic variables. The preprocessor variables are string variables that are used by the edit features of the preprocessor to prepare the input for the compiler part of FORM. The algebraic objects are the expressions and the various algebraic variables like the symbols, functions, vectors etc. There existed however very few possibilities to communicate from the algebraic level to the decision taking at the preprocessor level. This has changed dramatically with version 3 and the introduction of the dollar variables.

Dollar variables are basically (little) expressions that can be used to store various types of information. They can be used both as preprocessor objects as well as algebraic objects. They can also be defined and given contents both by the preprocessor and during execution on a term by term basis. Dollar variables are kept in memory. Hence it is important not to make them too big, because in that case performance might suffer.

What is a legal name for a dollar variable? Dollar variables have a name that consists of a dollar sign (\$) followed by an alphabetic character and then potentially more alphanumeric characters. Hence `$a` and `$var` and `$r4t78y0` are legal names and `$1a` is not a legal name. The variables do not have to be declared. However FORM will complain if a dollar variable is being used, before it has encountered a statement or an instruction in which the variable has been given a value. Hence giving a variable a value counts at the same time as a declaration.

What can be stored in a dollar variable?

- Algebraic expressions as in `$var = (a+b)^2;`
- Individual objects like indices, numbers, symbols.
- Zero.
- Parts of a term.
- Argument fields that consist of zero, one or more arguments.

Actually, the parts of a term are treated as a complete term and hence as a special case of an algebraic expression. Internally they are stored slightly differently for speed, but at the user level this should not be noticeable. Actually, with the exception of the argument fields, FORM can convert one type into the other and will try so, depending on the use that is made of the specific dollar variable. In the case that a variable is used in a way that should not be possible (like the content of a variable is a symbol, but it is used in a position where an index is expected) there will be a runtime error.

How is a variable used?

- As a preprocessor variable. This is done by putting the variable between a pair of ‘`’` as in ‘`$var`’. In this case the regular print routines of FORM make a textual representation of the variable as it exists at the moment that the preprocessor encounters this object, and this string is then substituted by the preprocessor as if it were the contents of a preprocessor variable.
- Like an expression during execution time. This would be the case in the statement

```
id x = y + $var;
```

in which `$var` is substituted in a way that is similar to the substitution of a local expression `F` in the statement

```
id x = y + F;
```

except for that the dollar variable is always stored in the CPU memory.

- As an algebraic object during execution time. This could be the case with any value of the variable that is not an expression. An example would be

```
id f(?a) = f(?a,$var);
```

in which the dollar variable contains an argument field.

- As an algebraic object in a delayed substitution of a pattern or a special statement. This may need some clarification. If we have the statement

```
id f($var) = anything;
```

the compiler does not substitute the current value of `$var`. The reason is that `$var` could have a different value for each term that runs into this statement, while the compiler compiles the statement only once. Hence FORM will substitute the value of `$var` only at the moment that it will attempt the pattern matching. This is called delayed substitution. If one likes the compiler to substitute a value, one can basically let the preprocessor take care of this by typing

```
id f('$var') = anything;
```

A similar delayed substitution takes place in statements of the type `Trace,$var;.`

How does one give a value to a dollar variable?

- In the preprocessor. This is done with an instruction of the type `#$var = 0;.` This is an instruction that can run over more than one line. The r.h.s. can be any algebraic expression. Specifically it can contain dollar variables or local/global expressions. Such expressions are worked out during the preprocessing. Hence this variable acquires a value immediately.
- During execution when control reaches a statement of the form `$var = expression;.` Again the r.h.s. can contain any normal algebraic expression including dollar variables and local/global expressions. The r.h.s. will be evaluated and the value will be assigned to `$var`. In the case that `$var` had already a value, the old value will be deleted and the new value will be ‘installed’.

- During execution when the dollar variable is assigned the value of a wildcard as in

```
id f(x?$var) = whatever;
```

If the function `f` occurs more than once in a term, `$var` will have the value of the last match. In the case that the value of the first match is needed one can use the option ‘once’ in the id-statement as in

```
id,once,f(x?$var) = whatever;
```

In general one can paste the dollar variable to the end of any wildcard description. Hence one can use `id f(x?{1,2,3}$var) = ...;` and

```
id f(x?set[n?$var1]$var2) = ...;
```

Note the difference between `#$a = 0;` and `$a = 0;`. One CANNOT make a wildcard construction for dollar variables themselves as in `id f($var?) = ...;`

Dollar variables CANNOT have arguments as in `$var(2)` or something equivalent. There is however a solution at the preprocessor level for this by defining individual variables `$var1` to `$varn` and then using `$var‘i’` or `‘$var‘i’` for some preprocessor variable `i`. The exception is the indication of factors when a dollar variable has been factorized (see the `#factdollar` instruction 3.29 and the `factdollar` statement 7.55). This is explained later in this chapter and in the chapter about polynomials 11.

Printing dollar variables:

- In the preprocessor one can use the `#write` instruction (see 3.65).
- During execution one can use the `Print` statement (see 7.114).

In both cases one should use the format string. The syntax is described in the chapters on these statements. The format descriptor of a dollar variable is `%%$` and this looks after the format string for the next dollar variable. Of course one can also use the dollar variable as a preprocessor variable when printing/writing in the preprocessor.

Examples.

Counting terms:

```
S a,b;
Off statistics;
L F = (a+b)^6;
#$a = 0;
$a = $a+1;
Print "      >> After %t we have %%$ term(s)", $a;
#write "      ># $a = ‘$a’"
># $a = 0
.sort
>> After + a^6 we have 1 term(s)
>> After + 6*a^5*b we have 2 term(s)
>> After + 15*a^4*b^2 we have 3 term(s)
>> After + 20*a^3*b^3 we have 4 term(s)
>> After + 15*a^2*b^4 we have 5 term(s)
```

```

    >> After + 6*a*b^5 we have 6 term(s)
    >> After + b^6 we have 7 term(s)
#write "      ># $a = '$a'"
># $a = 7
.end

```

Maximum power of x in an expression:

```

S x,a,b;
Off statistics;
L F = (a+b)^4+a*(a+x)^3;
.sort
#$a = 0;
if ( count(x,1) > $a ) $a = count_(x,1);
Print "      >> After %t the maximum power of x is %$", $a;
#write "      ># $a = '$a'"
># $a = 0
.sort
>> After + 3*x*a^3 the maximum power of x is 1
>> After + 3*x^2*a^2 the maximum power of x is 2
>> After + x^3*a the maximum power of x is 3
>> After + 4*a*b^3 the maximum power of x is 3
>> After + 6*a^2*b^2 the maximum power of x is 3
>> After + 4*a^3*b the maximum power of x is 3
>> After + 2*a^4 the maximum power of x is 3
>> After + b^4 the maximum power of x is 3
#write "      ># $a = '$a'"
># $a = 3
.end

```

Starting with version 4, FORM has the capability to factorize polynomials (see the chapter on polynomials 11). One type of objects that can be factorized is the dollar variables. The immediate question here is how to access the factors. As we mentioned before in this chapter, normally there is no direct way to use arguments for dollar variables. For the factors however we have a way of indexing the dollar variables as in `$var[1],...,$var[n]` when there are `n` factors. The number of factors can be obtained as `$var[0]`. In the index field can only be (nonnegative integer) numbers, dollar variables or factors of dollar variables that evaluate into (nonnegative integer) numbers.

```

Symbol x,y;
CFunction f1,f2;
Local F = f1(x^2+2*x*y+y^2)+f1(x^4-y^4);
id f1(x?$x) = f2(x);
FactDollar,$x;
Do $i = 1,$x[0];
    Print "In %t factor %$ is %$", $i,$x[$i];
Enddo;
.end
In + f2(y^2 + 2*x*y + x^2) factor 1 is y + x
In + f2(y^2 + 2*x*y + x^2) factor 2 is y + x
In + f2(- y^4 + x^4) factor 1 is - 1

```

```

In  + f2( - y^4 + x^4) factor 2 is y - x
In  + f2( - y^4 + x^4) factor 3 is y + x
In  + f2( - y^4 + x^4) factor 4 is y^2 + x^2

```

One thing to note is that the use of

```
f(<$x[1]>, ..., <$x[$x[0]]>)
```

is illegal. `$x[0]` will be inserted during execution time, while the expansion of the triple dot operator is done by the preprocessor. Hence we should use '`$x[0]`' but then `$x` must be known and factorized already at compile time.

6.1 Dollar variables in a parallel environment

When FORM is used for parallel processing, either by means of PARFORM or by means of TFORM, there can be a problem with the dollar variables as in principle there is a central administration and dollar variables that are defined during running will in general have the last assigned value. In a parallel environment this can be nondeterministic. Look for instance at the following example:

```

S    x,a,b;
CF   f;
L    F = f(a+b) + f(a+2*b);
.sort
id   f(x?$x) = f(x);
Multiply,$x;
Print;
.end

```

Usually this program will give the 'correct' answer, but in principle one thread could define `$x` and then the next thread could overwrite this value before the first thread has used it. This is serious. Hence FORM will veto the use of multiple threads/processors for modules in which dollar variables obtain values during the execution of the program, unless the user can give FORM more information about the use of the dollar variables. In the above case the value of `$x` will be local to each term and hence to each thread. The value in previous terms is unimportant. We can tell this to FORM with a variety of the moduleoption statement (see 7.89). This would be:

```

S    x,a,b;
CF   f;
L    F = f(a+b) + f(a+2*b);
.sort
id   f(x?$x) = f(x);
Multiply,$x;
Print;
ModuleOption,local,$x;
.end

```

In this case FORM makes at the start of the execution of the module a copy of whatever value `$x` has at that moment for each thread/processor (in this case no value yet and hence it gets set to zero) and then each thread/processor uses its own copy during execution. After the module has

been completed the local copies are removed and the original global value is accessible again. This way execution will be safe in a parallel environment.

There are more cases that FORM can handle in a parallel environment. These are also options in the moduleoption statement:

```
ModuleOption,maximum,$a;
ModuleOption,minimum,$b;
ModuleOption,sum,$c;
```

Here we say that \$a is accumulating a maximum numerical value, \$b collects a minimum numerical value and \$c is a numerical sum. In all three cases there is a central administration and the use of the variables has to be blocked for other threads/processors during the updating of the values. Sometimes that can be efficient, but in other programs that may actually make them slower. One should experiment. A sample program is given below:

```
S  a1,...,a10;
L  F = (a1+...+a10)^3;
.sort
#$c = 0;
Print +f "<%w> %t";
Multiply,(a1+...+a10);
$c = $c+1;
ModuleOption,sum,$c;
.sort
#message $c = '$c'
#$max = 0;
#$min = 10;
if ( count(a1,1) > $max ) $max = count_(a1,1);
if ( count(a4,1) < $min ) $min = count_(a4,1);
ModuleOption,maximum,$max;
ModuleOption,minimum,$min;
.sort
#message $max = '$max'
#message $min = '$min'
.end
```

The print statement is showing which thread is dealing with which term.

Chapter 7

Statements

7.1 abrackets, antibrackets

Type Output control statement
Syntax `ab[rackets][+][-] <list of names>;`
 `antib[rackets][+][-] <list of names>;`
See also bracket (7.11) and the chapter on brackets (9)

This statement does the opposite of the bracket statement (see 7.11). In the bracket statement the variables that are mentioned are placed outside brackets and inside the brackets are all other objects. In the antibracket statement the variables in the list are the only objects that are not placed outside the brackets. For the rest of the syntax, see the bracket statement (section 7.11).

7.2 also

Type Executable Statement
Syntax `a[lso] [options] <pattern> = <expression>;`
See also identify (7.68), idold (7.70)

The also statement should follow either an id statement or another also statement. The action is that the pattern matching in the also statement takes place immediately after the pattern matching of the previous id statement (or also statement) and after possible matching patterns have been removed, but before the r.h.s. expressions are inserted. It is identical to the idold statement (see 7.70). Example:

```
id      x = cosphi*x-sinphi*y;
also    y = sinphi*x+cosphi*y;
```

The options are explained in the section on the id statement (see 7.68).

7.3 antiputinside

Type Executable statement
Syntax antiputinside <name of function> [,<antibracket information>];
See also PutInside (7.120)

This statement puts all parts of the term with the exception of the variables in the antibracket information inside a function argument. The function must be a regular function (hence no tensor or table which are special types of functions). The antibracket information should adhere to the syntax of the bracket statement (7.11, 7.1) and all occurrences of all variables with the exception of the antibracket variables will be put inside the function. The coefficient will also be put inside the function.

7.4 antisymmetrize

Type Executable statement
Syntax an[tisymmetrize] {<name of function/tensor> [<argument specifications>];}
See also symmetrize (7.142), cyclesymmetrize (7.28), rcyclesymmetrize (7.122)

The argument specifications are explained in the section on the symmetrize statements (see 7.142).

The action of this statement is to anti-symmetrize the (specified) arguments of the functions that are mentioned. This means that the arguments are brought to ‘natural order’ in the notation of FORM and each exchange of arguments or groups of arguments results in a minus sign in the coefficient of the term. The ‘natural order’ may depend on the order of declaration of the variables. If two arguments or groups of arguments that are part in the anti-symmetrization are identical, the function is replaced by zero.

7.5 apply

Type Executable statement
Syntax apply [”<tablename(s)>”];
See also tablebases (12), apply (12.2)

This statement is explained in the chapter on tablebases.

7.6 argexplode

Type Executable statement
Syntax argexplode [<list of functions>]
See also argimplode (7.7)

See the description of the ArgImplode 7.7 statement.

7.7 argimplode

Type Executable statement
Syntax `argimplode [<list of functions>]`
See also `argexplode` (7.6)

This is a rather specialized statement. It converts one notation of indices, used for harmonic sums, harmonic polylogarithms and multiple zeta values into its alternative notation. The two notations are:

$Z(0,0,0,1,0,0,-1)$
 $Z(4,-3)$

In the first notation the indices can only be 0, 1 and -1. In the second notation there can be no zeroes. The ‘ArgImplode,Z;’ statement would be equivalent to the statement

```
repeat id Z(?a,0,x?!{0,0},?b) = Z(?a,x+sig_(x),?b);
```

and takes one from the first notation to the second. The ‘ArgExplode,Z;’ statement is equivalent to the statement

```
repeat id Z(?a,x?!{1,0,-1},?b) = Z(?a,0,x-sig_(x),?b);
```

and takes one from the second notation to the first. The reason that these statements have been built in lies in the fact that for many indices the repeat statements started to become very time-consuming.

For the harmonic sums, the harmonic polylogarithms and the multiple zeta values one can use the `summer6` and the `harmpol` packages in the FORM distribution. They are described in the papers

J. A. M. Vermaseren, *Harmonic sums, Mellin transforms and integrals*, *Int. J. Mod. Phys. A***14** (1999) 2037, <http://arxiv.org/abs/hep-ph/9806280>.

E. Remiddi and J. A. M. Vermaseren, *Harmonic polylogarithms*, *Int. J. Mod. Phys. A***15** (2000) 725, <http://arxiv.org/abs/hep-ph/9905237>.

7.8 argtoextrasymbol

Type Executable statement
Syntax `argtoextrasymbol [tonumber] [<argument specifications>];`
See also `topolynomial` (7.149) and `extrasymbols` (7.53, 2.11).

Converts function arguments into extra symbols. An argument will be replaced with an extra symbol. The arguments that have been encountered before are replaced with the same extra symbols. Unlike the `topolynomial` statement (7.149), the replacement occurs even for arguments consisting only of numbers and symbols (including extra symbols).

The `tonumber` option requests that function arguments are converted to positive integers corresponding to extra symbols. This provides an efficient mapping from any expression (stored as a function argument) to a number.

The function arguments to be converted can be specified in the same way as the `argument` statement (see 7.9).

7.9 argument

Type Executable statement
Syntax argument [<argument specifications>]
 {<name of function/set> [<argument specifications>]};
See also endargument (7.43)

This statement starts an argument environment. Such an environment is terminated by an endargument statement (see 7.43). The statements between the argument and the endargument statements will be applied only to the function arguments as specified by the remaining information in the argument statement. This information is given by:

- No further information: the statements are applied to all arguments of all functions.
- A series of numbers: the statements are applied to the given arguments of all functions.
- A function name (or a set of functions), possibly followed by a series of numbers: the statements are applied to the numbered arguments of the function specified. If a set of functions was specified, all the functions in the set will be taken. If no numbers are specified, all arguments of the function (or elements of the set) are taken.

The combination of a function (or set) possibly followed by numbers of arguments, can occur as many times as needed. The generic numbers of arguments that refer to all functions work in addition to the numbers specified for individual functions. Example

Argument 2,f,1,{f,f1},3,4;

This specifies the second argument of all functions. In addition the first argument of **f** will be taken and then also the third and fourth arguments of **f** and **f1** will be taken.

Argument/endargument constructions can be nested.

7.10 auto, autodeclare

Type Declaration statement
Syntax autodeclare <variable type> <list of variables to be declared>;
 auto <variable type> <list of variables to be declared>;

The variable types are

s[symbol]	Declaration of symbols. For options see 7.141.
v[ector]	Declaration of vectors. For options see 7.160.
i[index]	Declaration of indices. For options see 7.74.
i[ndices]	Declaration of indices. For options see 7.74.
f[unctions]	Declaration of noncommuting functions. For options see 7.95.
nf[unctions]	Declaration of noncommuting functions. For options see 7.95.
cf[unctions]	Declaration of commuting functions. For options see 7.14.

<code>co[mmuting]</code>	Declaration of commuting functions. For options see 7.14.
<code>t[ensors]</code>	Declaration of commuting tensors. For options see 7.145.
<code>nt[ensors]</code>	Declaration of noncommuting tensors. For options see 7.103.
<code>ct[ensors]</code>	Declaration of commuting tensors. For options see 7.27.

The action of the `autodeclare` statement is to set a default for variable types. In a statement of the type

```
AutoDeclare Symbol a,bc,def;
```

all undeclared variables of which the name starts with the character `a`, the string `bc` or the string `def` will be interpreted as symbols and entered in the name tables as such. In the case there are two statements as in

```
AutoDeclare CFunction b,d;
AutoDeclare Symbol a,bc,def;
```

all previously undeclared variables of which the name starts with `a`, `bc` or `def` will be declared as symbols. All other previously undeclared variables of which the name starts with a `b` or a `d` will be declared as commuting functions. This is independent of the order of the `autodeclare` statements. FORM starts looking for the most detailed matches first. Hence the variable `defi` will match with the string `def` first.

It is also allowed to use the properties of the various variables in the `autodeclare` statement:

```
AutoDeclare Index i=4,i3=3,i5=5;
```

This declares all previously undeclared variables of which the name starts with an `i` to be four dimensional indices, unless their names start with `i3` in which case they will be three dimensional indices, or their names start with `i5` in which case they will be five dimensional indices.

7.11 bracket

Type	Output control statement
Syntax	<code>b[rackets][+][-] <list of names>;</code>
See also	<code>antibracket</code> (7.1), <code>keep</code> (7.80), <code>collect</code> (7.19) and the chapter on brackets (9)

This statement causes the output to be reorganized in such a way that all objects in the ‘list of names’ are placed outside brackets and all remaining objects inside brackets. This grouping will remain till the next time that the expression is active and is being manipulated. Hence the brackets can survive `skip` (see 7.133), `hide` (see 7.67) and even `save` (see 7.128) and `load` (see 7.83) statements. The bracket information can be used by the `collect` (see 7.19) and `keep` (see 7.80) statements, as well in r.h.s. expressions when the contents of individual brackets of an expression can be picked up (see 9).

The list of names can contain names of symbols, vectors, functions, tensors and sets. In addition it can contain dotproducts. There should be only one bracket or antibracket (see 7.1) statement in each module. If there is more than one, only the last one has an effect. The presence of a set

has the same effect as having all the symbolic elements of the set declared in the (anti)bracket statement.

The presence of a + or - after the bracket (or anti bracket) refers to potential indexing of the brackets. Usually FORM has the information inside the terms in an expression. If it needs to search for a particular bracket it does so by starting at the beginning of that expression. This can be slow. If one likes to access individual brackets, it may be faster to tell FORM to make an index by putting the + after the bracket or antibracket keyword. For more information, see the chapter on brackets (see 9). A - indicates that no index should be made. Currently this is the default and hence there is no need to use this option. It is present just in case the default might be changed in a future version of FORM (in which FORM might for instance try to determine by itself what seems best. This option exists for case that the user would like to overrule such a mechanism).

See also the antibracket statement in 7.1.

7.12 break

Type	Executable statement
Syntax	break;

See also case (7.13), switch (7.140), default(7.30), endswitch (7.49).

When a break statement is reached in a switch construction the next statement to be executed is the first statement after the corresponding endswitch statement.

7.13 case

Type	Executable statement
Syntax	case,number;

See also switch (7.140), break (7.12), default(7.30), endswitch (7.49).

The cases in a switch construction are marked by a number. This number must be an integer that can be represented inside a FORM word. On a 64-bit processor this would be an integer in the range -2^{31} to $2^{31} - 1$. If the dollar variable in the switch statement has the same value as the integer in the case statement, the next statement to be executed is the first statement after the case statement. Usually cases are terminated by break statements, but if there is no break statement 'fall through' may occur in which execution continues with the first statement after the next case statement or default statement.

7.14 cfunctions

Type	Declaration statement
Syntax	c[functions] <list of functions to be declared>;
See also	functions (7.62), nfunctions (7.95)

This statement declares commuting functions. The name of a function can be followed by some information that specifies additional properties of the preceding function. These can be (name indicates the name of the function to be declared):

name#r	The function is considered to be a real function (default).
name#c	The function is considered to be a complex function. This means that internally two spaces are reserved. One for the variable name and one for its complex conjugate name#.
name#i	The function is considered to be imaginary.
name(s[ymmetric])	The function is totally symmetric. This means that during normalization FORM will order the arguments according to its internal notion of order by trying permutations. The result will depend on the order of declaration of variables.
name(a[ntisymmetric])	The function is totally antisymmetric. This means that during normalization FORM will order the arguments according to its internal notion of order and if the resulting permutation of arguments is odd the coefficient of the term will change sign. The order will depend on the order of declaration of variables.
name(c[yclesymmetric])	The function is cycle symmetric in all its arguments. This means that during normalization FORM will order the arguments according to its internal notion of order by trying cyclic permutations. The result will depend on the order of declaration of variables.
name(r[cyclesymmetric])	The function is reverse cycle symmetric in all its arguments. This means that during normalization FORM will order the arguments according to its internal notion of order by trying cyclic permutations and/or a complete reverse order of all arguments. The result will depend on the order of declaration of variables.
name(r[cyclic])	
name(r[eversecyclic])	

The complexity properties and the symmetric properties can be combined. In that case the complexity properties should come first as in

```
CFunction f1#i(antisymmetric);
```

7.15 chainin

Type	Executable statement
Syntax	Chainin,name of function;
See also	chainout (7.16)

Has the same effect as the statement

```
repeat id f(?a)*f(?b) = f(?a,?b);
```

if `f` is the name of the function specified. The `chainin` statement is just a faster shortcut.

7.16 chainout

Type Executable statement
 Syntax Chainout,name of function;
 See also chainin (7.15)

Has the same effect as the statement

```
repeat id f(x1?,x2?,?a) = f(x1)*f(x2,?a);
```

if `f` is the name of the function specified. The `chainout` statement is just a much faster shortcut.

7.17 chisholm

Type Executable statement
 Syntax chisholm [options] <spline indices>;
 See also trace4 (7.153) and the chapter on gamma algebra (14)

This statement applies the identity

$$\gamma_a \gamma_\mu \gamma_b \text{Tr}[\gamma_\mu S] = 2\gamma_a (S + S^R) \gamma_b$$

in order to contract traces. S is here a string of gamma matrices and S^R is the reverse string. This identity is particularly useful when the matrices $\gamma_6 = 1 + \gamma_5$ and/or $\gamma_7 = 1 - \gamma_5$ are involved. The `spline` index refers to which trace should be eliminated this way. The options are

symmetrize	If there is more than one contraction with other gamma matrices, the answer will be the sum of the various contractions, divided by the number of different contractions. This will often result in a minimization of the number of γ_5 matrices left in the final results.
nosymmetrize	The first contraction encountered will be taken. No attempt is made to optimize with respect to the number of γ_5 matrices left.

IMPORTANT: the above identity is only valid in 4 dimensions. For more details, see chapter 14 on gamma algebra.

7.18 cleartable

Type Declaration statement
 Syntax ClearTable [<list of tables>]

This statement clears the tables that are mentioned. Sometimes (sparse) tables can take so much space that there is no room for new elements, while old elements are not needed any longer. In that case one can clear the table and start all over again with filling it. It is also useful when one wants to reuse a table, but now with a different content.

7.19 collect

Type	Specification statement
Syntax	collect <name of function>; collect <name of function> <name of other function>; collect <name of function> <name of other function> <percentage>;
See also	bracket (7.11), antibracket (7.1) and the chapter on brackets (9)

Upon processing the expressions (hence expressions in hide as well as skipped expressions do not take part in this) the contents of the brackets (if there was a bracket or antibracket statement in the preceding module) are collected and put inside the argument of the named function. Hence if the expression F is given by

```
F =
  a*(b^2+c)
+ a^2*(b+6)
+ b^3 + c*b + 12;
```

the statement

```
Collect cfun;
```

will change F into

```
F = a*cfun(b^2+c)+a^2*cfun(b+6)+cfun(b^3+c*b+12);
```

The major complication occurs if the content of a bracket is so long that it will not fit inside a single term. The maximum size of a term is limited by the setup parameter `maxtermsize` (see 17). If this size is exceeded, FORM will split the bracket contents over more than one term, in each of which it will be inside the named function. It will issue a warning that it has done so.

If a second function is specified (the alternative collect function) and if a bracket takes more space than can be put inside a single term, the bracket contents will be split over more than one term, in each of which it will be inside the alternative collect function. In this case there is no need for a warning as the user can easily check whether this has occurred by checking whether the alternative function is present in the expression.

If additionally a percentage is specified (an integer in the range of 1 to 99) this determines how big the argument must be as compared to `MaxTermSize` (see chapter 17 on the setup) before use is made of the alternate collect function.

7.20 commuteinset

Type Declaration statement
Syntax `commuteinset <{list of noncommuting functions/tensors}>;`
See also functions (7.62)

This statement allows one or more sets of noncommuting functions and or tensors for its argument(s). The functions inside each set will commute with each other. It is allowed to have the same function inside more than one set. For a function to commute with itself (with for instance different arguments) it needs to be specified twice inside the same set. In that case it is more efficient to have a separate set with only two arguments. Example:

```
I  i1,...,i10;
F  A1,...,A10;
CommuteInSet{A1,A3,A5},{A1,g_},{A1,A1};
L  F = A5*A1*A5*A1*A5*A2*A3*A5*A1*A5*A3*A1;
L  G = g_(2,i1)*g_(2,i2,i3)*A1(i2)*g_(1,i4)*g_(1,5_,i5,i6)
      *A1(i1)*A1(i3)*g5_(1)*A3(i5)*A3(i4)*g5_(1);

Print +f +s;
.end

F =
+ A1*A1*A5*A5*A5*A2*A1*A1*A3*A3*A5*A5;
G =
+ g_(1,i4,i5,i6)*g_(2,i1,i2,i3)*A1(i1)*A1(i2)*A1(i3)*
  A3(i5)*A3(i4)*g_(1,5_);
```

7.21 commuting

Type Declaration statement
Syntax `co[mmuting] <list of functions to be declared>;`
See also cfunctions (7.14), functions (7.62)

This statement is completely identical to the cfunction statement (see 7.14).

7.22 compress

Type Declaration statement
Syntax `comp[ress] <on/off>;`
See also on (7.108), off (7.107)

This statement is obsolete. The user should try to use the compress option of the on (see 7.108) or the off (see 7.107) statements.

7.23 contract

Type Executable statement
Syntax `contract [<argument specifications>];`

Statement causes the contraction of pairs of Levi-Civita tensors $\mathbf{e}_-$ (see also 8) into combinations of Kronecker delta's. If there are contracted indices, and if their dimension is identical to the number of indices of the Levi-Civita tensors, the regular shortcuts are taken. If there are contracted indices with a different dimension, the contraction treats these indices temporarily as different and lets the contraction be ruled by the contraction mechanism of the Kronecker delta's. In practise this means that the dimension will enter via $\delta_\mu^\mu \rightarrow \dim(\mu)$.

In FORM there are no upper and lower indices. Of course the user can emulate those. The `contract` statement always assumes that there is a proper distribution of upper and lower indices if the user decided to work in a metric in which this makes a difference. Note however that due to the fact that the Levi-Civita tensor is considered to be imaginary, there is usually no need to do anything special. This is explained in the chapter on functions (see 8).

There are several options to control which contractions will be taken. They are

<code>Contract;</code>	Here only a single pair of Levi-Civita tensors will be contracted. The pair that is selected by FORM is the pair that will give the smallest number of terms in their contraction.
<code>Contract <number>;</code>	This tells FORM to keep contracting pairs of Levi-Civita tensors until there are <number> or <number>+1 Levi-Civita tensors left. A common example is <code>Contract 0;</code> which will contract as many pairs as possible.
<code>Contract:<number>;</code>	Here the number indicates the number of indices in the Levi-Civita tensors to be contracted. Only a single pair will be contracted and it will be the pair that gives the smallest number of terms.
<code>Contract:<number> <number>;</code>	The First number refers to the number of indices in the Levi-Civita tensors to be contracted. The second number refers to the number of Levi-Civita tensors that should be left (if possible) after contraction.

Note that the order in which FORM selects the contractions is by looking at which pair will give the smallest number of terms. This means that usually the largest buildup of terms is at the end. This is not always the case, because there can be a complicated network of contracted indices.

7.24 copyspectator

Type Executable statement
Syntax `copyspectator <exprname = spectator>;`

See chapter20 on spectators.

7.25 createspectator

Type Executable statement
Syntax `createspectator <spectatorname, "filename">;`

See chapter20 on spectators.

7.26 ctable

Type Declaration statement
Syntax `ctable <options> <table to be declared>;`
See also functions (7.62), table (7.143), ntable (7.102)

This statement declares a commuting table and is identical to the table command (see 7.143) which has the commuting property as its default.

7.27 ctensors

Type Declaration statement
Syntax `ct[ensors] <list of tensors to be declared>;`
See also functions (7.62), tensors (7.145), ntensors (7.103)

This statement declares commuting tensors. It is equal to the tensor statement (see 7.145) which has the commuting property as its default.

7.28 cyclesymmetrize

Type Executable statement
Syntax `cy[clesymmetrize] {<name of function/tensor> [<argument specifications>];}`
See also symmetrize (7.142), antisymmetrize (7.4), rcyclesymmetrize (7.122)

The argument specifications are explained in the section on the symmetrize statements (see 7.142).

The action of this statement is to cycle-symmetrize the (specified) arguments of the functions that are mentioned. This means that the arguments are brought to ‘natural order’ in the notation of FORM by trying cyclic permutations of the arguments or groups of arguments. The ‘natural order’ may depend on the order of declaration of the variables.

7.29 deallocateTable

Type	Declaration
Syntax	DeallocateTable,name(s) of sparse table(s);
See also	table (7.143), fill (7.57), table bases (12)

Works only for sparse tables. Deallocates all definitions of elements as obtained with ‘Fill’ statements as if there never were any ‘Fill’ statements for the given tables.

This statement exists because sometimes cleaning up big tables is needed when they take too much memory. This can be the case when a big tablebase has been used.

7.30 default

Type	Executable statement
Syntax	default;

See also case (7.13), break (7.12), switch(7.140), endswitch (7.49).

This is the default case in a switch construction.

7.31 delete

Type	Specification statement
Syntax	delete storage;
See also	save (7.128), load (7.83)
Syntax	delete extrasymbols;
Syntax	delete extrasymbols>number;
See also	extrasymbols (7.53)

This statement has currently two varieties. The delete storage clears the complete storage file and reduces it to zero size. The effect is that all stored expressions are removed from the system. Because it is impossible to remove individual expressions from the store file (there is no mechanism to fill the resulting holes) it is the only way to clean up the storage file. If some expressions should be excluded from this elimination process, one should copy them first into active global expressions, then delete the storage file, after which the expressions can be written to storage again with a .store instruction.

The delete extrasymbols variety removes extra symbols from the list. The default is that all extra symbols are removed, but one can also remove the symbols above a given number as in

```
#$es = 'extrasymbols_';
ToPolynomial;
....some code....
.sort
* now the new extra symbols are not needed anymore
Delete extrasymbols>'$es';
```

7.32 denominators

Type Executable statement
Syntax `denominators functionname;`

This statement allows the user to rename all occurrences of the built-in denominator function. This built-in function is kind of an oddity inside FORM. Denominators are presented by a very special function which doesn't really have a name and hence is rather hard to address. In addition there are special rules connected to denominators. Hence it is usually better to collect denominators inside functions that have been defined by the user and hence allow the user to manipulate them at will. Yet, objects can end up inside denominator functions, especially when output from other programs is read in. Hence this statement allows all occurrences of the denominator function to be renamed into the function that is given in the statement. This function will work well together with the `PolyRatFun` statement in which we define a `PolyFun` with two arguments of which the second acts as a denominator and the first as a numerator:

```
PolyRatFun, rat;  
Denominators, den;  
id den(x?) = rat(1,x);
```

For more about this one should consult the part on the `PolyRatFun` statement (7.112) and the chapter on polynomials (still to be included because the current version can handle only polynomials in a single variable and is also not optimized for many occurrences that have identical denominators).

7.33 dimension

Type Declaration statement
Syntax `d[imension] <number or symbol>;`
See also index (7.74)

Sets the default dimension. This default dimension determines the dimension of the indices that are being declared without dimension specification as well as the dimension of all dummy indices. At the moment an index is declared and there is no dimension specification, FORM looks for the default dimension and uses that. This index will then have this dimension, even when the default dimension is changed at a later moment. The dummy indices always have the dimension of the current default dimension. If the default dimension is changed the dimension of all dummy indices changes with it. Varieties:

`Dimension <number>;` Declares the number to be the default dimension. The number must be smaller than 32768 on 32bit architectures or 2147483648 on 64bit architectures. Negative numbers are not allowed. If one wants to work with negative dimensions, the practical workaround is to use a symbolic dimension and later replace that symbol appropriately.

`Dimension <symbol>;` Symbol must be the name of a symbol, either previously declared or declarable because of an auto-declaration (see 7.10). Declares the symbol to be the default dimension.

Dimension The symbols must be the names of symbols, either previously declared <symbol>:<symbol>; or declarable because of an auto-declaration (see 7.10). The first symbol will be the default dimension. The second symbol will be the first symbol minus 4. It will be used as such in the trace contractions. See also 7.154 and 7.74.

Examples:

```
Dimension 3;
Dimension n;
Dimension n:[n-4];
```

The default dimension in FORM is 4.

7.34 discard

Type Executable statement
Syntax dis[card];

This statement discards the current term. It can be very useful in statements of the type

```
if ( count(x,1) > 5 ) Discard;
```

which eliminates all terms that have more than five powers of x.

7.35 disorder

Type Executable statement
Syntax disorder <pattern> = <expression>;
See also identify (7.68)

This statement is identical to the disorder option of the id statement (see 7.68). It is just a shorthand notation for ‘id disorder’.

7.36 do

Type Executable statement
Syntax do \$loopvar = lowvalue,highvalue{,increment};
See also enddo (7.44)

The syntax is the typical syntax for do-loops. The loop variable has to be a dollar variable. For parallel performance this variable can be declared local in a moduleoption (see 7.89) statement, unless it is also used in other ways in the current module. The loop parameters should either be (short) integers or dollar variables or factors of dollar variables provided they evaluate at run time

to (short) integers. The `enddo` statement should be in the same module as the `do` statement. In addition it should be properly nested with `if`, `repeat`, `while` and argument constructions. The `do-loop` facility is in principle superfluous, because the `repeat` (7.126), `if` (7.71) and the pattern matcher can basically do everything the `do-loop` can do. Sometimes however the `do-loop` is easier to program and gives more readable code as shown here:

```
do $i = 1,5;
  id,only,x^$i = f(F[factor_^$i]);
enddo;
```

versus

```
id,only,x^n?{1,2,3,4,5} = ff(n);
repeat id ff(n?pos_) = ff(n-1)*f(F[factor_^n]);
id ff(n?neg0_) = 1;
```

One should note that the `do-loop` is evaluated at run time. Hence the dollar variables need to be evaluated at run time as well. Therefore, if it is possible, the preprocessor variety (see 3.19) is almost always faster in execution as in

```
#do i = 1,5
  id,only,x^'i' = f(F[factor_^'i']);
#enddo
```

This can of course not be done in constructions like

```
id f1(x?$x) = f2(x);
FactDollar,$x;
Do $i = 1,$x[0];
  Multiply f($i,$x[$i]);
Enddo;
```

because here `$x` and its factors are only known at run time and may be different for each term.

7.37 drop

Type	Specification statement
Syntax	<code>drop;</code> <code>drop <list of expressions>;</code>
See also	<code>ndrop</code> (7.93)

In the first variety this statement eliminates all expressions from the system. In the second variety it eliminates only the expressions that are mentioned from the system. All expressions that are to be dropped can still be used in the r.h.s. of other expressions inside the current module. Basically the expressions to be dropped are not treated for execution and after the module has finished completely they are removed. See also the `ndrop` statement 7.93.

7.38 dropcoefficient

Type Executable statement
Syntax DropCoefficient;

This statement replaces the coefficient of the current term by one. In principle it has the same effect as

```
Multiply 1/coeff_;
```

but there is always the philosophical issue what is the coefficient once one enters function arguments. Inside an Argument/EndArgument environment this statement would drop the coefficient of the terms inside the argument.

7.39 dropsymbols

Type Executable statement
Syntax DropSymbols;

This statement removes all symbols from a term. It has the same effect as

```
id,many,x?^n? = 1;
```

(x and n are symbols) except for that it is much faster.

7.40 else

Type Executable statement
Syntax else;
See also if (7.71), elseif (7.41), endif (7.45)

To be used in combination with an if statement (see 7.71). The statements following the else statement until the matching endif statement (see 7.45) will be executed for the current term if the conditions of the matching proceeding if statement and/or all corresponding elseif statements (see 7.41) are false. If any of the conditions of the matching proceeding if or elseif statements are true the statements following the else statement will be skipped.

7.41 elseif

Type Executable statement
Syntax elseif (<condition>);
See also if (7.71), else (7.40), endif (7.45)

Should be preceded by an if statement (see 7.71) and followed at least by a matching endif statement (see 7.45). If the conditions of the proceeding matching if statement and all proceeding matching elseif statements are false the condition of this elseif statement will be evaluated. If it is true, the statements following it until the next matching elseif, else or endif statement will be executed. If not, control is passed to this next elseif, else or endif statement. The syntax for the condition is exactly the same as for the condition in the if statement.

7.42 emptyspectator

Type Executable statement
Syntax emptyspectator <spectator>;

See chapter20 on spectators.

7.43 endargument

Type Executable statement
Syntax endargument;
See also argument (7.9)

Terminates an argument environment (see 7.9). The argument statement and its corresponding endargument statement must belong to the same module. Argument environments can be nested with all other environments.

7.44 enddo

Type Executable statement
Syntax enddo;
See also do (7.36)

See the do statement (7.36).

7.45 endif

Type Executable statement
Syntax endif;
See also if (7.71), elseif (7.41), else (7.40)

Terminates an if construction (see 7.71, 7.41 and 7.40). It should be noted that if constructions can be nested.

7.46 endinexpression

Type Executable statement
Syntax endinexpression;
See also inexpression(7.75)

Only to be used in combination with the inexpression statement. The combination

```
InExpression,expr;  
    Statements;  
EndInExpression;
```

is a more readable version of the construction

```
if ( expression(expr) );  
    Statements;  
endif;
```

7.47 endinside

Type Executable statement
Syntax endinside;
See also inside (7.77) and the chapter on \$-variables (6)

Terminates an ‘inside’ environment (see 7.77) which is used to operate on the contents of \$-variables (see 6).

7.48 endrepeat

Type Executable statement
Syntax endrepeat;
See also repeat (7.126), while (7.161)

Ends the repeat environment. The repeat environment is started with a repeat statement (see 7.126). The repeat and its matching endrepeat should be inside the same module. Repeat environments can be nested with all other environments (and other repeat environments).

7.49 endswitch

Type Executable statement
Syntax endswitch;

See also case (7.13), break (7.12), default(7.30), switch (7.140).

Ends a switch construction. It collects the various cases, puts them in order and decides whether the lookup of cases should be done by means of a jumptable, or by binary searching. The ratio (spread in cases)/(number of cases) determines whether a jumptable is constructed. The default value below which a jumptable is constructed is 4. This value can be changed in the setups (see the section on the setups 17) with the variable jumpratio.

7.50 endterm

Type Executable statement
 Syntax endterm;
 See also term (7.146), sort (7.134)

Terminates a term environment (see 7.146). Term environments can be nested with other term environments and with other environments in general. The whole environment should be part of one single module. See also 7.134.

7.51 endwhile

Type Executable statement
 Syntax endwhile;
 See also while (7.161), repeat (7.126)

Terminates a while environment (see 7.161). The while statement and its corresponding endwhile statement must be part of the same module.

7.52 exit

Type Executable statement
 Syntax exit ["<string>"];
 See also setexitflag (7.131)

Causes execution to be aborted immediately. The string will be printed in the output. This can be used to indicate where FORM ran into the exit statement.

7.53 extrasymbols

Type Declaration statement
 Syntax extrasymbols,array|vector|underscore,name;
 See also ToPolynomial (7.149), FromPolynomial (7.61), ArgToExtraSymbol (7.8)
 and extra symbols (2.11).

Starting with version 4.0 of FORM some built in operations or statements can only deal with symbols and numbers. Examples of this are factorization (7.54) (which uses the topolynomial facilities automatically) and output simplification (see the Format statement 7.60). The ToPolynomial statement takes each term, looks for objects that are not symbols to positive powers and replaces them by symbols. If the object has been encountered before the same symbol will be used, otherwise a new symbol will be defined. The object represented by the ‘extra symbol’ is stored internally and can be printed if needed with the %X option in the #write instruction (3.65). The representation of the extra symbols is by default the name Z followed by a number and an underscore character. If another name is desired this should be specified in an ‘ExtraSymbols’ statement. The name given may contain only alphabetic characters! Because some compilers do not like the underscore character, there is an alternative notation for the extra symbols. This is just for cosmetic reasons and one cannot feed these symbols into the compiler this way. This is with an array notation. The statement

```
ExtraSymbols,array,Ab;
```

would cause the second extra symbol to be printed as **Ab(2)**. The total number of defined extra symbols is given by the built in symbol extrasymbols_. The option vector in the ExtraSymbols statement is identical to the option array and the option underscore reverts the notation back to the default notation with the trailing underscore.

7.54 factarg

Type	Executable statement
Syntax	factarg options {<name of function/set> [<argument specifications>]};
See also	splitarg (7.135)

Splits the indicated function arguments into individual factors. The argument specifications are as in the splitarg statement (see 7.135). There are a few extra options:

- (0) Eliminates the coefficient of the term in the argument. Similar to Normalize,(0),....
- (1) The coefficient of the term and its sign are pulled out separately.
- (-1) The coefficient is pulled out with its sign.

In the case of the above options only the coefficient is treated. When these options are not used the whole term is treated as in:

```
Symbols a,b,c;
CFunctions f,f1,f2,f3;
Local F = f(-3*a*b)+f(3*a*b)
          +f1(-3*a*b)+f1(3*a*b)
          +f2(-3*a*b)+f2(3*a*b)
          +f3(-3*a*b)+f3(3*a*b);
FactArg,f;
Factarg,(0),f1;
Factarg,(1),f2;
Factarg,(-1),f3;
Print;
```

```
.end
```

```
F =  
  f(a,b,-1,3) + f(a,b,3) + 2*f1(a*b) + f2(a*b,-1,3) + f2(a*b,3)  
  + f3(a*b,-3) + f3(a*b,3);
```

When no extra options are used, starting with version 4.0, the whole argument is factorized over the rationals. This means that

```
f(x^2+2*x*y+y^2) --> f(y + x,y + x,1)
```

It should be noticed that FORM can although the internal algorithms can only factorize expressions with numbers and symbols, FORM redefines all non-symbol objects temporarily into symbols and at the end substitutes them back. This is done with a mechanism that is similar to that of the ToPolynomial statement.

See also the On OldfactArg; and Off OldFactArg statements for a compatibility mode with versions before version 4.0.

7.55 factdollar

Type	Executable statement
Syntax	factdollar <name of dollar variable>;
See also	the chapter on polynomials 11.

The FactDollar statement will factorize a dollar expression. If the dollar expression was already factorized the old factors will be removed first. Unlike expressions (see 7.56) where only either the expanded or the factorized version exists, with dollar expressions we have both versions simultaneously. This means that one can refer to the complete dollar in its unfactorized form and its factors. The factors are indicated between braces as in $\$x[1]$ which would be the first factor. The number of factors of $\$x$ is given by $\$x[0]$. One can also obtain the number of factors of a dollar variable with the numfactors_ function (see 8.49).

The index indicating the number of the factor can be a nonzero integer, no greater than the number of factors, or (a factor of) a dollar variable that evaluates into such a number. Composite expressions are not allowed. They should be worked out first in a separate dollar variable, after which this dollar variable can then be used as a factor indicator.

7.56 factorize

Type	Output control statement
Syntax	factorize {<name of expression(s)>};
See also	the chapter on polynomials 11.

If no expressions are mentioned all expressions will be affected by the action of this statement. One may exclude certain expressions with the nfactorize statement (see 7.94). If one or more expressions are mentioned they will be added to the list of expressions that will be affected.

The statement causes the output expression(s) that is/are marked as such to be factorized after they have been processed and already written to the output. This means that each expression, after having been written, is read again and factorized. Then the factorized result is written over the original output. After that FORM will start executing the statements of the current module on the next expression, sort it, write it to output, and if necessary read it again and factorize it. Expressions never exists in two varieties as the dollar variable that have been factorized. It is either unfactorized (default) or factorized. An expression remains factorized until an UnFactorize statement is encountered that mentions that this expression should be brought to unfactorized representation (see also UnFactorize 7.157 and NunFactorize 7.104). One should realize that factorization of complicated expressions can be a rather costly operation.

7.57 fill

Type Declaration statement
 Syntax fill <tableelement> = <expression> [, <moreexpressions>];
 See also table (7.143), fillexpression (7.58), printtable (7.116)

The standard way to define elements of a table. In the left hand side one specifies the table element without the extra function arguments that could potentially occur (see 7.143). In the right hand side one specifies what the table element should be substituted by. Example:

```
Table tab(1:2,1:2,x?);
Fill tab(1,1) = x+y;
Fill tab(2,1) = (x+y)^2;
Fill tab(1,2) = tab(1,1)+y;
Fill tab(2,2) = tab(2,1)+y^2;
```

The first fill statement is a bit like a continuous attempt to try the substitution

```
id tab(1,1,x?) = x+y;
```

The last two fill statements show that one could use the table recursively. If a real loop occurs the program may terminate due to stack overflow.

It is possible to define several table elements in one statement. In that case the various elements are separated by commas. The last index is the first one to be raised. This means that in the above example one could have written:

```
Table tab(1:2,1:2,x?);
Fill tab(1,1) = x+y, tab(1,1)+y, (x+y)^2, tab(2,1)+y^2;
```

One warning is called for. One should avoid using expressions in the right hand side of fill statements:

```
Table B(1:1);
Local dummy = 1;
.sort
Fill B(1) = dummy;
```

```

Drop dummy;
.sort
Local F = B(1);
Print;
.end

```

In the example a crash will result, because when we use the table element the expression dummy doesn't exist anymore. In a fill statement the r.h.s. is not expanded. Hence it keeps the reference to the expression dummy. When the table element is used the reference to the expression dummy is inserted and expanded. Hence one obtains the contents of dummy that exist at the moment of use. This is illustrated in the following example:

```

Table B(1:1);
Local dummy = 1;
.sort
Fill B(1) = dummy;
.sort
Local F = B(1);
Print;
.sort
Drop;
.sort
Local dummy = 2;
.sort
Local F = B(1);
Print;
.end

```

The final value of F will be 2, not 1.

A way to get around this problem is to force the evaluation of the table definition by using dollar variables:

```

Table B(1:1);
Local dummy = 1;
.sort
#$value = dummy;
Fill B(1) = '$value';
Drop dummy;
.sort
Local F = B(1);
Print;
.end

```

Here we use the character representation of the contents of the dollar variable to obtain an expression that doesn't need any further evaluation. If we would put

```
fill B(1) = $value;
```

a reference to the dollar variable would be inserted and it would only be evaluated at use again. In principle this could cause similar problems.

Not dropping the expression dummy can sometimes give the correct result, but is potentially still unsafe.

```

Table B(1:1);
Local u = 2;
Local dummy = 1;
.sort
Fill B(1) = dummy;
Drop dummy;
.sort
Local v = 5;
Local F = B(1);
Print;
.end

```

Here the answer will be 5, because after `u` has been dropped the expressions will be renumbered. Hence now `dummy` becomes the first expression, and eventually `v` becomes the second expression. The references in the table elements are not renumbered. Hence the r.h.s. of `B(1)` keeps pointing at the second expression, which at the moment of application has the value 5. One can see now also why the original example crashes. First `dummy` was the first expression and at the moment of application `F` is the first (existing) expression. Hence the substitution of `B(1)` causes a self reference and hence an infinite loop. Eventually some buffer will overflow.

7.58 fillexpression

Type	Declaration statement
Syntax	<code>fillexpression <table> = <expression>(<x1>,...,<xn>);</code> <code>fillexpression <table> = <expression>(<funname>);</code>
See also	<code>table</code> (7.143), <code>fill</code> (7.57) and the <code>table_</code> function (8.68)

Used to dynamically load a table during runtime. When there are n symbols (here called `x1` to `xn`) it is assumed that the table is n -dimensional. The expression must previously have been bracketed in these symbols and each of the brackets has the effect of a fill statement in which the powers of the `x1` to `xn` refer to the table elements. Brackets that do not have a corresponding table element are skipped.

In the case that only a function name is specified the arguments of the function refer to the table elements.

7.59 fixindex

Type	Declaration statement
Syntax	<code>fi[xindex] {<number>:<value>;}</code>
See also	<code>index</code> (7.74) and chapter 15.

Defines `d_(number,number) = value` in which `number` is the number of a fixed index (hence a positive short integer with a value less than `ConstIndex` (see 17)). The value should be a short integer, i.e. its absolute value should be less than 2^{15} on 32 bit computers and less than 2^{31} on

64 bit computers. One can define more than one fixed index in one statement. Before one would like to solve problems involving the choice of a metric with this statement, one should consult the chapter on the use of a metric (chapter 15).

7.60 format

Type Output control statement
 Syntax fo[rmat] <option>;
 See also print (7.114)

Controls the format for the printing of expressions. There is a variety of options.

<number>	Output will be printed using the indicated number of characters per line. The default is 72. Numbers outside the range 1-255 are corrected to 72. Positive numbers less than 39 are corrected to 39.
float [<number>]	Numbers are printed in floating point notation, even though internally they remain fractions. This is purely cosmetic. If no number is specified the precision of the output will be 10 digits. If a number is specified it indicates the number of digits to be used for the precision.
rational	Output format is switched back to rational numbers (in contrast to floating point output). This is the default.
nospaces	The output is printed without the spaces that make the output slightly more readable. This gives a more compact output.
spaces	The output is printed with extra spaces between the terms and around certain operators to make it slightly more readable. This is the default.
O0	FORM will turn off output optimization. See the section on output optimization 10
O1[options]	FORM will use level 1 output optimization. See the section on output optimization 10
O2[options]	FORM will use level 2 output optimization. See the section on output optimization 10
O3[options]	FORM will use level 3 output optimization. See the section on output optimization 10.
fortran	The output is printed in a way that is readable by a fortran compiler. This includes continuation characters and the splitting of the output into blocks of no more than 15 continuation lines. This number can be changed with the setup parameter ContinuationLines (see 17). In addition dotproducts are printed with the ‘dotchar’ in the place of the period between the vectors. This dotchar can be set in the setup file (see 17). Its default is the underscore character.

doublefortran	Same as the fortran mode, but fractions are printed with double floating point numbers, because some compilers convert numbers like 1. into 1.E0. With this format FORM will force double precision by using 1.D0.
quadruplefortran	Same as the fortran mode, but fractions are printed with quadruple floating point numbers, because some compilers convert numbers like 1. into 1.E0. With this format FORM will force quadruple precision by using 1.Q0.
quadfortran	Same as quadruplefortran.
fortran90	Similar to the fortran option, but prints the continuation lines according to the syntax of Fortran 90. If the fortran90 option is followed by a comma and a string that does not contain white space or other comma's, this string is attached to all numbers in coefficients of terms. Example: <pre>Format Fortran90,.0_ki;</pre> which would give in the printout: <pre>+23.0_ki/32.0_ki*a**2& & +34.0_ki/1325.0_ki*a**3</pre> When there is no string attached it defaults to a period as in the regular Fortran option.
C	Output will be C compatible. The exponent operator ($\wedge$) is represented by the function pow. It is the responsibility of the user that this function will be properly defined. Dotproducts are printed with the 'dotchar' in the place of the period between the vectors. This dotchar can be set in the setup file (see 17). Its default is the underscore character.
maple	Output will be as much as possible compatible with Maple format. It is not guaranteed that this is perfect.
mathematica	Output will be as much as possible compatible with Mathematica format. It is not guaranteed that this is perfect.
reduce	Output will be as much as possible compatible with Reduce format. It is not guaranteed that this is perfect.
The last few formats	have not been tried out extensively. The author is open for suggestions.
normal	Will return to the regular FORM formatting mode.

If the statement has no arguments the formatting will be reset to the mode it was in when the program started.

7.61 frompolynomial

Type Executable statement
Syntax `frompolynomial`
See also `factarg` (7.54), `ToPolynomial` (7.149) and `ExtraSymbols` (7.53, 2.11).

Starting with version 4.0 of FORM some built in operations or statements can only deal with symbols and numbers. Examples of this are factorization (7.54) and output simplification (still to be implemented). Whereas the `ToPolynomial` statement takes each term, looks for objects that are not symbols to positive powers and replaces them by symbols the `FromPolynomial` does the opposite: it replaces the newly defined extra symbols and replaces them back by their original meaning.

7.62 functions

Type Declaration statement
Syntax `f[unctions] <list of functions to be declared>;`
See also `cfunctions` (7.14), `tensors` (7.145), `ntensors` (7.103),
`table` (7.143), `ntable` (7.102), `ctable` (7.26)

Used to declare one or more functions. The functions declared with this statement will be noncommuting. For commuting functions one should use the `cf[unctions]` statement (see 7.14). Functions can have a number of properties that can be set in the declaration. This is done by appending the options to the name of the function. These options are:

<code>name#r</code>	The function is considered to be a real function (default).
<code>name#c</code>	The function is considered to be a complex function. This means that internally two spaces are reserved. One for the variable name and one for its complex conjugate <code>name#</code> .
<code>name#i</code>	The function is considered to be imaginary.
<code>name(s[ymmetric])</code>	The function is totally symmetric. This means that during normalization FORM will order the arguments according to its internal notion of order by trying permutations. The result will depend on the order of declaration of variables.
<code>name(a[ntisymmetric])</code>	The function is totally antisymmetric. This means that during normalization FORM will order the arguments according to its internal notion of order and if the resulting permutation of arguments is odd the coefficient of the term will change sign. The order will depend on the order of declaration of variables.
<code>name(c[yclesymmetric])</code>	The function is cycle symmetric in all its arguments. This means that during normalization FORM will order the arguments according to its internal notion of order by trying cyclic permutations. The result will depend on the order of declaration of variables.

name(r[cyclesymmetric])	The function is reverse cycle symmetric in all its arguments. This means that during normalization FORM will order the arguments according to its internal notion of order by trying cyclic permutations and/or a complete reverse order of all arguments. The result will depend on the order of declaration of variables.
name(r[cyclic])	
name(r[eversecyclic])	
name;number	The function has a restriction on the number of arguments. If the number of arguments of an occurrence of the function is not fulfilling the condition during normalization FORM will set the term equal to zero.
name;=number	
name;number	
name;=number	

The complexity properties, the symmetric properties and the number of arguments restrictions can be combined. In that case the complexity properties should come first and the argument restrictions should come last as in

```
Function f1#i(symmetric)>=4<8;
Function f1#i<=8;
```

7.63 funpowers

Type	Declaration statement
Syntax	funpowers <on/off>;
See also	on (7.108), off (7.107)

This statement is obsolete. The user should try to use the funpowers option of the on (see 7.108) or the off (see 7.107) statements.

7.64 gfactorized

Type	Definition statement
Syntax	g[lobal]factorized <option>;
See also	the chapter on polynomials 11, the factorize statement 7.56 and the LocalFactorized statement 7.82.

The syntax is like the syntax of the LocalFactorized (or LFactorized) statement 7.82. The only difference is that now the expression defined by the statement will become a global expression (see the Global statement 7.65).

7.65 global

Type	Definition statement
Syntax	g[lobal] <name> = <expression>; g[lobal] <names of expressions>;
See also	local (7.84)

Used to define a global expression. A global expression is an expression that remains active until the first `.store` instruction. At that moment it is stored into the ‘storage file’ and stops being manipulated. After this it can still be used in the right hand side of expressions and id statements (see 7.69). Global expressions that have been put in the storage file can be saved to a disk file with the `save` statement (see 7.128) for use in later programs.

There are two versions of the global statement. In the first the expression is defined and filled with a right hand side expression. The left hand side and the right hand side are separated by an `=` sign. In this case the expression can have arguments which will serve as dummy arguments after the global expression has been stored with a `.store` instruction. Note that this use of arguments can often be circumvented with the `replace_` function (see 8.59) as in

```
Global F(a,b) = (a+b)^2;
.store
Local FF = F(x,y);
Local GG = F*replace_(a,x,b,y);
```

because both definitions give the same result.

The second version of the global statement has no `=` sign and no right hand side. It can be used to change a local expression into a global expression.

7.66 goto

Type	Executable statement
Syntax	<code>go[to] <label>;</code>
See also	label (7.81)

Causes processing to proceed at the indicated label statement (see 7.81). This label statement must be in the same module.

7.67 hide

Type	Specification statement
Syntax	<code>hide;</code> <code>hide <list of expressions>;</code>
See also	<code>nhide</code> (7.96), <code>unhide</code> (7.158), <code>nunhide</code> (7.105), <code>pushhide</code> (7.119), <code>pophide</code> (7.113)

In the first variety this statement marks all currently active expressions for being put in hidden storage. In the second variety it marks only the specified active expressions as such.

If an expression is marked for being hidden, it will be copied to the ‘hide file’, a storage which is either in memory or on file depending on the combined size of all expressions being hidden. If this size exceeds the size of the setup parameter `scratchsize` (see 17) the storage will be on file. If it is less, the storage will be in memory. An expression that has been hidden is not affected by the statements in the modules as long as it remains hidden, but it can be used inside other expressions in the same way skipped expressions (see 7.133) or active expressions can be used. In particular

all its bracket information (see 7.11) is retained and can be accessed, including possible bracket indexing.

The hide mechanism is particularly useful if an expression is not needed for a large number of modules. It has also advantages over the storing of global expressions after a .store instruction (see 4), because the substitution of global expressions is slower (name definitions may have changed and have to be checked) and also a possible bracket index is not maintained by the .store instruction.

Expressions can be returned from a hidden status into active expressions with the unhide statement (see 7.158). One might want to consult the nhide statement (7.67) as well.

When an expression is marked to be hidden it will remain just marked until execution starts in the current module. When it is the turn of the expression to be executed, it is copied to the hide file instead.

Note that a .store instruction will simultaneously remove all expressions from the hide system.

7.68 identify

Type	Executable statement
Syntax	id[entify] [<options>] <pattern> = <expression>;
See also	also (7.2), idnew (7.69), idold (7.70)

The statement tries to match the pattern. If the pattern matches one or more times, it will be replaced by the expression in the r.h.s. taking the possible wildcard substitutions into account. For the description of the patterns, see chapter 5.

The options are

multi	This option is for combinations of symbols and dotproducts only and it does not use wildcard powers. FORM determines how many times the pattern fits in one pattern matching action. Then the r.h.s. is substituted to that power. It is the default for these kinds of patterns.
many	This is the default for patterns that contain other objects than symbols and dotproducts. The pattern is matched and taken out. Then FORM tries again to match the pattern in the remainder of the term. This is repeated until there is no further match. Then for each match the r.h.s. is substituted (with its own wildcard substitutions).
select	This option should be followed by one or more sets. After the sets the pattern can be specified. The pattern will only be substituted if none of the objects mentioned in the sets will be left after the pattern has been taken out. This holds only for objects 'at ground level'; i.e. the pattern matcher will not look inside function arguments for this. Note that this is a special case of the option 'only'.

once	The pattern is matched only once, even if it occurs more than once in the term. The first match that FORM encounters is taken. When wildcards are involved, this may depend on the order of declaration of variables. It could also be installation dependent. Also the setting of properorder (see 7.108 and 7.107) could be relevant. Try to write programs in such a way that the outcome does not depend on which match is taken.
only	The pattern will match only if there is an exact match in the powers of the symbols and dotproducts present.
ifmatch->	This option should be followed by the name (or number) of a label. If the pattern matches, the replacement will be made after which the execution continues at the label.
ifnomatch->	This option should be followed by the name (or number) of a label. If the pattern does not match, execution continues at the label.
disorder	This option is used for products of noncommuting functions or tensors. The match will only take place if the order of the functions in the match is different from what FORM would have made of it if the functions would be commuting. Hence if the functions in the term are in the order that FORM would give them if they would be commuting (which depends on the order of declaration) there will be no match. This can be rather handy when using wildcards as in $F(a?)*F(b?)$.
all	This option is rather special in that it generates all possible matches one by one. Normally, when there are many possible matches, FORM takes the first one it encounters. In the case of the all option it will run through all possible matches and produce all of them. There are however severe restrictions. First of all, other options are not allowed simultaneously, although ifmatch-> and ifnomatch-> are allowed because technically they are no options that concern the pattern matching. In addition it is not allowed to be in an idold/also statement, and it cannot be followed by such a statement. Most severely: it can have only functions in the left hand side. These functions can have all kinds of arguments, but outside the functions symbols, vectors, dotproducts etc. are not allowed. This is due to the fact that the backtracking when a wildcard combination fails, does not include such objects and it is this backtracking mechanism that is used to generate all matches. For the purpose of the all option tensors and unsubstituted tables count as functions. It should also be known that the all option cannot be used in the if(match()) construction. It would not make sense there anyway.

Example:

```

Vector Q,p1,...,p5,q1,...,q5;
Cfunction V(s),replace;
Format 60;
*   This is a t1 topology:
L   F = V(Q,p1,p4)*V(p1,p2,p5)*
      V(p2,p3,Q)*V(p3,p4,p5);
$t = term_;
id,all,$t*replace_(<p1,p1?>,...,<p5,p5?>) =

```

```

    $t*replace(<p1,q1>,...,<p5,q5>);
Print +s;
.end

F =
+ V(Q,p1,p4)*V(Q,p2,p3)*V(p1,p2,p5)*V(p3,p4,p5)*
replace(p1,q1,p2,q2,p3,q3,p4,q4,p5,q5)
+ V(Q,p1,p4)*V(Q,p2,p3)*V(p1,p2,p5)*V(p3,p4,p5)*
replace(p2,q1,p1,q2,p4,q3,p3,q4,p5,q5)
+ V(Q,p1,p4)*V(Q,p2,p3)*V(p1,p2,p5)*V(p3,p4,p5)*
replace(p3,q1,p4,q2,p1,q3,p2,q4,p5,q5)
+ V(Q,p1,p4)*V(Q,p2,p3)*V(p1,p2,p5)*V(p3,p4,p5)*
replace(p4,q1,p3,q2,p2,q3,p1,q4,p5,q5)
;

```

This program produces all renumberings of the momenta in the t1 topology that produce the same topology. The interesting thing here is that one does not have to know the topology to produce all topologically equivalent terms.

There are two options in the id,all statement:

all(n[ormalize]) Here the final answer is divided by the number of matches. In the all(<number>) example above that would be 4.

The number between the parentheses will be the maximum number of matches allowed. This means that once this number is reached, no further matches are produced.

7.69 idnew

Type Executable statement
Syntax idn[ew] [<options>] <pattern> = <expression>;
See also identify (7.68), also (7.2), idold (7.70)

This statement and its options are completely identical to the regular id or identify statement (see 7.68).

7.70 idold

Type Executable statement
Syntax ido[ld] [<options>] <pattern> = <expression>;
See also identify (7.68), also (7.2), idnew (7.69)

This statement and its options are completely identical to the regular also statement (see 7.2). The options are described with the id or identify statement (see 7.68).

7.71 if

Type	Executable statement
Syntax	if (<condition>); if (<condition>) <executable statement>
See also	elseif (7.41), else (7.40), endif (7.45)

Used for executing parts of code only when certain conditions are met. Works together with the else statement (see 7.40), the elseif statement (see 7.41) and the endif statement (see 7.45). There are two versions. In the first the if statement must be accompanied by at least an endif statement. In that case the statements between the if statement and the endif statement will be executed if the condition is met. It is also possible to use elseif and else statements to be more flexible. This is done in the same way as in almost all computer languages.

In the second form the if statement does not terminate with a semicolon. It is followed by a single regular statement. No endif statement should be used. The single statement will be executed if the condition is met.

The condition in the if statement should be enclosed by parentheses. Its primary components are:

count() Returns an integer power counting value for the current term. Should have arguments that come in pairs. The first element of the pair is a variable. The second is its integer weight. The types of variables that are allowed are symbols, dotproducts, functions, tensors, tables and vectors. The weights can be positive as well as negative. They have to be short integers (Absolute value $< 2^{15}$ on 32 bit computers and $< 2^{31}$ on 64 bit computers). The vectors can have several options appended to their name. This is done by putting a + after the name of the vector and have this followed by one or more of the following letters:

- v Loose vectors with an index are taken into account.
- d Vectors inside dotproducts are taken into account.
- f Vectors inside tensors are taken into account.
- ?set The set should be a set of functions. Vectors inside the functions that are members of the set are taken into account. It is assumed that those functions are linear in the given vector

When no options are specified the result is identical to +vfd.

match() The argument of the match condition can be any left hand side of an id statement, including options as once, only, multi, many and select (see 7.69). The id of the id statement should not be included. FORM will invoke the pattern matcher and see how many times the pattern matches. This number is returned. In the case of once or only this is of course at most one.

expression() The argument(s) of this condition is/are a list of expressions. In the case that the current term belongs to any of the given expressions the return value is 1. If it does not belong to any of the given expressions the return value is 0.

occurs() The argument(s) of this condition is/are a list of variables. In the case that any of the variables occurs inside the current term (including inside function arguments) the return value is 1. Otherwise the return value is zero.

findloop()	The arguments are as in the replaceloop statement (see 7.127) with the exception of the outfun which should be omitted. If FORM detects an index loop in the current term that fulfils the specified conditions the return value is 1. It is 0 otherwise.
multipleof()	The argument should be a positive integer. This object is to be compared with a number (could be obtained from a condition) and if this number is an integer multiple of the argument there will be a match. It should be obvious that such a compare only makes sense for the == and != operators.
<integer>	To be compared either with another number, the result of a condition or a multipleof object.
coefficient	Represents the coefficient of the current term.
\$-variable	Will be evaluated at runtime when the if statement is encountered. Should evaluate into a numerical value. If it does not, an error will result.

All the above primary components result in numerical objects. Such objects can be compared to each other in structures of the type <obj1> <operator> <obj2>. The result of such a compare is either true (or 1) or false (or 0). The operators are:

>	Results in true if object 1 is greater than object 2.
<	Results in true if object 1 is less than object 2.
=	Same as ==.
==	Results in true if both objects have the same value.
>=	Results in true if object 1 is greater than or equal to object 2.
<=	Results in true if object 1 is less than or equal to object 2.
!=	Results in true if object 1 does not have the same value as object 2.

If the condition for true is not met, false is returned. Several of the above compares can be combined with logical operators. For this it is necessary to enclose the above compares within parentheses. This forces FORM to interpret the hierarchy of the operators properly. The extra logical operators are

	The or operation. True if at least one of the objects 1 and 2 is true (or nonzero). False or zero if both are false or zero.
&&	The and operation. True if both the objects 1 and 2 are true (or nonzero). False or zero if at least one is false or zero.

Example:

```

if ( ( match(f(1,x)*g(?a)) && ( count(x,1,v+d,1) == 3 ) )
    || ( expression(F1,F2) == 0 ) );
    some statements
endif;
if ( ( ( match(f(1,x)*g(?a)) == 0 ) && ( count(x,1,v+d,1) == 3 ) )
    || expression(F1,F2) );
    some statements
endif;

```

We see that `match()` is equivalent to `( match() != 0 )` and something similar for `expression()`. This shorthand notation can make a program slightly more readable.

Warning! The if-statement knows only logical values as the result of operations. Hence the answer to anything that contains parenthesis (which counts as the evaluation of an expression) is either true (1) or false (0). Hence the object (5) evaluates to true.

7.72 ifmatch

Type	Executable statement
Syntax	<code>ifmatch-> <label> <pattern> = <expression>;</code>
See also	identify (7.68)

This statement is identical to the ifmatch option of the id statement (see 7.68). Hence

```
ifmatch-> ....
```

is just a shorthand notation for

```
id ifmatch-> ....
```

7.73 ifnomatch

Type	Executable statement
Syntax	<code>ifnomatch-> <label> <pattern> = <expression>;</code>
See also	identify (7.68)

This statement is identical to the ifnomatch option of the id statement (see 7.68). Hence

```
ifnomatch-> ....
```

is just a shorthand notation for

```
id ifnomatch-> ....
```

7.74 index, indices

Type	Declaration statement
Syntax	<code>i[ndex] <list of indices to be declared>;</code> <code>i[ndices] <list of indices to be declared>;</code>
See also	dimension (7.33), fixindex (7.59)

Declares one or more indices. In the declaration of an index one can specify its dimension. This is done by appending one or two options to the name of the index to be declared:

name=dim	The dimension is either a nonnegative integer or a previously declared symbol. If the dimension is zero this means that no dimension is attached to the index. The consequence is that the index cannot be summed over and index contractions are not performed for this index. If no dimension is specified the default dimension will be assumed (see the dimension statement 7.33).
name=dim:ext	The dimension is a symbol as above. Ext is an extra symbol which indicates the value of dim-4. This option is useful when traces over gamma matrices are considered (see 7.153 and 7.154).

7.75 inexpression

Type	Executable statement
Syntax	inexpression,name(s) of expression(s);
See also	endinexpression (7.46)

The combination

```
InExpression,expr;
    Statements;
EndInExpression;
```

is a more readable version of the construction

```
if ( expression(expr) );
    Statements;
endif;
```

7.76 inparallel

Type	Specification statement
Syntax	inparallel; inparallel <list of expressions>;
See also	NotInParallel (7.99), ModuleOption (7.89)

This statement is only active in the context of TFORM and PARFORM. It causes (small) expressions to be executed side by side. Normally the terms of expressions are distributed over the processors and the expressions are executed one by one. This isn't very efficient for small expressions because there is a certain amount of overhead. When there are many small expressions, this statement can cause each expression to be executed by its own processor. A consequence is that the expressions now can finish in a semi-random order and hence may end up in the output in a order that is different from when this statement isn't used. The proper order is restored in the first module that comes after and that doesn't use this option. One should be careful using this statement for big expressions, because in that case the sorting may need sort files and the output

may temporarily need scratch files and the simultaneous use of many files can slow execution down significantly.

In the case that no expressions are mentioned, all active expressions will be affected. When there is a list of expressions, only those mentioned will be affected, provided they are active. Several of these statements will work cumulatively. This statement doesn't affect expressions that are still to be defined inside the current module. If it is needed to affect such expressions inside the current module, one should use the `InParallel` option of the `ModuleOption 7.89` statement. This statement works independently of the 'On Parallel;' 7.108 and 'Off Parallel;' 7.107 statements.

7.77 `inside`

Type	Executable statement
Syntax	<code>inside <list of \$-variables>;</code>
See also	<code>endinside</code> (7.47) and the chapter on \$-variables (6)

works a bit like the argument statement (see 7.9) but with \$-variables instead of with functions. An `inside` statement should be paired with an `endinside` statement (see 7.47) inside the same module. The statements in-between will then be executed on the contents of the \$-variables that are mentioned. One should pay some attention to the order of the action. The \$-variables are treated sequentially. Hence, after the first one has been treated its contents are substituted by the new value. Then the second one is treated. If it uses the contents of the first variable, it will use the new value. If the first variable uses the contents of the second variable it will use its old value. Redefining any of the listed \$-variables in the range of the 'inside-environment' is very dangerous. It is not specified what FORM will do. Most likely it will be unpleasant.

7.78 `insidefirst`

Type	Declaration statement
Syntax	<code>insidefirst <on/off>;</code>
See also	<code>on</code> (7.108), <code>off</code> (7.107)

This statement is obsolete. The user should try to use the `insidefirst` option of the `on` (see 7.108) or the `off` (see 7.107) statements.

7.79 `intohide`

Type	Specification statement
Syntax	<code>intohide;</code> <code>intohide <list of expressions>;</code>
See also	<code>hide</code> (7.67)

In the first variety this statement marks all currently active expressions for being put in hidden

storage at the end of the module, after it has been processed. In the second variety it marks only the specified active expressions as such.

The difference with the `hide` (7.67) statement is that in the `hide` statement the expression is copied immediately into the hide system and it will not be processed in the current module, while in the `intohide` statement the expression is first processed and its final output in this module is sent to the hide system rather than to the regular scratch system. The effect is the same as not putting the `intohide` statement in the current module and putting a `hide` statement in the next, but it saves one copy operation and it is possibly a bit more economical with the disk space.

Note that a `.store` instruction will simultaneously remove all expressions from the hide system.

7.80 `keep`

Type Specification statement

Syntax `keep brackets;`

See also `bracket` (7.11), `antibracket` (7.1) and the chapter on brackets (9)

The effect of this statement is that during execution of the current module the contents of the brackets are not considered. The statements only act on the ‘outside’ of the brackets. Only when the terms are considered finished and are ready for the sorting are they multiplied by the contents of the brackets. At times this can save much computer time as complicated pattern matching and multiplications of function arguments with large fractions have to be done only once, rather than for each complete term separately (assuming that each bracket contains a large number of terms). There can be some nasty side effects. Assume an expression like:

```
F = f(i1,x)*(g(i1,y)+g(i1,z));
B f;
.sort
Keep Brackets;
sum i1;
```

the result will be

```
F = f(N1_?,x)*g(i1,y)+f(N1_?,x)*g(i1,z);
```

because at the moment of summing over `i1` FORM is not looking inside the brackets and hence it never sees the second occurrence of `i1`. There are some beneficial applications of the `keep` statement in the ‘mincer’ package that comes with the FORM distribution. In this package the most costly step was made faster by a significant factor (depending on the problem) due to the `keep brackets` statement.

7.81 `label`

Type Executable statement

Syntax `la[bel] <name of label>;`

See also `goto` (7.66)

Places a label at the current location. The name of the label can be any name or positive number. Control can be transferred to the position of the label by a goto statement (see 7.66) or the ifmatch option of an id statement (see 7.68). The only condition is that the goto statement and the label must be inside the same module. Once the module is terminated all existing labels are forgotten. This means that in a later module a label with the same name can be used again (this may not improve readability though but it is a good thing when third party libraries are used).

7.82 lfactorized

Type Definition statement
 Syntax l[ocal]factorized <name> = <expression>;
 See also the chapter on polynomials 11 and the factorize statement 7.56.

Used to define a local expression in factorized notation and keep it that way. The factors are recognized by multiplication and division signs at lowest bracket level. For the rest the expression is treated as a regular local expression. Example:

```
Symbols x,y,z;
LocalFactorized F1 = 3*(x+y)*(y+z)*((x+z)*(2*x+1));
LocalFactorized F2 = 3*(x+y)*(y+z)+((x+z)*(2*x+1));
Print;
.end

F1 =
  ( 3 )
  * ( y + x )
  * ( z + y )
  * ( z + x + 2*x*z + 2*x^2 );

F2 =
  ( z + 3*y*z + 3*y^2 + x + 5*x*z + 3*x*y + 2*x^2 );
```

As one can see in the second expression, the plus at ground level makes that there is only one factor. In the first expression the last factor is seen as a single factor and not two factor2 because of the extra parentheses. Only parentheses at ground level are used to recognize factors. If one needs those factors anyway, one should either leave away those parentheses or use an extra Factorize statement to have FORM refactorize the expression.

7.83 load

Type Declaration statement
 Syntax loa[d] <filename> [<list of expressions>];
 See also save (7.128), delete (7.31)

Loads a previously saved file (see 7.128). If no expressions are specified all expressions in the file are put in the storage file and obtain the status of stored global expressions. If a list of expressions is specified all those expressions are loaded and possible other expressions are ignored. If a specified expression is not present, an error will result. If one does not know exactly what expressions are present in a file one could load the file without a list of expressions, because FORM will list all expressions that it encountered.

7.84 local

Type Definition statement
 Syntax l[ocal] <name> = <expression>;
 l[ocal] <names of expressions>;
 See also global (7.65)

Used to define a local expression. A local expression is an expression that will be dropped when a .store instruction is encountered. If this is not what is intended one should use global expressions (see 7.65). The statement can also be used to change the status of a global expression into that of a local expression. In that case there is no = sign and no right hand side.

7.85 makeinteger

Type Executable statement
 Syntax makeinteger [<argument specifications>
 {<name of function/set> [<argument specifications>]}];
 See also normalize (7.98)

Normalizes the indicated argument of the indicated functions(s) in such a way that all terms in this argument have integer coefficients with a their greatest common divider being one. This still leaves the possibility that the first term of this argument may be negative. If this is not desired one can first normalize the argument and then make its coefficients integer. The overall factor that is needed to make the coefficients like described is taken from the overall factor of the complete term. Example:

```
S    a,b,c;
CF   f;
L    F = f(22/3*a+14/5*b+18/7*c);
MakeInteger,f;
Print +f;
.end
F =
    2/105*f(135*c + 147*b + 385*a);
```

Note that this feature can be used to make outputs look much more friendly. It can be used in combination with the AntiBracket statement (7.1) and the function dum_ (8.14) to imitate a smart extra level of brackets and make outputs shorter.

It is possible to introduce a scale factor when extracting the coefficient and multiplying it into the complete term.

MakeInteger, $\wedge < n >, f$; The number n must be an integer (may be negative) and if the coefficient that is extracted is c the whole term is multiplied by the factor c^n .

7.86 many

Type Executable statement
Syntax many <pattern> = <expression>;
See also identify (7.68)

This statement is identical to the many option of the id statement (see 7.68). Hence

`many ....`

is just a shorthand notation for

`id many ....`

7.87 merge

Type Executable statement
Syntax merge,functionname;
merge,once,functionname;
See also shuffle (7.132)

This statement is exactly the same as the shuffle statement (see 7.132).

7.88 metric

Type Declaration statement
Syntax metric <option>;

Remark: statement is inactive. Should have no effect.

7.89 moduleoption

Type Module control statement
Syntax moduleoption <option>[,<value>;
See also polyfun (7.111), processbucketsize (7.117), dollar variables (6.1)

Used to set a mode for just the current module. It overrides the normal setting and will revert to this normal setting after this module. The settings are:

parallel	Allows parallel execution of the current module if all other conditions are right. This is the default.
noparallel	Vetoes parallel execution of the current module.
inparallel	This option is more or less equivalent to the InParallel 7.76 statement. The difference is that because this statement comes at the end of the module, its effects include also the expressions that have been defined inside the current module. This is not the case for the InParallel statement. The InParallel option can be followed by the names of expressions. If no such names are present, all active expressions are affected. Otherwise only the expressions that are mentioned are affected. Once this option is mentioned no more options can be used inside the same ModuleOption statement. This is to avoid potential confusion that could arise when expressions are used with a name identical to the name of one of the options.
notinparallel	This option is more or less equivalent to the NotInParallel 7.99 statement. The difference is that because this statement comes at the end of the module, its effects include also the expressions that have been defined inside the current module. This is not the case for the NotInParallel statement. The NotInParallel option can be followed by the names of expressions. If no such names are present, all active expressions are affected. Otherwise only the expressions that are mentioned are affected. Once this option is mentioned no more options can be used inside the same ModuleOption statement. This is to avoid potential confusion that could arise when expressions are used with a name identical to the name of one of the options.
polyfun	Possibly followed by the name of a ‘polyfun’. Is similar to the polyfun statement (see 7.111) but only valid for the current module.
polyratfun	Possibly followed by the name of a ‘polyratfun’. Is similar to the polyfun statement (see 7.112) but only valid for the current module. If there is second name, it refers to the inverse polyratfun. More complicated options of the polyratfun statement cannot be used here.
processbucketsize	Followed by a number. Similar to the processbucketsize statement (see 7.117) but only valid for the current module.
local	Should be followed by a list of \$-variables. Indicates that the contents of the indicated \$-variables are not relevant once the module has been finished and neither is the term by term order in which the \$-variables obtain their value. In practise each processor/thread will work with its own copy of this variable.
maximum	Should be followed by a list of \$-variables. Indicates that of the contents of the indicated \$-variables the maximum is the only thing that is relevant once the module has been finished. The term by term order in which the \$-variables obtain their value is not relevant.

minimum	Should be followed by a list of \$-variables. Indicates that of the contents of the indicated \$-variables the minimum is the only thing that is relevant once the module has been finished. The term by term order in which the \$-variables obtain their value is not relevant.
sum	Should be followed by a list of \$-variables. Indicates that the indicated \$-variables are representing a sum. The term by term order in which the \$-variables obtain their value is not relevant.

The options ‘local’, ‘maximum’, ‘minimum’ and ‘sum’ are for parallel versions of FORM. The presence of \$-variables can be a problem when the order of processing of the terms is not well defined. These options tell FORM what these \$-variables are used for. In the above cases FORM can take the appropriate action when gathering information from the various processors. This will allow parallel execution of the current module. If \$-variables are used in a module and they are defined on a term by term basis, the normal action of FORM will be to veto parallel execution unless it is clear that no confusion can occur. See also chapter 18 on the parallel version and section 6.1 on the dollar variables.

7.90 modulus

Type	Declaration statement
Syntax	m[odulus] [option(s)] <value>;

Defines all calculus to be modulus the given integer value, provided this number is positive. The modulus calculus extends itself to fractions. This means that if the value is not a prime number division by zero could result. It is the responsibility of the user to avoid such problems. When the value in the modulus statement is either 0 or 1 the statement would be meaningless. It is used as a signal to FORM that modulus calculus should be switched off again.

The options are

NoFunctions Modulus calculus is not performed inside function arguments.

AlsoFunctions Modulus calculus is also performed inside function arguments.

CoefficientsOnly Modulus calculus is neither performed inside function arguments nor on powers of symbols.

PlusMin The values of numbers are reduced to the range $(-value + 1)/2$ to $(value - 1)/2$.

Positive The values of numbers are reduced to the range 0 to $value - 1$.

NoDollars The modulus calculus is not performed inside dollar variables.

AlsoDollars The modulus calculus is performed also inside dollar expressions.

InverseTable To speed up calculations all inverses are computed by means of a table. If the modulus value is very big, this table may be too big for the memory. That would result in an error message.

NoInverseTable No Table of Inverses is constructed. They are calculated whenever needed.

AlsoPowers Reduction is also used on powers of symbols with the relation $x^{\text{mod}} = x$ if mod is the given value

NoPowers No reduction on powers is done.

PrintPowersOf The proper syntax is here `printpowersof(generator)` in which generator is supposed to be a generator for calculus modulus the given value, which means that all numbers will be written as a power of the generator. If the number turns out not to be a proper generator an error will be given. Note that finding the powers is done by means of the construction of a table. Hence, if the modulus value is very big the table might not fit inside memory. This will result in an error message.

The default mode is `NoFunctions`, `Positive`, `NoInverseTable`, `NoDollars`, `NoPowers`.

The current syntax (version 4.0 and later) differs slightly from the previous syntax. As however there were many bugs in the old implementation we suspect that a slight change of the options does not inconvenience any many users.

7.91 multi

Type Executable statement
Syntax `multi <pattern> = <expression>;`
See also `identify` (7.68)

This statement is identical to the `multi` option of the `id` statement (see 7.68). Hence

```
multi ....
```

is just a shorthand notation for

```
id multi ....
```

7.92 multiply

Type Executable statement
Syntax `mu[ltiply] [<option>] <expression>;`

Statement multiplies all terms by the given expression. It is advisable to use the options when noncommuting variables are involved. They are:

`left` Multiplication is from the left.

`right` Multiplication is from the right.

There is no guarantee as to what the default is with respect to multiplication from the left or from the right. It is up to FORM to decide what it considers to be most efficient when neither option is present.

Note that one should not abbreviate this command to ‘`multi`’, because there is a separate `multi` command (see 7.91).

7.93 ndrop

Type Specification statement
Syntax `ndrop;`
 `ndrop <list of expressions>;`
See also `drop` (7.37)

In the first variety this statement cancels all drop plans. This means that all expressions scheduled for being dropped will be restored to their previous status of local or global expressions. In the second variety this happens only to the expressions that are specified. Example:

```
Drop;  
Ndrop F1,F2;
```

This drops all expressions, except for the expressions F1 and F2.

7.94 nfactorize

Type Output control statement
Syntax `nfactorize {<name of expression(s)>;}`
See also the chapter on polynomials 11 and 7.56.

When one uses a `factorize` (see 7.56) statement without arguments all expressions will be marked for factorization. If one would like to exclude a few expressions this can be done with the `NFactorize` statement. There should be at least one expression mentioned as in:

```
Factorize;  
NFactorize expr12,expr29;
```

One can also use the `Factorize` statement with a number of expressions after which the `NFactorize` statement can remove some from the list again as in:

```
Factorize expr1,...,expr100;  
NFactorize expr12,expr29;
```

7.95 nfunctions

Type Declaration statement
Syntax `n[functions] <list of functions to be declared>;`
See also `functions` (7.62), `cfunctions` (7.14)

This statement declares noncommuting functions. It is equal to the `function` statement (see 7.62) which has the noncommuting property as its default.

7.96 nhide

Type Specification statement
Syntax nhide;
 nhide <list of expressions>;
See also hide (7.67), unhide (7.158), nunhide (7.105), pushhide (7.119), pophide (7.113)

In its first variety this statement undoes all hide plans that exist thus far in the current module. In the second variety it does this only for the specified active expressions. See the hide statement in 7.67. Example:

```
Hide;  
Nhide F1,F2;
```

Here all active expressions will be transferred to the hide file except for the expressions F1 and F2.

7.97 nintohide

Type Specification statement
Syntax nintohide;
 nintohide <list of expressions>;
See also hide (7.67), unhide (7.158), nunhide (7.105), pushhide (7.119), pophide (7.113), intohide (7.79)

In its first variety this statement undoes all intohide plans that exist thus far in the current module. In the second variety it does this only for the specified active expressions. See the intohide statement in 7.79. Example:

```
Intohide;  
Nintohide F1,F2;
```

Here all active expressions will be transferred to the hide file at the end of the current module, except for the expressions F1 and F2.

7.98 normalize

Type Executable statement
Syntax normalize options {<name of function/set> [<argument specifications>]};
See also argument (7.9), splitarg (7.135), makeinteger (7.85)

Normalizes the indicated arguments of the indicated functions. Normalization means that the argument will be multiplied by the inverse of its coefficient (provided it is not zero). This holds for single term arguments. For multiple term arguments the inverse of the coefficient of the first term of the argument is used. The options and the argument specifications are as in the SplitArg statement (see 7.135). Under normal circumstances the coefficient that is removed from the argument(s) is multiplied into the coefficient of the term. This can be avoid with the extra option (0). Hence

Normalize,f; changes $f(2*x+3*y)$ into $2*f(x+3/2*y)$ but
 Normalize,(0),f; changes $f(2*x+3*y)$ into $f(x+3/2*y)$.

A more flexible way to extract the coefficient of the (first) term is by providing a scale factor as in

Normalize, $\wedge < n >$,f; The number n must be an integer (may be negative) and if the coefficient of the first term was c the whole term is multiplied by the factor c^n .

7.99 notinparallel

Type Specification statement
 Syntax notinparallel;
 notinparallel <list of expressions>;
 See also InParallel (7.76), ModuleOption (7.89)

This statement is only active in the context of TFORM and PARFORM. It vetoes (small) expressions to be executed side by side. For a complete explanation of this type of running one should look at the InParallel 7.76 statement. Because the default is that expressions are executed one by one, the major use of this statement is in constructions like:

```
InParallel;
NotInParallel F1,F25;
```

which would first mark all expressions to be executed in simultaneous mode and then make an exception for F1 and F25.

7.100 nprint

Type Output control statement
 Syntax np[rint] <list of names of expressions>;
 See also print (7.114)

Statement is used to take expressions from the list of expressions to be printed. When a print statement is used (see 7.114) without specification of expressions, all active expressions are marked for printing. With this statement one can remove a number of them from the list.

7.101 nskip

Type Specification statement
 Syntax nskip;
 nskip <list of expressions>;
 See also skip (7.133)

In the first variety it causes the cancellation of all skip plans (see 7.133) for expressions. The status of these expressions is restored to their previous status (active local or global expressions). In the second variety this is done for the specified expressions only. Example:

```
Skip;
Nskip F1,F2;
```

This causes all active expressions to be skipped except for the expressions F1 and F2.

7.102 ntable

Type Declaration statement
 Syntax ntable <options> <table to be declared>;
 See also functions (7.62), table (7.143), ctable (7.26)

This statement declares a noncommuting table. For the rest it is identical to the table command (see 7.143) which has the commuting property as its default.

7.103 ntensors

Type Declaration statement
 Syntax nt[ensors] <list of tensors to be declared>;
 See also functions (7.62), tensors (7.145), ctensors (7.27)

This statement declares noncommuting tensors. For the rest it is equal to the tensor statement (see 7.145) which has the commuting property as its default.

The options that exist for properties of tensors are the same as those for functions (see 7.62).

7.104 nunfactorize

Type Output control statement
 Syntax nunfactorize {<name of expression(s)>};
 See also the chapter on polynomials 11 and 7.157.

When one uses an UnFactorize (see 7.157) statement without arguments all expressions will be marked for being unfactorized. If one would like to exclude a few expressions this can be done with the NUnFactorize statement. There should be at least one expression mentioned as in:

```
UnFactorize;
NUnFactorize expr12,expr29;
```

One can also use the UnFactorize statement with a number of expressions after which the NUnFactorize statement can remove some from the list again as in:

```
UnFactorize expr1,...,expr100;
NUnFactorize expr12,expr29;
```

7.105 nunhide

Type Specification statement
 Syntax nunhide;
 nunhide <list of expressions>;
 See also hide (7.67), nhide (7.96), unhide (7.158), pushhide (7.119), pophide (7.113)

In its first variety this statement undoes all unhide (see 7.158 and 7.67) plans that the system has in the current module. In its second variety this happens only with the specified expressions. Example:

```
Unhide;
Nunhide F1,F2;
```

All expressions are taken from the hide system, except for the expressions F1 and F2.

7.106 nwrite

Type Declaration statement
 Syntax nw[rite] <keyword>;
 See also on (7.108), off (7.107)

This statement is considered obsolete. All its varieties have been taken over by the off statement (see 7.107) and the on statement (see 7.108). The current version of FORM will still recognize it, but the user is advised to avoid its usage. In future versions of FORM it is scheduled to be used for a different kind of writing and hence its syntax may change considerably. The conversion program conv2to3 should help in the conversion of programs that have been written for version 2. For completeness we still give the syntax and how it should be converted. The keywords are:

stats	Same as: Off stats;
statistics	Same as: Off statistics;
shortstats	Same as: Off shortstats;
shortstatistics	Same as: Off shortstatistics;
warnings	Same as: Off warnings;

allwarnings	Same as: Off allwarnings;
setup	Same as: Off setup;
names	Same as: Off names;
allnames	Same as: Off allnames;
shortstats	Same as: Off shortstats;
highfirst	Same as: Off highfirst;
lowfirst	Same as: Off lowfirst;
powerfirst	Same as: Off powerfirst;

7.107 off

Type	Declaration statement
Syntax	off <keyword>; off <keyword> <option>;
See also	on (7.108)

Statement to control settings during execution. Many of these settings replace older statements. The settings and their keywords are:

allnames	Turns the allnames mode off. The default.
allwarnings	Turns off the printing of all warnings.
backtrace	Disables the printing of a stack trace on termination. This is the default for normal form builds.
checkpoint	Deactivates the checkpoint mechanism. See 4.1.
compress	Turns compression mode off.
finalstats	Turns off the last line of statistics that is normally printed at the end of the run (introduced in version 3.2).
flint	Disables the interface to FLINT; use FORM's built-in polynomial routines.

highfirst	Puts the sorting in a low first mode.
humanstatistics	Disables the printing of human-readable numbers and units in the statistics. This is the default.
humanstats	Same as ‘Off humanstatistics;’
insidefirst	Not active at the moment.
lowfirst	Leaves the default low first mode and puts the sorting in a high first mode.
names	Turns the names mode off. This is the default.
nospacesinnumbers	Allows very long numbers to be printed with leading blank spaces at the beginning of a new line. The numbers are usually broken up by placing a backslash character at the end of the line and then continuing at the next line. For cosmetic purposes FORM puts usually a few blank spaces at the beginning of the line. FORM itself can read this but some programs cannot. This option can be turned off by the ‘on nospacesinnumbers;’ statement. The printing of the blank characters can be restored by turning this variable off. See also page 17 for a corresponding variable in the setup file.
oldfactarg	Switches the use of the FactArg statement 7.54 to the new mode of version 4 or later in which expressions in the argument of the mentioned function are completely factored over the rationals. The default is off.
parallel	Disallows the running of the program in parallel mode (only relevant for parallel versions of FORM).
powerfirst	Puts the sorting back into ‘highfirst’ mode.
processstats	Turns the process by process printing of the statistics in PARFORM off. Only the master process will be printing statistics. Other versions of FORM will ignore this option.
propercount	Turns the propercounting mode off. This means that for the generated terms in the statistics not only the ‘ground level’ terms are counted but also terms that were generated inside function arguments.
properorder	Turns the properorder mode off. This is the default.
setup	Switches off the mode in which the setup parameters are printed. This is the default.
stats	Same as ‘Off statistics’.

statistics	Turns off the printing of statistics.
shortstats	Same as ‘Off shortstatistics’.
shortstatistics	Takes the writing of the statistics back from shorthand mode to the regular statistics mode in which each statistics messages takes three lines of text and one blank line.
sortreallocate	Turns off the reallocation of the small and large buffer at the end of each module.
threadloadbalancing	Disables the loadbalancing mechanism of TFORM in parallel mode. In other versions of FORM this option is ignored.
threads	Disallows multithreaded running in TFORM. In other versions of FORM this option is ignored.
threadstats	Turns off the thread by thread printing of the statistics in TFORM. Only the master thread will be printing statistics. Other versions of FORM will ignore this option.
totalsize	Switches the totalsize mode off. For a more detailed description of the totalsize mode, see the ”On TotalSize;” command 7.108.
warnings	Turns off the printing of warnings.
wtimstats	Disables the wall-clock time in the timing information in the statistics on the master.

If a description is too short, one should also consult the description in the on statement (see 7.108).

7.108 on

Type	Declaration statement
Syntax	on <keyword>; on <keyword> <option>;
See also	off (7.107)

New statement to control settings during execution. Many of these settings replace older statements. The settings and their keywords are:

allnames	Same as ‘On names’ but additionally all system variables are printed as well. Default is off.
allwarnings	Puts the printing of warnings in a mode in which all warnings, even the very unimportant warnings are printed.

backtrace	Attempt to print a stack trace on termination, to assist with debugging. This is off by default for normal form builds and on by default for the debug binaries (vorm, tvorm, parvorm). For best results eu-addr2line or addr2line should be installed on the system.
checkpoint	Activates the checkpoint mechanism that allows for the recovery of a crashed FORM session. See 4.1 for detailed information.
compress	Turns compression mode on. This compression is a relatively simple compression that hardly costs extra computer time but saves roughly a factor two in disk storage. The old statement was ‘compress on’ but this should be avoided in the future. This setting is the default if FORM has not been compiled with gzip or zstd support.
compress,gzip	This setting is the default, if FORM has been compiled with gzip support but not with zstd support. This option may be followed by a comma or a space and a single digit. It activates the gzip compression for the sort file. This compression can make the intermediate sort file considerably shorter at the cost of some CPU time. The digit indicates the compression level. Zero means no compression and 9 is the highest level. The default level is 6. Above that the compression becomes very slow and doesn’t gain very much extra.
compress,zstd	This setting is the default, if FORM has been compiled with zstd support. This option may be followed by a comma or a space and a single digit. It activates the zstd compression for the sort file. This compression performs better than gzip, both in terms of compression/ decompression rates and compression ratio. The digit indicates the compression level. Zero means no compression and 9 is the highest level. The default level is 6. Above that the compression becomes very slow and doesn’t gain very much extra.
fewerstatistics	Determines how many of the statistics FORM prints when a small buffer is full. The keyword can be followed by a positive integer in which case one out of that many of these statistics will be printed. If no number is given the default value of 10 is used. When the number that follows is zero, statistics are never printed when sorting the small buffer.
fewerstats	Same as the above fewerstatistics.
finalstats	Determines whether FORM prints a final line of run time statistics at the end of the run. Default is on.
flint	Enables the interface to FLINT for polynomial operations. This is the default if FORM has been compiled with FLINT support. Currently this supports PolyRatFun (7.112), FactArg (7.54), FactDollar (3.29, 7.55), div_ (8.13), rem_ (8.58), mul_ (8.45), gcd_ (8.30), and inverse_ (8.34). FORM still uses its built-in routines for Factorize (7.56) and when using Modulus (7.90) mode. Note that occasionally the overall sign of gcd_ may differ from that of the built-in routine.
highfirst	In this mode polynomials are sorted in a way that high powers come before low powers.

humanstatistics	If enabled, human-readable numbers and units are included in the statistics printing. Disabled by default.
humanstats	Same as the above humanstatistics.
insidefirst	Not active at the moment.
lowfirst	In this mode polynomials are sorted in a way that low powers come before high powers. This is the default.
names	Turns on the mode in which at the end of each module the names of all variables that have been defined by the user are printed. This is an inspection mode for debugging by the user. Default is off.
nospacesinnumbers	Makes that very long numbers are printed with no leading blank spaces at the beginning of a new line. The numbers are usually broken up by placing a backspace character at the end of the line and then continuing at the next line. For cosmetic purposes FORM puts usually a few blank spaces at the beginning of the line. FORM itself can read this but some programs cannot. Hence this printing of the blank characters can be omitted by turning this variable on. See also page 17 for a corresponding variable in the setup file.
oldfactarg	Switches the use of the FactArg statement 7.54 to the old mode from before version 4. This is a compatibility mode to allow oldprograms that rely on a specific working of the FactArg statement to still run. The default is off.
parallel	Allows the running of the program in parallel mode unless other problems prevent this. This is of course only relevant for parallel versions of FORM. The default is on.
powerfirst	In this mode polynomials are sorted in a way that high powers come before low powers. The most relevant is however the combined power of all symbols.
processstats	Only active for PARFORM. It determines whether all processes print their run time statistics or only the master process does so. Default is on.
propercount	Sets the counting of the terms during generation into ‘propercount’ mode. This means that only terms at the ‘ground level’ are counted and terms inside functions arguments are not counted in the statistics. This setting is the default.
properorder	Turns the properorder mode on. The default is off. In the properorder mode FORM pays particular attention to function arguments when bringing terms and expressions to normal form. This may cost a considerable amount of extra time. In normal mode FORM is a bit sloppy (and much faster) about this, resulting sometimes in an ordering that appears without logic. This concerns only function arguments! This mode is mainly intended for the few moments in which the proper ordering is important.

setup	Causes the printing of the current setup parameters for inspection. Default is off.
shortstatistics	Puts the writing of the statistics in a shorthand mode in which the complete statistics are written on a single line only.
shortstats	Same as ‘On shortstatistics’.
sortreallocate	Reallocate the small and large buffer (also on the worker threads) at the end of every module. In some cases this can significantly reduce FORM’s memory usage as measured by “resident set size”. For programs which consist of a large number of very quickly-running modules, this can incur a noticable performance penalty. See also #sortreallocate (3.57) for a single-module version of this feature.
statistics	Turns the writing of runtime statistics on. This is the default. It is possible to change this default with one of the setup parameters in the setup file (see 17).
stats	Same as ‘On statistics’.
threadloadbalancing	Causes the load balancing mechanism in TFORM to be turned on or off. Default is on. Ignored by other versions of FORM.
threads	Allows the running of the program in multithreaded mode unless other problems prevent this. This is of course only relevant for TFORM. Other versions of FORM ignore this. The default is on.
threadstats	Only active for TFORM. It determines whether all threads print their run time statistics or only the master thread does so. Default is on.
totalsize	Puts FORM in a mode in which it tries to determine the maximum space occupied by all expressions at any given moment during the execution of the program. This space is the sum of the input/output/hide scratch files, the sort file(s) and the .str file. This maximum is printed at the end of the program. The same can be obtained with the "TotalSize ON" command in the setup (see 17) or the -T option in the command tail when FORM is started (see 1).
warnings	Turns on the printing of warnings in regular mode. This is the default.
wtimestats	Prints the wall-clock time in the timing information in the statistics. The wall-clock time is indicated by ‘WTime’ instead of ‘Time’ in the normal statistics with ‘shortstatistics’ turned off. For parallel versions, it affects the statistics only on the master, and does not change those on the workers. The same can be obtained with the -W option in the command line options of FORM (see 1) or ‘WTimeStats ON’ in the setup (see 17). Default is off.

7.109 **once**

Type Executable statement
Syntax `once <pattern> = <expression>;`
See also `identify` (7.68)

This statement is identical to the `once` option of the `id` statement (see 7.68). Hence

```
once ....
```

is just a shorthand notation for

```
id once ....
```

7.110 **only**

Type Executable statement
Syntax `only <pattern> = <expression>;`
See also `identify` (7.68)

This statement is identical to the `only` option of the `id` statement (see 7.68). Hence

```
only ....
```

is just a shorthand notation for

```
id only ....
```

7.111 **polyfun**

Type Declaration statement
Syntax `polyfun <name of function>;`
`polyfun;`
See also `moduleoption` (7.89)

Declares the specified function to be the ‘polyfun’. The polyfun is a function of which the single argument is considered to be the coefficient of the term. If two terms are otherwise identical the arguments of their polyfun will be added during the sorting, even if these arguments are little expressions. Hence

```
PolyFun acc;  
Local F = 3*x^2*acc(1+y+y^2)+2*x^2*acc(1-y+y^2);
```

will result in

```
F = x^2*acc(5+y+5*y^2);
```

Note that the external numerical coefficient is also pulled inside the polyfun.

If the polyfun statement has no argument, FORM reverts to its default mode in which no polyfun exists. This does not change any terms. If one would like to remove the polyfun from the terms one has to do that ‘manually’ as in

```
PolyFun;
id  acc(x?) = x;
```

in which we assume that previously the function acc had been declared to be the ‘polyfun’.

7.112 polyratfun

Type	Declaration statement
Syntax	polyratfun <name of function>; polyratfun <name of function>,<name of function>; polyratfun;
See also	polyfun (7.111), moduleoption (7.89)

Declares the specified function to be the ‘polyratfun’. The polyratfun is a function with two arguments which together form a rational polynomial that acts as the coefficient of the term. If two terms are otherwise identical the arguments of their polyratfun will be added during the sorting, even if these arguments are little nontrivial. Hence

```
PolyRatFun acc;
Local F = 3*x^2*acc(1+y+y^2,1-y)+2*x^2*acc(1-y+y^2,1+y);
```

will result in

```
F = x^2*acc(-y^3-10*y^2-2*y-5,y^2-1);
```

Note that the external numerical coefficient is also pulled inside the polyratfun.

If the polyratfun statement has no argument, FORM reverts to its default mode in which no polyratfun exists. This does not change any terms.

The polyratfun has many similarities with the polyfun (see 7.111). At any moment there can only be at most either one polyfun or one polyratfun. Occurrences of the polyfun or the polyratfun with the wrong number or the wrong type of arguments are treated as regular functions.

There is a fundamental difference between the polyfun and the polyratfun. The last one is far more restrictive. It can have only numbers and symbols for its arguments. Also the ordering of the terms in the arguments can be different. In the polyratfun the terms are always sorted with the highest power first. In the polyfun the ordering is as with the regular terms. By default the lowest powers come first as one usually likes for power series expansions.

When two functions are specified, the first will be the PolyRatFun, and the second will be its inverse as in

```
PolyRatFun rat,RAT;
```

in which case

```
RAT(x1,x2) = rat(x2,x1)
```


This can be handy when one needs to solve systems of equations by manual interference. In that case exchanging numerators and denominators can be rather messy, while just changing a name is far less error-prone.

In many cases it may be very wasteful to keep full track of the complete rational polynomial. An example is the reduction of a complicated 4-loop massless propagator diagram for which the rational polynomials can easily have hundreds of powers of the dimension parameter $D = 4 - 2\epsilon$. In the end one has to expand in terms of ϵ although it is not known in advance to how many powers. For this there are two extra options in the `polyratfun` statement. The first is

```
PolyRatFun rat(divergence,x);
```

in which `x` is the name of the symbol of interest. In this case the `polyratfun` keeps only its most divergent term in this variable `x` and gives it the coefficient one. The result is that terms will never cancel and at the end of the calculation one can see how many poles in `x` were maximally present, and hence how far one has to expand in `x`. Because the contents of the `polyratfun` are extremely simple, the expensive rational arithmetic is completely absent and things should go rather fast.

In the second option one can specify how far one should expand:

```
PolyRatFun rat(expand,x,power);
```

In this case the denominator can only be a polynomial in the variable `x`. It will be expanded and multiplied by the numerator and eventually all terms with powers of `x` that are greater than '`power`' will be discarded. The remaining incidence of the function `rat` will then have only one argument, like the `polyfun` (see 7.111). The advantage is that now the addition of two coefficients is a simple and straightforward operation that does not need the expensive polynomial GCD computations. Of course one can program such expansions externally and maybe better suited for the problem at hand, but using this option of the `polyratfun` is much faster and gives fewer chances of mistakes.

7.113 pophide

Type Specification statement

Syntax `pophide;`

See also `hide` (7.67), `nhide` (7.96), `unhide` (7.158), `nunhide` (7.105), `pushhide` (7.119)

Undoes the action of the most recent `pushhide` statement (see 7.119). If there is no matching `pushhide` statement an error will result.

7.114 print

Type Print statement

Syntax `Print [<options>];`

`Print { [<options>] <expression>;`

`Print [<options>] "<format string>" [<objects>;`

See also `print[]` (7.115), `nprint` (7.100), `printtable` (7.116)

General purpose print statement. It has three modes. In the first two modes flags are set for the printing of expressions after the current module has been finished. The third mode concerns printing during execution. This allows the printing of individual terms or $\$$ -variables on a term by term basis. It should be considered as a useful debugging device.

In the first mode all active expressions are scheduled for printing. The options are

<filename> The results will be printed to the file. The < and > are mandatory for this option. This option can only be used in the version that prints out individual terms, i.e. the version with the format string.

+f Printing will be only to the log file.

-f Printing will be both to the screen and to the log file. This is the default.

+s Each term will start a new line. This is called the single term mode.

+ss Each term will start a new line. In addition each internal group will start a new line. A group is either a single function or all symbols together, or all dotproducts together, or all vectors together, or all Kronecker delta's together.

+sss Like the +ss option but now each symbol and its power will start a new line. The same for individual dotproducts (and their power), vectors and Kronecker delta's.

-s Regular term mode. There can be more terms in a line. Linebreaks are placed when the line is full. The line size is set in the format statement (see 7.60). This is the default.

-ss Lowers the single term mode to -s. If one would like to switch off the single term mode altogether, -s suffices.

-sss Lowers the single term mode to -ss. If one would like to switch off the single term mode altogether, -s suffices.

In the second mode one can specify individual expressions to be printed. The options hold for all the expressions that follow them until new options are specified. The options are the same as for the first mode.

In the third mode there is a format string as for the printf command in the C programming language. Of course the control characters are not exactly the same as for the C language because the objects are different. The special characters are:

%t The current term will be printed at this position including its sign, even if this is a plus sign.

%T The current term will be printed at this position. If its coefficient is positive no leading plus sign is printed.

%w The number of the current thread will be printed. This is for TFORM only. In the sequential version this combination is skipped. The number zero refers to the master thread.

%W The number of the current thread and its CPU-time at the moment of printing. This is for TFORM only. In the sequential version this combination is skipped. The number zero refers to the master thread.

%%\$ A dollar expression will be printed at this position. The name(s) of the dollar expression(s) should follow the format string in the order in which they are used in the format string.

%% The character %.

% If this is the last character of the string no linefeed will be printed at the end of the print command.

\n A linefeed.

Each call is terminated with a linefeed. Example:

```
Symbols a,b,c;
Local F = 3*a+2*b;
Print "> %T";
id a = b+c;
Print ">> %t";
Print;
.end
> 3*a
>> + 3*b
>> + 3*c
> 2*b
>> + 2*b
```

```
F =
5*b + 3*c;
```

In the third mode one can also use the $+/-f$ options of the first mode. This should be placed before the format string as in

```
Print +f "(%$) %t", $var;
```

Because of the mixed nature of this statement it can occur in more than one location in the module.

7.115 print[]

Type	Output control statement
Syntax	print[] {[<options>] <name>};
See also	print (7.114)

Print statement to cause the printing of expressions at the end of the current module. Is like the first two modes of the regular print statement (see 7.114), but when printing FORM does not print the contents of each bracket, only the number of terms inside the bracket. Is to be used in combination with a bracket or an antibracket statement (see 7.11 and 7.1). Apart from this the options are identical to those of the first two modes of the print statement.

7.116 printtable

Type Print statement
Syntax `printtable [<options>] <tablename>;`
 `printtable [<options>] <tablename> > <filename>;`
 `printtable [<options>] <tablename> >> <filename>;`
See also `print` (7.114), `table` (7.143), `fill` (7.57), `fillexpression` (7.58),
 and the `table_` function (8.68)

Almost the opposite of a `FillExpression` statement (see 7.58). Prints the contents of a table according to the current format (see 7.60). The output can go to standard output, the log file or a specified file. The elements of the table that have been defined and filled are written in the form of fill statements (see 7.57) in such a way that they can be read in a future program to fill the table with the current contents. This is especially useful when the `fillexpression` statement has been used to dynamically extend tables based on what FORM has encountered during running. This way those elements will not have to be computed again in future programs.

The options are

+f Output is to the logfile and not to the screen.
-f Output is both to the logfile and to the screen. This is the default.
+s Output will be in a mode in which each new term starts a new line.
-s Output will be in the regular mode in which new terms continue to be written on the same line within the limits of the number of characters per line as set in the format statement. Default is 72 characters per line. This can be changed with the format statement (see 7.60).

If redirection to a file is specified output will be only to this file. The +f option will be ignored. There are two possibilities:

> filename The old contents of the file with name 'filename' will be overwritten.
>> filename The table will be appended to the file with the name 'filename'. This allows the writing of more than one table to a file.

7.117 processbucketsize

Type Declaration statement
Syntax `processbucketsize <value>;`
See also `moduleoption` (7.89), `setup` (17)

Sets the number of terms in the buckets that are sent to the secondary processors in PARFORM, one of the parallel versions of FORM (see chapter 18). In all other versions this statement is ignored. See also the `moduleoption` (7.89) statement and the corresponding parameter for the `setup` (17).

7.118 propercount

Type Declaration statement
Syntax propercount <on/off>;
See also on (7.108), off (7.107)

This statement is obsolete. The user should try to use the propercount option of the on (see 7.108) or the off (see 7.107) statements.

7.119 pushhide

Type Specification statement
Syntax pushhide;
See also hide (7.67), nhide (7.96), unhide (7.158), nunhide (7.105), pophide (7.113)

Hides all currently active expressions (see 7.67). The pophide statement (see 7.113) can bring them back to active status again.

7.120 putinside

Type Executable statement
Syntax putinside <name of function> [, <bracket information>];
See also AntiPutInside (7.3)

This statement puts the complete term inside a function argument. The function must be a regular function (hence no tensor or table which are special types of functions). If there is bracket information, this information should adhere to the syntax of the bracket statement (7.3) and only occurrences of the bracket variables will be put inside the function. The coefficient will also be put inside the function.

7.121 ratio

Type Executable statement
Syntax ratio <symbol1> <symbol2> <symbol3>;

This statement can be used for limited but fast partial fractioning. In the statement

ratio a,b,c;

in which **a**, **b** and **c** should be three symbols FORM will assume that $c = b - a$ and then make the substitutions

$$\frac{1}{a^m} \frac{1}{b^n} = \sum_{i=0}^{m-1} (-1)^i \binom{n-1+i}{n-1} \frac{1}{a^{m-i}} \frac{1}{c^{n+i}} + \sum_{i=0}^{n-1} (-1)^m \binom{m-1+i}{m-1} \frac{1}{b^{n-i}} \frac{1}{c^{m+i}}$$

$$\frac{b^n}{a^m} = \sum_{i=0}^n \binom{n}{i} \frac{c^i}{a^{m-n+i}} \quad m \geq n$$

$$\frac{b^n}{a^m} = \sum_{i=0}^{m-1} \binom{n}{i} \frac{c^{n-i}}{a^{m-i}} + \sum_{i=0}^{n-m} \binom{m-1+i}{m-1} c^i b^{n-m-i} \quad m < n$$

Of course, such substitutions can be made also by the user in a more flexible way. This statement has however the advantage of the best speed.

Actually the ratio statement is a leftover from the Schoonschip inheritance. For most simple partial fractioning one could use

```
repeat id 1/[x+a]/[x+b] = (1/[x+a]-1/[x+b])/[b-a];
repeat id [x+a]/[x+b] = 1-[b-a]/[x+b];
repeat id [x+b]/[x+a] = 1+[b-a]/[x+a];
```

or similar constructions. This does not give the speed of the binomials, but it does make the program more readable and it is much more flexible.

7.122 rcyclesymmetrize

Type Executable statement
 Syntax rc[yclesymmetrize] {<name of function/tensor> [<argument specifications>];}
 See also symmetrize (7.142), cyclesymmetrize (7.28), antisymmetrize (7.4)

The argument specifications are explained in the section on the symmetrize statement (see 7.142).

The action of this statement is to reverse-cycle-symmetrize the (specified) arguments of the functions that are mentioned. This means that the arguments are brought to ‘natural order’ in the notation of FORM by trying cyclic and reverse cyclic permutations of the arguments or groups of arguments. The ‘natural order’ may depend on the order of declaration of the variables.

7.123 redefine

Type Executable statement
 Syntax r[edefine] <preprocessor variable> "<string>";
 See also preprocessor variables in the chapter on the preprocessor (3)

This statement can be used to change the contents of preprocessor variables. The new contents can be used after the current module has finished execution and the preprocessor becomes active again for further translation and compilation. This termwise adaptation of the value of a preprocessor variable can be very useful in setting up multi module loops until a certain condition is not met any longer. Example:

```
#do i = 1,1
```

```

statements;
if ( condition ) redefine i "0";
.sort
#enddo

```

As long as there is a term that fulfils the condition the loop will continue. This defines effectively a while loop (see 7.161) over various modules. Note that the .sort instruction is essential. Note also that a construction like

```

if ( count(x,1) > 3 ) redefine i "'i'+1";

```

is probably not going to do what the user intends. It is not going to count terms with more than three powers of x. The preprocessor will insert the compile time value of the preprocessor variable i. If this is 0, then each time a term has more than three powers of x, i will get the string value 0+1. If one would like to do such counting, one should use a dollar variable (see 6).

7.124 removespectator

Type Executable statement
 Syntax removespectator <spectator>;

See chapter20 on spectators.

7.125 renumber

Type Executable statement
 Syntax renumber <number>;
 See also sum (7.139)

Renumbers the dummy indices. Dummy indices are indices of the type N1_?. Normally FORM tries to renumber these indices to make the internal representation of a term 'minimal'. It does not try exhaustively though. Especially interference with symmetric or antisymmetric functions is far from perfect. This is due to considerations of economy. With the renumber statement the user can force FORM to do better. The allowable options are:

- 0 All exchanges of one pair of dummy indices are tried until all pair exchanges yield no improvements. This is the default if no option is specified.
- 1 If there are N sets of dummy indices all N! permutations are tried. This can be very costly when a large number of indices is involved. Use with care!

7.126 repeat

Type Executable statement
Syntax repeat;
 repeat <executable statement>
See also endrepeat (7.48), while (7.161)

The repeat statement starts a repeat environment. It is terminated with an endrepeat statement (see 7.48). The repeat statement and its matching endrepeat statement should be inside the same module.

The statements inside the repeat environment should all be executable statements (or print statements) and if any of the executable statements inside the environment has changed the current term, the action of the endrepeat statement will be to bring control back to the beginning of the environment. In that sense the repeat/endrepeat combination acts as

```
do
  executable statements
while any action due to any of the statements
```

The second form of the statement is a shorthand notation:

```
repeat;
  single statement;
endrepeat;
```

is equivalent to

```
repeat single statement;
```

Particular attention should be given to avoid infinite loops as in

```
repeat id a = a+1;
```

A more complicated infinite loop is

```
repeat;
  id S(x1?)*R(x2?) = T(x1,x2,x2-x1);
  id T(x1?,x2?,x3?pos_) = T(x1,x2-2,x3-1)*X(x2);
  id T(x1?,x2?,x3?) = S(x1)*R(x2);
endrepeat;
```

If the current term is $S(2)*R(2)$, the statements in the loop do not change it in the end. Yet the program goes into an infinite loop, because the first id statement will change the term (action) and the third statement will change it back. FORM does not check that the term is the same again. Hence there is action inside the repeat environment and hence the statements will be executed again. This kind of hidden action is a major source of premature terminations of FORM programs.

Repeat environments can be nested with all other environments (and of course also with other repeat/endrepeat combinations).

7.127 replaceloop

Type Executable statement
Syntax `replaceloop <parameters>;`
See also the `findloop` option of the `if` statement (7.71)

This statement causes the substitution of index loops. An index loop is a sequence of contracted indices in which the indices are arguments of various instances of the same function and each contracted index occurs once in one instance of the function and once in another instance of the function. Such a contraction defines a connection and if a number of such connections between occurrences of the function form a loop this structure is a candidate for replacement. Examples of such loops are:

```
f(i1,i2,j1)*f(i2,i1,j2)
f(i1,i2,j1)*f(i2,i3,j2)*f(i1,i3,j3)
f(i1,k1,i2,j1)*f(k2,i2,i3,j2)*f(i1,k3,i3,j3)
```

The first term has a loop of two functions or vertices and the other two terms each define a loop of three vertices. The parameters are:

<code><name></code>	The name of the function that defines the ‘vertices’. This must always be the first parameter.
<code>arguments=number</code>	Only occurrences of the vertex function with the specified number of arguments will be considered. The specification of this parameter is mandatory.
<code>loopsize=number</code>	Only a loop with this number of vertices will be considered.
<code>loopsize=all</code>	All loop sizes will be considered and the smallest loop is substituted.
<code>loopsize<number</code>	Only loops with fewer vertices than ‘number’ will be considered and the smallest loop will be substituted.
<code>outfun=<name></code>	Name of an output function in which the remaining arguments of all the vertex functions will be given. This parameter is mandatory.
<code>include-<name></code>	Name of a summable index that must be one of the links in the loop. This parameter is optional.

The `loopsize` parameter is mandatory. Hence one of its options must be specified. The order of the parameters is not important. The only important thing is that the name of the vertex function must be first. The names of the keywords may be abbreviated as in

```
ReplaceLoop f,a=3,l=all,o=ff,i=i2;
```

although this does not improve the readability of the program. Hence a more readable abbreviated version might be

```
ReplaceLoop f,arg=3,loop=all,out=ff,inc=i2;
```

The action of the statement is to remove the vertex functions that constitute the loop and replace them by the output function. This `outfun` will have the arguments of all the vertex functions minus the contracted indices that define the loop. The order of the arguments is the order in which they are encountered when following the loop. The order of the arguments in the `outfun` depends however on the order in which FORM encounters the vertices. Hence the `outfun` will often be cyclesymmetric (see 7.62 and 7.28). If FORM has to exchange indices to make a ‘proper loop’

(i.e. giving relevance to the first index as if it is something incoming and the second index as if it is something outgoing) and if the vertex function is antisymmetric, each exchange will result in a minus sign. Examples:

```
Functions f(antisymmetric),ff(cyclesymmetric);
Indices i1,...,i8;
Local F = f(i1,i4,i2)*f(i5,i2,i3)*f(i3,i1,i6)*f(i4,i7,i8);
ReplaceLoop f,arg=3,loop=3,out=ff;
```

would result in

```
-f(i4,i7,i8)*ff(i4,i5,i6)
```

and

```
Functions f(antisymmetric),ff(cyclesymmetric);
Indices i1,...,i9;
Local F = f(i1,i4,i2)*f(i5,i2,i3)*f(i3,i1,i6)*f(i4,i7,i8)
          *f(i6,i7,i8);
ReplaceLoop f,arg=3,loop=all,out=ff;
```

would give

```
-f(i1,i4,i2)*f(i5,i2,i3)*f(i3,i1,i6)*ff(i4,i6)
```

because the smallest loop will be taken. A number of examples can be found in the package ‘color’ for group theory invariants that is part of the FORM distribution.

A related object is the findloop option of the if statement (see 7.71). This option just probes whether a loop is present but makes no replacements.

7.128 save

Type	Declaration statement
Syntax	sa[ve] <filename> [<names of global expressions>];
See also	load (7.83)

Saves the contents of the store file (all global expressions that were stored in .store instructions) to a file with the indicated name. If a list of expressions is provided only those expressions are saved and the others are ignored.

Together with the load statement (see 7.83) the save statement provides a mechanism to transfer data in internal notation from one program to another. It is the preferred method to keep results of a lengthy job for further analysis without the need for the long initial running time.

In order to avoid confusion .sav is the preferred extension of saved files.

7.129 select

Type Executable statement
Syntax `select <list of sets> <pattern> = <expression>;`
See also `identify` (7.68)

This statement is identical to the `select` option of the `id` statement (see 7.68). Hence

```
select ....
```

is just a shorthand notation for

```
id select ....
```

7.130 set

Type Declaration statement
Syntax `set <set to be declared>[(option)]:<element> [<more elements>];`

Declares a single set and specifies its elements. Sets have a type of variables connected to them. There can be sets of symbols, sets of functions, sets of vectors, sets of indices and sets of numbers. For the purpose of sets tensors and tables count as functions.

There can also be mixed sets of indices and numbers. When a number could be either a fixed index or just a number FORM will keep the type of the set unfixed. This can change either when the next element is a symbolic index or a number that cannot be a fixed index (like a negative number). If the status does not get resolved the set can be used in the wildcarding of both symbols and indices. Normally sets of numbers can be used only in the wildcarding of symbols.

Currently the only option is the ordered set, indicated by

```
Set name(ordered):x4,x3,x1,x6,x2;
```

which would be stored as `x1,x2,x3,x4,x6` if that would be the order of declaration.

7.131 setexitflag

Type Executable statement
Syntax `setexitflag;`
See also `exit` (7.52)

Causes termination of the program after execution of the current module has finished.

7.132 shuffle

Type Executable statement
Syntax `shuffle,functionname;`
`shuffle,once,functionname;`
See also `stuffle` (7.138)
`merge` (7.87)

This statement is exactly the same as the merge statement. It takes two occurrences of the mentioned function and outputs terms, each with one function in which the two argument lists have been merged in all different ways, keeping the relative ordering of the two lists preserved. It is the opposite of the `distrib_` function (see 8.12). Hence

```
Local F = f(a,b)*f(c,d);  
shuffle,f;
```

will result in

$$+f(a,b,c,d)+f(a,c,b,d)+f(a,c,d,b)+f(c,a,b,d)+f(c,a,d,b)+f(c,d,a,b)$$

One can also obtain the same result with the statements

```
Multiply,ff;  
repeat id f(x1?,?a)*f(x2?,?b)*ff(?c) =  
    +f(?a)*f(x2,?b)*ff(?c,x1)  
    +f(x1,?a)*f(?b)*ff(?c,x2);  
id f(?a)*f(?b)*ff(?c) = f(?c,?a,?b);
```

but the advantage of the `shuffle` statement is that it also does a certain amount of combinatorics when there are identical arguments. Unfortunately the combinatorics doesn't extend over groups of arguments that are identical as in

```
CF f;  
L F = f(0,1,0,1,0,1)*f(0,1,0,1,0,1);  
Shuffle,f;  
.end
```

Time =	0.00 sec	Generated terms =	141
	F	Terms in output =	32
		Bytes used =	892

It does get the combinatorics between two zeroes or two ones, but it cannot handle the groups. The explicit method above however doesn't do any combinatorics and generates 924 terms.

One of the applications of this statement is in the field of harmonic sums, harmonic polylogarithms and multiple zeta values. Its twin brother is the `stuffle` statement (see 7.138).

When the option `once` is mentioned, only one pair will be contracted this way. Without this option all occurrences of the function inside a term will be treated till there are only terms with a single occurrence of the function.

7.133 skip

Type Specification statement
Syntax skip;
 skip <list of expressions>;
See also nskip (7.101)

In the first variety this statement marks all active expressions that are in existence at the moment this statement is compiled, to be skipped. In the second variety this is done only to the active expressions that are specified. If an expression is skipped in a given module, the statements in the module have no effect on it. Also it will not be sorted again at the end of the module. This means that any bracket information (see 7.11) in the expression remains the way it was. Consult also the nskip statement in 7.101.

Skipped expressions can be used in the expressions in the r.h.s. of id statements (see 7.68) or multiply statements (see 7.92), etc.

7.134 sort

Type Executable statement
Syntax sort;
See also term (7.146), endterm (7.50)

Statement to be used inside the term environment (see 7.146 and 7.50). It forces a sort in the same way as a .sort instruction forces a sort for entire expressions.

7.135 splitarg

Type Executable statement
Syntax splitarg options {<name of function/set> [<argument specifications>]};
See also splitfirstarg (7.136), splitlastarg (7.137), factarg (7.54)

Takes the indicated argument of a function and if such an argument is a subexpression that consists on more than one term, all terms become single arguments of the function as in

`f(a+b-5*c*d) --> f(a,b,-5*c*d)`

The way arguments are indicated is rather similar to the way this is done in the argument statement (see 7.9). One can however indicate only a single group of functions in one statement. Additionally there are other options. All options are in the order that they should be specified:

(term)	Only terms that are a numerical multiple of the given term are split off. The terms that are split off will trail the remainder.
((term))	Only terms that contain the given term will be split off. The terms that are split off will trail the remainder.

The statement is terminated with a sequence of functions or sets of functions. The splitting action will apply only to the specified functions or to members of the set(s). If no functions or sets of functions are specified all functions will be treated, including the built in functions.

The argument specifications consist of a list of numbers, indicating the arguments that should be treated. If no arguments are specified, all arguments will be treated.

7.136 splitfirstarg

Type Executable statement

Syntax splitfirstarg {<name of function/set> [<argument specifications>]};

See also splitarg (7.135), splitlastarg (7.137)

A little bit like the SplitArg statement (see 7.135). Splits the given argument(s) into its first term and a remainder. Then replaces the argument by the remainder, followed by the first term.

The statement is terminated with a sequence of functions or sets of functions. The splitting action will apply only to the specified functions or to members of the set(s). If no functions or sets of functions are specified all functions will be treated, including the built in functions.

The argument specifications consist of a list of numbers, indicating the arguments that should be treated. If no arguments are specified all arguments will be treated.

7.137 splitlastarg

Type Executable statement

Syntax splitlastarg {<name of function/set> [<argument specifications>]};

See also splitarg (7.135), splitfirstarg (7.136)

A little bit like the SplitArg statement (see 7.135). Splits the given argument(s) into its last term and a remainder. Then replaces the argument by the remainder, followed by the last term.

The statement is terminated with a sequence of functions or sets of functions. The splitting action will apply only to the specified functions or to members of the set(s). If no functions or sets of functions are specified all functions will be treated, including the built in functions.

The argument specifications consist of a list of numbers, indicating the arguments that should be treated. If no arguments are specified all arguments will be treated.

7.138 stuffle

Type Executable statement
 Syntax stuffle,functionname+;
 stuffle,functionname-;
 stuffle,once,functionname+;
 stuffle,once,functionname-;

See also shuffle (7.132)

This statement takes two occurrences of the mentioned function and outputs terms, each with one function in which the two argument lists have been merged according to the rules for nested sums. The plus and minus signs refer to ones favorite definition for nested sums. In the case of the plus sign, the definition is

$$\sum_{i=1}^N \sum_{i=1}^N = \sum_{i=1}^N \sum_{j=1}^{i-1} + \sum_{j=1}^N \sum_{i=1}^{j-1} + \sum_{i=j=1}^N \quad (7.4)$$

while in the case of the minus the definition is

$$\sum_{i=1}^N \sum_{i=1}^N = \sum_{i=1}^N \sum_{j=1}^i + \sum_{j=1}^N \sum_{i=1}^j - \sum_{i=j=1}^N \quad (7.5)$$

It is assumed that we have harmonic sums (see the summer library in the FORM distribution). For such sums we expect functions with lists of nonzero integer arguments. Example:

```
CF S,R;
Symbols N,n;
L F = S(R(1,-3),N)*S(R(-5,1),N);
id S(R(?a),n?)*S(R(?b),n?) = S(?a)*S(?b)*R(n);
Stuffle,S-;
id S(?a)*R(n?) = S(R(?a),n);
Print +s;
.end
```

```
Time =          0.00 sec      Generated terms =          12
          F          Terms in output =          12
          Bytes used      =          462
```

```
F =
+ S(R(-6,-4),N)
- S(R(-6,-3,1),N)
- S(R(-6,1,-3),N)
- S(R(-5,1,-4),N)
+ S(R(-5,1,-3,1),N)
+ 2*S(R(-5,1,1,-3),N)
- S(R(-5,2,-3),N)
- S(R(1,-5,-4),N)
+ S(R(1,-5,-3,1),N)
+ S(R(1,-5,1,-3),N)
```

```

+ S(R(1,-3,-5,1),N)
- S(R(1,8,1),N)
;

```

The above program is equivalent to the basis procedure in the summer library. As with the shuffle statement (see 7.132) a certain amount of combinatorics has been built in.

When the option once is mentioned, only one pair will be contracted this way. Without this option all occurrences of the function inside a term will be treated till there are only terms with a single occurrence of the function.

The stuffle command takes also the effect of roots of unity 7.141 into account in the same way that the signs of alternating sums are taken into account. This means that the sum indices don't have to be integers, but could be multiples of a single symbol that has been declared to be a root of unity 7.141.

7.139 sum

Type Executable statement
 Syntax sum <list of indices>;
 See also renumber (7.125)

The given indices will be summed over. There are two varieties. In the first the index is followed by a sequence of nonnegative short integers. In that case the summation means that for each of the integers a new instance of the term is created in which the index is replaced by that integer. In the second variety the index is either the last object in the statement or followed by another index. In that case the index is replaced by an internal dummy index of the type N1_? (or with another number instead of the 1). Such indices have the current default dimension and can be renamed at will by FORM to bring terms into standard notation. For example:

```
f(N2_?,N1_?)*g(N2_?,N1_?)
```

will be changed into

```
f(N1_?,N2_?)*g(N1_?,N2_?) .
```

The user can use these dummy indices in the left hand side of id statements.

7.140 switch

Type Executable statement
 Syntax switch,\$-variable;

See also case (7.13), break (7.12), default(7.30), endswitch (7.49).

The argument of the switch statement should be a dollar variable which evaluates into an integer that first inside a FORM word. On a 64-bit processor this would be an integer in the range -2^{31} to $2^{31} - 1$. The switch statement should be paired with an endswitch statement. Between the two

there will be a number of cases, each marked by an integer. If the value of the dollar variable corresponds to the value of one of these cases, execution will continue with the first statement after the corresponding case statement. Example:

```
id  f(x?$x) = f(x);
switch $x;
  case -1;
    some statements
  break;
  case 3;
    more statements
  break;
  case 4;
  case 5;
    and a few more
  break;
  default;
    and the default action
  break;
endswitch;
```

In principle the action is the same as in any computer language that has a switch construction, including the fall-through between case 4 and case 5. Whether the selection of the cases goes by binary search in a sorted list or by jumtable is determined by the endswitch statement.

7.141 symbols

Type Declaration statement

Syntax s[symbols] <list of symbols to be declared>;

Declares one or more symbols. Each symbol can be followed by a number of options. These are (assuming that x is the symbol to be declared):

- x#r The symbol is real. This is the default.
- x#c The symbol is complex. This means that two spaces are reserved for this symbol, one for x and one for x# (the complex conjugate).
- x#i The symbol is imaginary.
- x#=number The symbol is a number-th root of unity This means that the number-th power of the symbol will be replaced by one and half this power (if even) by -1. Negative powers will be replaced by corresponding positive powers.
- x(:5) The symbol has the maximum power 5. This means that x^6 and higher powers are automatically eliminated during the normalization of a term. Of course any other number, positive or negative, is allowed.

- x(-3:) The symbol has the minimum power -3. This means that x^{-4} and lower powers are automatically eliminated during the normalization of a term. Of course any other number, positive or negative, is allowed. Note that when the minimum power is positive, terms that have no power of x should technically be eliminated, but FORM will not do so. Such an action can be achieved at any moment with a combination of the count option of an if statement (see 7.71) and a discard statement (see 7.34).
- x(-3:5) The combination of a maximum and a minimum power restriction (see above).

Complexity properties and power restrictions can be combined. In that case the complexity properties come first and then the power restrictions.

7.142 symmetrize

Type Executable statement
 Syntax `symm[etrize] {<name of function/tensor> [<argument specifications>];}`
 See also `antisymmetrize` (7.4), `cyclesymmetrize` (7.28), `rcyclesymmetrize` (7.122)

The arguments consist of the name of a function (or a tensor), possibly followed by some specifications. Hence we have the following varieties:

- <name> The function is symmetrized in all its arguments.
- <name><numbers> The function is symmetrized in the arguments that are mentioned. If there are fewer arguments than the highest number mentioned in the list or arguments, no symmetrization will take place.
- <name>:<number> Only functions with the specified number of arguments will be considered. Note: the number should follow the colon directly without intermediate space or comma.
- <name>:<number><numbers> If there is a number immediately following the colon, only functions with exactly that number of arguments will be considered. If the list of arguments contains numbers greater than this number, they will be ignored. If no number follows the colon directly, this indicates that symmetrization will take place, no matter the number of arguments of the function. If the list of arguments has numbers greater than the number of arguments of the function, these numbers will be ignored.
- <name>
 <(groups of numbers)> The groups are specified as lists of numbers of arguments between parenthesis. All groups must have the same number of arguments or there will be a compile error. The groups are symmetrized as groups. The arguments do not have to be adjacent. Neither do they have to be ordered. The symmetrization takes place in a way that the first elements of the groups are most significant, etc. If any argument number is greater than the number of arguments of the function, no symmetrization will take place.

<name>:<number>
<(groups of numbers)>

The groups are specified as lists of numbers of arguments between parenthesis. All groups must have the same number of arguments or there will be a compile error. The groups are symmetrized as groups. The arguments do not have to be adjacent. Neither do they have to be ordered. The symmetrization takes place in a way that the first elements of the groups are most significant, etc. If no number follows the colon directly symmetrization takes place no matter the number of arguments of the function. Groups that contain a number that is greater than the number of arguments of the function will be ignored. If a number follows the colon directly, only functions with that number of arguments will be symmetrized. Again, groups that contain a number that is greater than the number of arguments of the function will be ignored.

The action of this statement is to symmetrize the (specified) arguments of the functions that are mentioned. This means that the arguments are brought to ‘natural order’ in the notation of FORM by trying permutations of the arguments or groups of arguments. The ‘natural order’ may depend on the order of declaration of the variables.

Examples:

```
Symmetrize Fun;  
Symmetrize Fun 1,2,4;  
Symmetrize Fun:5;  
Symmetrize Fun: 1,2,4;  
Symmetrize Fun:5 1,2,4;  
Symmetrize Fun (1,6),(7,3),(5,2);  
Symmetrize Fun:8 (1,6),(7,3),(5,2);  
Symmetrize Fun: (1,6),(7,3),(5,2);
```

7.143 table

Type	Declaration statement
Syntax	table <options> <table to be declared>;
See also	functions (7.62), ctable (7.26), ntable (7.102), fill (7.57)

The statement declares a single table. A table is a very special instance of a function. Hence it can be either commuting or noncommuting. The table statement declares its function to be commuting. A noncommuting table is declared with the ntable statement (see 7.102). A table has a number of table indices (in the case of zero indices the table has to be sparse) and after that it can have a number of regular function arguments with or without wildcarding. The table indices can come in two varieties: matrix like or sparse. In the case of a matrix like table, for each of the indices a range has to be specified. FORM then reserves a location for each of the potential elements. For a sparse table one only specifies the number of indices. Sparse tables take less space, but they require more time searching whether an element has been defined. For a matrix like table FORM can look directly whether an element has been defined. Hence one has a tradeoff between space and speed. A zero-dimensional (sparse) table has of course only a single element.

Table elements are defined with the fill statement (see 7.57). Fill statements for table elements cannot be used before the table has been declared with a table or ntable statement.

When FORM encounters an unsubstituted table it will look for its indices. Then it can check whether the table element has been defined. If not, it can either complain (when the ‘strict’ option is used) or continue without substitution. Note that an unsubstituted table element is a rather expensive object as FORM will frequently check whether it can be substituted (new elements can be defined in a variety of ways....). If the indices match a defined table element, FORM will check whether the remaining arguments of the table will match the function-type arguments given in the table declaration in the same way regular function arguments are matched. Hence these arguments can contain wildcards and even argument field wildcards. If a match occurs, the table is replaced immediately.

The options are

check	A check is executed on table boundaries. An element that is outside the table boundaries (regular matrix type tables only) will cause an error message and execution will be halted.
relax	Normally all elements of a table should be defined during execution and an undefined element will give an error message. The relax option switches this off and undefined elements will remain as if they are regular functions.
sparse	The table is considered to be sparse. In the case of a sparse table only the number of indices should be specified. Ranges are not relevant. Each table element is stored separately. Searching for table elements is done via a balanced tree. This takes of course more time than the matrix type search with is just by indexing. A matrix like table is the default.
strict	If this option is specified all table elements that are encountered during execution should be defined. An undefined table element will result in an error and execution is halted. Additionally all table elements should be properly defined at the end of the module in which the table has been defined.
zerofill	Any undefined table element is considered to be zero.
onefill	Any undefined table element is considered to be one.

The defaults are that the table is matrix like and table elements that cannot be substituted will result in an error.

Ranges for indices in matrix like tables are indicated with a colon as in

```
Symbol x;
Table t1(1:3,-2:4);
Table t2(0:3,0:3,x?);
Table sparse,t3(4);
```

The table **t1** is two dimensional and has 21 elements. The table **t2** is also two dimensional and has 16 elements. In addition there is an extra argument which can be anything that a wildcard symbol will match. The table **t3** is a sparse table with 4 indices.

If the computer on which FORM runs is a 32 bit computer no table can have more than $2^{15} = 32768$ elements. On a 64 bit computer the limit is 2^{31} , but one should take into account that each element declared causes some overhead.

If the wildcarding in the declaration of a table involves the definition of a dollar variable (this is allowed! See 6) parallel execution of the entire remainder of the FORM program is switched off. This is of course only relevant for parallel versions of FORM. But if at all possible one should try to find better solutions than this use of dollar variables, allowing future parallel processing of the program.

In some cases tables are built up slowly during the execution of a program and used incrementally. This means that more and more CPU memory is needed. Eventually this can cause a crash by lack of memory. In the case that the earlier elements of the table aren't needed anymore, one could use the ClearTable 7.18 statement.

7.144 tablebase

This statement is explained in the chapter on tablebases (12).

7.145 tensors

Type	Declaration statement
Syntax	t[ensors] <list of tensors to be declared>;
See also	functions (7.62), ctensors (7.27), ntensors (7.103)

A tensor is a special function that can have only indices for its arguments. If an index is contracted with the index of a vector Schoonschip notation is used. This means that the vector is written as a pseudo argument of the tensor. It should always be realized that in that case in principle the actual argument is a dummy index. Tensors come in two varieties: commuting and noncommuting. The tensor statement declares a tensor to be commuting. In order to declare a tensor to be noncommuting one should use the ntensor statement (see 7.103).

The options that exist for properties of tensors are the same as those for functions (see 7.62).

7.146 term

Type	Executable statement
Syntax	term;
See also	endterm (7.50), sort (7.134)

Begins the term environment. This environment is terminated with the endterm statement (see 7.50). The action is that temporarily the current term is seen as a little expression by itself. The statements inside the environment are applied to it and one can even sort the results with the sort statement (see 7.134) which should not be confused with the .sort instruction that terminates a

module. Inside the term environment one can have only executable statements and possibly term-wise print statements (see 7.114). When the end of the term environment is reached, the results are sorted (as would be done with an expression at the end of a module) and execution continues with the resulting terms. This environment can be nested.

7.147 **testuse**

Type Executable statement
 Syntax `testuse [”<tablename(s)>”];`
 See also `tablebases` (12), `testuse` (12.10)

This statement is explained in the chapter on `tablebases`.

7.148 **threadbucketsize**

Type Declaration
 Syntax `ThreadBucketSize,number;`
 See also the section on `TFORM` (18.1)

This statement is only active in `TFORM`. In all other versions of `FORM` it is ignored. It sets the size of the buckets that the master thread prepares for treatment by the workers. Bigger buckets means less overhead in signals, but when the buckets are too big the workers may have to wait too long before getting tasks. The best bucket size is usually between 100 and 1000, although this depends very much on the problem. The default value is currently 500. For more ways to set this variable one should consult the section on `TFORM` (18.1). To find out what its value is, use the ‘`ON,setup;`’ statement (7.108 and 17).

7.149 **topolynomial**

Type Executable statement
 Syntax `topolynomial[,OnlyFunctions[,<list of functions>]];`
 See also `factarg` (7.54), `FromPolynomial` (7.61), `ArgToExtraSymbol` (7.8)
 and `ExtraSymbols` (7.53, 2.11).

Starting with version 4.0 of `FORM` some built in operations or statements can only deal with symbols and numbers. Examples of this are factorization (7.54) and output simplification (still to be implemented). The `ToPolynomial` statement takes each term, looks for objects that are not symbols to positive powers and replaces them by symbols. If the object has been encountered before, the same symbol will be used, otherwise a new symbol will be defined. The object represented by the ‘extra symbol’ is stored internally and can be printed if needed with the `%X` option in the `#write` instruction (3.65). Note that negative powers of symbols will also be replaced.

In some cases one would like to do this only for a subset of objects. It is possible to do this only for functions, using the `OnlyFunctions` option. If no functions are specified, all functions will be replaced by extra symbols. If a list of functions is specified, only those functions will be replaced.

7.150 `tospectator`

Type Executable statement
 Syntax `tospectator <spectator>;`

See chapter20 on spectators.

7.151 `totensor`

Type Executable statement
 Syntax `totensor [nosquare] [functions] [!<vector or set>] <vector> <tensor>;`
`totensor [nosquare] [functions] [!<vector or set>] <tensor> <vector>;`
 See also `tovector` (7.152)

Looks for multiple occurrences of the given vector, either inside dotproducts, contracted with a tensor, as argument of a function or as a loose vector with an index. In all occurrences in which the vector has been contracted a dummy index is introduced to make the contraction apparent. Then all these vectors with their indices are replaced by the specified tensor with all the indices of these vectors. To make this clearer:

$$p^{\mu_1} p^{\mu_2} p^{\mu_3} \rightarrow t^{\mu_1 \mu_2 \mu_3}$$

and hence

```
p.p1^2*f(p,p1)*p(mu)*tt(p1,p,p2,p)
```

gives after `totensor p,t;`

```
f(N1_?,p1)*tt(p1,N2_?,p2,N3_?)*t(p1,p1,mu,N1_?,N2_?,N3_?)
```

The options are

<code>nosquare</code>	Dotproducts with twice the specified vector (square of the vector) are not taken into account.
<code>functions</code>	Vectors that are arguments of regular functions will also be considered. By default this is not done.
<code>!vector</code>	Dotproducts involving the specified vector are not treated.
<code>!set</code>	The set should be a set of vectors. All dotproducts involving a vector of the set are not treated.

7.152 tovector

Type Executable statement
Syntax tovector <tensor> <vector>;
 tovector <vector> <tensor>;
See also totensor (7.151)

The opposite of the totensor statement. The tensor is replaced by a product of the given vectors, each with one of the indices of the tensor as in:

$$t^{\mu_1\mu_2\mu_3} \rightarrow p^{\mu_1}p^{\mu_2}p^{\mu_3}$$

7.153 trace4

Type Executable statement
Syntax trace4 [<options>] <index>;
See also tracen (7.154), chisholm (7.17), unittrace (7.159)
 and the chapter on gamma algebra (14)

Takes the trace of the gamma matrices with the given trace line index. It assumes that the matrices are defined in four dimensions, hence it uses some relations that are only valid in four dimensions. For details about these relations and other methods used, consult chapter 14 on gamma matrices. The options are:

contract	Try to use the Chisholm identity to eliminate this trace and contract it with other gamma matrices. See also 7.17. This is the default.
nocontract	Do not use the Chisholm identity to eliminate this trace and contract it with other gamma matrices. See also 7.17.
nosymmetrize	When using the Chisholm identity to eliminate this trace and contract it with other gamma matrices, do not do it in the symmetric fashion, but use the first contraction encountered. See also 7.17.
notrick	The final stage of trace taking, when all indices are different and there are no contractions with identical vectors, as well as no γ_5 matrices present, is done with n-dimensional methods, rather than with 4-dimensional tricks.
symmetrize	When using the Chisholm identity to eliminate this trace and contract it with other gamma matrices, try to do it in the symmetric fashion. See also 7.17.
trick	The final stage of trace taking, when all indices are different and there are no contractions with identical vectors is done using the 4-dimensional relation $\gamma^a\gamma^b\gamma^c = \epsilon^{abcd}\gamma_5\gamma^d + \gamma^a\delta^{bc} - \gamma^b\delta^{ac} + \gamma^c\delta^{ab}$ This gives a shorter result for long traces. It is the default.

7.154 **tracen**

Type Executable statement
Syntax `tracen <index>;`
See also `trace4` (7.153), `chisholm` (7.17), `unittrace` (7.159)
and the chapter on gamma algebra (14)

Takes the trace of the gamma matrices with the spin line indicated by the index. It is assumed that the trace is over a symbolic number of dimensions. Hence no special 4-dimensional tricks are used. The presence of γ_5 , γ_6 or γ_7 is not tolerated. When indices are contracted FORM will try to use the special symbol for the dimension-4 if it has been defined in the declaration of the index (see 7.74). This results in relatively compact expressions. For more details on the algorithm used, see chapter 14 on gamma matrices.

7.155 **transform**

Type Executable statement
Syntax `transform,function(s),<one or more transformations>;`

Statement to manipulation function arguments and fields of arguments. Allows speedy transformations without the need of multiple statements or repeat loops.

The function(s) is/are indicated as individual, comma or blank space separated, functions or sets of functions.

If there is more than one transformation, the transformations are separated by comma's (or blanks, unless the blank space would not induce a comma).

Each transformation consists of its keyword, indicating its type, followed by a range of arguments that is enclosed by parentheses. After that specific information may follow. The range is as in

```
(1,4)
(3,last)
(last-6,last-2)
```

hence two indicators, separated by a comma. If the first number is bigger than the second the arguments will be processed in reverse order whenever this is relevant. In the descriptions below we will indicate the range by (r1,r2). The numbers in the above examples may be also dollar variables, provided they evaluate into numbers at the time of execution. Hence

```
($x,$y)
($x,last)
(last-$x,last-2)
```

are potentially legal ranges. One may not use `$x+2` or other expressions that still need evaluation.

The transformations that are allowed currently are:

replace	<code>replace(r1,r2)=(from1,to1,from2,to2,...,fromn,ton)</code> in which the from-to pairs are as in the <code>replace_</code> function. Here however there are more options than in the <code>replace_</code> function as we can specify (small) numbers as well as in <code>replace(1,last)=(0,1,1,0)</code> which would replace arguments that are zero by one and arguments that are one by zero. Generic arguments are indicated by the new variables <code>xarg_</code>, <code>iarg_</code>, <code>parg_</code> and <code>farg_</code> as in <code>replace(1,last)=(xarg_,2xarg_+1,p)</code> which would replace <code>f(2,a)</code> by <code>f(5, 2a+1,p)</code> if <code>a</code> is a symbol and <code>p</code> a vector. To catch <code>p</code> one would need to use <code>parg_</code>.
encode	<code>encode(r1,r2):base=number</code> will interpret the arguments as the digits in a base 2 number system, compute the complete number and replace the arguments by a single argument that is that number. The number must fit inside a single FORM word and so must each of the original arguments. They should actually be smaller than the number of the base.
decode	<code>decode(r1,r2):base=number</code> will do the opposite of <code>encode</code>. It will take a single argument (the smallest of the two given) and expand it into digits in a number system given by the base. It will create the specified number of digits and replace the original number by the given number of arguments representing these digits. If <code>r2</code> is less than <code>r1</code> the digits will be in reverse order.
tosumnotation	<code>tosumnotation(r1,r2)</code> or <code>implode(r1,r2)</code> realizes an encoding in which zeroes are absorbed as extra values in the first nonzero argument that is following. This is used when dealing with harmonic sums and harmonic polylogarithms. An example is that <code>(0,0,1,0,a,0,0,0,-1)</code> (which is in integral notation) goes into <code>(3,2*a,-4)</code> (which is in sum notation). Currently only a single symbol is allowed and the numbers should be (small) integers because otherwise the reverse operation (<code>explode</code>) would generate too many arguments. Instead of “<code>tosumnotation</code>” one may also use the word “<code>implode</code>” in accordance with the <code>argimplode</code> statement.
tointegralnotation	<code>tointegralnotation(r1,r2)</code> or <code>explode(r1,r2)</code> undoes what <code>implode</code> might have done. Hence each integer with an absolute value n generates $n - 1$ zeroes and leaves something with absolute value one. Instead of “<code>tointegralnotation</code>” one may also use the word “<code>explode</code>” in accordance with the <code>argexplode</code> statement.
permute	<code>permute(1,3,5)(2,6)</code> will permute the arguments according to the cycles indicated. The cycles are executed in order and may overlap. Their number is not restricted. In the above example <code>f(a1,a2,a3,a4,a5,a6,a7) → f(a3,a6,a5,a4,a1,a2,a7)</code>. It is allowed to use <code>\$</code>-variables in the cycles, including <code>\$</code>-variables that are obtained by matching argument field wildcards.
reverse	<code>reverse(r1,r2)</code> reverses the order of the arguments in specified range.
dedup	<code>dedup(r1,r2)</code> removes duplicates from the arguments in the range, keeping the first.
cycle	<code>cycle(r1,r2)=+/-number</code> will perform a cyclic permutation of the indicated range of arguments. If the number is preceded by a <code>-</code> the cycling is to the left. If there is a plus sign the cycling is to the right. Note that either the plus or the minus sign is mandatory. The number following the <code>+/-</code> sign is also allowed to be a dollar variable provided it evaluates to a legal number during execution.

islyndon	islyndon(r1,r2)=(yes,no) will test whether the indicated range of arguments forms a Lyndon word according to the ordering of arguments in FORM. The yes and no arguments are what the main term will be multiplied by when the range forms a Lyndon word or does not respectively. Because the definition of a Lyndon word is the unique minimal cyclic permutation of the arguments, and because often we may need the unique maximal cyclic permutation there are varieties: for the minimum one may also use islyndon<(r1,r2)=(yes,no) or islyndon-(r1,r2)=(yes,no), while for the maximum one may use islyndon>(r1,r2)=(yes,no) or islyndon+(r1,r2)=(yes,no).
tolyndon	tolyndon(r1,r2)=(yes,no) will permute the given range in a cyclic manner till it is (if possible) a Lyndon word according to the ordering of arguments in FORM. The yes and no arguments are what the main term will be multiplied by when afterwards the range forms a Lyndon word or does not respectively. Because the definition of a Lyndon word is the unique minimal cyclic permutation of the arguments, and because often we may need the unique maximal cyclic permutation there are varieties: for the minimum one may also use tolyndon<(r1,r2)=(yes,no) or tolyndon-(r1,r2)=(yes,no), while for the maximum one may use tolyndon>(r1,r2)=(yes,no) or tolyndon+(r1,r2)=(yes,no). If the output is not a Lyndon word, this will be due to that it is a minimum or maximum that is not unique.
addargs	addargs(r1,r2) replaces the indicated range of arguments by their sum. This is effectively the inverse of the SplitArg statement.
mulargs	mulargs(r1,r2) replaces the indicated range of arguments by their product. This is effectively the inverse of the FactArg statement.
dropargs	dropargs(r1,r2) removes the indicated range of arguments.
selectargs	selectargs(r1,r2) removes all arguments with the exception of the indicated range of arguments.

Some Examples. Assume that we have some Multiple Zeta Values (see the papers on harmonic sums, harmonic polylogarithms and the MZV data mine) in the sum notation, but for calculational reason we want to use a binary encoding (as used in the MZV programs). We could have

```

Symbol x,x1,x2;
CF H,H1;
Off Statistics;
L F = H(3,4,2,6,1,1,1,2);
repeat id H(?a,x?!{0,1},?b) = H(?a,0,x-1,?b);
Print;
.sort

F =
H(0,0,1,0,0,0,1,0,1,0,0,0,0,0,1,1,1,1,0,1);

Multiply H1;
repeat id H(x?,?a)*H1(?b) = H(?a)*H1(?b,1-x);
id H1(?a)*H = H(?a);
Print;
.sort

```

```

F =
  H(1,1,0,1,1,1,0,1,0,1,1,1,1,1,0,0,0,0,1,0);

repeat id H(x1?,x2?,?a) = H(2*x1+x2,?a);
Print;
.end

F =
  H(907202);

```

The new version of the same program would be

```

Symbol x,x1,x2;
CF H,H1;
Off Statistics;
L F = H(3,4,2,6,1,1,1,2);
Transform,H,explode(1,last),
      replace(1,last)=(0,1,1,0),
      encode(1,last):base=2;

Print;
.end

F =
  H(907202);

```

It should be clear that this is simpler and faster. On a 64-bits computer it is faster by more than a factor 100.

7.156 tryreplace

Type Executable statement
 Syntax tryreplace {<name> <replacement>};
 See also the replace_ function (8.59)

The list of potential replacements should be similar to the arguments of the replace_ function (see 8.59). FORM will make a copy of the current term, try the replacement and if the replacement results in a term which, by the internal ordering of FORM, comes before the current term, the current term is replaced by the new variety.

7.157 unfactorize

Type Output control statement
 Syntax unfactorize {<name of expression(s)>};
 See also the chapter on polynomials 11 and the factorize statement 7.56.

Without arguments the statement causes all expressions that were factorized to be 'unfactorized'. This means that all factors are multiplied and the expression is replaced by this new version. Like the factorize statement this statement is an output control statement, which means that it takes effect after an expression has been processed in the current module (see also the factorize 7.56 statement).

Because an immediate multiplication of all factors is sometimes far from optimal, FORM uses a binary scheme to combine factors. After each step there will be a sort operation. This means that when statistics are printed, there may be several statistics for this step.

When the statement has arguments, these arguments should be names of expressions. In that case the unfactorization is applied only to the expressions that are specified.

If one likes to unfactorize all expressions except for a few ones, one can use the unfactorize statement without arguments and then exclude the few expressions that should not be treated with the nunfactorize statement (see 7.104).

7.158 unhide

Type	Specification statement
Syntax	unhide; unhide <list of expressions>;
See also	hide (7.67), nhide (7.96), nunhide (7.105), pushhide (7.119), pophide (7.113)

In its first variety this statement causes all statements in the hide file to become active expressions again. In its second variety only the specified expressions are taken from the hide system and become active again. An expression that is made active again can be manipulated again in the module in which the unhide statement occurs. For more information one should look at the hide statement in 7.67.

Note that if only a number of expressions is taken from the hide system, the hide file may be left with 'holes', i.e. space between the remaining expressions that contain no relevant information any longer. FORM contains no mechanism to use the space in these holes. Hence if space is at a premium and many holes develop one should unhide all expressions (this causes the hide system to be started from zero size again) and then send the relevant expressions back to the hide system.

7.159 unittrace

Type	Declaration statement
Syntax	u[nittrace] <value>;
See also	trace4 (7.153), tracen (7.154), chisholm (7.17) and the chapter on gamma algebra (14).

Sets the value of the trace of the unit matrix in the Dirac algebra (i.e. the object **g1_(n)** for trace line **n**). The parameter **value** can be either a short positive number or any symbol with the exception of **i_**. See also chapter 14.

7.160 vectors

Type Declaration statement
Syntax `v[ectors] <list of vectors to be declared>;`

Used for the declaration of vectors. Example:

```
Vectors p,q,q1,q2,q3;
```

7.161 while

Type Executable statement
Syntax `while ( condition );`
See also `endwhile` (7.51), `repeat` (7.126), `if` (7.71)

This statement starts the while environment. It should be paired with an `endwhile` statement (see 7.51) which terminates the while environment. The statements between the `while` and the `endwhile` statements will be executed as long as the condition is met. For the description of the condition one should consult the `if` statement (see 7.71). The `while/endwhile` combination is equivalent to the construction

```
repeat;  
  if ( condition );  
  
  endif;  
endrepeat;
```

If only a single statement is inside the environment one can also use

```
while ( condition ) statement;
```

Of course one should try to avoid infinite loops. In order to maximize the speed of FORM not all internal stacks are protected and hence the result may be that FORM may crash. It is also possible that FORM may detect a shortage of buffer space and quit with an error message.

For each term for which execution reaches the `endwhile` statement, control is brought back to the `while` statement. For each term that reaches the `while` statement the condition is checked and if it is met, the statements inside the environment are executed again on this term. If the condition is not met, execution continues after the `endwhile` statement.

7.162 write

Type Declaration statement
Syntax `w[rite] <keyword>;`
See also `on` (7.108), `off` (7.107)

This statement is considered obsolete. All its varieties have been taken over by the on statement (see 7.108) and the off statement (see 7.107). The current version of FORM will still recognize it, but the user is advised to avoid its usage. In future versions of FORM it is scheduled to be used for a different kind of writing and hence its syntax may change considerably. The conversion program conv2to3 should help in the conversion of programs written for version 2. For completeness we still give the syntax and how it should be converted. The keywords are:

allnames	Same as: On allnames;
allwarnings	Same as: On allwarnings;
highfirst	Same as: On highfirst;
lowfirst	Same as: On lowfirst;
names	Same as: On names;
powerfirst	Same as: On powerfirst;
setup	Same as: On setup;
shortstatistics	Same as: On shortstatistics;
shortstats	Same as: On shortstats;
statistics	Same as: On statistics;
stats	Same as: On stats;
warnings	Same as: On warnings;

Chapter 8

Functions

Functions are objects that can have arguments. There exist several types of functions in FORM. First there is the distinction between commuting and noncommuting functions. Commuting functions commute with all other objects. This property is used by the normalization routines that bring terms into standard form. Noncommuting functions do not commute necessarily with other noncommuting functions. They do however commute with objects that are considered to be commuting, like symbols, vectors and commuting functions. Various instances of the same noncommuting function but with different arguments do not commute either.

The next subdivision of the category of functions is in regular functions, tensors and tables. Tensors are special functions that can have only indices or vectors for their arguments. If an argument is a vector, it is assumed that this vector is there as the result of an index contraction. Tables are functions with automatic substitution rules. A table must have at least one table index. Each time during normalization FORM will check whether an instance of a table can be substituted. This means that undefined table elements will slow the program down somewhat.

All the various types of functions are declared with their own declaration statements. These are described in the chapter for the statements (see chapter 7).

One of the useful properties of functions is the wildcarding of their arguments during pattern matching. The following argument wildcards are possible:

- x? Here x is a symbol. This symbol can match either a symbol, any numerical argument, or a complete subexpression argument that is not vectorlike or indexlike.
- i? Here i is an index. This index can match either an index, a vector (actually the dummy index of the vector that was contracted), or a complete subexpression that is vector like (again actually the contracted dummy index).
- v? Here v is a vector. This vector can match either a vector or a complete subexpression that is vector like.
- f? Here f is any functiontype. This function can match any function. It is the responsibility of the user to avoid problems in the right-hand side if f happens to match a tensor.
- ?a This is an argument field wildcard. This can match a complete set of arguments. The set can be empty. Argument field wildcards have a name that starts with a question mark followed by a name. They do not have to be declared as there cannot be confusion.

In addition to the above syntax FORM knows a number of special functions with well defined properties. All these functions have a name that ends in an underscore. In addition the names of these built in objects are case insensitive. This means for instance that the factorial function can

be referred to as `fac_`, `Fac_` or `FAC_` or whatever the user considers more readable. The built in functions are:

8.1 `abs_`

With one argument that is numerical it evaluates into the absolute value of the argument.

8.2 `bernoulli_`

If it has one nonzero integer argument n , it evaluates into the n -th coefficient in the power series expansion of $x/(1 - e^{-x})$.

8.3 `binom_`

`binom_(n,i) =  $n!/(i!(n-i)!)$` . If the arguments are non integer or negative, no substitution is made.

8.4 `conjg_`

Currently not doing anything.

8.5 `content_`

This function expects the name of a single expression or a dollar variable for its argument. If it finds this the content of this expression or dollar variable is returned. The content is defined as a term that has

- for its numerator the GCD of the numerators of all terms in the expression.
- for its denominator the LCM of the denominators of all terms in the expression.
- all the common subexpressions in all terms of the expression.
- the most negative powers of all symbols and dotproducts with negative powers in the terms of the expression.

When there are no negative powers and no denominators in the coefficients, this definition of the content coincides with the classical definition of the content of a polynomial over the integers. Our content has the property that if we divide the expression by it, we are left with an expression of which the coefficients are all integer, there are no negative powers and the GCD of all terms combined is one.

This function has one limitation. It will not consider noncommuting objects. Neither will it consider denominator functions.

Caveat: this function is evaluated each time it is encountered. Therefore the best thing is to evaluate it once in the definition of a dollar variable or an expression as in

```
##$x = content_(F);
Local G = (a+b)^10*$x;
```

Here the content is computed only once. In

```
Local G = (a+b)^10*content_(F);
```

11 terms are generated and the content is only worked out when the terms are normalized. This means that it will be evaluated 11 times. If one does not like dollar variables and still wants to evaluate the content only once the code would be

```
Local G = ab^10*content_(F);
id ab = a+b;
```

because now the term will be normalized before the substitution makes it into eleven terms. This assumes of course that the content does not contain the variable ab.

8.6 count_

Similar to the count object in the if statement (see 7.71). This function expects the same arguments as the count object and returns the corresponding count value for the current term.

8.7 d_

The kronecker delta. Should have two indices for arguments. Often indicated as $\delta^{\mu\nu}$. In automatic summation over the indices the d_ often vanishes again as in $d_{\mu\nu} p_{\mu} q_{\nu} \rightarrow p \cdot q$ and similar replacements. Internally this object is treated in a rather special way. Hence it will not match a function wildcard.

8.8 dd_

This is a combinatorics function. The tensor dd_ with an even number of indices is equal to the totally symmetric tensor built up from products of kronecker delta's. Each term in this symmetric combination is normalized to one. In principle there are $n!/(2^{n/2}(n/2)!)$ terms in this combination. The profit comes when some or all the indices are contracted with vectors and some of these vectors are identical. In that case FORM will use combinatorics to generate only different terms, each with the proper prefactor. This can result in great time and space savings.

8.9 delta_

With one numerical argument the result is one if the argument is zero and zero otherwise. With two arguments the result is one if the arguments are numerical and identical. If they are numerical and they differ the result is zero. In all other cases nothing is done.

8.10 deltap_

If one argument and it is numerical the result is zero if the argument is zero and one otherwise. If two arguments, the result is zero if the arguments are numerical and identical. If they are numerical and they differ the result is one. In all other cases nothing is done.

8.11 denom_

Internal function to describe denominators. Has a single argument. `den(a+b)` is printed as $1/(a+b)$.

8.12 distrib_

This is a combinatorics function. It should have at least five arguments. If we have

```
distrib_(type,n,f1,f2,x1,...,xm)
```

with type and n integers, f1 and f2 functions and then a number of arguments there can be action if $-2 \leq \text{type} \leq 2$. The typical action is that the arguments $x_1, \dots, x_m$ will be divided over the two functions in all possible ways. For each possibility a new term is generated. The relative order of the arguments is kept. If type is negative it is assumed that the collection of x-arguments is antisymmetric and hence the number of permutations needed to make the split will determine whether there will be a minus sign on the resulting term. When type is zero all possible divisions are generated. Hence there will be 2^m divisions. The second argument is then not relevant. If type is 1 or -1 the second parameter says that the first function should obtain n arguments. The remaining arguments go to the second function. If type is 2 or -2 the second function should obtain n arguments. Example:

```
Symbols x1,...,x4;
CFunctions f,f1,f2;
Local F = f(x1,...,x4);
id f(?a) = distrib_(-1,2,f1,f2,?a);
Print +s;
.end
```

```
F =
+ f1(x1,x2)*f2(x3,x4)
- f1(x1,x3)*f2(x2,x4)
+ f1(x1,x4)*f2(x2,x3)
+ f1(x2,x3)*f2(x1,x4)
- f1(x2,x4)*f2(x1,x3)
+ f1(x3,x4)*f2(x1,x2)
;
```

When adjacent x-arguments are identical FORM uses combinatorics to avoid generating more terms than necessary.

8.13 div_

`div_(x1,x2)` is replaced by the quotient of the arguments. The arguments can be any valid subexpressions, provided the whole function fits inside a term. When an argument is only an active expression or a \$-expression it is only expanded during the division. This way the contents of such expressions can exceed the maximum term size. One should however realize that in that case the operation takes place in allocated memory. This function replaces the experimental function `polydiv_` that existed in version 3.

8.14 dum_

Special function for printing virtual brackets. `dum_(a+b)` is printed as `(a+b)`: the name of this function is not printed!

8.15 dummy_

For internal use only.

8.16 dummyten_

For internal use only.

8.17 e_

The Levi-Civita tensor. It is a totally antisymmetric tensor with well defined contraction rules (see 7.23).

8.18 exp_

Internal function with two arguments. Represents argument1 to the power argument2. Of course it is printed in the standard power notation.

8.19 exteuclidean_

This is a number function. It expects two positive integer arguments. It then computes the Greatest Common Divider of these arguments with the use of the extended Euclidean algorithm. The answer will be in the same function but now there will be four arguments as in:

```
Symbols x1,x2,x3,x4;
Local F = exteuclidean_(54,84);
Print;
.sort

F =
    exteuclidean_(54,84,-3,2);

id exteuclidean_(x1?,x2?,x3?,x4?) = x1*x3+x2*x4;
Print;
.end

F =
    6;
```

We can see that we obtain the GCD with the relation that is characteristic for the extended Euclidean algorithm. When the two arguments are relative prime, one obtains the so-called mod-inverses of these numbers:

```
Symbols x1,x2,x3,x4,a,b;
Local F = exteuclidean_(97,101);
Print;
.sort
```

```

F =
    exteuclidean_(97,101,25,-24);

id exteuclidean_(x1?,x2?,x3?,x4?) = x1*x3+x2*x4
    +a*mod2_(1/97,101)+b*mod2_(1/101,97);
Print;
.end

F =
    1 - 24*b + 25*a;

```

Here 25 is the inverse of 97 when we calculate modulus 101 and -24 is the inverse of 101 when we calculate modulus 97.

This function can be very handy when a calculation has been done modulus various prime numbers and one would like to know the result modulus the product of these numbers. This combination is done with the aid of the Chinese remainder theorem:

```

#procedure ChineseRemainder(NAME,NAME1,NAME2,M1,M2,PAR)
*
*   Assumes that NAME1 is an expression mod $M1
*   Assumes that NAME2 is an expression mod $M2
*   Creates $ch1r and $ch2r with the property that
*   the expression NAME = NAME1*$ch1r+NAME2*$ch2r
*   is the corresponding equation mod $M1*$M2
*
Modulus 0; * we need to switch off previous settings.
#$ch1r = exteuclidean_('$M1','$M2');
#inside $ch1r;
    id exteuclidean_(xxx1?,xxx2?,xxx3?,xxx4?) = xxx2*xxx4;
#endinside;
#$ch2r = exteuclidean_('$M1','$M2');
#inside $ch2r;
    id exteuclidean_(xxx1?,xxx2?,xxx3?,xxx4?) = xxx1*xxx3;
#endinside;
$MM12 = '$M1'*'$M2';
Modulus,plusmin,'$MM12';
Local 'NAME' = 'NAME1i'*$ch1r+'NAME2i'*$ch2r;
.sort
*
#endprocedure

```

8.20 extrasymbol_

This function expects a single argument. This argument can be a number or an extra symbol(see 2.11). In either case the function is replaced by the expression that the corresponding extra symbol stands for.

If there are more arguments or the argument does not represent a legal extra symbol, no substitution is made.

8.21 fac_

The factorial function. If it has a single nonzero integer argument n it is replaced by $n!$ but if the result is bigger than the maximum allowable number an error will result.

8.22 factorin_

When the argument is a single $\$$ -variable or an expression the function is replaced by the common factor in the terms of that $\$$ -variable or expression. This common factor consists in the first place of all symbolic objects that occur in all terms. In addition the numerical factor consists of the GCD of all numerators and the LCM of all denominators. Hence if the $\$$ -variable or expression is divided by the result of factorin_ all coefficients become integer.

8.23 farg_

For internal use only.

8.24 firstbracket_

In the case that there is a single argument and this single argument is the name of an expression, this function is replaced by the part that is outside brackets in the first term of the expression. If there are no brackets the function is replaced by one.

8.25 firstterm_

This function expects the name of an expression or a dollar variable for its (single) argument. It will return the first term in this expression or dollar variable. When it has to obtain the first term of an expression, FORM uses the expression in the representation in which it was stored at the end of the previous module. If the expression did not exist in the previous module, it will attempt to use the expression as defined and processed in the current expression. If the expression has only been defined in the current module and has not yet been processed (as is the case when referring to the first term in the current expression) the answer will be unspecified. This use is considered illegal, even though it does not generate an error message.

8.26 g5_

The γ_5 Dirac gamma matrix. We assume here that it anticommutes with the other Dirac gamma matrices. Anybody who does not like that should program private libraries (this should not be too difficult with the cycle symmetric functions (see 7.62)). There should be a single index to indicate the spinline.

8.27 g6_

There should be a single index to indicate the spinline. As in Schoonschip we use $\gamma_6 = 1 + \gamma_5$.

8.28 `g7_`

There should be a single index to indicate the sinline. As in Schoonschip we use $\gamma_7 = 1 - \gamma_5$.

8.29 `g_`

The Dirac gamma matrix. Its first argument should be an index (either symbolic or numeric). Then follow zero, one or more indices to indicate a string of gamma matrices that belong together. Gamma matrices with the same first index are considered to belong together, but as long as the indices are symbolic no assumptions are made about whether they go together or not. Hence no commutation or anticommutation properties are applied for different spin lines unless the sinline indices are both numeric.

8.30 `gcd_`

`gcd_(x1,...,xn)` is replaced by the greatest common divisor of the arguments. The arguments can be any valid subexpressions, provided the whole function fits inside a term. When an argument is only an active expression or a $\$$ -expression it is only expanded during evaluation of the GCD. This way the contents of such expressions can exceed the maximum term size. One should however realize that in that case the operation takes place in allocated memory. This function replaces the experimental function `polygcd_` that existed in version 3.

8.31 `gi_`

The unit Dirac gamma matrix. Should have a single index to indicate its spin line. Its is identical to a regular gamma matrix with no Lorenz indices: $\text{gi}_n = g_n$

8.32 `id_`

This function is a crossbreed between the `replace_` 8.59 function and the `id` statement 7.68. To become active it needs an even number of arguments. The odd numbered arguments can be anything of the types:

- a single symbol, possibly to an integer power.
- a single dotproducts, possibly to an integer power.
- a single function, possibly with any number and type of arguments.

When FORM encounters an `id_` function the last step of normalizing a term is to replace the `id` function by a number substitutions in which the odd arguments are replaced by the following even arguments. These are not wildcard substitutions as in the `replace_` function, but substitutions as in regular `id` statements. The matching of the odd arguments is done in a single step as in an `id-al` construction 7.2. Hence

`id_(x^2,y+z,y,u+v,x,z+u)`

effectively becomes

```

id x^2 = y+z;
al y = u+v;
al x = z+u;

```

FORM treats multiple occurrences of the `id_` function one at a time. It takes the leftmost occurrence first, takes the patterns from the term, expands the right hand sides, tries to normalize the resulting terms and only then continues with the next `id_` function. For this reason the `id_` function is noncommuting.

8.33 integer_

This is a rounding function. It should have either one or two arguments. If there is a single argument and it is numeric, it will be rounded down to become an integer. If there are two arguments of which the first is numeric and the second is either 1, 0 or -1, the result will be the rounded value of the first argument. If the second argument is 1, the rounding will be down, when it is -1, the rounding will be up and when it is zero the rounding will be towards zero. In all other cases nothing is done.

8.34 inverse_

`inverse_(x1,x2)` expects two arguments which are polynomials in the same single variable. The return expression x_3 has the property that x_1x_3 divided by x_2 has remainder 1. Or in other words: x_3 is the inverse of x_1 modulus x_2 . The arguments can be any valid subexpressions, provided the whole function fits inside a term. When an argument is an active expression or a $\$$ -expression it is only expanded during the division. This way the contents of such expressions can exceed the maximum term size. One should however realize that in that case the operation takes place in allocated memory.

8.35 invfac_

One divided by the factorial function. If it has a single nonzero integer argument n , it is replaced by $1/n!$, but if this results in a number bigger than the maximum allowable number an error will result.

8.36 makerational_

This function takes two arguments. Both are integers. We assume calculus modulus the second argument. The function is then replaced by a fraction of which both elements are less than the square root of the second argument and that, in calculus modulus this second number would give the same result as the first number modulus the second number. Example:

```

#$m = prime_(1);
#write <> "The prime number is %$", $m
The prime number is 2147483587
L F = MakeRational_(12345678,$m);
Print;
.sort

```



```

F =
    9719/38790;

Modulus '$m';
Print;
.end

F =
    12345678;

```

This function can be used to reconstruct fractions when calculus has been done modulus one or more prime numbers.

8.37 `match_`

Currently not active. Replaced automatically by 1.

8.38 `max_`

If all its arguments are numeric, this function returns the maximum value of these arguments.

8.39 `maxpowerof_`

If this function has a single argument that is a symbol, it returns the maximum power restriction of this symbol. If none was given it will be the installation dependent value MAXPOWER which is 10000 on 32 bit machines and 500000000 on 64 bit machines.

8.40 `min_`

If all its arguments are numeric, this function returns the minimum value of these arguments.

8.41 `minpowerof_`

If this function has a single argument that is a symbol, it returns the minimum power restriction of this symbol. If none was given it will be the installation dependent value -MAXPOWER which is -10000 on 32 bit machines.

8.42 `mod_`

If there are two integer arguments and the second argument is a positive short integer (less than 2^{15} on 32 bit computers and less than 2^{31} on 64 bit computers) the return value is the first argument modulus the second. Note that if the second argument is not a prime number and the first argument contains a denominator, division by zero could occur. It is up to the user to avoid such cases. See also the `mod2_` function 8.43 and the `rem_` function 8.58.

The function has one peculiarity: when the second argument is one, the function is left untouched.

8.43 mod2_

This gives basically the same action as the `mod_` function (see 8.42), but the answer will be in the range $-\lceil(p-1)/2\rceil$ to $\lceil(p+1)/2\rceil$.

8.44 moebius_

The Moebius function. It has the property:

- `moebius_(1) = 1`
- `moebius_(n) = 0` if `n` can be divided by the square of a prime number
- `moebius_(n) = 1` if `n` contains an even number of different prime factors
- `moebius_(n) = -1` if `n` contains an odd number of different prime factors

The number `n` must be a positive integer for the evaluation to take place. The argument should be no bigger than can be allowed by the biggest primes in Form. This means for a 64-bit computer arguments above 2^{31} may not give a correct result with the current algorithms that try for prime factors. There is a second algorithm that does not suffer from this problem:

```
id moebius_(n) = 1-sum_(j,1,n-1,integer_(n/j)*moebius_(j));
```

but this algorithm needs all lower values and hence is inherently quadratic. Therefore we did not use it.

8.45 mul_

`mul_(x,y)` is replaced by `x*y`, but internally the multiplication is performed via polynomial routines introduced in FORM version 4. This can be faster than the normal way of multiplications for big polynomials: e.g., `mul_($x,$y)` where the `$`-variables `$x` and `$y` store big polynomials.

A drawback is, because the polynomial routines accept only symbols, all non-symbolic objects in the operands are temporarily translated to (commuting) extra symbols. This process breaks the ordering of non-commutative objects in the result.

8.46 nargs_

Is replaced by an integer indicating the number of arguments that the function has.

8.47 node_

For a description of this function, please see the section on diagrams 21.

8.48 nterms_

If this function has only one argument it is replaced by the number of terms inside this argument.

8.49 numfactors_

This function returns the number of factors in a factorized expression (see the chapter on polynomials 11) or dollar variable 6. It expects a single argument which should be the name of an expression or a dollar variable. If the expression or dollar variable has not been factorized, the function returns zero.

8.50 partitions_

This function generates all partitions of a list of arguments into n parts. Each part consists of a function name and a size. This function exploits symmetries of the arguments to make sure that no argument is generated twice. Instead, a combinatorial prefactor is computed.

The syntax distinguishes three cases:

- 1] `partitions_(n,[function,n1,]_1,...,[function,nn,]_n,arguments)`
- 2] `partitions_(n,[function,n1,]_1,...,[function,0],arguments)`
- 3] `partitions_(0,function,n1,arguments)`

In the first case, the first entry specifies the number of partitions n . It should be followed by n parts, defined by a function name and the number of arguments for that function. The final entries are the arguments that will be distributed over the functions. The number of arguments should be the same as the sum of all the function argument sizes. There are no restrictions on the type of arguments.

The second case is the same as the first, except that the last partition has a 0 for the size. This means that any leftover arguments are collected in this term. Thus `partitions_(2,f1,3,f2,0,arguments)` yields the same as `distrib_(1,3,f1,f2,arguments)`.

The third case, determined by a 0 for the number of partitions followed by one part, spreads the arguments over a repeated instance of that part. Thus `partitions_(0,f1,2,arguments)` is similar to `dd_(arguments)`.

In case of a deviation from the above rules, no action will be taken. Some examples are given below:

```
partitions_(2,f1,2,f2,1,x1,x1,x3) =
    + f1(x1,x1)*f2(x3) + 2*f1(x1,x3)*f2(x1)
    ;
partitions_(3,f1,2,f2,1,f3,0,x1,x1,x1,x2,x2,x2) =
    + 3*f1(x1,x1)*f2(x1)*f3(x2,x2,x2)
    + 9*f1(x1,x1)*f2(x2)*f3(x1,x2,x2)
    + 18*f1(x1,x2)*f2(x1)*f3(x1,x2,x2)
    + 18*f1(x1,x2)*f2(x2)*f3(x1,x1,x2)
    + 9*f1(x2,x2)*f2(x1)*f3(x1,x1,x2)
    + 3*f1(x2,x2)*f2(x2)*f3(x1,x1,x1)
    ;
partitions_(0,f1,3,x1,x1,x1,x4,x5,x6) =
    + f1(x1,x1,x1)*f1(x4,x5,x6)
    + 3*f1(x1,x1,x4)*f1(x1,x5,x6)
    + 3*f1(x1,x1,x5)*f1(x1,x4,x6)
    + 3*f1(x1,x1,x6)*f1(x1,x4,x5)
    ;
```

8.51 pattern_

Currently not active. Replaced automatically by 1.

8.52 perm_

Generates all permutations of the arguments, with exception of the first argument which should be the name of a function. This function will then have the permuted arguments as in:

```
CFunction f;  
Symbols x1,...,x3;  
Local F = perm_(f,x1,x2,x3);  
Print +s;  
.end
```

```
F =  
+ f(x1,x2,x3)  
+ f(x1,x3,x2)  
+ f(x2,x1,x3)  
+ f(x2,x3,x1)  
+ f(x3,x1,x2)  
+ f(x3,x2,x1)  
;
```

The permutations are generated with an algorithm that takes subsequent cyclic permutations. If one puts a nonzero integer before the function argument the output terms will be multiplied by -1 when the permutation is odd.

When the function name is the only argument the answer will be just this function without arguments. One could argue that technically the answer should be zero, but this way the attention of the user may be attracted to the occurrence which might not be the case when the term 'just vanishes'. It is however rather simple to add a statement that makes such a function zero.

8.53 poly_

This was an experimental function in version 3. It was for internal use with a whole category of other experimental functions of which the functionality has been replaced by better working functions that are more general. This category included the functions polyadd_, polydiv_, polygcd_, polyintfac_, polymul_, polynorm_, polyrem_ and polysub_. See also the chapter on polynomials 11 and the functions gcd_ 8.30, div_ 8.13 and rem_ 8.58.

8.54 prime_

For a number of internal operations FORM needs prime numbers that are neither very large nor very small. Hence it generates, when needed prime numbers that still fit inside a single FORM word, but are maximal within that limitation. Hence for a 64-bits computer in which the largest positive 'small' integer in FORM is $2^{31} - 1$, it works its way down from there. Once it has determined that a number is prime it stores it in a list. The function prime_ gives access to this list. The single argument n (n a positive integer) makes that **prime_(n)** will be replaced by the n-th

member of the list. There is a limitation to the size of the list which is implementation dependent. The number will anyway never be smaller than the maximum power that is allowed for symbols. Example:

```
Symbols x1,x2,x3,x4;
ON highfirst;
Local F = x1*prime_(1)+x2*prime_(2)
          +x3*prime_(3)+x4*prime_(4);
Print;
.end
```

```
F =
2147483587*x1 + 2147483579*x2 + 2147483563*x3 + 2147483549*x4;
```

This function is useful when calculations generate very large intermediate coefficients, but in the end the answer is relatively simple again. In that case one can do the calculation modulus one or more prime numbers. If more prime numbers are used the Chinese remainder theorem can be used (see the `exteuclidean_` function 8.19 to combine the results and the `makerational_` function 8.36 can be used if fractions have to be reconstructed). An example of this kind of use is given in the simple Groebner basis procedure that is in the packages library in the FORM site.

8.55 `putfirst_`

This function allows one to select a given argument by its number. The syntax is:

```
putfirst_(functionname,numberofargument,arguments....);
```

It will select the indicated argument in the argument field indicated by arguments and output this as the first argument in the indicated function. This argument will then be followed by the remaining arguments. Example:

```
S  a,a1,...,a10;
CF f,g;
L  F = g(a,a1,...,a10);
id g(?a) = putfirst_(f,4,?a);
Print;
.end
F =
f(a3,a,a1,a2,a4,a5,a6,a7,a8,a9,a10);
```

8.56 `random_`

A random number generator. When the function has a single positive integer argument, the function will return a pseudo random number in the range of one to that number inclusive. Hence one can imitate a die roll with the call `random_(6)`. The program uses a random number generator as described in vol 2 of the "Art of computer programming, vol2" by D. Knuth with the parameters set at 89,38 to give as long a cycle as possible. For very large numbers the program pastes several random numbers together. The generator can be initialized with the preprocessor `#setrandom 3.54` instruction. When running with TFORM or PARFORM each worker runs an independent generator with its own seed. The seeds of the workers are derived from the seed of the master and the

number of the worker in a non-trivial way. It should be noted however that with workers it may be impossible to reproduce previous runs as it is non-deterministic which term ends up in which worker.

8.57 ranperm_

Generates a random permutation of the arguments, with exception of the first argument which should be the name of a function. This function will then have the permuted arguments as in:

```
CFunction f;
Symbols x1,...,x5;
Local F = ranperm_(f,1,2,3,4,5,6)
          +ranperm_(f,x1,x2,x3+x1,x4,x5);
Print +s;
.end
```

```
F =
  + f(x5,x1,x3 + x1,x4,x2)
  + f(3,1,6,2,4,5)
;
```

The permutation is generated with the same random number generator that is used by the function `random_ 8.56` and hence is susceptible to the same initialization procedure that can be executed with the `#setrandom 3.54` instruction.

8.58 rem_

`rem_(x1,x2)` is replaced by the remainder of the division of x_1 by x_2 . The arguments can be any valid subexpressions, provided the whole function fits inside a term. When an argument is only an active expression or a $\$$ -expression it is only expanded during the division. This way the contents of such expressions can exceed the maximum term size. One should however realize that in that case the operation takes place in allocated memory. This function replaces the experimental function `polyrem_` that existed in version 3.

8.59 replace_

This function defines a rather general purpose replacement mechanism. It should have pairs of arguments. Each pair consists of a single symbol, index, vector or function, followed by what this object should be replaced by in the entire term. Functions can only be replaced by functions, indices only by indices. A vector can be replaced by a single vector or by a vector like expression. A symbol can be replaced by a single symbol, a numerical expression or a complete subexpression that is not index like or vector like. This mechanism is sometimes needed to make replacements in ways that are very hard with the `id` statements because those do not make replacements automatically inside function arguments (see 7.69). It also allows to exchange two variables as the replacements are executed simultaneously by the wildcard substitution mechanism.

```
Multiply replace_(x,y,y,x);
```

will exchange `x` and `y`. Because there is no definite order in which multiple `replace_` functions are treated, one should not use more than a single one at the same time inside a term. At times multiple `replace_` functions may lead to confusion inside FORM.

8.60 `reverse_`

Can only occur as an argument of a function. Is replaced by the reversed string of its own arguments.

8.61 `root_`

If we have `root_(n,x)` and `n` is a positive integer and `x` is a rational number and `y` is a rational number with $y^n = x$ (no imaginary numbers are considered and negative numbers are avoided if possible. Only one root is given) then `root_(n,x)` is replaced by `y`. This function was originally intended for internal use. Do not hold it against the author that `root_(2,1)` is replaced by `1`. In the case that it is needed the user should manipulate the sign or the complexity properties externally.

8.62 `setfun_`

Currently not active.

8.63 `sig_`

Is replaced by the sign of the (numerical) argument, i.e. by `-1` if there is a single negative argument and by `+1` if there is a single numerical argument that is greater or equal to zero.

8.64 `sign_`

`sign_(n)` is replaced by $(-1)^n$ if `n` is an integer.

8.65 `sizeof_`

If there is a single argument and this argument is the name of an active (or previously active during the current job) expression, the function is replaced by the number of FORM words in this expression. Stored expressions that were entered via a load statement (see 7.83) are excluded from this because for them this information is not readily available.

8.66 `sum_`

General purpose sum function. The first argument should be the summation parameter (a symbol). The second argument is the starting point of summation, the third argument the ‘upper’ limit and a potential fourth argument the increment. These numbers should all be integers. Summation stops when the summation parameter obtains a value that has passed the upper limit. The last argument is the summand, the object to be summed over. It can be any subexpression. If it contains the summation parameter, it will be replaced by its value for each generated term. Examples:

```
sum_(j,1,4,sign_(j)*x^j/j)
sum_(i,1,9,2,sign_((i-1)/2)*x^i*invfac_(i))
```

8.67 sump_

Special sum function. Its arguments are like for the `sum_` function, but each new term is the product of the previously generated term with the last argument in which the current value of the summation parameter has been substituted. The first term is always one. Example:

```
Symbol i,x;
Local F = sump_(i,0,5,x/i);
Print;
.end
```

```
F =
1 + x + 1/2*x^2 + 1/6*x^3 + 1/24*x^4 + 1/120*x^5;
```

This function is a leftover from the Schoonschip days. The ordinary `sum_` function is much more readable.

8.68 table_

For action the arguments should be the name of a table and then either the name of a function or one symbol for each dimension of the table. In the case of the list of symbols the return value will be a monomial in the given symbols in which the powers of the symbols correspond to the table indices of the defined table elements with the coefficients the table contents corresponding to those indices. In the case of a function name the return value will be a sum over terms in which the table elements are indicated by arguments in the given function while these functions are then multiplied by the corresponding table elements. This is one way to put a complete table inside an expression and store it (with the `save` statement of 7.128) in a binary way for a future run in which the table can be filled again with the `fillexpression` (see 7.58) statement. Note that for obvious reasons one should avoid using symbols or functions that also occur inside the table definitions.

8.69 tbl_

This function is the ‘table stub function’ as used by the `tablebase` construction. This is explained in chapter 12. It is mainly for internal use, but it could occur in the output.

8.70 term_

This function has no arguments. It is replaced by the current term. It can be used to load the current term into a dollar variable as in

```
$x = term_;
```


8.71 termsin_

If there is a single argument and this argument is the name of an active (or previously active during the current job) expression, the function is replaced by the number of terms in this expression. Stored expressions that were entered via a load statement (see 7.83) are excluded from this because for them FORM would have to actually count the terms.

8.72 termsinbracket_

If there is no argument, or the single argument is zero, the function is replaced by the number of terms in the current bracket, provided the expression has been bracketed at its last sort and a keep brackets statement (see 7.80) has been used. Note that the terms have to be counted. Hence this is a relatively expensive command. More options will be implemented in the future.

8.73 theta_

If there is a single numerical argument x the function is replaced by one if $x \geq 0$ and by zero if $x < 0$. If there are two numerical arguments x_1 and x_2 the function is replaced by one if $x_1 = x_2$ or if the arguments are in natural order (if theta_ would be a symmetric function there would be no reason to exchange the arguments) and by zero if the arguments are not in natural order (they would be exchanged in a symmetric function). In all other cases nothing is done.

8.74 thetap_

If there is a single numerical argument x the function is replaced by one if $x > 0$ and by zero if $x \leq 0$. If there are two numerical arguments x_1 and x_2 the function is replaced by zero if $x_1 = x_2$ or if the arguments are not in natural order. If the arguments are in natural order the function is replaced by one. In all other cases nothing is done.

8.75 topologies_

For a description of this function, please see the section on diagrams 21.

8.76 Extra reserved names

In addition there are some names that have been reserved for future use. At the moment these functions do not do very much. It is hoped that in the future some simplifications of the arguments can be implemented. These functions are:

sqrt_	The regular square root.
ln_	The natural logarithm.
sin_	The sine function.
cos_	The cosine function.
tan_	The tangent function.
asin_	The inverse of the sine function.

<code>acos_</code>	The inverse of the cosine function.
<code>atan_</code>	The inverse of the tangent function.
<code>atan2_</code>	Another inverse of the tangent function.
<code>sinh_</code>	The hyperbolic sine function.
<code>cosh_</code>	The hyperbolic cosine function.
<code>tanh_</code>	The hyperbolic tangent function.
<code>asinh_</code>	The inverse of the hyperbolic sine function.
<code>acosh_</code>	The inverse of the hyperbolic cosine function.
<code>atanh_</code>	The inverse of the hyperbolic tangent function.
<code>li2_</code>	The dilogarithm function.
<code>lin_</code>	The polylogarithm function.

The user is allowed to use these functions, but it could be that in the future they will develop a nontrivial behaviour. Hence caution is required.

Chapter 9

Brackets

At times one would like to order the output in a specific way. In an expression which is for instance a polynomial in terms of the symbol x , one might want to make this behaviour in terms of x more apparent by printing the output in such a way, that all powers of x are outside parentheses, and the whole rest is inside parentheses. This is done with the bracket statement:

```
Bracket x;
```

or in short notation

```
B x;
```

One can specify more than one object in the bracket statement, but only a single bracket statement (the last one) is considered. Bracket statements belong to the module in which they occur. Hence they are forgotten after the next end-of-module.

If a vector is mentioned in a bracket statement, all occurrences of this vector as a loose vector, a vector with any index, inside a dotproduct, or inside a tensor are taken outside brackets. If the vector occurs inside a non-commuting tensor, all other non commuting objects that are to the left of this tensor will also be taken outside the parentheses.

When a function or tensor is mentioned in a bracket statement, it is not allowed to have any arguments in the bracket statement. All occurrences of this function will be pulled outside brackets. If the function is non-commuting, all other functions and/or tensors that are non-commuting and are to the left of the specific function(s) or tensor(s) will also be outside parentheses.

The opposite of the bracket statement is the antibracket statement:

```
AntiBracket x;
```

or

```
ABracket x;
```

or

```
AB x;
```

This statement causes also brackets in the output, but now everything is put outside brackets, except for powers of x and coefficients. This way one can make the x -dependence apparent differently.

Because the bracket statement causes a different ordering of the terms when storing the expression, one can use this ordering in the next module. There are various ways to do this.

One can use the contents of a given bracket in a r.h.s. expression as in

```

Symbols a,b,c,x;
L F = a*x^2+b*x+c;
B x;
.sort
L Discriminant = F[x]^2-4*F[x^2]*F[1];
Print;
.end

```

The outside of the bracket is placed between braces after the name of the expression. The bracket that has nothing outside is referred to with the number 1. If a bracket is empty, its contents will be represented by the value zero.

The regular algorithm by which FORM finds brackets in an expression, is to start from the beginning and inspect each term until it finds the appropriate bracket. This is fully in the spirit of the sequential treatment of expressions in FORM. This can however be rather slow in big expressions that reside on a disk. Hence there is the bracket index feature. It is invoked by putting a `+`-sign after the bracket (or `B`) statement as in

```
Bracket+ x;
```

or

```
B+ x;
```

This option causes FORM to build a tree of (disk) positions for the different brackets, with the condition that the whole storage of this tree of brackets does not exceed a given maximum space, named 'bracketindexsize' (see chapter 17 on the setup parameters). If the index would need more space FORM will start skipping brackets in the index. This means that it will have to look for the bracket in a sequential fashion, but starting from the position indicated by the previous bracket in the index. This will still be very fast, provided the index is not very small.

When the bracket index option is used, FORM will not compress the expressions that use such an index with the zlib compression, even if the user asked for this in an earlier statement. The use of the index indicates that the brackets are going to be used intensively, and hence the continuous decompression that would result would destroy most of the profit that comes from the index. If the brackets are only for cosmetics in the output, it is better not to use the index option. It does use resources to construct the index tree. Also when brackets are only used sequentially as in the features discussed below, the presence of the index is not beneficial. It should only be applied when contents of brackets are used in the above way (like with the discriminant).

There are several statements that make use of the bracket ordering:

- **Keep Brackets;** This statement takes from the input one term at a time as usual, but then it takes the part outside the brackets, executes the statements of the module only on that part of the term, and then, when all statements of the module have had their effect, the resulting term(s) is/are multiplied by the full content of the bracket. The next term taken from the input will be the first term of the next bracket. This way one can hide part of the terms for the pattern matcher. Also one can avoid that the same matching will occur many times, as in an expression of the type

$$+ f(y)*(x+x^2+x^3+x^4+1)$$

If we would want to make a replacement of the type

```
Keep Brackets;
id f?{f1,f2,f3}(u?) = f(u+1)/u;
```

the pattern matching and the substitution would have to be done only once, rather than 5 times, as would be the case if the Keep bracket statement would not be used.

- Collect FunctionName; The contents of the various brackets will be placed inside a function with the given name. Hence

```
+ f(y)*(x+x^2+x^3+x^4+1)
+ f(y^2)*(x+2*x^2+3*x^3+4*x^4+1)
```

with

```
Collect h;
```

would result in:

```
+ f(y)*h(x+x^2+x^3+x^4+1)
+ f(y^2)*h(x+2*x^2+3*x^3+4*x^4+1)
```

This can be very useful to locate x -dependence even further, because bracketing the new expression in terms of h could make very clear whether a given polynomial in x would factor the whole expression, or which factors are occurring. To bring $h(x+1)$ and $h(2*x+2)$ to multiples of the same objects one should consult the pages on the `normalize` (7.98) and `makeinteger` (7.85) statements.

The `Collect` statement, together with the `PolyFun` statement, can also be very useful, if the variable x (or other variables) is temporarily not playing much of a role in the pattern matching. It can make the program much faster.

For more information on the `collect` statement one should consult section 7.19.

Restrictions: The bracket index can only be used with active expressions. Hence the access of specific brackets in stored expressions will always be of the slow variety. To make it faster, one can copy the expression into a local expression with indexed brackets, use it, and drop the expression when it is not needed any longer.

The brackets can also be used to save space on the disk in problems in which the expressions become rather large. Let us assume the following simple problem:

```
Symbols x1,...,x12;
Local F = (x1+...+x12)^10;
.sort
id x1 = x4+x7;
.end
```

If the program is run like this the expression F contains 352716 terms after the `sort` and after the `id` the sorting in the `.end` results in a final stage sort of which the statistics are:

Time =	46.87 sec			
	F	Terms active	=	504240
		Bytes used	=	13462248
Time =	52.09 sec	Generated terms	=	646646
	F	Terms in output	=	184756
		Bytes used	=	4883306

We see, that the intermediate sort file still contains more than 500000 terms and more than 13 Mbytes, while the final result contains less than 5 Mbytes. Why is this? When the terms in FORM are sorted first come the powers of x_1 , because this is the variable that was declared first. Hence the terms that do not have powers of x_1 come much later in the input and will not be compared with the terms generated by the substitution of for instance a single power of x_1 until very late in the sorting. What can we do about this? We can try to group the terms in the first sort such that after the substitution like terms will be ‘very close’ to each other and hence will add quickly. This is done in the program

```
Symbols x1,...,x12;
Local F = (x1+...+x12)^10;
AntiBracket x1,x4,x7;
.sort
id x1 = x4+x7;
.end
```

Now all powers of the mentioned variables will be inside the brackets and all other variables will be outside. Because the terms inside the brackets are all following each other in the input of the second module, terms that will add will be generated closely together. The result is visible in the final statistics:

Time =	47.23 sec			
	F	Terms active	=	184761
		Bytes used	=	4928008
Time =	48.40 sec	Generated terms	=	646646
	F	Terms in output	=	184756
		Bytes used	=	4883306

Now the final step of the sorting has already almost the proper number of terms. The difference is due to brackets that are half in one ‘patch’ on the disk and half in the next ‘patch’ (for the meaning of the patches, one should read the part about sorting in chapter 17 on the setup file. It should be rather clear now that this saves disk space and the corresponding amount of time. These early cancellations can also be seen in the first statistics message of the second module. In the first case it is

Time =	19.76 sec	Generated terms	=	10431
	F 5216	Terms left	=	8065
		Bytes used	=	239406

and in the second case it is

Time =	22.82 sec	Generated terms	=	10124
	F 5835	Terms left	=	3186
		Bytes used	=	96678

This also causes a more efficient use of the large buffer and again a better use of the disk. There have been cases in which this ‘trick’ was essential to keep the sort file inside the available disk space.

Chapter 10

Output optimization

One of the uses of symbolic programs is to prepare formulas for further numerical processing. Technically speaking such processing is not part of computer algebra, although some packages may provide facilities for this. In FORM such facilities, such as Monte Carlo integration, do not exist at the moment, but, starting with version 4.1, FORM does provide statements to construct outputs in C or Fortran that are highly optimized with respect to the number of arithmetic operations that are needed for their evaluation. The algorithms used for this are described in the papers

- Code Optimization in FORM - <https://arxiv.org/abs/1310.7007>
- Improving multivariate Horner schemes with Monte Carlo tree search - <https://arxiv.org/abs/1207.7079>
- Combining Simulated Annealing and Monte Carlo Tree Search for Expression Simplification - <https://arxiv.org/abs/1312.0841>
- Why Local Search Excels in Expression Simplification - <https://arxiv.org/abs/1409.5223>

In short, an optimal Horner scheme is constructed after which common subexpressions are eliminated. The methods for finding the optimal scheme can use a simple heuristic, Monte Carlo Tree Search, or a Stochastic Local Search approach such as Simulated Annealing

In this section the precise format of the commands that concern the optimizations will be described. In optimized output FORM needs temporary variables. In order to avoid conflicts with user defined objects FORM uses the extra symbols 7.53 for these variables. This means that the user can control their output representation in the standard way. In addition there are preprocessor variables that tell how many of these extra symbols were needed:

optimminvar _ The number of extra symbols before the optimization process started.

optimmaxvar _ The number of extra symbols after the optimization process finished.

Each new optimization will remove the old optimization results and start the extra symbols from the number there were before the optimization started. Because this may cause interference with the functioning of the extrasymbol statement, regular printing with output optimization and the extrasymbol statement cannot occur inside the same module. Such occurrence would result in an error message.

Because the output optimization is done for expressions that contain only symbols, FORM has to convert all non-symbols and negative powers of symbols to extra symbols before it starts the optimization. This is another reason why interference between the extrasymbol 7.53 statement

and output optimizations is forbidden. When the results are printed, the definition of the extra symbols that are introduced this way are printed as well.

FORM has two ways to perform optimizations. The first and easiest is in the regular output. If one asks for optimization (by specifying the proper format for this) and follows this by a print statement, the output printed will be in optimized form. This is however just a representation of the expression and the next module will obtain the original expression for its input.

The more useful way to obtain an optimized output is with the `#optimize` instruction. To use this instruction properly one should understand what FORM does when it optimizes an expression. The whole process of optimization takes place inside the memory. Hence, FORM cannot optimize expressions that do not fit inside the CPU memory. The notation is however fairly compact and FORM needs far less space than for instance the compiler (and gives better results). The result of the optimization is stored inside a buffer. There is only a single optimization buffer and the preprocessor variables `optimminvar_` and `optimmaxvar_` refer to the contents of this buffer. When the `#optimize` instruction is used it loads this buffer and the contents stay around until either a `#clearoptimize` instruction is used or a new `#optimize` instruction is issued.

The `#optimize` instruction changes the original expression to its optimized shape in which it is usually a very short expression that refers to one or more extra symbols. The optimization information is automatically erased, and with it the expression that was optimized, when a second `#optimize` instruction is issued. Clearing the optimization buffer means that the information of the first expression is irretrievably lost and the contents of the first expression become meaningless, because its extra symbols have been erased. Hence if the user still needs this expression it is necessary to make a copy of it before optimization.

The optimization buffers, and the optimized expression, can be removed by the user with the `#clearoptimize` instruction. This is mandatory before the use of a `ToPolynomial 7.149` statement, because that may introduce new extra symbols.

The contents of the optimization buffer can be written with the `%O` combination in the format string in the `#write` instruction. This means that it is easy to write this output to file. Consider for instance the following program:

```
CF  f;
S   a,b,c;
L   H = f(a)+f(b)+(a+b+c)^2;
L   G = f(c)+(a+b+c)^3;
Format 02;
Print +f;
.sort
ExtraSymbols,array,w;
Format Fortran;
#optimize G
#write <outg.f> "      REAL*8 w('optimmaxvar_')"
#write <outg.f> "%O"
#write <outg.f> "      G = %e",G
#clearoptimize
.sort
#optimize H
#write <outh.f> "      REAL*8 w('optimmaxvar_')"
#write <outh.f> "%O"
#write <outh.f> "      H = %e",H
.end
```

This program shows the two different methods and shows what is left of the expressions G and H. It also shows that we have to deal with the expressions one by one when we use the `#optimize` instruction, while in the regular printing of the output this is not needed because the expression itself remains in its unoptimized version.

10.0.1 Optimization options of the `Format` statement

The `Format` statement has a number of options to control the code optimization. The easiest to use are the following:

- O0** Switches off all optimizations and prints the output the normal FORM way. This is the default.
- O1** Activates the lowest level of optimization. It is very fast, i.e., linear in the size of the expression, and gives reasonably efficient code.
- O2** Activates the medium level of optimization. This is slower than the previous setting, but usually gives better results.
- O3** Activates the highest level of optimization using MCTS. It can be rather slow, but usually gives even better results.
- O4** Activates the highest level of optimization using Local Stochastic Search. It is usually much faster than MCTS and may give better results.

Below we show how to use O4 and how it compares to O2:

```
#-
S  a,b,c,d,e,f,g,h,i,j,k,l,m,n;
L  G = (4*a^4+b+c+d + i^4 + g*n^3)^10 +
      (a*h + e + f*i*j + g + h)^8 + (i + j + k + l + m + n)^12;
L  H = G;
Format O2;
.sort
#optimize G
#write "Optimized with O2:"
#write "Optimized with Horner scheme: 'optimscheme_'"
#write "Number of operations in output: 'optimvalue_'"
#clearoptimize
.sort
Format O4,saIter=1000; * use 1000 iterations for optimization
#optimize H
#write "Optimized with O4:"
#write "Optimized with Horner scheme: 'optimscheme_'"
#write "Number of operations in output: 'optimvalue_'"
.end
```

which gives the output:

```
Optimized with O2:
Optimized with Horner scheme: i,n,j,m,l,k,g,a,d,c,b,h,f,e
```

Number of operations in output: 2578

Optimized with O4:

Optimized with Horner scheme: m,h,k,a,l,e,n,g,j,c,f,b,i,d

Number of operations in output: 1937

The preprocessor variable `optimscheme_` gives the best Horner scheme that the program found and the preprocessor `optimvalue_` gives the number of arithmetic operations in the resulting expression.

These levels of optimization refer to some default settings of all controlling parameters. These default values are in Tab. 10.1. It is also possible to set each parameter individually to fine-tune the optimization process. The parameters that can be set are divided in several categories. First, it is possible to set which Horner schemes are tried:

Horner=(Occurrence | MCTS | SA) Determines whether an occurrence order Horner scheme is used, or whether MCTS, or Stochastic Local Search is employed to find Horner schemes.

HornerDirection=(Forward | Backward | ForwardOrBackward | ForwardAndBackward) Forward makes that the MCTS search in the O3 option will determine the outermost variables in the multivariate Horner scheme first and then work its way inward. In the case of backward, the tree search determines the innermost variable first. In some cases this can give much better results when there are many common subexpressions involving a limited number of variables. ForwardOrBackward tries both of these schemes. ForwardAndBackward fills the order from both sides simultaneously, resulting in more options, but also a much larger search tree. If there are many variables, it could make the search tree too large to obtain good results.

When the option `Horner=Occurrence` is used the option `backward` will switch to something called ‘anti-occurrence’ which means that the most frequent variable corresponds to the innermost brackets.

In the case of MCTS there are various parameters that can control the search process:

MCTSConstant=<value> This sets the constant C_P in the UCT formula that governs the Monte Carlo tree search. It is supposed to be given as a real number with a decimal point (no floating point notation that includes powers).

MCTSNumExpand=<value> The number of times the tree is traversed and hence the number of times that a Horner scheme is constructed.

MCTSNumKeep=<value> During the MCTS procedure FORM only tries to construct a proper ordering for the Horner scheme, followed by a common subexpression elimination in the style of the O1 option. The best ‘value’ schemes are remembered and for those a common subexpression elimination in the style of the O2 option is done afterward. This second style elimination is far more costly. In nearly all cases the best O2-style scheme is in the very few top O1-style schemes.

MCTSNumRepeat=<value> Sometimes it is more advantageous to run a new tree search several times, each with a smaller number of expansions. This parameter tells how many times we will run with a new tree. The total number of tree traversals is the product of `MCTSNumRepeat` and `MCTSNumExpand`.

MCTSNumExpand=<*value1***value2*> Makes FORM to run ‘value1’ trees, each with ‘value2’ Horner scheme constructions. Hence this option is equivalent to the combination **MCTSNumRepeat**=<*value1*>, **MCTSNumExpand**=<*value2*>.

MCTSTimeLimit=<*value*> The maximum time in seconds that is used when searching through the tree.

MCTSDecayMode=<*value*> Determines how the C_P parameter in the UCT formula decreases:

value	effect
0	no decay
1	linear decay with iteration number
2	faster decay for the final iterations
3	decrease with iteration number and with node depth

For Stochastic Local Search the following parameters can be set:

saIter=<*value*> Number of optimization steps that will be performed. This has the most influence on the quality of the simplification. The default value is 1000.

saMaxT=<*value*> Maximum temperature used in Simulated Annealing. The higher the temperature, the more exploration occurs. The default value is 2000.

saMinT=<*value*> Minimum temperature used in Simulated Annealing. The lower the temperature, the more exploitation occurs. The default value is 1.

The cooling rate from saMaxT to saMinT is exponential in saIter. More information can be found in the research papers.

The Horner methods generate a number of Horner schemes: one or two in the case of occurrence order schemes, depending of the direction parameter, and a number equal to MCTSNumKeep in the case of MCTS. Next, for each stored Horner scheme other optimizations are performed as determined by the following parameter:

Method=(None | CSE | Greedy | CSEGreedy) Determines what method is used for optimizing the generated Horner schemes. CSE performs a simple common subexpression elimination and Greedy performs greedy optimizations (see the paper for more explanations) which are more sophisticated versions of CSE’s. CSEGreedy performs CSE followed by greedy optimizations; usually this is somewhat faster than just greedy optimizations, but it gives slightly worse results. The option None does nothing after applying the Horner scheme and is only useful for debugging purposes.

When the method of greedy optimizations is used, repeatedly all potential optimizations are determined and a few of them are performed. The following parameters are used to tune the greedy method:

GreedyMaxPerc The percentage of the possible optimizations that is performed.

GreedyMinNum The minimum number of possible optimizations that is performed.

GreedyTimeLimit The maximum time in seconds that is spent in the process of greedy optimization.

There are also two more general settings:

Stats=(On | Off) This parameter determines whether statistics of the optimization are shown. Statistics are printed in the format

```
*** STATS: original 1P 16M 5A : 23
```

```
*** STATS: optimized 0P 10M 5A : 15
```

in which P indicates power operations (at least a third power), M the number of multiplications and A the number of additions/subtractions. The last number is the total number of operations in which an n -th power counts as $n - 1$ operations.

TimeLimit=<value> This set both the MCTSTimeLimit and the GreedyTimeLimit to half of the given value.

Finally there are some parameters that are of a rather specialized nature. They can be used for debugging purposes or in the case that one knows already what is the best Horner scheme. Their default values are Off.

DebugFlag=(On | Off) In the case that the value is On, the list of temporary variables is printed in reverse order with the string "id " in front. This makes them into a set of FORM substitutions that undo the optimizations. One can use this for instance to make sure that the optimized code is identical to the original.

PrintScheme=(On | Off) This option (when On) will print the Horner scheme. That is the order in which the variables were taken outside parentheses.

Scheme=(list of symbols) The list should be enclosed by parentheses and the symbols should be separated by either blanks or comma's. This option will fix the Horner scheme to be used. One could for instance use the output of the PrintScheme option for this to avoid a lengthy search when a good order of the variables is already known. Things become a bit tricky when extra symbols are involved. One should make sure that their labelling is identical to when the scheme was created! When extra symbols are used in their array/vector notation, one needs to separate them by comma's, because blank spaces next to parentheses are eliminated by the preprocessor. If one specifies the wrong number of variables, the results can be quite unpredictable. At the moment of compilation FORM does not know the variables that are actually used. The safe thing is to verify the actual variables with a testrun using the PrintScheme option in the O1 mode.

All options should be specified in a single format statement and be separated either by commas or blank spaces. When **Format Optimize** is used, first the default settings are taken and then the options that are specified overwrite them. It is allowed to have the O1, O2, O3, O4 optimization specifications followed by options. In that case the program first sets the values of those specifications and then modifies according to what it encounters in the rest of the statement.

	O1	O2	O3 (default)	O4 (default)
Horner	occurrence	occurrence	MCTS	SA
HornerDirection	OR	OR	OR	OR
MCTSConstant	—	—	1.0	—
MCTSNumExpand	—	—	1000	—
MCTSNumKeep	—	—	10	—
MCTSNumRepeat	—	—	1	—
MCTSTimeLimit	—	—	0	—
MCTSDecayMode	—	—	1	—
saIter	—	—	—	1000
saMinT	—	—	—	1
saMaxT	—	—	—	2000
Method	cse	greedy	greedy	greedy
GreedyMinNum	—	10	10	10
GreedyMaxPerc	—	5	5	5
GreedyTimeLimit	—	0	0	0
Stats	off	off	off	off
TimeLimit	0	0	0	0

Table 10.1: Values for the various parameters in the predefined optimization levels. OR stands for ForwardOrBackward.

Chapter 11

Polynomials and Factorization

Starting with version 4, FORM is equipped with powerful handling of rational polynomials and with factorization capabilities. Because this creates many new possibilities, it brings a whole new category of commands with it. We will list most of these here.

First there are the rational polynomials. These work a bit like the PolyFun 7.111, but now with two arguments: a numerator and a denominator. Instead of PolyFun the function is designated as PolyRatFun 7.112 as in the example below:

```
Symbol x,y;  
CFunction rat;  
PolyRatFun rat;  
L   F = rat(x+y,x-y)+rat(x-y,x+y);  
Print;  
.end
```

```
F =  
  rat(2*x^2 + 2*y^2,x^2 - y^2);
```

Dealing with a PolyRatFun can be very handy, but one should realize that there is a limit to the size of the arguments, because the PolyRatFun with its arguments is part of a term and hence is limited by the maximum size of a term 17. One should also take into account that the manipulation of multivariate polynomials, and in particular the GCD operation, can be rather time consuming. The PolyRatFun has one limitation as compared to the regular PolyFun: in its arguments one may use only symbols. Of course FORM is equipped with a mechanism to replace other objects by extra internally generated symbols 7.53. One could imagine FORM to automatically convert these objects to symbols, do the polynomial arithmetic and then convert back. This is done with factorization and the gcd_ 8.30, div_ 8.13 and rem_ 8.58 functions. But because the addition of PolyRatFun's is such a frequent event, this would be very costly in time. Hence it is better that the user does this once in a controlled way.

The PolyFun and PolyRatFun declarations are mutually exclusive. The PolyRatFun is considered a special type of PolyFun and there can be only one PolyFun at any moment. If one wants to switch back to a mode in which there is neither a PolyFun nor a PolyRatFun one can use

```
PolyRatFun;
```

to indicate that after this there is no function with that status.

When a PolyRatFun has only a single argument, this argument is interpreted as the numerator of a fraction. FORM will add automatically a second argument which has the value 1.

The second important polynomial facility is factorization. This is not necessarily something trivial. First of all, with very lengthy multivariate input, this can be unpractically slow. Second of all, there are various types of objects that we may factorize and each has its special needs. One of those needs is access to the factors, which is different for the factors of function arguments, of \$-expressions or even complete expressions. In addition \$-expressions should be factorizable either from the preprocessor or on a term by term basis. Let us start with function arguments.

One can factorize function arguments with the FactArg statement 7.54. The factors are each represented by a separate argument as in

```
Symbol x,y;
CFunction f1,f2;
Local F = f1(x^4-y^4)+f2(3*y^4-3*x^4);
FactArg,f1,f2;
Print;
.end

F=
  f1(y-x,y+x,y^2+x^2,-1)+f2(y-x,y+x,y^2+x^2,3);
```

Overall constants and overall signs are taken separately as one can see. If one wants the factors in separate functions one can use the ChainOut 7.16 command as in

```
Symbol x,y;
CFunction f1,f2;
Local F = f2(3*y^4-3*x^4);
FactArg,f2;
Print;
.sort

F=
  f2(y-x,y+x,y^2+x^2,3);

ChainOut,f2;
id f2(x?number_) = x;
Print;
.end

F=
  3*f2(y-x)*f2(y+x)*f2(y^2+x^2);
```

Factorization of expressions is a bit more complicated. Clearly this cannot be a command at the term level. Hence we had two options on how to implement this. One would have been as a preprocessor instruction, which we did not select, and the other is as some type of format statement, which is what we did opt for. In the case we factorize an expression, the original unfactorized expression is replaced by the factorized version. After that we keep the factorized version only and that may bring some restrictions with it. Of course, in the same way one can factorize an expression, one can unfactorize it. The corresponding statements are Factorize 7.56, NFactorize 7.94, UnFactorize 7.157 and NUnFactorize 7.104. These statements are used at the end of the module in the same place as one might use the bracket statement 7.11. It should be noticed however that a factorized expression will never apply the bracket mechanism. They are

mutually exclusive, because internally we use the bracket mechanism with a built in symbol factor_ to indicate the factors. Here is an example:

```
Symbol x,y;
Local F = x^4-y^4;
Print;
.sort
```

Time =	0.00 sec	Generated terms =	2
	F	Terms in output =	2
		Bytes used =	64

```
F=
-y^4+x^4;
```

```
Print;
Factorize F;
.end
```

Time =	0.00 sec	Generated terms =	2
	F	Terms in output =	2
		Bytes used =	64

Time =	0.00 sec	Generated terms =	7
	F	Terms in output =	7
	factorize	Bytes used =	288

```
F=
(-1)
*(y-x)
*(y+x)
*(y^2+x^2);
```

We have printed the statistics in this example to show that the factorization prints its own statistics. This factorization is executed after the expression has been completed and before manipulations on the next expression start. This way it is possible to overwrite the first output by the factorized output and we do not loose disk space unnecessarily.

The next question is of course how to find out how many factors an expression has and how to access individual factors. There is a function numfactors_ which gives the number of factors in an expression:

```
Symbol x,y;
Local F1 = x^4-y^4;
Local F2 = 0;
Local F3 = 1;
Local F4 = x^4-y^4;
Print;
Factorize F1,F2,F3;
.sort
```

```

F1=
  (-1)
  *(y-x)
  *(y+x)
  *(y^2+x^2);

F2=0;

F3=
  (1);

F4=
  -y^4+x^4;
  #do i = 1,4
  #${n}'i' = numfactors_(F'i');
  #message expression F'i' has '${n}'i' factors
~~~expression F1 has 4 factors
  #enddo
~~~expression F2 has 1 factors
~~~expression F3 has 1 factors
~~~expression F4 has 0 factors
  .end

```

As we see, an expression that is zero still gives one factor when it is factorized. When the expression is not factorized it will return 0 in all cases. The factors can be accessed easily once one knows that the factors are stored by means of the bracket mechanism and the n-th factor is the bracket with the n-th power of the symbol factor_ outside the bracket:

```

Symbol x,y;
Local F = x^4-y^4;
Factorize F;
.sort
#${n} = numfactors_(F);
#do i = 1,'${n}'
Local F'i' = F[factor_'^i'];
#enddo
Print;
.end

F=
  (-1)
  *(y-x)
  *(y+x)
  *(y^2+x^2);

F1=
  -1;

```

```

F2=
  y-x;

F3=
  y+x;

F4=
  y^2+x^2;

```

It is also possible to put an expression in the input in a factorized format. For this we have the LocalFactorized 7.82 and GlobalFactorized 7.64 commands. These commands can be abbreviated to LFactorized, GFactorized or even LF and GF. One should notice that these commands do not execute a factorization. They accept the factors as the user provides them:

```

Symbol x,y;
LocalFactorize E = -(x+1)*(x+2)*((x+3)*(x+4));
Print;
.end

E =
  ( - 1 )
  * ( 1 + x )
  * ( 2 + x )
  * ( 12 + 7*x + x^2 );

```

This can go to some extremes when we feed in expressions containing powers and expressions that are potentially already factorized:

```

Symbol x,y;
LocalFactorize E = -(x+1)*(x+2)*((x+3)*(x+4));
Local F = -(x+1)*(x+2)*((x+3)*(x+4));
Print;
.sort

E=
  (-1)
  *(1+x)
  *(2+x)
  *(12+7*x+x^2);

F=
  -24-50*x-35*x^2-10*x^3-x^4;

LF G = (x-1)*(x+2)^2*E^2*F^2;
Print G;
.end

G=
  (-1+x)
  *(2+x)

```

```

*(2+x)
*(-1)
*(1+x)
*(2+x)
*(12+7*x+x^2)
*(-1)
*(1+x)
*(2+x)
*(12+7*x+x^2)
*(-24-50*x-35*x^2-10*x^3-x^4)
*(-24-50*x-35*x^2-10*x^3-x^4);

```

To put some order in this one may factorize the new expression again:

```

Symbol x,y;
LocalFactorize E = -(x+1)*(x+2)*((x+3)*(x+4));
Local F = -(x+1)*(x+2)*((x+3)*(x+4));
.sort
LF G = (x-1)*(x+2)^2*E^2*F^2;
Print G;
Factorize G;
.end

```

```

G=
(-1+x)
*(1+x)
*(1+x)
*(1+x)
*(1+x)
*(2+x)
*(2+x)
*(2+x)
*(2+x)
*(2+x)
*(2+x)
*(3+x)
*(3+x)
*(3+x)
*(3+x)
*(4+x)
*(4+x)
*(4+x)
*(4+x);

```

In this case all constants are multiplied, all factors are factorized, and all factors in the new format are sorted.

The case that one or more factors are zero is special. In principle the zero factors are kept as in:

```

Symbol x,y;
LocalFactorize E = -0*(x+1)*(x+2)*0*((x+3)*(x+4));

```

```

Print;
.end

E=
  (-1)
  *(0)
  *(1+x)
  *(2+x)
  *(0)
  *(12+7*x+x^2);

```

This way one can see what has happened when a substitution makes a factor zero. When we factorize this expression again however the whole expression becomes zero. If this is not intended and one would like to continue with the factors that are nonzero we have the keepzero option in the factorize statement as in:

```

Symbol x,y;
Format Nospaces;
LocalFactorize E = -0*3*(x+1)*(x+2)/2*0*((x+3)*(x+4));
Print;
.sort

```

```

E=
  (-1)
  *(0)
  *(3)
  *(1+x)
  *(2+x)
  *(1/2)
  *(0)
  *(12+7*x+x^2);
Print;
Factorize(keepzero) E;
.end

```

```

E=
  (0)
  *(-3/2)
  *(1+x)
  *(2+x)
  *(3+x)
  *(4+x);

```

We see here that first all constants are separate factors and the new factorization combines them. The keepzero option does the same with the factors that are zero. The zero factor will always be the first. Hence it is rather easy to test for whether the total expression should actually be zero. We just have to look whether `E[factor_]` is zero.

The unfactorize 7.157 statement is the opposite of the factorize statement. It takes the factorized expression and multiplies out the factors. It also uses the current brackets for formatting the output.

```

Symbol x,y;
LFactorized F = (x+1)*(x+y)*(y+1);
Print;
.sort

```

```

F=
  (1+x)
  *(y+x)
  *(1+y);

Print;
Bracket x;
UnFactorize F;
.end

```

```

F=
  +x*(1+2*y+y^2)
  +x^2*(1+y)
  +y+y^2;

```

In principle there are various models by which the unfactorization can be done in an efficient way. In addition it would be less efficient when the master would do all the work as is the case with the factorize statement. Currently this statement is still being developed internally. It is possible to make ones own emulation of it. Here we give the 'brute force' way:

```

Symbol x,y;
LFactorized F = (x+1)*(x+y)*(y+1);
Print;
.sort

```

```

F=
  (1+x)
  *(y+x)
  *(1+y);

#$num = numfactors_(F);
Local G = <F[factor_^1]>*. . .*<F[factor_-'$num']>;
Bracket x;
Print;
.end

```

```

F=
  (1+x)
  *(y+x)
  *(1+y);

G=
  +x*(1+2*y+y^2)
  +x^2*(1+y)

```

```
+y+y^2;
```

Factorization of $\$$ -expressions is yet a different thing. The $\$$ -expressions do not have a bracket mechanism. Hence we need different ways of storing the factors. In the case of expressions we have to work in a way that is potentially disk based. With $\$$ -expressions we work in allocated memory. Hence we also store the factors in allocated memory. In that case we can keep both the original and the factors. The factors are accessed by referring to their number between braces. The number zero refers to the number of factors:

```
Symbol x,y;
CFunction f;
Off Statistics;
#$a = x^4-y^4;
Local F = f(x^4-y^4)+f(x^6-y^6);
Print;
.sort

F=
  f(-y^4+x^4)+f(-y^6+x^6);
#factdollar $a;
#do i = 1,$a[0]
  #write <> "Factor 'i' of '$a' is '$a['i']'"
Factor 1 of -y^4+x^4 is -1
  #enddo
Factor 2 of -y^4+x^4 is y-x
Factor 3 of -y^4+x^4 is y+x
Factor 4 of -y^4+x^4 is y^2+x^2
  id f(x?$b) = f(x);
  FactDollar $b;
  do $i = 1,$b[0];
    Print "Factor %$ of %$ is %$", $i, $b, $b[$i];
  enddo;
Print;
.end
Factor 1 of -y^4+x^4 is -1
Factor 2 of -y^4+x^4 is y-x
Factor 3 of -y^4+x^4 is y+x
Factor 4 of -y^4+x^4 is y^2+x^2
Factor 1 of -y^6+x^6 is -1
Factor 2 of -y^6+x^6 is y-x
Factor 3 of -y^6+x^6 is y+x
Factor 4 of -y^6+x^6 is y^2-x*y+x^2
Factor 5 of -y^6+x^6 is y^2+x*y+x^2

F=
  f(-y^4+x^4)+f(-y^6+x^6);
```

We see here a variety of new features. The preprocessor can factorize $\$a$ with the `#FactDollar` instruction. We do indeed pick up the number of factors in the preprocessor as `'$a[0]'` and the

factors themselves as '\$a[1]' etc. For the \$-variable that needs to be manipulated during running time things as a bit more complicated. We define \$b as part of a wildcard pattern matching. This is still rather normal. Then we use the FactDollar statement. Notice that for each term we will have a different \$b. To access the factors we cannot use the preprocessor methods because those are only available at compile time. Hence we cannot use the preprocessor #do instruction and therefore we need an execution time do statement. The loop parameter will have to be a \$-variable as well. The do statement and the print statement show now how one can use the factors. In the output one can see that indeed we had two different contents for \$b. And the arguments of the function f remain unaffected.

One may also ask for the number of factors in a \$-expression with the numfactors_ function as in:

```
Symbol x,y;
CFunction f;
Format Nospaces;
#$a = x^4-y^4;
#factdollar $a;
Local F = f(numfactors_($a))
    +f(<$a[1]>,...,<$a['$a[0]']>);
Print;
.end
```

```
F=
    f(-1,y-x,y+x,y^2+x^2)+f(4);
```

Note that in the second case we need to use the construction '\$a[0]' because the preprocessor needs to substitute the number immediately in order to expand the triple dot operator. This cannot wait till execution time.

Some remarks.

The time needed for a factorization depends strongly on the number of variables used. For example factorization of $x^{60} - 1$ is much faster than factorization of $x^{60} - y^{60}$. One could argue that the second formula can be converted into the first, but there is a limit to what FORM should do and what the user should do.

```
Symbol x,y;
Format NoSpaces;
On ShortStats;
Local F1 = x^60-1;
Local F2 = y^60-x^60;
Factorize F1,F2;
Print;
.end
```

0.00s	1>	2-->	2:	52 F1
0.07s	1>	51-->	51:	1524 F1 factorize
0.07s	1>	2-->	2:	64 F2
1.17s	1>	51-->	51:	1944 F2 factorize

```
F1=
    (-1+x)
    *(1-x+x^2)
```



```

*(1-x+x^2-x^3+x^4)
*(1-x+x^3-x^4+x^5-x^7+x^8)
*(1+x)
*(1+x+x^2)
*(1+x+x^2+x^3+x^4)
*(1+x-x^3-x^4-x^5+x^7+x^8)
*(1-x^2+x^4)
*(1-x^2+x^4-x^6+x^8)
*(1+x^2)
*(1+x^2-x^6-x^8-x^10+x^14+x^16);

```

F2=

```

(y-x)
*(y+x)
*(y^2-x*y+x^2)
*(y^4-x*y^3+x^2*y^2-x^3*y+x^4)
*(y^4+x*y^3+x^2*y^2+x^3*y+x^4)
*(y^2+x*y+x^2)
*(y^2+x^2)
*(y^8-x*y^7+x^3*y^5-x^4*y^4+x^5*y^3-x^7*y+x^8)
*(y^8+x*y^7-x^3*y^5-x^4*y^4-x^5*y^3+x^7*y+x^8)
*(y^8-x^2*y^6+x^4*y^4-x^6*y^2+x^8)
*(y^4-x^2*y^2+x^4)
*(y^16+x^2*y^14-x^6*y^10-x^8*y^8-x^10*y^6+x^14*y^2+x^16);

```

When one has a factorized expression and one uses the multiply statement, all terms in the factorized expression are multiplied the specified amount. This may lead to a counterintuitive result:

```

Symbols a,b;
LF F = (a+b)^2;
multiply 2;
Print;
.end

```

F =

```

( 2*b + 2*a )
* ( 2*b + 2*a );

```

This is a consequence of the way we store the factors. This way each factor will be multiplied by two. If one would like to add a factor one can do this by the following simple mechanism:

```

Symbols a,b;
LF F = (a+b)^2;
.sort
LF F = 2*F;
Print;
.end

```

F =

```

( 2 )

```

```
* ( b + a )  
* ( b + a );
```

In version 3 there were some experimental polynomial functions like `polygcd_`. These have been removed as their functionality has been completely taken over by the new functions `gcd_` 8.30, `div_` 8.13 and `rem_` 8.58 and some statements like `normalize` 7.98, `makeinteger` 7.85 and `factarg` 7.54. Unlike regular functions, the functions `gcd_`, `div_` and `rem_` have the peculiarity that if one of the arguments is just an expression or a `$`-expression, this expression is not evaluated until the function is evaluated. This means that the evaluated expression does not have to fit inside the maximum size reserved for a single term. In some cases, when the `gcd_` function is invoked with many arguments, the expression may not have to be evaluated at all! The GCD of the other arguments may be one already.

Chapter 12

The TableBase

The tablebase statement controls a database-like structure that allows FORM to control massive amounts of data in the form of tables and table elements. The contents of a tablebase are formed by one or more table declarations and a number of fill statements. These fill statements however are not immediately compiled. For each fill statement a special fill statement is generated and compiled that is of the form

```
Fill tablename(indices) = tbl_(tablename,indices,arguments);
```

The function `tbl_` is a special function to make a temporary table substitution. It indicates that the corresponding element can be found in a tablebase that has been opened. At a later stage one can tell FORM to see which table elements are actually needed and then only those will be loaded from the tablebase and compiled.

Tablebases have a special internal structure and the right hand sides of the fill statements are actually stored in a compressed state. These tablebases can be created with special statements and uploaded with any previously compiled table. Hence one can prepare a tablebase in a previous job, to be used at a later stage, without the time penalty of loading the whole table at that later stage.

Assume we have a file named `no11fill.h` that looks like

```
Symbols ...;
Table,sparse,no11fill(11,N?);
Fill no11fill(-3,1,1,1,1,1,1,1,0,0,0) = ....
Fill no11fill(-2,1,1,1,1,1,1,1,0,0,0) = ....
etc.
```

It should be noted that only sparse tables can be stored inside a tablebase. The right hand sides could be typically a few kilobytes of formulas and there could be a few thousand of these fill statements. To make this into a tablebase one would use the program

```
#-
#include no11fill.h
#+
TableBase "no11.tbl" create;
TableBase "no11.tbl" addto no11fill;
.end
```

The include instruction makes that FORM reads and compiles the table. Then the first tablebase statement creates a new tablebase file by the name `no11.tbl`. If such a file existed already, the old

version will be lost. If one would like to add to an existing tablebase, one should use the ‘open’ keyword. The second tablebase statement adds the table `no11fill` to the tablebase file `no11.tbl`. This takes care of declaring the table, making an index of all elements that have been filled and putting their right hand sides, in compressed form, into the tablebase. The compression is based on the `zlib` library, provided by Jean-loup Gailly and Mark Adler (version 1.2.3, July 18, 2005) and it strikes a nice balance between speed and compression ratio.

The tablebase can be loaded in a different program as in

```
TableBase "no11.tbl" open;
```

This loads the main index of the file into memory. To protect against accidentally adding additional table to an existing tablebase or in case tablebases are stored in shared directories where not all users have write access, they can be opened using

```
TableBase "no11.tbl" open, readonly;
```

Trying to add tables to a tablebase opened in read-only mode results in an error.

If one would like to compile the short version of the fill statements (the normal action at this point) one needs to use the `load` option. Without any names of tables it will read the index of all tables. If tables are specified, only the index of those tables is taken and the proper `tbl_` fill statements are generated:

```
TableBase "no11.tbl" open;  
TableBase "no11.tbl" load no11fill;
```

If one would like to compile the complete tables, rather than just the shortened versions, one can use the `enter` option as in:

```
TableBase "no11.tbl" open;  
TableBase "no11.tbl" enter no11fill;
```

Let us assume we used the `load` option. Hence now an occurrence of a table element will be replaced by the stub-function `tbl_`. In order to have this replaced by the actual right hand side of the original fill statement we have to do some more work. At a given moment we have to make FORM look which elements are actually needed. This is done with the `TestUse` statement as in

```
TestUse no11fill;
```

This does nothing visible. It just marks internally which elements will be needed and have not been entered yet.

The actual entering of the needed elements is done with the `use` option:

```
TableBase "no11.tbl" use;
```

If many elements are needed, this statement may need some compilation time. Note however that this is time at a moment that it is clear that the elements are needed, which is entirely different from a fixed time at the startup of a program when the whole table is loaded as would have to be done before the tablebase statement existed. Usually however only a part of the table is needed, and in the extreme case only one or two elements. In that case the profit is obvious.

At this point the proper elements are available inside the system, but because we have two versions of the table (one the short version with `tbl_`, the other the complete elements) we have to tell FORM to apply the proper definitions with the ‘`apply`’ statement.

Apply;

Now the actual rhs will be inserted.

One may wonder why this has to be done in such a ‘slow’ way with this much control over the process. The point is that at the moment the table elements are recognized, one may not want the rhs yet, because it may be many lines. Yet one may want to take the elements away from the main stream of action. Similarly, having a table element recognized at a certain stage, may not mean automatically that it will be needed. The coefficient may still become zero during additional manipulations. Hence the user is left with full control over the process, even though that may lead to slightly more programming. It will allow for the fastest program.

For the name of a tablebase we advise the use of the extension .tbl to avoid confusion.

Note that the above scheme may need several applications, if table elements refer in their definition to other table elements. This can be done with a construction like:

```
#do i = 1,1
  TestUse;
  .sort
  TableBase "basename.tbl" use;
  Apply;
  if ( count(tbl_,1) ) Redefine i "0";
  .sort
#enddo
```

It will stay in the loop until there are no more tbl_ functions to be resolved.

The complete syntax (more is planned):

12.1 addto

Syntax:

```
TableBase "file.tbl" addto tablename;
TableBase "file.tbl" addto tablename(tableelement);
```

See also open (12.9) and create (12.4).

Adds the contents of a (sparse) table to a tablebase. The base must be either an existing tablebase (made accessible with an open statement) or a new tablebase (made available with a create statement). In the first version what is added is the collection of all fill statements that have been used to define elements of the indicated table, in addition to a definition of the table (if that had not been done yet). In the second version only individual elements of the indicated table are added. These elements are indicated as it should be in the left hand side of a fill statement.

One is allowed to specify more than one table, or more than one element. If one likes to specify anything after an element, it should be realized that one needs to use a comma for a separator, because blank spaces after a parenthesis are seen as irrelevant.

Examples:

```
TableBase "no11.tbl" open;
TableBase "no11.tbl" load;
TableBase "no11.tbl" addto no11filb;
TableBase "no11.tbl" addto no11fill(-3,1,1,1,1,2,1,1,0,0,0),
                             no11fill(-2,1,1,2,1,1,1,1,0,0,0);
```

12.2 apply

Syntax:

Apply [number] [tablename(s)];

See also testuse (12.10) and use (12.11).

The actual application of fill statements that were taken from the tablebases. If no tables are specified, this is done for all tables, otherwise only for the tables whose names are mentioned. The elements must have been registered as used before with the application of a testuse statement, and they must have been compiled from the tablebase with the use option of the tablebase statement. The number refers to the maximum number of table elements that can be substituted in each term. This way one can choose to replace only one element at a time. If no number is present all occurrences will be replaced. This refers also to occurrences inside function arguments. If only a limited number is specified in the apply statement, the occurrences inside function arguments have priority.

12.3 audit

Syntax:

TableBase "file.tbl" audit;

See also open (12.9)

Prints a list of all tables and table elements that are defined in the specified tablebase. This tablebase needs to be opened first. As of the moment there are no options for the audit. Future options might include formatting of the output.

12.4 create

Syntax:

TableBase "file.tbl" create;

See also open (12.9)

This creates a new file with the indicated name. This file will be initialized as a tablebase. If there was already a file with the given name, its old contents will be lost. If one would like to add to an existing tablebase, one should use the 'open' option.

12.5 enter

Syntax:

TableBase "file.tbl" enter;

TableBase "file.tbl" enter tablename(s);

See also open (12.5) and load (12.6).

Scans the specified tablebase and (in the first variety) creates for all elements of all tables in the tablebase a fill statement with its full contents. This is at times faster than reading the fill statements from a regular input file, because the tablebase has its contents compressed. Hence this costs less file access time. When table names are specified, only the tables that are mentioned have their elements treated this way.

The tablebase must of course be open for its contents to be available.

If one would like FORM to only see what elements are available and load that information one should use the load option.

12.6 load

Syntax:

```
TableBase "file.tbl" load;  
TableBase "file.tbl" load tablename(s);
```

See also open (12.9) and enter (12.5).

Scans the index of the specified tablebase and (in the first variety) creates for all elements of all tables in the tablebase a fill statement of the type

```
Fill tablename(indices) = tbl_(tablename,indices,arguments);
```

This is the fill statement that will be used when elements of one of these tables are encountered. The function `tbl_` is called the (table)stub function. When table names are specified, only the tables that are mentioned have their elements treated this way.

The tablebase must of course be open for its contents to be available.

If one would like to actually load the complete fill statements, one should use the enter option.

12.7 off

Syntax:

```
TableBase "file.tbl" off subkey;
```

See also addto (12.1) and off (12.8).

Currently only the subkey 'compress' is recognized. It makes sure that no compression is used when elements are being stored in a tablebase with the addto option. This could be interesting when the right hand sides of the fill statements are relatively short.

12.8 on

Syntax:

```
TableBase "file.tbl" on subkey;
```

See also addto (12.1) and off (12.7).

Currently only the subkey 'compress' is recognized. It makes sure that compression with the gzip algorithms is used when elements are being stored in a tablebase with the addto option. This is the default.

12.9 open

Syntax:

```
TableBase "file.tbl" open;  
TableBase "file.tbl" open, readonly;
```

See also create (12.4)

This opens an existing file with the indicated name. It is assumed that the file has been created with the 'create' option in a previous FORM program. It gives the user access to the contents of the tablebase. In addition it allows the user to add to its contents. If the read-only option is set, no contents can be added.

Just like with other files, FORM will look for the file in in current directory and in all other directories mentioned in the environment variable 'FORMPATH' (see for instance the `#call` (3.10) and the `#include` (3.34) instructions).

12.10 testuse

Syntax:

```
TestUse;
```

```
TestUse tablename(s);
```

See also use (12.11).

Tests for all elements of the specified tables (if no tables are mentioned, this is done for all tables) whether they are used in a stub function tbl_. If so, this indicates that these elements must be compiled from a tablebase, provided this has not been done already. The compilation will have to be done at a time, specified by the user. This can be done with the use option. All this statement does is set some flags in the internals of FORM for the table elements that are encountered in the currently active expressions.

12.11 use

Syntax:

```
TableBase "file.tbl" use;
```

```
TableBase "file.tbl" use tablename(s);
```

See also testuse (12.10) and apply (12.2).

Causes those elements of the specified tables to be compiled, that a previous testuse statement has encountered and that have not yet been compiled before. If no tables are mentioned this is done for all tables. The right hand sides of the definition of the table elements will not yet be substituted. That is done with an apply statement.

Chapter 13

Dictionaries

At times one would like to manipulate the output to facilitate further processing. A standard example is that the output formula should be included in a $\text{\LaTeX}$ file. Also the use of terms in the output as patterns with wildcards in the LHS of an id-statements needs textual translation. Another example is the representation of fractions in a numerical program that works with floating point numbers. Complete solutions for such problems are not included in FORM, but with the partial solution of ‘dictionaries’ one can do quite a lot already.

In FORM a dictionary is a collection of ‘words’ together with their translation. The word can be a number, a variable, a function with its arguments or a special output token like a multiplication sign or a power indicator. The translation can be any string. Generic patterns have not been implemented. That would be more like grammar and involves special complications. As shown later, currently there is one exception to this rule.

A dictionary is defined with the preprocessor instruction

```
#opendictionary name
```

in which ‘name’ is the name of the dictionary. There can be more dictionaries, provided they have different names. It is allowed to open already existing dictionaries. Only one dictionary can be open at a given time. Dictionaries are closed with the instruction

```
#closedictionary
```

and because there can be only one open dictionary, it is clear which dictionary should be closed.

A dictionary is opened to add words to it. This is done with the `#add` instruction as in

```
#add x1: "x_1"  
#add *: "\ "  
#add mu: "\mu"
```

which would tell the system that when the dictionary is in use, the variable `x1` should be printed as the string `x_1` and a multiplication sign should become a backslash character followed by a blank space. The (index) `mu` would be printed as the string `\mu`.

A dictionary can be used with the

```
#usedictionary name <(options)>
```

instruction. At the moment a dictionary is being used there cannot be any open dictionaries. Hence we can stop using a dictionary with the

```
#closedictionary
```

instruction without running into inconsistencies. The options control partial use of a dictionary, as for instance only for individual variables, or only for numbers. They can also control whether translations should be made inside function arguments or inside dollar variables (when used as preprocessor variables).

What words are allowed?

variable This can be the name of a symbol, a vector, an index or a function (this includes commuting functions, non-commuting functions, tensors and tables).

number This must be a positive integer number.

fraction This must be a positive rational number.

special character Currently this can be the multiplication sign (*), or the power sign (^ or **).

a range Indicated between parentheses, this is a range of extra symbols. There can be more than one range.

a function with arguments This would be a complete function subterm.

The options in the #usedictionary should be enclosed between parentheses and separated by comma's. They can be:

allnumbers All numbers will be looked up in the dictionary.

integersonly Only integer numbers will be looked up.

nonumbers Numbers will not be looked up.

numbersonly Only numbers will be looked up.

novariables Loose variables will not be looked up.

variablesonly Only loose variables will be looked up.

nospecials Specials (multiplication signs and power signs) will not be looked up.

specialonly Only specials (multiplication signs and power signs) will be looked up.

nofunwithargs Functions with arguments will not be looked up.

funwithargsonly Only functions with arguments will be looked up.

warnings Warnings concern the look up of numbers. If a fortran or C format is being used and the dictionary cannot be used in such a way that floating point notation and/or decimal points can be avoided, a warning will be given.

nowarnings No floating point warnings are given.

infunctions Substitutions are also made inside function arguments.

notinfunctions No substitutions are made inside function arguments.

\$ Substitutions are made also when dollar variables are expanded. The default is that this is not done.

The defaults are that all potential objects are looked up (also inside function arguments) and no warnings are given.

The use is best illustrated with a few examples.

```
Symbols x1,y2,z3,N;
Indices mu,nu,ro,si;
Tensor tens;
CFunction S,R,f;
ExtraSymbols array w;
#OpenDictionary test
  #add x1: "x_1"
  #add y2: "y^{(2)}"
  #add z3: "{\cal Z}"
  #add *: " "
  #add S(R(1),N): "S_1(N)"
  #add S(R(2),N): "S_2(N)"
  #add S(R(1,1),N): "S_{1,1}(N)"
  #add f: "\ln"
  #add mu: "\mu"
  #add nu: "\nu"
  #add ro: "\rho"
  #add si: "\sigma"
  #add tens: "T"
#CloseDictionary
Local F = x1*y2*z3
      + S(R(1),N) + S(R(1,1),N) + S(R(2),N)
      + tens(mu,nu,ro,si) + f(x1+1);
#usedictionary test
Print +s;
.end
```

This program gives for its output

```
F =
  + x_1 y^2 {\cal Z}
  + T(\mu,\nu,\rho,\sigma)
  + S_1(N)
  + S_{1,1}(N)
  + S_2(N)
  + \ln(1 + x_1)
;
```

Of course, there is nothing here that could not have been done with a good text editor, but having this inside the FORM program makes that if there are changes in the FORM program, it will be less work to implement them in the eventual L^AT_EX files.

Things become different when numerical output is involved. Take for instance the fraction 1/3 inside a FORTRAN program. Using the option

```
Format Fortran;
```

one would obtain

1./3.

and with

```
Format DoubleFortran;
```

one would obtain

1.D0/3.D0

while using

```
Format QuadFortran;
```

one would obtain

1.Q0/3.Q0

which means that one might have three varieties of the same program, depending on the precision in which one would like run it. It would be far better to have a single version and only determine in the make file what the precision should be. The FORTRAN code for such a program could look like

```
REAL one,three,third
PARAMETER (one=1,three=3,third=one/three)
```

after which one should either use the name 'third' or a construction like 'one/three'. Let us take a simple program like

```
Symbol x,n;
Format DoubleFortran;
Local F = (1+x)^7/7;
id x^n? = x*x^n/(n+1);
Print;
.end
F =
& 1.D0/7.D0*x + 1.D0/2.D0*x**2 + x**3 + 5.D0/4.D0*x**4 + x**5 + 1.D
& 0/2.D0*x**6 + 1.D0/7.D0*x**7 + 1.D0/56.D0*x**8
```

If we define a dictionary we can make this into

```
Symbol x,n;
Format DoubleFortran;
#OpenDictionary numbers
  #add 2: "TWO"
  #add 5: "FIVE"
  #add 7: "SEVEN"
#CloseDictionary
Local F = (1+x)^7/7;
id x^n? = x*x^n/(n+1);
#UseDictionary numbers
Print;
.end
F =
& 1/SEVEN*x + 1/TWO*x**2 + x**3 + FIVE/4*x**4 + x**5 + 1/TWO*x**6
& + 1/SEVEN*x**7 + 1.D0/56.D0*x**8
```

one can see that some of the numbers have been replaced by text strings. In particular these are the numbers 2, 5 and 7. The output is now presented in such a way that the compiler can do the rest, provided we do this with all numbers that occur, and we feed the proper information to the compiler.

One can also replace complete fractions as in

```
Symbol x,n;
Format DoubleFortran;
#OpenDictionary numbers
  #add 2: "TWO"
  #add 5: "FIVE"
  #add 7: "SEVEN"
  #add 1/2: "HALF"
#CloseDictionary
Local F = (1+x)^7/7;
id x^n? = x*x^n/(n+1);
#UseDictionary numbers
Print;
.end
F =
& 1/SEVEN*x + HALF*x**2 + x**3 + FIVE/4*x**4 + x**5 + HALF*x**6 +
& 1/SEVEN*x**7 + 1.D0/56.D0*x**8
```

because the fractions take precedence.

The next question is how one makes sure to have all numbers that need replacement? For that one can use the warnings option:

```
Symbol x,n;
Format DoubleFortran;
#OpenDictionary numbers
  #add 2: "TWO"
  #add 5: "FIVE"
  #add 7: "SEVEN"
  #add 1/2: "HALF"
#CloseDictionary
Local F = (1+x)^7/7;
id x^n? = x*x^n/(n+1);
#UseDictionary numbers (warnings)
Print;
.end
```

```
Time =          0.00 sec      Generated terms =          8
          F          Terms in output =          8
          Bytes used      =          204
```

```
F =
& 1/SEVEN*x + HALF*x**2 + x**3 + FIVE/4*x**4 + x**5 + HALF*x**6 +
>>>>>>>Could not translate coefficient with dictionary numbers<<<<<<<<
<<<
& 1/SEVEN*x**7 + 1.D0/56.D0*x**8
```

In this case the line after the warning contains a fraction that was not substituted. This allows one to add either 56 or 1/56 to the dictionary. This gives the program

```
Symbol x,n;
Format DoubleFortran;
#OpenDictionary numbers
  #add 2: "cd2"
  #add 5: "cd5"
  #add 7: "cd7"
  #add 56: "cd56"
  #add 1/2: "c1d2"
  #add 5/4: "c5d4"
#CloseDictionary
Local F = (1+x)^7/7;
id x^n? = x*x^n/(n+1);
#UseDictionary numbers (warnings)
Print;
.end
F =
& 1/cd7*x + c1d2*x**2 + x**3 + c5d4*x**4 + x**5 + c1d2*x**6 + 1/
& cd7*x**7 + 1/cd56*x**8
```

Here we have selected a different notation that allows extension easily. A good way to do this now is to put the dictionary in a file numbers.hh and the corresponding FORTRAN definitions in a file numbers.h and then include these files in the proper places. The numbers.hh file would be

```
#OpenDictionary numbers
  #add 2: "cd2"
  #add 5: "cd5"
  #add 7: "cd7"
  #add 56: "cd56"
  #add 1/2: "c1d2"
  #add 5/4: "c5d4"
#CloseDictionary
```

and the numbers.h file would be

```
REAL cd2,cd5,cd7,cd56,c1d2,c5d4
PARAMETER (cd2=2,cd5=5,cd7=7,cd56=56,c1d2=1/cd2,c5d4=cd5/4)
```

and when the dictionary file is updated one may update the FORTRAN file simultaneously.

Setting the precision of the declaration REAL can be done by compiler options. These may depend on the compiler. One should consult the manpages.

Printing the extra symbols (7.53) may be a bit trickier. A range is indicated with a pair of parentheses enclosing one or two (positive) numbers. If there are two numbers, they should be separated by a comma. There can be more than one range. In the substitution one can use the wildcards %# and %@ to indicate the number of the extra symbol. The first wildcard indicates the number of the symbol and the second starts it counting with 1 from the beginning of the range.

```
Symbol x;
CFunction f;
```

```

#OpenDictionary ranges
  #add (1,2): "w(%#)"
  #add (3): "ww(%#)"
  #add (4,6): "www(%@)"
#CloseDictionary
Local F = <f(1)*x^1>+...+<f(6)*x^6>;
ToPolynomial;
Print;
.sort

F =
  x*Z1_ + x^2*Z2_ + x^3*Z3_ + x^4*Z4_ + x^5*Z5_ + x^6*Z6_;

#UseDictionary ranges
Print;
.end

F =
  x*w(1) + x^2*w(2) + x^3*ww(3) + x^4*www(1) + x^5*www(2) + x^6*www(3);

```

The use of the dictionaries in dollar variables can best be shown with an example that has much in common with graph theory. Assume we have an expression that contains all topologies we are interested in, with a notation for the momenta. The function vx represents a vertex and we use it as a symmetric function. Here we show two topologies from massless two-loop propagators:

$$\begin{aligned}
& +vx(p_0,p_1,-p_4)*vx(-p_1,p_2,p_5)*vx(q_0,-p_2,-p_3)*vx(p_4,p_3,-p_5)*topo(1) \\
& +vx(p_0,p_1,p_2)*vx(-p_1,p_3,p_4)*vx(q_0,-p_2,-p_3,-p_4)*topo(2)
\end{aligned}$$

where the q_0 momentum is taken to be $-p_0$. The problem is what happens when in a diagram of topology one, one of the lines is removed. If for instance the p_1 line is removed, we will end up with the second topology, but the question is: how should we relabel the momenta to obtain the notation of topology 2. Taking out p_1 gives us:

$$+vx(p_0,-p_4,p_2,p_5)*vx(q_0,-p_2,-p_3)*vx(p_4,p_3,-p_5)*topo(1)$$

and to see what renaming we need is usually a major source of errors. We can do this automatically if we can substitute the second topology into the remainder of the first using proper wildcards and storing the matches in dollar variables. This can be done with a dictionary:

```

#OpenDictionary match
  #add p0: "p0?{p0,q0}$p0"
  #add q0: "q0?{p0,q0}$q0"
  #do i = 1,5
    #add p'i': "p'i'?$p'i'"
  #enddo
#CloseDictionary

```

We put the various candidate topologies that could match, one by one, into the variable \$child as in (after using brackets on the expression with the topologies):

```
#$child = Topologies[topo(2)];
```

but generating an id-statement from it would be very laborious without the dictionaries:

```
id '$Orig' = 1;
```

would result in:

```
id vx(-p2,-p3,q0)*vx(-p4,p0,p2,p5)*vx(-p5,p3,p4) = 1;
```

but with the dictionary activated as in

```
#inside $child
#UseDictionary match($)
id '$Orig' = 1;
#CloseDictionary
#endinside
```

the generated code is

```
id vx(-p2?$p2,-p3?$p3,q0?{p0,q0}$q0)*vx(-p4?$p4,p0?{p0,q0}$p0,
p2?$p2,p5?$p5)*vx(-p5?$p5,p3?$p3,p4?$p4) = 1;
```

and from the dollar variables we can generate a statement with the the renumbering

```
id topo(1) = topo(2)*replace_(p0,-p0,p1,q1,p2,-p2,p3,-p1,p4,p3,p5,-p4);
```

We used $p_1 \rightarrow q_1$ as initialization before the pattern matching and $p_0 = q_0$ we can replace by $p_0 = p_0$. The q_1 should be replaced by means of momentum conservation, but that goes beyond the scope of this example.

It should be clear from the above that the dictionaries are the beginning of a new development. One should expect more capabilities in the future and suggestions are highly appreciated, provided they lead to something that can be implemented in a reasonable amount of time. Hence, for instance, there will not be a complete L^AT_EX output format that can take line length into account.

Chapter 14

Dirac algebra

For its use in high energy physics FORM is equipped with a built-in class of functions. These are the gamma matrices of the Dirac algebra which are generically denoted by g_- . The gamma matrices fulfill the relations:

$$\begin{aligned} \{g_-(j1, \mu), g_-(j1, \nu)\} &= 2 * d_-(\mu, \nu) \\ [g_-(j1, \mu), g_-(j2, \nu)] &= 0 \quad j1 \text{ not equal to } j2. \end{aligned}$$

The first argument is a so-called spin line index. When gamma matrices have the same spin line, they belong to the same Dirac algebra and commute with the matrices of other Dirac algebra's. The indices μ and ν are over space-time and are therefore usually running from 1 to 4 (or from 0 to 3 in Bjorken & Drell metric). The totally antisymmetric product $\epsilon_-(m1, m2, \dots, mn) g_-(j, m1) \times \dots g_-(j, mn)/n!$ is defined to be $\text{gamma}5$ or $g5_-(j)$. The notation 5 finds its roots in 4 dimensional space-time. The unit matrix is denoted by $g_i_-(j)$. In four dimensions a basis of the Dirac algebra can be given by:

$$\begin{aligned} g_i_-(j) \\ g_-(j, \mu) \\ [g_-(j, \mu), g_-(j, \nu)]/2 \\ g5_-(j) * g_-(j, \mu) \\ g5_-(j) \end{aligned}$$

In a different number of dimensions this basis is correspondingly different. We introduce the following notation for convenience:

$$\begin{aligned} g6_-(j) &= g_i_-(j) + g5_-(j) && \text{(from Schoonschip)} \\ g7_-(j) &= g_i_-(j) - g5_-(j) \\ g_-(j, \mu, \nu) &= g_-(j, \mu) * g_-(j, \nu) && \text{(from Reduce)} \\ g_-(j, \mu, \nu, \dots, ro, si) &= \\ &g_-(j, \mu, \nu, \dots, ro) * g_-(j, si) \\ g_-(j, 5_) &= g5_-(j) \\ g_-(j, 6_) &= g6_-(j) \\ g_-(j, 7_) &= g7_-(j) \end{aligned}$$

The common operation on gamma matrices is to obtain the trace of a string of gamma matrices. This is done with the statement:

trace4, j	Take the trace in 4 dimensions of the combination of all gamma matrices with spin line j in the current term. Any non-commuting objects that may be between some of these matrices are ignored. It is the users responsibility to issue this statement only after all functions of the relevant matrices are resolved. The four refers to special tricks that can be applied in four dimensions. This allows for relatively compact expressions. For the complete syntax, consult 7.153.
tracen, j	Take the trace in an unspecified number of dimensions. This number of dimensions is considered to be even. The traces are evaluated by only using the anticommutation properties of the matrices. As the number of dimensions is not specified the occurrence of a g5_(j) is a fatal error. In general the expressions that are generated this way are longer than the four dimensional expressions. For the complete syntax, consult 7.154.

It is possible to alter the value of the trace of the unit matrix gi_(j). Its default value is 4, but by using the statement (see 7.159)

```
unittrace value;
```

it can be altered. Value may be any positive short number ($< 2^{15}$ on 32 bit machines and $< 2^{31}$ on 64 bit machines) or a single symbol with the exception of the symbol i_.

There are several options for the 4-dimensional traces. These options find their origin in the Chisholm relation that is valid in 4 dimensions but not in a general number of dimensions. This relation can be found in the literature. It is given by:

$$\gamma_\mu Tr[\gamma_\mu S] = 2(S + S^R) \quad (14.1)$$

in which S is a string of gamma matrices with an odd number of matrices (γ_5 counts for an even number of matrices). S^R is the reversed string. This relation can be used to combine traces with common indices. The use of this relation is the default for trace4. If it needs to be switched off, one should add the extra option ‘nocontract’:

```
trace4,nocontract,j;
```

The option ‘contract’ is the default but it can be used to enhance the readability of the program. The second option that refers to this relation is the option ‘symmetrize’. Often it happens that there are two or more common indices in two spin lines. Without the symmetrize option (or with the ‘nosymmetrize’ option) the first of these indices is taken and the relation is applied to it. With the ‘symmetrize’ option the average over all possibilities is taken. This means of course that if there are two common indices the amount of work is doubled. There is however a potentially large advantage. In some traces that involve the use of γ_5 the use of automatic algorithms results often in an avalanche of terms with a single Levi-Civita tensor, while symmetry arguments can show that these terms should add up to zero. By working out the traces in a more symmetric fashion FORM is often capable of eliminating all or nearly all of these Levi-Civita tensors. Normally such an elimination is rather complicated. It involves relations that have so far defied proper implementation, even though people have been looking for such algorithms already for a long time. Hence the use of the symmetry from the beginning seems at the moment the best bet.

It is possible to only apply the Chisholm identity without taking the trace. This is done with the chisholm statement (see 7.17).

The n dimensional traces can use a special feature, when the declaration of the indices involved will allow it. When an index has been declared as n-dimensional and the dimension is followed by a second symbol as in

```
symbols n,nn;
index mu=n:nn;
```

and if the index `mu` is a contracted index in a single n -dimensional trace, then the formula for this trace can be shortened by using `nn` (one term) instead of the quantity $(n - 4)$ (two terms). This can make the taking of the n -dimensional traces significantly faster.

Algorithms:

FORM has been equipped with several built in rules to keep the number of generated terms to a minimum during the evaluation of a trace. These rules are:

rule 0 Strings with an odd number of matrices (gamma5 counts for an even number of matrices) have a trace that is zero, when using `trace4` or `tracen`.

rule 1 A string of gamma matrices is first scanned for adjacent matrices that have the same contractable index, or that are contracted with the same vector. If such a pair is found, the relations

$$\begin{aligned} g_{-}(1, \mu, \mu) &= g_{i-}(1) * d_{-}(\mu, \mu) \\ g_{-}(1, p1, p1) &= g_{i-}(1) * p1.p1 \end{aligned}$$

are applied.

rule 2 Next there is a scan for a pair of the same contractable indices that has an odd number of other matrices in between. This is done only for 4 dimensions (`trace4`) and the dimension of the indices must be 4. If found, the Chisholm identity is applied:

$$g_{-}(1, \mu, m1, m2, \dots, mn, \mu) = -2 * g_{-}(1, mn, \dots, m2, m1)$$

rule 3 Then (again only for `trace4`) there is a search for a pair of matrices with the same 4 dimensional index and an even number of matrices in between. If found, one of the following variations of the Chisholm identity is applied:

$$\begin{aligned} g_{-}(1, \mu, m1, m2, \mu) &= 4 * g_{i-}(1) * d_{-}(m1, m2) \\ g_{-}(1, \mu, m1, m2, \dots, mj, mn, \mu) &= \\ &\quad 2 * g_{-}(1, mn, m1, m2, \dots, mj) \\ &\quad + 2 * g_{-}(1, mj, \dots, m2, m1, mn) \end{aligned}$$

rule 4 Then there is a scan for pairs of matrices that have the same index or that are contracted with the same vector. If found, the identity:

$$\begin{aligned} g_{-}(1, \mu, m1, m2, \dots, mj, mn, \mu) &= \\ &\quad 2 * d_{-}(\mu, mn) * g_{-}(1, \mu, m1, m2, \dots, mj) \\ &\quad - 2 * d_{-}(\mu, mj) * g_{-}(1, \mu, m1, m2, \dots, mn) \\ &\quad \dots \\ &\quad -/+ 2 * d_{-}(\mu, m2) * g_{-}(1, \mu, m1, \dots, mj, mn) \\ &\quad +/- 2 * d_{-}(\mu, m1) * g_{-}(1, \mu, m2, \dots, mj, mn) \\ &\quad -/+ d_{-}(\mu, \mu) * g_{-}(1, m1, m2, \dots, mj, mn) \end{aligned}$$

is used to 'anticommute' these identical objects till they become adjacent and can be eliminated with the application of rule 1. In the case of an n -dimensional trace and when μ is an index (it might also be a vector in the above formula) for which the definition of the dimension involved two symbols, there is a shorter formula. In that case the last three terms can be combined into two terms:

$$\begin{aligned} & -/+(n-4)*g_{-}(1,m1,m2,\dots,mj,mn) \\ & -/+4*d_{-}(m1,m2)*g_{-}(1,m3,m4,\dots,mj,mn) \end{aligned}$$

It should be clear now that this formula is only superior, when there is a single symbol to represent $(n-4)$. After this all gamma matrices that are left have a different index or are contracted with different vectors. These are treated using:

rule5 Traces in 4 dimensions for which all gamma matrices have a different index, or are contracted with a different four-vector are evaluated using the reduction formula

$$\begin{aligned} g_{-}(1,\mu,\nu,\rho) = & \\ & g_{-}(1,5_{-},si)*e_{-}(\mu,\nu,\rho,si) \\ & +d_{-}(\mu,\nu)*g_{-}(1,\rho) \\ & -d_{-}(\mu,\rho)*g_{-}(1,\nu) \\ & +d_{-}(\nu,\rho)*g_{-}(1,\mu) \end{aligned}$$

For `tracen` the generating algorithm is based on the generation of all possible pairs of indices/vectors that occur in the gamma matrices in combination with their proper sign. When the dimension is not specified, there is no shorter expression.

Remarks:

When an index is declared to have dimension n and the command `trace4` is used, the special 4 dimensional rules 2 and 3 are not applied to this index. The application of rule 1 or 4 will then give the correct results. The result will nevertheless be wrong due to rule 5, when there are at least 10 gamma matrices left after the application of the first 4 rules, as the two algorithms in rule 5 give a difference only, when there are at least 10 gamma matrices. For counting gamma matrices the γ_5 counts for 4 matrices with respect to this rule. The result is unpredictable, when both indices in four dimensions and indices in n dimensions occur in the same string of gamma matrices. Therefore one should be very careful, when using the four dimensional trace under the condition that the results need to be correct in n dimensions. This is sometimes needed, when a γ_5 is involved. The `tracen`-statement will not allow the presence of a γ_5 . In general it is best to emulate n -dimensional traces with a γ_5 separately. The eventual trace, with all matrices with a different index, can be generated with the use of the 'distrib_' function:

```
*
*   Symmetric trace of a gamma5 and 12 regular matrices
*
I   m1,...,m12;
F   G5,g1,g2;
L   F = G5(m1,...,m12);
id  G5(?a) = distrib_(-1,4,g1,g2,?a);
id  g1(?a) = e_(?a);
id  g2(?a) = g_(1,?a);
```

```

tracen,1;
.end

```

Time =	1.07 sec	Generated terms =	51975
	F	Terms in output =	51975
		Bytes used =	919164

This rather symmetric result is in contrast to the 4-dimensional result which is much shorter, but it is very unsymmetric:

```

*
*   Regular trace of a gamma5 and 12 regular matrices
*
I   m1,...,m12;
L   F = g_(1,5_,m1,...,m12);
trace4,1;
.end

```

Time =	0.02 sec	Generated terms =	1053
	F	Terms in output =	1029
		Bytes used =	20284

The precise workings of the distrib_ function is given in 8.12.

One should be careful when using projection operators of spinors. The sloppy way is to write

$$(g_{-}(1,p)+m)$$

but technically this is not correct. The correct way is

$$(g_{-}(1,p)+m*gi_{-}(1))$$

to avoid the possibility that in the end a trace will be taken over a term that does not have any gamma matrix. If the projection operator is however multiplied by other gamma matrices, it makes no difference whether the unit matrix is present. That is why the sloppy notation will almost always give the correct result. Almost always....

Chapter 15

A few notes on the use of a metric

When FORM was designed, it was decided to make its syntax more or less independent of a choice of the metric. Hence statements and facilities that programs like Schoonschip or REDUCE provide but which depend on the choice of a metric have been left out. Instead there are facilities to implement any choice of the metric, when the need really arises. When one makes a proper study of it, it turns out that one usually has to do very little or nothing.

First one should realize that FORM does not know any specific metric by itself. Dotproducts are just objects of manipulation. It is assumed that when a common index of two vectors is contracted, this works out properly into a scalar object. This means that if one has a metric with upper and lower indices, one index is supposed to be an upper index and the other is supposed to be a lower index. If the user does not like this, it is his/her responsibility to force the system into a different action. This is reflected in the fact that FORM does not have an internal metric tensor $\eta_{\mu\nu}$. It has only a Kronecker delta $\delta_{\mu\nu} = d_(\text{mu}, \text{nu})$ with $p(\text{mu}) * d_(\text{mu}, \text{nu}) * q(\text{nu}) \rightarrow p.q$ when mu and nu are summable indices.

The dependency of a metric usually enters with statements like $p^2 = \pm m^2$, which the user should provide anyway, because FORM does not have such knowledge. Connected to this is the choice of a propagator as either $\gamma_\mu p_\mu + m$ or $\gamma_\mu p_\mu + i m$. This is also something the user should provide. The only objects that FORM recognizes and that could be considered as metric-dependent are the gamma matrices and the Levi-Civita tensor ϵ_- . Because the trace of a γ_5 involves a Levi-Civita tensor, the two are intimately connected. The anticommutator of two gamma matrices is defined with the Kronecker delta. Amazingly enough that works out well, provided that, if such Kronecker delta's survive in the output, they are interpreted as a metric tensor. This should be done with great care, because at such a point one does something that depends of the metric; one may have to select whether the indices are upper or lower indices. One should check carefully that the way the output is interpreted leads indeed to the results that are expected. This is anyway coupled to how one should interpret the input, because in such a case one would also have an input with 'open' indices and give them a proper interpretation. The rule is that generally one does not have to do anything. The upper indices in the input will be upper indices in the output and the same for lower indices.

The contraction of two Levi-Civita tensors will give products of Kronecker delta's. This means that formally there could be an error of the sign of the determinant of the metric tensor, if one would like the Kronecker delta to play the role of a metric tensor. Hence it is best to try to avoid such a situation.

In FORM the γ_5 is an object that anticommutes with the γ_μ and has $\gamma_5 \gamma_5 = 1$. Its properties in

the trace are

$$Tr[\gamma_5 \gamma_{m_1} \gamma_{m_2} \gamma_{m_3} \gamma_{m_4}] = 4\epsilon_{\mu_1 \mu_2 \mu_3 \mu_4}$$

This has a number of interesting consequences. The V-A and V+A currents are represented by $\gamma_7 \gamma_\mu = (1 - \gamma_5) \gamma_\mu$ and $\gamma_6 \gamma_\mu = (1 + \gamma_5) \gamma_\mu$ respectively. Under conjugation we have to replace γ_5 by $-\gamma_5$ as is not uncommon.

There was a time that a conjugation operation was planned in FORM. As time progressed, it was realized that this would introduce problems with some of the internal objects. Hence some objects have the property that they are considered imaginary. In practise FORM does not do anything with this. Neither does it do anything with the declarations real, complex and imaginary. If ever a way is found to implement a conjugation operator that will make everybody happy, it may still be built in.

The above should give the user enough information to convert any specific metric to what is needed to make FORM do what is expected from it. Afterwards one can convert back, provided no metric specific operations are done. Such metric specific things are for instance needed in some types of approximations in which one substitutes objects by (vector)components halfway the calculation. In that case one cannot rely on that the conversions at the beginning and the end will be compensating each other. For this case FORM allows the user to define a private metric. All the tools exist to make this a success with the exception of a loss in speed of course. Let us have a look at the contraction of two Levi-Civita tensors in an arbitrary metric:

```
Indices m1,m2,m3,n1,n2,n3,i1,i2,i3;
Cfunction eta(symmetric),e(antisymmetric);
Off Statistics;
*
*   We have our own Levi-Civita tensor e
*
Local F = e(m1,m2,m3)*e(m1,m2,m3);
*
*   We write the contraction as
*
id  e(m1?,m2?,m3?)*e(n1?,n2?,n3?) =
      e_(m1,m2,m3)*e_(i1,i2,i3)*
      eta(n1,i1)*eta(n2,i2)*eta(n3,i3);
*
*   Now we can use the internal workings of the contract:
*
Contract;
Print +s;
.sort

F =
+ eta(i1,i1)*eta(i2,i2)*eta(i3,i3)
- eta(i1,i1)*eta(i2,i3)^2
- eta(i1,i2)^2*eta(i3,i3)
+ 2*eta(i1,i2)*eta(i1,i3)*eta(i2,i3)
- eta(i1,i3)^2*eta(i2,i2)
```

```

;

*
*   For specifying a metric we need individual components:
*
Sum i1,1,2,3;
Sum i2,1,2,3;
Sum i3,1,2,3;
Print +s;
.sort

F =
+ 6*eta(1,1)*eta(2,2)*eta(3,3)
- 6*eta(1,1)*eta(2,3)^2
- 6*eta(1,2)^2*eta(3,3)
+ 12*eta(1,2)*eta(1,3)*eta(2,3)
- 6*eta(1,3)^2*eta(2,2)
;

*
*   And now we can provide the metric tensor
*
id eta(1,1) = 1;
id eta(2,2) = 1;
id eta(3,3) = -1;
id eta(1,2) = 0;
id eta(1,3) = 0;
id eta(2,3) = 0;
Print +s;
.end

F =
- 6
;

```

This is the ultimate in flexibility of course. It can also be worked out in a different way. In this case we try to change the behaviour of the Kronecker delta a bit. This is dangerous and needs, in addition to a good understanding of what is happening, good testing to make sure that what the user wants is indeed what does happen. Here we use the `FixIndex` (7.59) statement. This one assigns specific values to selected diagonal elements of the Kronecker delta. Of course it is the responsibility of the user to make sure that the calculation will indeed run into those elements. This is by no means automatic, because when FORM uses formal indices it never writes them out in components. Moreover, it would not be defined what would be the components connected to an index. The index could run over 0, 1, 2, 3 or over 1, 2, 3, 4, or maybe even over 5, 7, 9, 11. And what does an n-dimensional index run over? In the above example it is the `sum` (7.139) statement that determines this. Hence this is fully under the control of the user. Therefore a proper way to deal with the above example would be

```
Indices i1,i2,i3;
```



```

FixIndex 1:1,2:1,3:-1;
Off Statistics;
*
Local F = e_(i1,i2,i3)*e_(i1,i2,i3);
Sum i1,1,2,3;
Sum i2,1,2,3;
Sum i3,1,2,3;
Print +s;
.sort

F =
    + 6*e_(1,2,3)*e_(1,2,3)
    ;

Contract;
Print +s;
.end

F =
    - 6
    ;

```

In the case that one would like to exchange the order of the summation and the contraction, while using the FixIndex mechanism, one needs to be more careful. In that case we have to prevent the indices from being summed over while they are indices of a Kronecker delta, because as long as the indices are symbolic, FORM will replace $d_{(i1,i1)}$ by the dimension of $i1$, and that is not what we want. Hence we have to declare the indices to be non-summable by giving them dimension zero:

```

Indices i1=0,i2=0,i3=0;
FixIndex 1:1,2:1,3:-1;
Off Statistics;
*
Local F = e_(i1,i2,i3)*e_(i1,i2,i3);
Contract;
Print +s;
.sort

F =
    + d_(i1,i1)*d_(i2,i2)*d_(i3,i3)
    - d_(i1,i1)*d_(i2,i3)*d_(i2,i3)
    - d_(i1,i2)*d_(i1,i2)*d_(i3,i3)
    + 2*d_(i1,i2)*d_(i1,i3)*d_(i2,i3)
    - d_(i1,i3)*d_(i1,i3)*d_(i2,i2)
    ;

Sum i1,1,2,3;
Sum i2,1,2,3;
Sum i3,1,2,3;
Print +s;

```

```

.end

F =
    - 6
;

```

As we can see, the automatic summation over the indices is not performed now and this gives us a chance to do the summation manually. After that the `fixindex` statement can have its effect.

It should be clear from the above examples that it is usually much easier to manipulate the input in such a way that the terms with two Levi-Civita tensors have the negative sign from the beginning. This would give programs that are less complicated and much faster.

Hence we are faced with the situation that in normal cases one does not do anything. If one wants to go beyond this and wants to interfere with the inner workings themselves by for instance inserting a factor i in front of the γ_5 and emulating the upper and lower indices of a favorite metric, this leads from one problem to the next. Extreme care is needed. This is usually done by people who have first worked with other programs in which things don't work as naturally as in FORM. By the time one has really figured out how to deal with the metric and how to make use of the internal algorithms of FORM, one usually does not have to do very much again.

As in the Zen saying:

To the beginning student mountains are mountains and water is water. To the advanced student mountains stop being mountains and water stops being water. To the master mountains are mountains again and water is water again.

Of course the modern master also checks that what he expects the system to do, is indeed what the system does.

Chapter 16

Sorting and statistics

The sorting system is a vital part of FORM and one of the main reasons why the speed of FORM compares so favorably with other systems. A good understanding of what happens during the sorting of expressions is essential if one wants to write efficient programs. In essence the sorting is done by a tree sort. However due to the nature of mathematical expressions there is a complication. When two terms are identical with the possible exception of their coefficient, we will add their coefficients, put this new coefficient in the place of the coefficient of the first term, and drop the second term. If the new coefficient happens to be zero, both terms are dropped. Hence the number of terms during the sort is not fixed. For a tree sort this is not a major complication. What is more annoying though is that the new coefficient may take more space inside the storage than either of the old coefficients. Let us have a look now at what happens in a FORM program. Much can be seen from the statistics.

```
S x1,...,x4;  
L F = (x1+...+x4)^4;  
.end
```

Time =	0.01 sec	Generated terms =	35
	F	Terms in output =	35
		Bytes used =	628

In this case the program generated 35 terms. Whenever a term is generated and FORM is done with it (no more statements will act on it), FORM will write it into a buffer which is called the small buffer. Additionally it stores a pointer to the location of this term inside the small buffer. Next it will continue generating terms. This process will be stopped by either of three conditions:

1. FORM is finished generating terms.
2. The last generated term does not fit inside the space remaining in the small buffer.
3. There is no space for a pointer to the last generated term inside the array of pointers.

In either of these three cases FORM will sort the contents of the small buffer. This sorting is done 'by pointers' and hence it is important that the whole small buffer fits inside the physical memory of the computer. If this would not be the case, some very inefficient swapping of memory might be the result. During this sorting FORM may run into the problem that the coefficient of two combined terms does not fit in the place of one of the two old coefficients. This means that the combined term will need more space, but because the old terms might be enclosed by other

terms, this space may not be available locally. To this end FORM has some spare space in the small buffer which is called the small extension. Actually the term SmallExtension is used for the combination of the small buffer and its extra space. The extra space is at least 1/6 times the size of the small buffer, but typically it will be about 1/3 the size of the small buffer. In some exceptional cases (with heavy use of a polynomial coefficient via the PolyFun command) bigger sizes might be useful.

In the case that the new combined term needs more space than each of the old terms, the new term is placed in the extension space. If, during the sort, the extension space becomes exhausted, FORM will make a garbage collection of the entire extended small buffer. This will always result in the extension space becoming empty again, because the notation of the terms in FORM is such the new combined term will at most occupy an amount of space equal to the sum of the spaces of the original two terms. In older versions of FORM this garbage collection was executed by means of a temporary disk file. In the new version it is done inside the memory by temporarily allocating a new buffer. Anyways such garbage collections are relatively rare.

In the above example, the sorting occurred because the generation of terms was finished. Hence the sorted output is written away in such a way that it can be used as input for a potential next module (or to be printed). Hence let us change the size of the small buffer:

```
#: SmallSize 300
S x1,...,x4;
L F = (x1+...+x4)^4;
.end
```

Time =	0.00 sec	Generated terms =	13
	F	1 Terms left =	13
		Bytes used =	236
Time =	0.00 sec	Generated terms =	26
	F	1 Terms left =	26
		Bytes used =	476
Time =	0.00 sec	Generated terms =	35
	F	1 Terms left =	35
		Bytes used =	632
Time =	0.00 sec	Generated terms =	35
	F	Terms in output =	35
		Bytes used =	628

Now the size of the small buffer will be only 300 bytes. As a result the 13-th term does not fit. We can see this in the statistics: the 13-th term has been generated and FORM sorts the small buffer. The output of the 12 sorted terms is written to another buffer, called the large buffer. Inside the large buffer the terms are lightly compressed. This compression is related to the fact that in each ‘patch’ the terms are already sorted and hence we may not have to repeat the identical beginnings of each term. Hence the amount of space used after this sort is less than the 300 bytes of the small buffer, even though the 13-th term gave an overflow for these 300 bytes. The small buffer fills up again at the 26-th term and again it is sorted and the results written to the large buffer. Finally, after 35 terms, the generation is finished. Hence the remains in the small buffer are also sorted and written as a third ‘patch’ into the large buffer. Then the large buffer is sorted. For

this a different sort technique is used. It is assumed that the large buffer is not always residing inside the physical memory. Hence parts of it may be swapped out temporarily. With the size of current days memories this may not happen very often, unless one sets the size of the buffer to something comparable to the memory size of the computer and several programs are running at the same time. Anyway, swapping will not affect the large buffer very much. FORM will merge the ‘patches’ by going sequentially through them with a method called ‘tree of losers’ in the book by Knuth (the art of computer programming, vol. 3). Because it goes sequentially through the patches, uses all the information it reads and never needs it again, this method is indeed rather well resistant to swapping.

The next complication is of course when the large buffer is full. This can be either because its byte space is full, or because the maximum number of patches is exceeded. Because the sorting method uses quite a few variables for each patch, there is a space allocated for them and hence there is a maximum number of patches. If we set this to 2 (just for demonstration purposes) we obtain:

```
#: SmallSize 200
#: LargePatches 2
S x1,...,x4;
L F = (x1+...+x4)^4;
.end
```

```
Time =      0.00 sec    Generated terms =          9
      F      1 Terms left   =          9
              Bytes used    =        164
```

```
Time =      0.00 sec    Generated terms =         17
      F      1 Terms left   =         17
              Bytes used    =        312
```

```
Time =      0.00 sec    Generated terms =         26
      F      1 Terms left   =         26
              Bytes used    =        478
```

```
Time =      0.00 sec
      F      Terms active   =         26
              Bytes used    =        474
```

```
Time =      0.00 sec    Generated terms =         35
      F      1 Terms left   =         35
              Bytes used    =        630
```

```
Time =      0.00 sec
      F      Terms active   =         35
              Bytes used    =        786
```

```
Time =      0.00 sec    Generated terms =         35
      F      Terms in output =         35
              Bytes used    =        628
```

We see that after the third small buffer has been sorted, the third patch cannot be written to the large buffer. Hence the large buffer is sorted (indicated by the special statistics involving the phrase ‘Terms active’). The result of this is written as a sorted patch to the sort file. This file is one of the temporary files that FORM can create. It has the extension .sor. Now the third patch can be written into the –by now empty– large buffer. At the end of term generation, the last small buffer is sorted, its results written into the large buffer, then that is sorted and its results written as the final patch into the sort file. Then, finally the patches in the sort file are merged in a method similar to the way the large buffer is sorted. This final sort is a disk to disk sort. Hence it can use the disk rather intensely and the use of the CPU may drop temporarily, although it is nothing so dramatic as when the computer is involved in heavy inefficient swapping as can be the case with many other algebra programs. Also, this is usually only a small fraction of the running time of the program. The exception may be when FORM is running several processes and they are all using disk sorts simultaneously. In that case some file systems may not be very good at handling the ensuing traffic jams.

Also the disk to disk sort will have a maximum number of patches that can be sorted simultaneously. If this number is exceeded there will be one or more extra stages in the sorting, all of which will be disk to disk sorts. It is advisable to tune the setup parameters in such a way that one can prevent this, because it involves usually needless use of resources. One can try to increase the parameter FilePatches, but the problem is that FORM uses a caching system to buffer the inputs from the sort file. The cache buffers have to have a size that is at least twice the maximum size of a term. For each patch it needs a buffer and all buffers together should fit inside the combination of the large buffer and the small extended buffer. This puts an upper limit on the number of file patches. Additionally this buffer (SortIOsize) should not be very small, because otherwise the disk IO operations are very inefficient. Hence it helps often to increase the size of the small buffer and the large buffer first. That gives fewer patches. Additionally it in turn can allow for more file patches that are not too small.

One thing that one can see now is that if terms are to cancel or to add, it is advantageous if this happens already in an early stage of the sorting. This means that it is most efficient if these terms will end up in the small buffer at the same time. This should explain the example given in the section on brackets. This way fewer terms are written to the large buffer and/or the sort file, which means that less disk space will be used.

The sizes of buffers involved can all be tuned to a given hardware. How this is done is explained in the chapter on the setup 17.

When FORM is dealing with the arguments of functions and if an argument is a multiterm subexpression, also such subexpressions need to be sorted. In older versions of FORM this was done inside the at that moment remaining space of the small buffer and its extension. The reason was that such subexpressions would be rather short (they would have to fit inside a function argument and were hence limited by the maximum size of a term) and buffer space was hard to come by in computers with small memories. In the new version of FORM other subexpression sorts were added: the sorting in the term environment (see 7.146) and the sorting of \$-expressions. Both sorts do not have the restriction of the maximum size of a term. They can result in expressions that are arbitrarily long (although that might not give efficient programs). Hence the sorting of subexpressions have now their own buffers. And more than one such set may be needed if for instance the term environment is used in a nested fashion. Of course the settings for the buffers of this ‘subsort’ are not quite as large as for the main buffers. And the user can of course also influence their settings as explained in the chapter on the setup 17. This chapter gives also all default values.

There is one restriction on the sorting of function arguments and \$-expressions: They are not

allowed to go into the stage4 sorting. Any such attempt will result in an error message and the suggestion to raise the size of the buffers for this type of sorting.

When FORM is running in parallel mode (either TFORM or PARFORM) each worker will need its own buffers. In PARFORM in which the processors each control their own memory, the size of each of these buffers are the same as for the master process. In TFORM with its shared memory the sizes that the user selects for the sort buffers and the scratch file caches refer to the buffers of the master thread. The workers each get basically buffers with $1/N$ times the size of the buffer of the master. They may be made a bit bigger when potential conflicts with MaxTermSize occur.

Chapter 17

The setup

When FORM is started, it has a number of settings built in that were determined during its installation. If the user would like to alter these settings, it is possible to either specify their desired values in a setup file or to do so at the beginning of the program file. There are two ways in which FORM can find a setup file. The first way is by having a file named 'form.set' in the current directory. If such a file is present, FORM will open it and interpret its contents as setup parameters. If this file is not present, one may specify a setup file with the -s option in the command tail. This option must precede the name of the input file. After the -s follow one or more blanks or tabs and then the full name of the setup file. FORM will try to read startup parameters from this file. If a file 'form.set' is present, FORM will ignore the -s option and its corresponding file name. This order of interpretation allows the user to define an alias with a standard setup file which can be overruled by a local setup file. If, in the beginning of the program file, before any other statements with the exception of the #- instruction and commentary statements, there are lines that start with #: the remaining contents of these lines are interpreted exactly like the lines in the setup file. The specifications in the program file take precedence over all other specifications. If neither of the above methods is used, FORM will use a built in set of parameters. Their values may depend on the installation and are given below.

The following is a list of parameters that can be set. The syntax is rather simple: The full word must be specified (case insensitive), followed by one or more blanks or tabs and the desired number, string or character. Anything after this is considered to be commentary. In the setup file lines that do not start with an alphabetic character are seen as commentary. The sizes of the buffers are given in bytes, unless mentioned otherwise. A word is 2 bytes for 32 bit machines and 4 bytes for 64 bit machines.

In FORM version 3.3 and later, it is also allowed to define preprocessor variables (see also 3.1) in the setup file. In addition one can use preprocessor variables in the setup, provided it is not in the name of the parameter/keyword.

bracketindexsize	Maximum size in bytes of any individual index of a bracketted expression. Each expression will have its own index. The index starts with a relatively small size and will grow if needed. But it will never grow beyond the specified size. If more space is needed, FORM will start skipping brackets and find those back later by linear search. See also chapter 9 and section 7.11.
CommentChar	This should be followed by one or more blanks and a single non-blank character. This character will be used to indicate commentary, instead of the regular * in column 1.

CompressSize	When compressing output terms, FORM needs a compression buffer. This buffer deals recursively with compression and decompression of terms that are either written or read. Its size will be at least MaxTermSize but when there is heavy use of expressions in the right hand side of definitions or substitution it would have to be considerably longer. It is hoped that in the future this parameter can be eliminated. CompressSize should be given in bytes.
ConstIndex	This is the number of indices that are considered to be constant indices like in fixed vector components (the so-called fixed indices). The size of this parameter is not coupled to any array space, but it should not go much beyond 1000 on a 32 bit machine. On a 64 bit machine it can go considerably further.
ContinuationLines	The maximum number of continuation lines, after which expressions will be broken apart when printed. The value of 0 means that no limit is imposed. The precise format of continuation lines depend on the current format (see 7.60) settings; #write instruction also allows for additional control (see 3.65).
Define	The syntax is as in the #define instruction in the preprocessor (see 3.1), with the remark that in the setup file there should be no leading # character as that would make the line into commentary. Example: <code>define MODULUS "31991"</code> which could be used at a later point in the program to activate a modulus statement (see 7.90).
DotChar	There should be a single character following this name (and the blank(s) after it). This character will be used instead of the <code>_</code> , when dotproducts are printed in Fortran output. This option is needed because some Fortran compilers do not recognize the underscore as a valid character. In the olden days one could use here the dollar character but nowadays many Fortran compilers do not recognize this character as belonging to a variable name.
FunctionLevels	The maximum number of levels that may occur, when functions have functions in their arguments.
HideSize	The size of the hide buffer. The size of this buffer is normally set equal to scratchsize (see below). If one uses the setting of HideSize after the setting of ScratchSize, one can give the hide buffer its own size. There are cases that this can make the program faster.
IncDir	Directory (or path of directories) in which FORM will look for files if they are not to be found in the current directory. This involves files for the #include and #call instructions. This variable takes precedence over the Path variable.
InsideFirst	Not having any effect at the moment.
JumpRatio	See the endswitch (7.49) statement.

MaxNumberSize	Allows the setting of the maximum size of the numbers in FORM. The number should be given in words. For 32 bit systems a word is two bytes and for 64 bit systems a word is 4 bytes. The number size is always limited by the maximum size of the terms (see MaxTermSize). Actually it has to be less than half of MaxTermSize because a coefficient contains both a numerator and a denominator. It is not always a good idea to have the number size at its maximum value, especially when MaxTermSize is large. In that case it could be very long before a runaway algorithm runs into limitations of size (arithmetic for very long fractions is not very fast due to the continuous need for computing GCD's)
MaxTermSize	This is the maximum size that an individual term may occupy in words. This size does not affect any allocations. One should realize however that the larger this size is the heavier the demand can be on the workspace, because the workspace acts as a heap during the execution and sometimes allocations have to be made in advance, before FORM knows what the actual size of the term will be. Consequently the evaluation tree cannot be very deep, when WorkSpace / MaxTermSize is not very big. MaxTermSize controls mainly how soon FORM starts complaining about terms that are too complicated. Its absolute maximum is 32568 on 32 bit systems and about 10^9 on 64 bit systems (of course the workspace would have to be considerably larger than that....).
MaxWildCards	The maximum number of wildcards that can be active in a single matching of a pattern. Under normal circumstance the default value of 100 should be more than enough.
NoSpacesInNumbers	Long numbers are usually spread over several lines by placing a backspace character at the end of each line and then continuing at the next line. For cosmetic purposes FORM puts usually a few blank spaces at the beginning of the new line. FORM itself can read this but some programs cannot. Hence one can put FORM in a mode in which these blanks are omitted. The values of the variable are ON or OFF. There is also a command to change this behaviour at runtime. See the on and off commands in sections 7.108 and 7.107.
NumStoreCaches	This number determines how many store caches (see the description of the SizeStoreCache setup parameter below) there will be. In the case of parallel processing this will be the number of caches per processor.
NwriteStatistics	When this word is mentioned, the default setting for the statistics is that no run time statistics will be shown. Ordinarily they will be shown.
NwriteThreadStatistics	This variable has the values ON or OFF. It controls for TFORM whether the statistics of the individual threads will be printed. The default value is ON.

OldOrder	A special flag (values ON/OFF) by which one can still select the old option of not checking for the order of statements inside a module. This should be used only in the case that it is nearly impossible to change a program to the new mode in which the order of the statements (declarations etc) is relevant. In the future this old mode may not exist.
Parentheses	The maximum number of nestings of parentheses or functions inside functions. The variable may be eliminated in a later version.
Path	Directory (or path of directories) in which FORM will look for files if they are not to be found in the current directory. This involves files for the <code>#include</code> and <code>#call</code> instructions. FORM will test this path after a potential path specified as <code>IncDir</code> .
ProcedureExtension	The extension that will be used by FORM for finding the procedures that are in separate files. Restrictions on the strings used are as explained in the preprocessor <code>#procedureextension</code> instruction in section 3.44.
ProcessBucketSize	For the parallel version PARFORM. It is ignored in other versions. Tells PARFORM how many terms there should be in the buckets that are being distributed over the secondary processors. See also 7.117.
ResetTimeOnClear	The value is ON or OFF. The default value is ON. This means that by default the clock is reset after each <code>.clear</code> (see chapter 4 on modules) instruction at the end of a module.
ScratchSize	The size of the input and the output buffers for the regular algebra processing. Terms are read in in chunks this size and are written to the output file using buffers of this size. There are either two or three of these buffers, depending on whether the hide facility is being used (see 7.67). These buffers must have a size that is at least as large as the <code>MaxTermSize</code> . These buffers act as caches for the files with the extension <code>.sc1</code> , <code>.sc2</code> and <code>.sc3</code> . See also the <code>HideSize</code> parameter above for the independent setting of the size of the hide buffer.
SizeStoreCache	The size of the caches that are used for reading terms when stored expressions are used in the r.h.s. of a statement. Typically there are several such caches and they make the reading much faster. In the case of parallel processing these caches become very important because without them the different processes may all want to read from the <code>.str</code> file at the same time and execution speed will suffer badly. The number of store caches is determined by the <code>NumStoreCaches</code> setup parameter which is described above. The size of these caches doesn't have to be very large as compared to some of the other buffers. It is recommended though to have them at least as large as <code>MaxTermSize</code> (see above).

SortType	Possible values are "lowfirst", "highfirst" and "powerfirst". "lowfirst" is the default. Determines the order in which the terms are placed during sorting. In the case of lowfirst, lower powers of symbols and dotproducts come before higher powers. In the case of highfirst it is the opposite. In the case of powerfirst the combined powers of all symbols together are considered and the highest combined powers come first. See also the on statement in 7.108.
TempDir	This variable should contain the name of a directory that is the directory in which FORM should make its temporary files. If the -t option is used when FORM is started, the TempDir variable in the setup file is ignored. FORM can create a number of different temporary files.
TempSortDir	This variable should contain the name of a directory that is the directory in which FORM should make its temporary sort files. If the -ts option is used when FORM is started, the TempSortDir variable in the setup file is ignored. If TempSortDir is not specified, then the value of TempDir is used also for sort files.
ThreadBucketSize	Only relevant for TFORM. The size of the number of terms sent to the workers simultaneously. For details see the chapter on the parallel version (18).
ThreadLoadBalancing	Only relevant for TFORM. Possible values are ON or OFF. For details see the chapter on the parallel version (18).
Threads	Only relevant for TFORM (see chapter on the parallel version). Specifies the default number of worker threads to be used. The values 0 and 1 will indicate that running will only be done by the master thread (18).
ThreadScratchOutSize	The size of the output scratch buffers for each of the worker threads. These buffers will be used by TFORM when the InParallel statement 7.76 is active. They are used to catch the output of the expressions as processed by the individual workers before they are copied to the output scratch buffer/file of the master. The output scratch buffer/file of each worker will never contain more than one expression at a time.
ThreadScratchSize	The size of the input scratch buffers for each of the worker threads. These buffers are only used when the main scratch buffers of the master process aren't sufficient and scratch files have been made. When the buffers of the master are big enough, the workers only use pointers to the buffer of the master. Once there are scratch files the buffer is used for caching the input from those files. In that case each worker has its own cache. For reading purposes it can actually be counter productive if these buffers are very large. This parameter sets the value for the input and the hide scratch files. The output scratch size for the workers is set with the ThreadScratchOutSize parameter.

TotalSize	Puts FORM in a mode in which it tries to determine the maximum space occupied by all expressions at any given moment during the execution of the program. This space is the sum of the input/output/hide scratch files, the sort file(s) and the .str file. This maximum is printed at the end of the program. The same can be obtained with the "On TotalSize" statement (see 7.108) or the -T option in the command tail when FORM is started (see 1).
WorkSpace	The size of the heap that is used by the algebra processor when it is evaluating the substitution tree. It will contain terms, half finished terms and other information. The size of the workspace may be a limitation on the depth of a substitution tree.
WTimeStats	Turns on the wall-clock time mode in the statistics. See the 'On wtimestats' statement 7.108.

Variables that take a path for their value expect a sequence of directories, separated by colon characters as in the UNIX way to define such objects.

The above parameters are conceptually relatively easy. The parameters that are still left are more complicated and are often restricted in their size by some relationships. Hence it is necessary to understand the sorting inside FORM a little bit before using them. On the other hand these parameters can influence the performance noticeably. See also chapter 16 for more details.

When terms are sent to 'output' by the main algebra engine, they are put inside a buffer. This buffer is called the 'small buffer'. Its size is given by the variable *SmallSize*. When this buffer is full, or when the number of terms in this buffer exceeds a given maximum, indicated by the variable *TermsInSmall*, the contents of the buffer are sorted. The sorting is done by pointers, hence it is important that the small buffer resides inside the physical memory. During the sorting it may happen that coefficients are added. The sum of two rational numbers can take more space than any of the individual numbers, so there will be a space problem. This has been solved by the construction of an extension to the small buffer. The variable *SmallExtension* is the size of the small buffer together with this extension. The value for *SmallExtension* will always be at least 7/6 times the value of *SmallSize*.

The result of the sorting of the small buffer is written to the 'large buffer' (with the size *LargeSize*) as a single object and the filling of the small buffer can resume. Whenever there is not enough room in the large buffer for the result of sorting the small buffer, or whenever there are already a given number of these sorted 'patches' in it (controlled by the variable *LargePatches*) the buffer will be sorted by merging the patches to make room for the new results. The output is written to the sort file as a single patch. Then the results from the small buffer can be written to the large buffer. This game can continue till no more terms are generated. In the end it will be necessary to sort the results in the intermediate sort file. This can be done with up to *FilePatches* at a time. Because file operations are notoriously slow the combination of the small buffer, the small extension and the large buffer is used for caching purposes. Hence this space can be split in 'FilePatches' caches. The limitation is that each cache should be capable to contain at least two terms of maximal size. This means that the sum of *SmallExtension* and *LargeSize* must be at least *FilePatches* times $2 * \text{MaxTermSize} * (\text{bytes in short integer})$. It is possible to set the size of these caches directly with the variable *SortIOsize*. If the variable is too large, the variable *FilePatches* may be adjusted by FORM. If there are more than *FilePatches* patches in the sort file, a second sort file is needed for the output of each 'superpatch'. When the first sort file has been treated, the second sort file can be treated in exactly the same way as its predecessor. This process will finish

eventually. When there are at most FilePatches patches in a sort file, the output of their merging can be written directly to the regular output. For completeness we give a list of all these variables:

FilePatches	The maximum number of patches that can be merged simultaneously, when the intermediate sort file is involved.
LargePatches	The maximum number of patches that is allowed in the large buffer. The large buffer may reside in virtual memory, due to the nature of the sort that is applied to it.
TermsInSmall	The maximum number of terms that is allowed in the small buffer before it is sorted. The sorted result is either copied to the large buffer or written to the intermediate sort file (when LargeSize is too small).
SmallSize	The size of the small buffer in bytes.
SmallExtension	The size of the small buffer plus its extension.
LargeSize	The size of the large buffer.
SortIOSize	The size of the buffer that is used to write to the intermediate sorting file and to read from it. It should be noted that if this buffer is not very large, the sorting of large files may become rather slow, depending on the operating system. Hence we recommend a potential fourth stage in the sorting over having this number too small to fit more filepatches in the combined small and large buffer. Setting the small and large buffers to a decent size may avoid all problems by a: making more space for the caching, b: creating fewer file patches to start with.

There is a second set of the above setup parameters for sorts of subexpressions as in function arguments or in the term environment (see 7.146). Because these things can happen with more than one level, whatever allocations have to be made (during runtime when needed) may have to be made several times. Hence one should be far more conservative here than with the global allocations. Anyway, those sorts should rarely involve anything very big. With the function arguments the condition is that the final result will fit inside a single term, but with the term environment no such restriction exists. The relevant variables here are subfilepatches, sublargepatches, sublargetsize, subsmallexension, subsmallsize, subsortiosize and subtermsinsmall. Their meanings are the same as for the variables without the sub in front.

When FORM is running in parallel mode (either TFORM or PARFORM) each worker will need its own buffers. In PARFORM in which the processors each control their own memory, the size of each of these buffers are the same as for the master process. In TFORM with its shared memory the above sizes refer to the buffers of the master thread. The workers each get basically buffers with 1/N times the size of the buffer of the master. This may get made a bit bigger when potential conflicts with MaxTermSize occur.

The (typically) largest buffers (the small and large buffers) may be reallocated at the end of a single module (see #sortreallocate (3.57)) or at the end of each module (see “On sortreallocate;” (7.108)). In some cases this can significantly reduce FORM’s memory usage as measured by “resident set size”. For programs which consist of a large number of very quickly-running modules, this can incur a noticable performance penalty if performed every module.

The default settings are

Variable	32-bits	64-bits	tform 64-bits
bracketindexsize	200000	200000	200000
commentchar	*	*	*
compresssize	90000	90000	90000
constindex	128	128	128
continuationlines	15	15	15
dotchar	.	.	.
filepatches	256	256	256
functionlevels	30	30	30
hidesize	50000000	50000000	50000000
incdir	.	.	.
insidefirst	ON	ON	ON
largepatches	256	256	256
largesize	50000000	800000000	1500000000
maxnumbersize	200	200	200
maxtermsize	10000	40000	40000
maxwildcards	100	100	100
nospacesinnumbers	OFF	OFF	OFF
numstorecaches	4	4	4
nwritefinalstatistics	OFF	OFF	OFF
nwritestatistics	OFF	OFF	OFF
nwritethreadstatistics	OFF	OFF	OFF
oldorder	OFF	OFF	OFF
parentheses	100	100	100
path	.	.	.
processbucketsize	1000	1000	1000
scratchsize	50000000	500000000	500000000
sizestorecache	32768	32768	32768
smallextension	20000000	300000000	600000000
smallsize	10000000	150000000	300000000
sortiosize	100000	100000	100000
sorttype	lowfirst	lowfirst	lowfirst
subfilepatches	64	64	64
sublargepatches	64	64	64
sublargesize	26880512	26880512	26880512
subsmallextension	3840032	3840032	3840032
subsmallsize	2560016	2560016	2560016
subsortiosize	32768	32768	32768
subtermsinsmall	10000	10000	10000
tempdir	.	.	.
tempsortdir	.	.	.
termsinsmall	100000	2000000	3000000
threadbucketsize	500	500	500
threadloadbalancing	ON	ON	ON
threads	0	0	0
threadsortfilesynch	OFF	OFF	OFF
threadscratchoutsize	2500000	2500000	2500000
threadscratchsize	100000	100000	100000
workspace	10000000	40000000	40000000

If one compares these numbers with the corresponding numbers for older versions one will notice that here we assume that the standard computer will have much more memory available than in the ‘old time’. Basically we expect that a serious FORM user has at least 64 Mbytes available. If it is considerably less one should define a setup file with smaller settings.

More recently a new notation for large numbers has been allowed. One can use the characters K, M, G and T to indicate kilo (three zeroes), mega (6 zeroes), giga (9 zeroes) and tera (12 zeros) as in 10M for 10000000.

To find out what the setup values are, one can use the ‘ON,setup;’ statement (7.108).

In version 3.3 and later one may use environment variables for the values of the setup parameters, either in the setup file or at the beginning of the .frm file. The environment variable is used as a preprocessor variable in the sense that its name is enclosed in a backquote-quote pair as in ‘**VARNAME**’. The variable will be looked for and if found it will be substituted. This can however not be done in a recursive way, because the regular routines that take care of the preprocessor variables are not active yet when the setups are read.

Chapter 18

The parallel version

FORM has two versions that can make use of several processors simultaneously. Which version can be used profitably depends very much on the architecture of the computer one is using. Each version has its own control commands which are ignored by the other version and the sequential version of FORM. The parallel versions are:

- PARFORM: This version runs on processors that have their own memory and preferably their own disk. Each processor gets a copy of the complete program and MPI is used for the communication. When the network connections are very fast one can also use PARFORM on computer clusters. PARFORM was developed at the university of Karlsruhe.
- TFORM: This version uses POSIX threads and runs on computers which have several processors with a shared memory. Data is kept as common data as much as possible and only when a worker thread gets a task a minimal amount of data is copied to its private buffers. Currently it seems to perform best on computers with two or four processors.

Both PARFORM and TFORM suffer from the same bottlenecks. At the beginning of a module there is a single expression, managed by a master process which then has to distribute the terms over the workers. At the end of the module the sorted results of the workers have to be gathered in by the master and merged into a single expression again. Efficiency depends critically on how fast the terms can be given to the workers, how well the load for the workers is balanced and how much time the master has to spend in the final stages of the sorting. Another factor is the complexity of the operations inside the module. If the module has very few and simple statements, the gain in performance will be much less than when the module has much work to do for each term.

The PARFORM and TFORM specific code is internally completely separated. This offers the possibility that sooner or later the two can be combined to allow efficient running on clusters of dual or quad processor machines. Whether this would give significant extra benefits needs to be investigated. When this project will be undertaken depends very much on the availability of such computers.

Because PARFORM uses MPI and because different MPI environments are normally not binary compatible, the port to a new machine requires a recompilation of the source code and a relinking to the MPI library. Hence we do not have executables in the distribution site. One needs to build PARFORM on one's computer. For TFORM the situation is much more favorable. Its treatment of the parallelization follows the standard for POSIX threads (or PThreads) for which the libraries are implemented on almost any UNIX system and many other systems.

The ideal of a parallel version of FORM is that it should execute nearly any regular FORM program, whether it was written for parallelization or not. And it should execute much faster

on several processors than the sequential version on a single processor. The performance is given by the improvement factor which is the execution time of the sequential version divided by the execution time of the parallel version as measured in real time (not CPU time) on a computer that has no other major tasks. The ideal would of course be that a computer with N processors would give an improvement factor of N . It should be easy to see that this ideal cannot be reached, due to the bottlenecks described above. Also the compilation takes place on a single processor and the instructions of the preprocessor are typically also tasks for a single thread/processor. Yet for small numbers of processors one can do rather well. Many old calculations, when repeated with TFORM would give improvement factors above 1.7 on a dual pentium machine and around 3 or a bit higher on a quad opteron machine. This was without modifying even a single statement in the programs. Of course these numbers depend very much on the type of the problem and the programming style used. As of yet there is very little experience with parallel versions of FORM. Hence people will have to discover what are good ways of getting the most out of their computer. It is expected that there will be much progress in the coming years.

First we will now discuss the running of the two versions. After that we will describe some common syntactic problems.

18.1 TFORM

Let us assume that the executable of TFORM is called `tform`. It is used exactly the same way as the sequential version of FORM (named `form`) is used with the exception of the possibility to specify the number of worker threads with the `-w` option. The command

```
tform -w4 calcdia
```

would execute the program in the file `calcdia.frm`, using 4 worker threads, in addition to the one master thread. When the `-w` option is not given or when only one worker thread is asked for, `tform` will run the whole program inside the master thread. Because `tform` always has some overhead this is usually a little bit slower than using `form`. Strange enough there are exceptions although this may have to do with the fact that measuring the time of a program doesn't always give the same numbers.

It is also possible to specify the number of worker threads in the setup file, using the line

```
Threads 4
```

for 4 threads. And as with all setup parameters one can pass this information also via the environment variable `FORM_threads` or with the line

```
#: Threads 4
```

at the beginning of the program file.

When the master passes terms to the workers, it has to signal the workers that there is some data. In their turn, each worker has to send the master a signal when it has completed its task and it is ready for more. Such signals cost time. Hence it is usually best to send terms in groups, called buckets. The optimal number of terms in a bucket depends very much on the problem and the size of the expression. Bigger buckets mean less overhead in signals. If the buckets are too big the workers may have to wait too much. Values between 100 and 1000 are usually rather good. There is a default bucket size which is typically around 500. The user can change this value in two ways: The first is with the `ThreadBucketSize` setup parameter in the `form.set` file (or at the startup of the program file, or with the `FORM_threadbucketsize` environment variable) and the second is

with the ThreadBucketSize statement (see 7.148) which is a declaration like Symbol or Dimension. The first terms in an expression will be sent in smaller buckets to get the workers something to do as soon as possible.

Usually the bigger buckets give a better performance, but they suffer from a nasty side-effect. Complicated terms that need much execution time have a tendency to stick together. Hence there can be one bucket with most of the difficult terms and at the end of the module all workers and the master have to wait for one worker to finish. This can be improved with a load balancing mechanism. The current version will take terms from the buckets of workers that take more time than the others. By default this mechanism is on, but it can be switched on or off with the ‘on ThreadLoadBalancing;’ and ‘off ThreadLoadBalancing;’ statements. It can also be set as one of the setup parameters in the form.set file with

```
ThreadLoadBalancing OFF
```

or

```
ThreadLoadBalancing ON
```

or at the start of the program or in the environment.

The LINUX operating system tries to cache files that are to be written to disk. Somehow, when several big files have to be written it gets all confused (it is not known in what way). This means that if tform produces 4 large sort files eventually the system becomes intolerably slow. At one time a test program was 4.5 times slower with 4 worker processors than with just the master running, even though the master had a single even bigger sort file. This has been improved by having the file-to-file sort of the threads changed into a file-to-masterbuffers-to-combined-output. Yet the writing and subsequent merging of the 4 files at the same time can be disastrous. Work is done to improve this, but it may not be easy to circumvent facilities of the operating system. Apparently the quality of the drivers is crucial here.

One can switch the parallel processing on or off (for the complete module) at any moment in the program with the statements

```
On Threads;  
Off Threads;
```

or using the moduleoption statement (7.89) that affects TFORM’s behaviour for just the current module:

```
ModuleOption Parallel;  
ModuleOption NoParallel;
```

Additionally one can switch the statistics per thread on or off with

```
On ThreadStats;  
Off ThreadStats;
```

When the thread statistics are switched off only the statistics of the master thread are printed which is usually only the final statistics for each of the expressions.

The timing information in the statistics is the CPU time spent by the thread that prints the statistics. Hence the total CPU time spent is the sum of the time of all workers and the time of the master. In good running the time of the master should be the smallest number. When the statistics per thread are switched off, only the statistics of the master process will be printed with this ‘small’ number. Hence it may look like the program isn’t progressing very much.

For debugging purposes the term by term print statement (see 7.114) is equipped with the %W and %w format strings. The first will cause the printing of the number of the current thread and the CPU-time used thus far in that thread. The second will only print the number of the current thread. The thread with the number zero is the master thread. Putting a statement like

```
Print +f "<%W> %t";
```

would show which thread is processing which term and when.

These are all the commands that specifically concern TFORM. When more experience is gained using TFORM, more parameters and commands may become available.

The fact that the threads need private data makes that TFORM will use more memory than FORM. Most of the buffers are not very large, but of course there are some buffers which need to be large, like the sort buffers and the scratch input/hide buffers. The sizes that the user specifies for these buffers are for the corresponding buffers of the master. The workers get each 1/N times the size for these buffers, when there are N workers. In the case that makes these buffers too small because of for instance MaxTermSize, the buffers may become larger.

18.2 ParFORM

Let us call the executable of PARFORM parform. The user must execute parform as an MPI application. In many MPI implementations, this is done by using the mpirun command:

```
mpirun -np 4 parform calcdia
```

This example executes the program in the file calcdia.frm, using 4 processes, in which one process is the master process and the other 3 processes are the worker processes. One has to keep in mind that in some MPI implementations environment variables will not be passed to an MPI application. Alternatively extra options are needed for passing them. If one wants to run PARFORM under a job scheduler on a computer cluster environment, one may need to write a job script, which depends to a great extent on the environment.

PARFORM uses MPI for communications between the master and workers. Actually terms are distributed by using point-to-point send/receive operations of MPI. Since there is some latency for establishing a connection between processes, especially between those running on different computers, it is best to send terms in groups, like buckets in TFORM. The default number of terms in a bucket is currently 1000 in PARFORM. It can be changed with the ProcessBucketSize statement (7.117) if this is deemed necessary. It can also be changed for the current module with the statement (7.89).

```
ModuleOption ProcessBucketSize number;
```

And finally it can also be changed in the setup, using the ProcessBucketSize (17) setup parameter. The first terms in an expression will be sent in smaller buckets to get the workers something to do as soon as possible.

One can switch the parallel processing on or off (for the complete module) at any moment in the program with the statements

```
On Parallel;
Off Parallel;
```

or using the moduleoption statement (7.89) that affects PARFORM's behaviour for just the current module:

```
ModuleOption Parallel;
ModuleOption NoParallel;
```

Additionally one can switch the statistics per process on or off with

```
On ProcessStats;
Off ProcessStats;
```

When the process statistics are switched off only the statistics of the master process are printed which are usually only the final statistics for each of the expressions.

As in TFORM, %W and %w in the term by term print statement (see 7.114) are available in PARFORM. They print the number of the current process and the CPU-time used thus far in that process.

In principle one can run all FORM or TFORM programs with PARFORM. In practice PARFORM is not so efficient for some problems, in which more data have to be synchronized between the master and the workers. The cases for which PARFORM needs to send data via MPI include:

- The redefine statements, which modify preprocessor variables on the workers.
- Modifying \$-variables in regular statements with a moduleoption statement (see 6.1, 7.89 and 18.3).
- Expression names appearing in right hand sides of definition or substitution statements.

The last case may need more explanation. Consider the following code:

```
Local G = F;
id a = F;
```

where the expression F is supposed to be already defined. The point is that these substitutions of the expression F are performed on the workers. The workers, however, do not know the contents of the expression F because it is stored on the master. Therefore, before executing this module PARFORM needs to make the master broadcast the expression F to the workers. This may be quite time-consuming because the expression could be very large.

18.3 Some problems

Both parallel versions share a number of problems which are inherent to running in an environment in which the order in which terms are processed isn't deterministic. Most of these problems concern \$-variables. They present a mix between private and common information. Consider the code

```
id f(x?$xvar) = g(x);
id .....
id a^n? = b^n*h($var);
```

Of course one could do this simple example differently, but we are discussing the principle. What we have here is that each term that passes the first statement will acquire its own value of \$var, to be used a bit later. It is clear that if we have a common administration of \$-variables we would have to 'lock' the value for a considerable amount of time, thereby spoiling much of the gains of parallel processing. Hence in this case it would be best that each worker maintains its own local value of \$var. But in the following example we have the opposite:

```
#$xmax = -1;
if ( count(x,1) > $xmax ) $xmax = count_(x,1);
```

Here we collect a maximum power in the variable `$xmax`. If each worker would have a local value of `$xmax`, the question is what to do with all these local values at the end of the module. A human will see that here we are collecting a maximum, but the computer cannot and should not see this. Hence the general rule in parallel processing is that when there are `$`-variables obtaining a value during the algebraic phase of a module the entire module is run sequentially, unless FORM has been helped with a `moduleoption` statement for each of the variables involved. Hence in the last example

```
ModuleOption Maximum $xmax;
```

would tell FORM how to combine the local values in PARFORM (PARFORM maintains local values of all `$`-variables). In TFORM it would put the value directly into the central administration, provided it is bigger than the previous value. Only during the update the variable would have to be locked.

There are several options in the `moduleoption` statement:

- Maximum: The variable must have a numerical value and the maximum is collected.
- Minimum: The variable must have a numerical value and the minimum is collected.
- Sum: The variable must have a numerical value and the sum is collected.
- Local: The value will be kept privately and no attempt is made to put it in the central administration, neither during the execution of the module, nor at the end. If there was already a variable by this name in the central administration it will keep the value it had before the module started execution. At the end of the module, all private values will be forgotten.

The `redefine` statement is a major inefficiency in a parallel environment. It redefines a preprocessor variable and there is only a single bookkeeping for such variables. This means that the variable has to be sent to the master process (PARFORM) or that a lock has to be placed to prevent other workers to write to the same storage simultaneously (TFORM). In addition the final value in the preprocessor variable will be determined by the last term processed in any of the workers. This may not be the same term in different runs. It is up to the user to write programs that still give correct results under such conditions. The best way around the inefficiency is using `$`-variables and preprocessor instructions. We show this in an example in which we construct the equivalent of a conditional repeat that includes a `.sort` instruction.

```
#do i = 1,1
  statements
  if ( count(x,1) > 0 ) redefine i "0";
  .sort
#enddo
```

To run this in parallel, it is better to use the following code.

```
#do i = 1,1
  #$i = 1;
  statements
```

```

    if ( count(x,1) > 0 ) $i = 0;
    ModuleOption minimum $i;
    .sort
    #redefine i "$i"
#enddo

```

In this program the centrally stored value of `$i` is updated at most once. Admittedly it is not as simple as the `redefine` statement, but it works in all versions of FORM starting with version 3.0.

It should be noted that when a new expression is defined in its defining module it starts out as a single term. Hence it cannot benefit from parallelization in that module. Therefore the code

```

#define MAX "200"
Symbols x0,...,x10;
Local F = (x0+...+x'MAX')^3;
id x1 = -x2-...-x'MAX';
.end

```

will execute inside a single worker while

```

#define MAX "200"
Symbols x0,...,x10;
Local F = (x0+...+x'MAX')^3;
.sort
id x1 = -x2-...-x'MAX';
.end

```

will make the first expansion inside a single worker and the more costly substitution can be made in parallel. A better load balancing algorithm in which at any node in the expansion tree tasks can be given to idle workers would solve this problem, but due to some complications this has not yet been implemented. The structure of FORM will however allow such an implementation.

Chapter 19

External communication

To communicate with other programs FORM is equipped with special commands. One set of commands is rather simple in nature: the `#pipe` (see section 3.39) and `#system` (see section 3.59) instructions allow FORM to run programs in the regular command shell. Sometimes however much more sophistication is needed because these instructions have a rather large overhead and need to start new processes each time they are executed. Hence a second more extensive set of instructions was developed that allows the start of an external process, keep it open and maintain a two way communication with it. Similarly it is possible to start FORM in such a way from other programs. Many details of the method of implementation and a number of examples are given in a separate paper which can also found in the FORM site (<http://www.nikhef.nl/~form>) under publications (look for the file `extform.ps` or `extform.pdf`). Here we will just show the essentials and the syntax.

The basic idea is to open (by means of the preprocessor) a number of external channels (there is no reason to be restricted to just one) by starting the corresponding program in a command shell. This program is kept running and a number is assigned to each channel. Next we can select a channel and communicate with it. To not run into syntactic problems, because the external program may have different ideas of what a formula should look like, one may have to install filters. These are additional programs that should be prepared before the FORM program is started that process the communication to convert from one notation to the other.

19.1 `#external`

Syntax:

```
#external ["prevar"] systemcommand
```

See also

Starts the command in the background, connecting to its standard input and output. By default, the external command has no controlling terminal, the standard error stream is redirected to `/dev/null` and the command is run in a subshell in a new session and in a new process group (see the preprocessor instruction `#setexternalattr`).

The optional parameter “prevar” is the name of a preprocessor variable placed between double quotes. If it is present, the “descriptor” (small positive integer number) of the external command is stored into this variable and can be used for references to this external command (if there is more than one external command running simultaneously).

The external command that is started last becomes the “current” (active) external command. All further instructions `#fromexternal` and `#toexternal` deal with the current external command.

19.2 #toexternal

Syntax:

```
#toexternal "formatstring" [,variables]
```

See also

Sends the output to the current external command. The semantics of the "formatstring" and the [,variables] is the same as for the #write instruction, except for the trailing end-of-line symbol. In contrast to the #write instruction, the #toexternal instruction does not append any newline symbol to the end of its output.

19.3 #fromexternal

Syntax:

```
#fromexternal[+-] "[ $\$$ ]varname" [maxlength]
```

Appends the output of the current external command to the FORM program. The semantics differ depending on the optional arguments. After the external command sends the prompt, FORM will continue with a next line after the line containing the #fromexternal instruction. The prompt string is not appended. The optional + or - sign after the name has influence on the listing of the content. The varieties are:

```
#fromexternal[+-]
```

The semantics is similar to the #include instruction but folders are not supported.

```
#fromexternal[+-] "[ $\$$ ]varname"
```

is used to read the text from the running external command into the preprocessor variable varname, or into the dollar variable \$varname if the name of the variable starts with the dollar sign "\$".

```
#fromexternal[+-] "[ $\$$ ]varname" maxlength
```

is used to read the text from the running external command into the preprocessor (or dollar) variable varname. Only the first maxlength characters are stored.

19.4 #prompt

Syntax:

```
#prompt [newprompt]
```

Sets a new prompt for the current external command (if present) and all further (newly started) external commands.

If newprompt is an empty string, the default prompt (an empty line) will be used.

The prompt is a line consisting of a single prompt string. By default, this is an empty string.

19.5 #setexternal

Syntax:

```
#setexternal n
```

Sets the "current" external command. The instructions #toexternal and #fromexternal deal with the current external command. The integer number n must be the descriptor of a running external command.

19.6 `#rmexternal`

Syntax:

```
#rmexternal [n]
```

Terminates an external command. The integer number `n` must be either the descriptor of a running external command, or 0.

If `n` is 0, then all external programs will be terminated.

If `n` is not specified, the current external command will be terminated.

The action of this instruction depends on the attributes of the external channel (see the `#setexternalattr` (section 19.5) instruction). By default, the instruction closes the commands' IO channels, sends a KILL signal to every process in its process group and waits for the external command to be finished.

19.7 `#setexternalattr`

Syntax:

```
#setexternalattr list_of_attributes
```

sets attributes for *newly started* external commands. Already running external commands are not affected. The list of attributes is a comma separated list of pairs `attribute=value`, e.g.:

```
#setexternalattr shell=noshell,kill=9,killall=false
```

Possible attributes are:

kill Specifies which signal is to be sent to the external command either before the termination of the FORM program or by the preprocessor instruction `#rmexternal`. By default this is 9 (SIGKILL). Number 0 means that no signal will be sent.

killall Indicates whether the KILL signal will be sent to the whole group or only to the initial process. Possible values are “`true`” and “`false`”. By default, the kill signal will be sent to the whole group.

daemon Indicates whether the command should be “daemonized”, i.e. the initial process will be passed to the init process and will belong to the new process group in the new session. Possible values are “`true`” and “`false`”. By default, “`true`”.

shell specifies which shell is used to run a command. (Starting an external command in a subshell permits to start not only executable files but also scripts and pipelined jobs. The disadvantage is that there is no way to detect failure upon startup since usually the shell is started successfully.) By default this is “`/bin/sh -c`”. If set `shell=noshell`, the command will be started by the instruction `#external` directly but not in a subshell, so the command should be a name of the executable file rather than a system command. The instruction `#external` will duplicate the actions of the shell in searching for an executable file if the specified file name does not contain a slash (/) character. The search path is the path specified in the environment by the PATH variable. If this variable isn't specified, the default path “`:/bin:/usr/bin`” is used.

stderr specifies a file to redirect the standard error stream to. By default it is “`/dev/null`”. If set `stderr=terminal`, no redirection occurs.

Only attributes that are explicitly mentioned are changed, all others remain unchanged. Note, changing attributes should be done with care. For example,

```
#setexternalattr daemon=false
```

starts a command in the subshell within the current process group with default attributes `kill=9` and `killall=true`. The instruction `#rmexternal` sends the KILL signal to the whole group, which means that also FORM itself will be killed.

19.8 An example

An example of the above instructions could be:

```
1  symbol a,b;
2
3  #external "n1" cat -u
4
5  #external "n2" cat -u
6
7  * cat simply repeats its input. The default prompt is an
8  * empty line. So we use "\n\n" here -- one "\n" is to finish
9  * the line, and the next "\n" is the prompt:
10 #toexternal "(a+b)^2\n\n"
11
12 #setexternal 'n1'
13 * For this channel the prompt will be "READY\n":
14 #toexternal "(a+b)^3\nREADY\n"
15
16 #setexternal 'n2'
17 * Set the default prompt:
18 #prompt
19 Local aPLUSbT02=
20 #fromexternal
21     ;
22
23 #setexternal 'n1'
24 #prompt READY
25 Local aPLUSbT03=
26 #fromexternal
27     ;
28
29 #rmexternal 'n1'
30 #rmexternal 'n2'
31
32 Print;
33 .end
```

Two external channels are opened in lines 3 and 5. The UNIX utility “cat” simply repeats its input. The option “-u” is used to prevent the output buffering. The option is ignored by the GNU cat utility but is mandatory for non-GNU versions of cat.

After line 5 the current external channel is ‘n2’. The default prompt is an empty line so in line 10 “\n\n” is used – one “\n” is to finish the line, and the next “\n” is the prompt.

Line 12 switches the current channel to ‘n1’. For this channel the prompt will be “READY”, see line 24, hence the expression is finished by “\nREADY\n”.

Line 16 switches to the ‘n2’ external channel and line 18 sets the default prompt (which is extra in this example since the default prompt was not changed up to now).

Results (just a literal repetition of the sent expressions) are read in lines 20 and 26.

Lines 29 and 30 close the external channels.

19.9 Embedding FORM in other applications

The external channel instructions permit FORM to swallow an external program. The same mechanism can be used in order to *embed* FORM in other applications.

There is a possibility to start FORM from another program providing one (or more) communication channels (see below). These channels will be visible from a FORM program as “pre-opened” external channels existing after FORM starts. There is no need to open them with the `#external` instruction. In this case, the preprocessor variable “PIPES_” is defined and is equal to the total number of the pre-opened external channels. Pre-opened external channel descriptors are contained in the preprocessor variables “PIPE1_”, “PIPE2_”, etc. For example, if ‘PIPES_’ is 3 then there are 3 pre-opened external channels with the descriptors ‘PIPE1_’, ‘PIPE2_’ and ‘PIPE3_’ so e.g. the following instruction could be used:

```
#setexternal 'PIPE2_'
```

without

```
#external "PIPE2_"
```

The external channel attributes make no sense for the pre-opened channel (see the `#setexternalattr` instruction (section 19.7)). Formally, they are as follows:

```
kill=0,  
killall=false,  
daemon=false,  
stderr=/dev/tty,  
shell=noshell
```

In order to activate the pre-opened external channels, the parent application must follow some standards. Here we describe a low-level protocol, the corresponding C-interface is available from the FORM distribution site under packages and then `externalchannels`.

Before starting FORM, the parent application must create one or more pairs of pipes. A pipe is a pair of file descriptors, one is for reading and another is for writing. In LINUX, see “man 2 pipe”. The read-only descriptor of the first pipe in the pair and the write-only descriptor of the second pipe must be passed to FORM as an argument of a command line option “-pipe” in ASCII decimal format. The argument of the option is a comma-separated list of pairs “r#,w#” where “r#” is a read-only descriptor and “w#” is a write-only descriptor; alternatively, an environment variable FORM_PIPES containing this list can be used (the command line option overrides the environment variable). For example, to start FORM with two pre-opened external channels the parent application has to create first four pipes. Lets us suppose the first pipe was created with the descriptors 5 and 6, the second pipe has the descriptors 7 and 8, the third pipe has the descriptors

9 and 10 and the fourth pipe has the descriptors 11 and 12. The descriptors 5 and 8 will be used by FORM as the input and the output for the first pre-opened external channel while the descriptors 9 and 12 will be used by FORM as the input and the output for the second pre-opened external channel.

Then the parent application must start FORM with the following command line option:

```
-pipe 5,8,9,12
```

Upon startup, FORM sends its PID (the Process Identifier) in ASCII decimal format with an appended newline character to the descriptor 8 and then FORM will wait for the answer from the descriptor 5. The answer must be two comma-separated integers in ASCII decimal format followed by a newline character. The first integer corresponds to the FORM PID while the second one is the parent process PID. If the answer is not obtained after some timeout, or if it is not correct (i.e. it is not a list of two integers or the first integer is not the FORM PID) then FORM fails. If everything is correct, FORM creates the pre-opened channel and puts its descriptor in the preprocessor variable "PIPE1_".

Then FORM processes the second pair of arguments, "9,12".

After all pairs have been processed FORM creates the preprocessor variable "PIPES_" and puts into this variable the total number of created pre-opened external channels.

The order of processing the pairs of numbers in the argument is fixed exactly as it was described above i.e. from the left to the right.

Chapter 20

Spectators

At times expressions contain many terms that will not be treated for many modules to come. For the actions in those modules they are considered spectator terms and they may consume much computer time due to their presence during the sorting. For this we have the spectator system in which we can send those terms to a special file, named a spectator file, in such a way that they can be picked up at a convenient time in the future. In short:

Spectators are expressions together with a filename. The file is for storage when the spectator becomes too big. Create a spectator with the statement

```
CreateSpectator Exprname,"filename";
```

The file will be made in the same directory where the temporary files are made. Example:

```
CreateSpectator Yintegrals,"Yfile.spec";
```

One may send terms to a spectator with the executable statement

```
ToSpectator Exprname;
```

An example would be

```
if ( count(Z,1) == 0 ) ToSpectator Yintegrals;
```

The terms are dumped into the file as they are at the moment the ToSpectator statement is executed. No brackets etc. In the future they may be compressed. At the moment they are not.

Recovery of the contents of the spectator is done with the CopySpectator statement as in

```
CopySpectator NewExp = Yintegrals;
```

Currently you can only read the spectators this way. You cannot make more complicated constructions. You can only do things with the terms of the spectator expression after the contents have been put in your new expression. The spectator file remains in existence. In later modules you can still add to it. You cannot read from and add to the same spectator in the same module. The CopySpectator command can be followed by executable statements in the same module. This may not be economical because the contents of the spectator have not been sorted. There could be identical terms that occur many times or even cancel. Better sort them first.

A spectator can be removed from the system with the statement

```
RemoveSpectator Yintegrals;
```

It is also possible to truncate a spectator down to zero length with

```
EmptySpectator Yintegrals;
```

You can have as many spectators as you like, but they all have some cache buffers. There may also be limitations in the file system on the maximum number of open files. The filename for the spectator is purely for the sake of the users administration and recognition. You cannot carry the file over to other programs. There is no variable administration in it as in the saved files.

The .global instruction makes a spectator file survive a .store. It is up to the user to make sure that all the variables in it also survive the .store. There is no checking!

One use of the spectator system would be when integrating many different terms by means of a very prolonged recursion system in which integrals of a given complexity are reduced to integrals of lower complexity, but each such reduction may take quite a few steps. One could have:

```
#do i = 'MAXCOMPLEXITY'-1,0,-1
  CreateSpectator complex'i',"complex'i'.spec";
#enddo
Local F'MAXCOMPLEXITY' = ....;
#do i = 'MAXCOMPLEXITY'-1,1,-1
  #do ii = 1,1;
*    routine for doing a part of the recursion level 'i'
    ....

    #do j = 'i'-1,0,-1
      if ( complexityofterm == 'i' ) ToSpectator complex'i';
    #enddo
    if ( notyetfinished ) redefine ii "0";
    .sort
  #enddo
  Drop F{'i'+1};
  CopySpectator F'i' = complex'i';
  .sort
  RemoveSpectator complex'i';
#enddo
*
* and finally, assuming that complexity zero means finished:
*
Drop F1;
CopySpectator F0 = complex0;
.sort
RemoveSpectator complex0;
```

Some remarks are called for here. If one works with the polyratfun concept and the rational polynomials become rather complicated, the sorting after the CopySpectator statement can have serious bottleneck problems in TFORM and PARFORM because the addition of those polynomials can be rather expensive and much of it ends up in the master processor. This can be made better with the following construction (assuming the function rat was declared as polyratfun at the moment of all the ToSpectator statements that wrote to the spectator):

```
PolyRatFun;
Drop F1;
CopySpactator F0 = complex0;
```

```

ABracket+ rat;
.sort
PolyRatFun rat;
RemoveSpectator complex0;
.sort

```

First we remove the declaration of rat as polyratfun. Then we read the spectator and sort it such that all terms that should be added eventually are grouped together. This sorting is very cheap as only identical terms are combined. Then we declare the polyratfun again and because of the way the terms are sorted nearly all additions take place inside the workers, hence at maximum parallelization efficiency.

The above method still contains one inefficiency: because the polyratfun is declared again, the contents of the rat function need to be 'normalized' again, while they were already normalized. This involves calculating a gcd of the numerator and the denominator, which is an expensive operation and is useless in this case. For this we have a special option in the polyratfun declaration:

```
PolyRatFun rat-;
```

This will skip the normalization on the input of the module. One should note however that if one uses this option under different conditions in which the input rat function might not be normalized, the program might crash or even give wrong answers. Hence this option should only be used with the highest degree of caution! This is an option for very experienced users only. No support is given concerning programs that run correctly without the use of this option and fail when using it.

It should be noted that in the sequential version of FORM this construction is not needed at all, because there is only one processor anyway.

Chapter 21

Diagram generation

Starting with version 5.0 FORM is equipped with the diagram generator of Toshiaki Kaneko "A Feynman graph generator for any order of coupling constants", Toshiaki Kaneko(Meiji Gakuin U.): Comput.Phys.Comm. 92 (1995) 127-152, e-Print: hep-th/9408107. Recently he has reprogrammed it in C++ and made it into a library, accompanied with a manual (to be published). FORM makes use of this library and has built its own syntax around it, based on many years of experience working with the QGRAF generator ("Automatic Feynman graph generation", Paulo Nogueira: J.Comput.Phys. 105 (1993) 279-289) and the problems encountered with the large number of diagrams used with the Mincer and Forcer programs. The whole has been set up in such a way that if more features are needed they should not be exceptionally hard to be implemented.

In version 4.3 the diagram generator was not complete yet. Effectively it only accepted one type of scalar particles, but one could have any number of particles at the vertices. Hence it could be used as a fast topology generator.

For the FORM implementation in version 5.0 and later two new types of variables are introduced, a number of new functions and a number of new preprocessor variables. In addition the special concept of models is new.

Let us start with models, particles and vertices:

Model Particles and vertices exist only inside the scope of a model. A model is a very special set of which the elements will be the particles and the vertices. It is defined by

```
Model name;
  Particle .....;
  .
  .
  Particle .....;
  Vertex .....;
  .
  .
  Vertex .....;
EndModel;
```

in which the particle and vertex statements are declarations that will be explained below. The order is important. All Particle declarations must come before the vertex declarations.

Particle A particle is a very special type of function. It is declared with

```
Particle particlename[,antiparticlename][,<sign><number>][,external];
```

A particle has an antiparticle assigned to it. If no name of an antiparticle is specified the particle will be its own antiparticle as is the case with for instance a photon or a gluon. The sign indicates whether a particle is a fermion (-) or a boson (+) and the number is the dimension of its SU(2) representation. Hence a scalar particle has the number 1, an electron the number 2, etc. If no spin information is provided it is assumed that the particle is a scalar. After the spin option there is the external option to indicate that particle will not be considered for the choice of propagators (edges). To avoid that a scalar particle has an antiparticle by the name external one should avoid

```
Particle MyScalar,external;
```

and use

```
Particle MyScalar,1,external;
```

instead.

Vertex A vertex is also a very special type of function. It is declared with

```
Vertex particle1,...,particlen:coupling;
```

for an npoint vertex. The coupling constant is either the number 1, or a (product of) symbol(s) to an integer power. If the symbols have not been declared before they will be declared during the reading of the vertex declaration.

Some examples:

```
Model PHI3;
  Particle phi,1;
  Vertex phi,phi,phi:g;
EndModel;
Model QCD;
  Particle qua,QUA,-2;
  Particle gho,GHO,-1;
  Particle glu,+3;
  Vertex qua,QUA,glu:g;
  Vertex gho,GHO,glu:g;
  Vertex glu,glu,glu:g;
  Vertex glu,glu,glu,glu:g^2;
EndModel;
```

It is possible to have several models simultaneously.

In some cases one might want to generate diagrams in which the internal (multi)loop propagators are represented as special edges. This requires the definition of more complicated models as in

```
#do i = 0,'LOOPS'
Model QCD'i';
  Particle qua,QUA,-2;
```

```

Particle gho,GH0,-1;
Particle glu,+3;
Vertex qua,QUA,glu:g;
Vertex gho,GH0,glu:g;
Vertex glu,glu,glu:g;
Vertex glu,glu,glu,glu:g^2;
#do j = 1,'i'-1
    Vertex qua,QUA:g^{2*'j'};
    Vertex gho,GH0:g^{2*'j'};
    Vertex glu,glu:g^{2*'j'};
#enddo
EndModel;
#enddo

```

in which the two-point vertices are the propagators with j loops. As one can see in this example, one can have vertices with identical particle contents but different powers of the coupling constants.

One can generate diagrams with the `diagrams_` function. Its syntax is

```

diagrams_(model,set_of_input_particles,set_of_output_particles,
    set_of_external_momenta,set_of_internal_momenta,
    number_of_loops_or_coupling_constants,options)

```

In the case of generating a boson propagator the external symmetry can be used most effectively by declaring both as input particles as in `glu,glu` and have the set of output particles empty. The output is given in terms of a few functions. These functions are

topo_ Its argument is the number of the topology. This numbering is consistent in the sense that whether one generates only topologies (with the option `TOPOLOGIESONLY_`) or the full diagrams makes no difference for the number of the topology of the diagrams. This allows relative simple programs to match topologies and renumberings to a predefined list of topologies, after which this renumbering can then be done later in an economical way for the full set of diagrams.

node_ Indicates a vertex. The first argument is the number of the vertex, the second the combined coupling constants of the vertex and after that the momenta of the particles if only the topologies are determined, or the particles with their momentum in their argument when the full diagrams are generated as in:

```

- 2*topo_(1)*node_(1,1,glu(-Q1))*node_(2,1,glu(-Q2))*node_(3,g,qua(-p1)
,QUA(-p2),glu(Q1))*node_(4,g,qua(-p3),QUA(p1),glu(Q2))*node_(5,g,qua(p2)
,QUA(-p4),glu(-p5))*node_(6,g,qua(p4),QUA(p3),glu(p5))

```

The external lines are connected by a vertex to the vacuum. Those are the nodes with the lowest numbers as in `node_(1,1,glu(-Q1))` and only a single particle. If we would have generated the topologies only we would have gotten for the first diagram

```

+ 1/4*topo_(1)*node_(1,1,-Q1)*node_(2,1,-Q2)*node_(3,g,Q1,-p1,-p2)*
node_(4,g,Q2,p1,-p3)*node_(5,g,p2,-p4,-p5)*node_(6,g,p3,p4,p5)

```

edge_ It is also possible to generate the diagrams in terms of edges, either with the nodes:

```

+ 1/4*topo_(1)*node_(1,1,-Q1)*node_(2,1,-Q2)*node_(3,g,Q1,-p1,-p2)*
node_(4,g,Q2,p1,-p3)*node_(5,g,p2,-p4,-p5)*node_(6,g,p3,p4,p5)*edge_(1,
Q1,1,3)*edge_(2,Q2,2,4)*edge_(3,p1,3,4)*edge_(4,p2,3,5)*edge_(5,p3,4,6)*
edge_(6,p4,5,6)*edge_(7,p5,5,6)

```

or without the nodes:

```

- 2*topo_(1)*edge_(1,glu(Q1),1,3)*edge_(2,glu(Q2),2,4)*edge_(3,QUA(p1),
3,4)*edge_(4,qua(p2),3,5)*edge_(5,QUA(p3),4,6)*edge_(6,qua(p4),5,6)*
edge_(7,glu(p5),5,6)

```

Edges are only given when the options ask for it.

block_ A block is a subdiagram that is connected to the remaining part of the diagram by a single vertex. These blocks are only given if the options ask for it.

Options are provided by means of the sum of a number of predefined preprocessor variables. They are

ONEPI_ One-particle irreducible subgraphs only.

WITHOUTINSERTIONS_ No internal higher order propagators.

NOTADPOLES_ No tadpoles.

SYMMETRIZE_ Symmetrize in terms of the external legs.

TOPOLOGIESONLY_ Generate only topologies.

NONODES_ Do not put the nodes in the output.

WITHEDGES_ Put the edges in the output.

WITHBLOCKS_ Put the blocks in the output.

NOSNAILS_ No snails in the output.

NOEXTSELF_ No external self-energy subgraphs.

WITHONEPISETS_ Print the function `onepi_` containing the one-particle irreducible subsets.

When using the **WITHOUTINSERTIONS_** option one can obtain these higher order subgraphs by adapting the definition of the model. Here we show this for QCD:

```

#do i = 0, 'LOOPS'
Model QCD 'i';
  Particle qua, QUA, -2;
  Particle gho, GH0, -1;
  Particle glu, +3;
  Vertex qua, QUA, glu: g;
  Vertex gho, GH0, glu: g;
  Vertex glu, glu, glu: g;
  Vertex glu, glu, glu, glu: g^2;
#do j = 1, 'i'-1

```

```

Vertex qua,QUA:g^{2*'j'};
Vertex gho,GHO:g^{2*'j'};
Vertex glu,glu:g^{2*'j'};
#enddo
EndModel;
#enddo

```

We see here a number of two-point vertices with a higher power of the coupling constants. This gives, omitting the quarks and the ghosts:

```

#define LOOPS "3"
#do i = 0,'LOOPS'
Model QCD'i';
Particle qua,QUA,-2;
Particle gho,GHO,-1;
Particle glu,+3;
Vertex qua,QUA,glu:g;
Vertex gho,GHO,glu:g;
Vertex glu,glu,glu:g;
Vertex glu,glu,glu,glu:g^2;
#do j = 1,'i'-1
EndModel;
#enddo
Vector Q,Q1,...,Q7,p,p0,...,p21;
Symbols x1,x2,n1,...,n14;
Indices j1,j2,i1,...,i21;
Set QQ:Q1,...,Q7;
Set pp:p1,...,p21;
Set empty;;
.global
L Fprop'LOOPS' = diagrams_(QCD'LOOPS',{glu,glu},empty,QQ,pp,'LOOPS',
    'OnePI_'+'NoTadpoles_'+'Symmetrize_'+'TopologiesOnly_');
Print +f +s;
.end

```

Time =	0.00 sec	Generated terms =	46
	Fprop3	Terms in output =	46
		Bytes used =	17788

```

Fprop3 =
+ 1/4*topo_(1)*node_(1,1,-Q1)*node_(2,1,-Q2)*node_(3,g,Q1,-p1,-p2)*
node_(4,g,Q2,p1,-p3)*node_(5,g,p2,-p4,-p5)*node_(6,g,p3,p4,-p6)*node_(7,
g,p5,-p7,-p8)*node_(8,g,p6,p7,p8)
+ ...
+ 1/2*topo_(10)*node_(1,1,-Q1)*node_(2,1,-Q2)*node_(3,g,-p1,-p2,-p3)*
node_(4,g,p1,-p4,-p5)*node_(5,g,Q1,p2,-p6)*node_(6,g,Q2,p4,p6)*node_(7,
g^2,p3,p5)
+ ...
+ 1/4*topo_(28)*node_(1,1,-Q1)*node_(2,1,-Q2)*node_(3,g,Q1,-p1,-p2)*

```

```

node_(4,g,Q2,-p3,-p4)*node_(5,g^2,p1,p3)*node_(6,g^2,p2,p4)
+ ...
+ 1/2*topo_(44)*node_(1,1,-Q1)*node_(2,1,-Q2)*node_(3,g,Q1,-p1,-p2)*
node_(4,g,Q2,p1,-p3)*node_(5,g^4,p2,p3)
+ 1/4*topo_(45)*node_(1,1,-Q1)*node_(2,1,-Q2)*node_(3,g^2,-p1,-p2)*
node_(4,g^2,Q1,p1,-p3,-p4)*node_(5,g^2,Q2,p2,p3,p4)
+ ...
;

```

With the quarks and the ghosts there are 202 diagrams. One can see in topo_(44) that we have a two-loop propagator, indicated by node_(5,...). In topo_(10) and topo_(45) we see a single one-loop propagator. In topo_(28) we see the diagram with two one-loop propagators. In we omit the TopologiesOnly_ option, we will obtain the full diagrams with the same topo_ indicators. A few of those diagrams are:

```

- topo_(28)*node_(1,1,glu(-Q1))*node_(2,1,glu(-Q2))*node_(3,g,qua(-p1),
QUA(-p2),glu(Q1))*node_(4,g,qua(-p4),QUA(-p3),glu(Q2))*node_(5,g^2,qua(
p3),QUA(p1))*node_(6,g^2,qua(p2),QUA(p4))
- topo_(28)*node_(1,1,glu(-Q1))*node_(2,1,glu(-Q2))*node_(3,g,gho(-p1),
GH0(-p2),glu(Q1))*node_(4,g,gho(-p4),GH0(-p3),glu(Q2))*node_(5,g^2,gho(
p3),GH0(p1))*node_(6,g^2,gho(p2),GH0(p4))
+ 1/2*topo_(28)*node_(1,1,glu(-Q1))*node_(2,1,glu(-Q2))*node_(3,g,glu(
Q1),glu(-p1),glu(-p2))*node_(4,g,glu(Q2),glu(-p3),glu(-p4))*node_(5,g^2,
glu(p1),glu(p3))*node_(6,g^2,glu(p2),glu(p4))

```

In version 4.3 there existed also the topologies_ and the phi_ functions to just generate topologies. This was before a complete implementation was made. For backward compatibility we have kept these functions, but in the future they may become obsolete.

For people who have programs that rely on the topologies_ function we give here nevertheless the syntax and a few examples

```

Vectors Q1,Q2,p1,...,p8;
Set QQ:Q1,Q2;
Set PP:p1,...,p8;
#define L00PS "2"
Local F = topologies_('L00PS',2,{3,},QQ,PP);
Print +f +s;
.end

```

```

F =
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,Q2,p1,-p3)*
node_(4,p2,-p4,-p5)*node_(5,p3,p4,p5)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,p1,-p3,-p4)*
node_(4,p2,p3,-p5)*node_(5,Q2,p4,p5)
;

```

Here the second parameter indicates the number of external legs, the third parameter is a set that tells, in this case, that 3-point vertices are allowed, QQ is the set of external momenta and PP is the set of internal momenta. The function node_ is a built-in function to indicate the vertices.

If one would like to allow more types of vertices one may change the third parameter:

```

Vectors Q1,Q2,p1,...,p8;
Set QQ:Q1,Q2;
Set PP:p1,...,p8;
#define LOOPS "2"
Local F = topologies_('LOOPS',2,{3,4},QQ,PP);
Print +f +s;
.end

```

```

F =
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,Q2,-p1,-p2)*node_(3,p1,p2,-p3,p3
)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,Q2,p1,-p3)*
node_(4,p2,p3,-p4,p4)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,Q2,p1,-p3)*
node_(4,p2,-p4,-p5)*node_(5,p3,p4,p5)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,Q2,-p3,-p4)*
node_(4,p1,p2,p3,p4)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,p1,-p3,-p4)*
node_(4,Q2,p2,p3,p4)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,p1,-p3,-p4)*
node_(4,p2,p3,-p5)*node_(5,Q2,p4,p5)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2,-p3)*node_(3,Q2,p1,p2,p3
)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,-p1,-p2,-p3)*node_(3,Q2,p1,-p4)*
node_(4,Q1,p2,p3,p4)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,-p1,-p2,-p3)*node_(3,p1,p2,-p4)*
node_(4,Q1,Q2,p3,p4)
;

```

and suddenly there are 9 topologies.

In the above configuration the program sees the external lines as different. If there is however a symmetry between the two external lines of a propagator-like diagram (as is the case with boson propagators) one can indicate this by putting a minus sign in front of the number 2:

```

Vectors Q1,Q2,p1,...,p8;
Set QQ:Q1,Q2;
Set PP:p1,...,p8;
#define LOOPS "2"
Local F = topologies_('LOOPS',-2,{3,4},QQ,PP);
Print +f +s;
.end

```

```

F =
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,Q2,-p1,-p2)*node_(3,p1,p2,-p3,p3
)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,Q2,p1,-p3)*
node_(4,p2,p3,-p4,p4)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,Q2,p1,-p3)*
node_(4,p2,-p4,-p5)*node_(5,p3,p4,p5)

```

```

+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,Q2,-p3,-p4)*
node_(4,p1,p2,p3,p4)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,Q2,-p3,-p4)*
node_(4,p1,p3,-p5)*node_(5,p2,p4,p5)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2)*node_(3,p1,-p3,-p4)*
node_(4,Q2,p2,p3,p4)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,Q1,-p1,-p2,-p3)*node_(3,Q2,p1,p2,p3
)
+ node_(0,-Q1)*node_(1,-Q2)*node_(2,-p1,-p2,-p3)*node_(3,p1,p2,-p4)*
node_(4,Q1,Q2,p3,p4)
;

```

Now the program assumes this symmetry and one notices only 8 topologies remaining.

In the case of the `topologies_` function there are no combinatorics factors. This will of course be different with the future function `diagrams_`.

That this function is very fast can be seen when one generates all topologies of a 6-loop propagator with only 3-point vertices:

```

Vectors Q1,Q2,p1,...,p17;
Set QQ:Q1,Q2;
Set PP:p1,...,p17;
#define LOOPS "6"
Local F = topologies_('LOOPS',-2,{3,},QQ,PP);
.end

```

Time =	0.85 sec	Generated terms =	2793
	F	Terms in output =	2793
		Bytes used =	2010976

This is very much faster than the program that is used most widely.

The program to do this with the `diagrams_` function is:

```

#define LOOPS "6"
Model PHI3;
Particle phi,+1;
Vertex phi,phi,phi:g;
EndModel;
Vector Q1,Q2,p1,...,p{3*'LOOPS'};
Set QQ:Q1,Q2;
Set pp:p1,...,p{3*'LOOPS'};
Set empty;;
Local F = diagrams_(PHI3,{phi,phi},empty,QQ,pp,'LOOPS',
'OnePI_'+ 'NoTadpoles_'+ 'Symmetrize_'+ 'TopologiesOnly_');
.end

```

Time =	0.85 sec	Generated terms =	2793
	F	Terms in output =	2793
		Bytes used =	2010976

and without the symmetry:


```

#define LOOPS "6"
Model PHI3;
    Particle phi,+1;
    Vertex phi,phi,phi:g;
EndModel;
Vector Q1,Q2,p1,...,p{3*‘LOOPS’};
Set QQ:Q1,Q2;
Set pp:p1,...,p{3*‘LOOPS’};
Set empty;;
Local F = diagrams_(PHI3,{phi,phi},empty,QQ,pp,‘LOOPS’,
    ‘OnePI_’+‘NoTadpoles_’+‘TopologiesOnly_’);
.end

```

Time =	0.15 sec	Generated terms =	4999
	F	Terms in output =	4999
		Bytes used =	3599296

Chapter 22

Floating point

Starting with version 5.0 FORM is equipped with arbitrary floating point capacity. The low level routines are part of the GMP and mpfr libraries which should be available on most systems. If not they can be picked up easily from the internet. The main commands involving the floating point system are

#startfloat This instruction is needed to startup the floating point system. Invoking it will allocate a number of arrays. The instruction has either one or two arguments:

```
#startfloat numberofbits <,maximumweight>
```

The first argument is mandatory. It tells how many bits of accuracy are required. FORM will round to at least this precision. Because the internal routines work with WORDs the precision will be rounded up to the nearest integer number of WORDs. The second argument is optional for when one wants to work with multiple zeta values (MZVs) or Euler sums. It specifies the maximum weight that will be used. The evaluation of the sums requires a number of auxiliary arrays. The default value is zero. If one would like to change the precision during a run, this is possible. The effect would be that the existing arrays are released and new arrays will be allocated.

#endfloat This instruction releases all arrays allocated for the floating point system.

tofloat Converts the rational coefficients at the ground level to floating point numbers in the precision specified in the **#startfloat** instruction. From this point on the coefficient at this level will be floating point. If one needs to convert numbers inside a function argument one should use the argument environment. This can be nested.

torational Tries to convert the floating point coefficients to rational numbers. To this end it uses repeated fractions as in

$$x \rightarrow n_0 + 1/(n_1 + 1/(n_2 + 1/(n_3 + \dots)))$$

with x a floating point number. The algorithm keeps track of the remaining precision and if $1/n_i$ is close to this precision it truncates the sequence at n_{i-1} . After that it works out the fraction. It could be that x cannot be expressed as a fraction within the given precision. This can usually be seen by that the fractions are ‘rather wild’, or that the result changes when the precision is increased. This statement can also be abbreviated to ‘torat’.

evaluate If this command has no arguments all floating point functions that FORM knows about will be evaluated. The currently allowed arguments are the functions `mzv_`, `euler_`, `sqrt_` and `mzvhalf_`. If any (or more than one) of these are specified only those functions will be evaluated.

In addition to the above commands there are the following functions that can be evaluated `sqrt_`, `ln_`, `li2_`, `gamma_`, `agm_`, `sin_`, `cos_`, `tan_`, `asin_`, `acos_`, `atan_`, `sinh_`, `cosh_`, `tanh_`, `asinh_`, `acosh_`, `atanh_`. For the functions `atan2_` and `lin_` there is currently no code. The `agm_` function is the arithmetic geometric mean of its two input values.

In addition to the above functions there are also the constant `pi_`, the basis of the natural logarithm `ee_` and the Euler-Mascheroni constant `em_`. These constants will also be expanded with the `evaluate` command. When given as an argument to `evaluate`, only the specified constants will be evaluated.

The following example shows some work with Multiple Zeta Values (MZV's):

```
#StartFloat 500,15
L F1 =
-mzv_(8,1,1,5)
+29056868/39414375*mzv_(2)^6*mzv_(3)
-47576/40425*mzv_(2)^5*mzv_(5)
-163291/18375*mzv_(2)^4*mzv_(7)
-4/105*mzv_(2)^3*mzv_(3)^3
-450797/11025*mzv_(2)^3*mzv_(9)
+7/5*mzv_(2)^2*mzv_(3)^2*mzv_(5)
+16/25*mzv_(2)^2*mzv_(3)*mzv_(5,3)
+454049/1400*mzv_(2)^2*mzv_(11)
-16/25*mzv_(2)^2*mzv_(5,3,3)
+3*mzv_(2)*mzv_(3)^2*mzv_(7)
+61/14*mzv_(2)*mzv_(3)*mzv_(5)^2
+2/7*mzv_(2)*mzv_(3)*mzv_(7,3)
+2172853/420*mzv_(2)*mzv_(13)
-2/7*mzv_(2)*mzv_(7,3,3)
+1/7*mzv_(2)*mzv_(5,5,3)
-33/4*mzv_(3)^2*mzv_(9)
-133/6*mzv_(3)*mzv_(5)*mzv_(7)
-25/9*mzv_(3)*mzv_(9,3)
-244/105*mzv_(5)^3
-359/105*mzv_(5)*mzv_(7,3)
+3/10*mzv_(7)*mzv_(5,3)
+89/18*mzv_(9,3,3)
+569/105*mzv_(7,3,5);
L F2 = mzv_(15);
Evaluate mzv_;
Print +f;
.sort

F1 =
9.1234206877960755900164875575406726239325002222490534540605137258846994\
916348297032751308227224952419629422497720599224543719959652966613231560\
```

```

6913925597e+0;

F2 =
1.0000305882363070204935517285106450625876279487068581775065699328933322\
671563422795730723343470175484943669684442492832530297757588781904321794\
4047700034253;

Skip F1,F2;
L X = F1/F2;
Torational;
Print +f;
.end

X =
229903169/25200;

```

0.08 sec out of 0.09 sec

The `#startfloat` initializes the floating point system and allocates arrays for 500 bits of precision. If there is a second number it indicates the maximum weight for MZVs and Euler sums. The functions are only evaluated when the proper command is given. In the second module we divide the numbers and convert the result to a rational. It is a good idea to try this with various precisions to see whether this is stable. With 100 bits the final answer would be

```
24136499874427167202968388241342/2645634679509635700210944273;
```

while at 150 bits we have already the same answer as with 500 bits. The fraction that is obtained by this program can be proven to be correct.

Index

..., 24
+...+, 24
+...-, 24
,..., 24
-...+, 24
-...-, 24
..., 24
.clear, 40, 52, 231
.end, 51
.global, 52
.prc, 27, 40
.sav, 134
.sc1, 231
.sc2, 231
.sc3, 231
.sor extension, 226
.sort, 31, 49, 51, 53, 131, 137, 145
.store, 52, 79, 96, 97, 107, 134
.str, 231
.tbl, 201
/.../, 24
:..., 24
?name, 58
#+, 20
#-, 20
#:, 20
#\$name, 20
#add, 205
#addseparator, 25
#append, 25, 47
#break, 26, 27, 29, 45
#breakdo, 26
#call, 26, 30, 203, 229, 231
#case, 27, 29, 45
#close, 28
#closedictionary, 205, 206
#commentchar, 28
#create, 28, 47
#default, 29, 45
#define, 27, 29, 40
#do, 25, 30, 32, 46, 47, 50
#else, 32, 34
#elseif, 32, 34
#enddo, 30, 32, 50
#endif, 32, 34, 50
#endinside, 33
#endprocedure, 33, 39, 50
#endswitch, 26, 27, 29, 33, 45, 50
#exchange, 33
#external, 34, 42, 244–246, 248
#factdollar, 34
#fromexternal, 34, 42, 244, 245
#if, 32, 34, 36, 46, 50
#ifdef, 32, 36
#ifndef, 32, 36
#include, 34, 37, 40, 45, 199, 203, 229, 231, 245
#inside, 33, 37
#message, 37
#name, 20
#opendictionary, 205
#pipe, 38, 244
#preout, 38
#printtimes, 39
#procedure, 26, 39, 50
#procedureExtension, 27
#procedureextension, 40, 231
#prompt, 40, 245
#redefine, 29, 40
#remove, 41
#rmexternal, 41, 43, 246, 247
#rmseparator, 42
#setexternal, 42, 245
#setexternalattr, 41, 42, 246, 248
#setrandom, 43, 169
#show, 43
#sortreallocate, 45
#switch, 26, 27, 29, 33, 45, 50
#system, 41, 46, 244
#terminate, 46
#toexternal, 34, 42, 47, 244, 245

- #undefine, 47
- #usedictionary, 205
- #write, 25, 28, 38, 47, 49, 245
- \$-variable, 33, 48, 60, 85, 104, 109, 110, 126, 131, 145, 162, 242
- %, 48
- 32 bits, 23, 91, 100, 145, 165, 214, 228–230
- 64 bits, 23, 92, 100, 145, 165, 214, 228–230

- aborted, 86
- abracquets, 67
- abs_, 157
- acos_, 174
- acosh_, 174
- active, 113, 126, 137, 153
- active expressions, 96, 105
- addargs, 151
- addition, 23
- addto, 201, 203
- Adler
 - Mark, 200
- algebra
 - Dirac, 153
- algorithms, 215
- all, 98
- also, 67, 99
- alternative, 75
- ambiguity, 59
- and, 23
- antibracket, 68, 75, 107, 127, 175
- antibrackets, 67
- anticommutate, 216
- antiputinside, 68
- antisymmetric, 73, 94, 134, 159, 160
- antisymmetrize, 68
- append, 128
- apply, 68, 200, 202, 204
- argexplode, 69
- argimplode, 69
- argument, 70, 83, 84, 104, 107, 113, 123, 124, 137
- argument field, 61, 144, 156
- argument field wildcard, 58
- argument statement, 137
- arguments of functions, 226
- arithmetic, 22, 23, 60
- arithmetic operations, 180
- array, 59

- asin_, 173
- asinh_, 174
- atan2_, 174
- atan_, 174
- atanh_, 174
- audit, 202
- auto, 70
- autodeclare, 12, 70

- backquote, 20, 39
- backslash, 26, 48
- bernoulli_, 157
- binom_, 157
- binomials, 130
- Bjorken & Drell metric, 213
- bottleneck, 237
- braces, 11, 176
- bracket, 46, 71, 97, 127, 129, 137, 159, 173, 175
 - curly, 22, 59
- bracket index, 97, 176
- bracketindexsize, 176, 228
- brackets, 71, 75, 91, 226
 - curly, 11
 - indexing, 72
 - keep, 105
- bucket, 146, 238
- buffer
 - large, 224, 233
 - procedure, 26
 - small, 223, 233
- bug, 18

- C, 23, 29, 47, 93, 126
- C-interface, 248
- cache, 233, 239
- caches, 231
- caching, 226
- calculator, 22, 26, 31
 - preprocessor, 11
- caveat, 15
- chainin, 73
- chainout, 74
- channel
 - external, 244
- checkpoints, 55
- Chinese remainder theorem, 161, 169
- Chisholm, 148, 214, 215
- chisholm, 74
- clock, 52

CMODULE_, 21
 coeff_, 6
 coefficient, 87, 101, 113, 123, 124
 coefficients
 integer, 107
 collect, 31, 75, 177
 colon, 9
 color package, 134
 combinatorics, 158, 159
 comma, 11, 23
 command shell, 244
 commentary, 20, 28, 37
 commentchar, 228
 Common subexpression elimination, 184
 communication, 237
 external, 244
 two way, 244
 commuteinset, 76
 commuting, 8, 57, 70, 73, 76, 78, 94, 98, 143, 145, 156
 compilation, 130
 compile, 200
 compiler, 20
 complex, 5, 8, 73, 94, 141, 219
 complicated, 10
 complication, 75, 223
 compress, 76, 202, 203
 compressed, 199
 compresssize, 229
 condition, 100
 conflicts, 12
 conjg_, 157
 conjugate
 complex, 5
 conjugation, 219
 complex, 5
 constindex, 91, 229
 content_, 157
 continuationlines, 229
 contract, 77, 214
 contracted indices, 133
 contraction, 218
 contractions
 trace, 81
 conv2to3, 116
 copyspectator, 250
 cos_, 173
 cosh_, 174
 count, 100, 142
 count_, 158
 CPU time, 239
 crash, 55
 crashed FORM program, 56
 create, 202, 203
 createspectator, 250
 CSE, 184
 curly bracket, 22
 cycle, 150
 cycle symmetric, 73, 94
 cyclesymmetric, 133
 cyclesymmetrize, 78
 Cygwin, 1

 d_, 158
 daemon, 42, 246
 dangerous, 220
 database, 199
 DATE_, 21
 dd_, 158
 deallocatetable, 79
 debugging, 126, 185
 declarations, 51
 decode, 150
 dedup, 150
 default dimension, 140
 define, 229
 definitions, 51
 delayed substitution, 29, 62
 delete, 79
 delta
 Kronecker, 7, 15, 77, 218, 220
 kronecker, 158
 delta_, 158
 deltap_, 158
 den_, 6
 denom_, 158
 denominators, 4, 80
 deterministic, 241
 dictionaries, 205
 dimension, 6, 7, 102
 default, 13, 80, 140
 Dirac, 153, 162, 163
 Dirac algebra, 213
 discard, 81, 142
 disk to disk sort, 226
 disorder, 81, 98

- distrib_, 136, 159, 217
- div_, 120, 159, 187
- division, 23
- do loop, 30, 32
- dollar, 90
 - printing, 63
- dotchar, 4, 93, 229
- dotproducts, 4, 229
- double quote, 47
- doublefortran, 93, 208
- drop, 82, 107, 112
- dropargs, 151
- dum_, 107, 159
- dummy, 131, 140, 156
- dummy arguments, 96
- dummy_, 160
- dummyindices_, 10
- dummyten_, 160
- dynamical loading, 91
- e_, 160
- edit, 20
- ee_, 6, 263
- efficient, 223
- Einstein, 7, 9, 16
- elements, 135
- else, 83, 84, 100
- else statement, 83, 84
- elseif, 83, 84, 100
- elseif statement, 84
- em_, 6, 263
- embed, 248
- emptyspectator, 250
- encode, 150
- end of module specifications, 51
- endargument, 70, 83, 84
- endif, 83, 84, 100
- endif statement, 83, 84
- endinexpression, 85
- endinside, 104
- endrepeat, 85, 132
- endterm, 86, 145
- endwhile, 86, 154
- enter, 202
- environment, 22, 236
 - argument, 70, 84
 - inside, 85
 - repeat, 85
 - term, 86, 137, 145
 - while, 86, 154
- environment variables, 2
- error, 18
 - runtime, 61
- error stream, 246
- even_, 10
- exclamation, 11
- executable statements, 51
- execution, 51, 135
- exists, 35
- exit, 86
- exp_, 160
- explode, 150
- exponent, 23
- exponent operator, 93
- expression, 4, 33, 48, 82, 96, 100, 162
- extension
 - .sav, 134
 - small, 224
- exteuclidean_, 160
- extform.pdf, 244
- extform.ps, 244
- extra symbols, 44, 79, 87, 180, 210
- extrasymbol_, 161
- extrasymbols, 48
- EXTRASYMBOLS_, 21
- extrasymbols_, 6
- fac_, 162
- factarg, 87, 118, 120, 121
- factdollar, 120
- factorial, 23, 162, 164
- factorin_, 162
- factorize, 120
- fall-through, 45
- farg_, 162
- file, 48
 - close, 28
 - disk, 96
 - hide, 96, 153
 - include, 45
 - input, 1, 202
 - intermediate, 51
 - log, 2, 126, 128
 - new, 202
 - program, 228
 - saved, 107

- setup, 3, 228
 - sort, 233, 239
 - storage, 52, 79, 96, 107
 - store, 134, 231
 - temporary, 3
- filename, 250
- filepatches, 226, 233, 234
- files
 - .sav, 15
- fill, 79, 89, 91, 128, 144, 199, 201–203
- fillexpression, 91, 128, 172
- filter, 244
- findloop, 101, 134
- firstbracket_, 162
- firstterm_, 162
- fixed index, 91
- fixed_, 10
- fixindex, 59, 91, 220
- flexibility, 220
- float, 92
- floating point, 92
- fold, 37
- folders, 245
- foldname, 37
- folds, 45
- form.set, 3, 228
- FORM_PIPES, 248
- format, 63, 92, 126, 128
 - C, 93
 - doublefortran, 93
 - float, 92
 - fortran, 92
 - fortran90, 93
 - maple, 93
 - mathematica, 93
 - normal, 93
 - nospaces, 92
 - optimize, 92
 - quadfortran, 93
 - quadruplefortran, 93
 - rational, 92
 - reduce, 93
 - spaces, 92
- format string, 47, 126
- FORMPATH, 3, 26, 37, 48, 203
- FORMSETUP, 3
- FORMTMP, 3
- FORMTMPSORT, 3
- Fortran, 229
- fortran, 4, 48, 49, 92, 207
- fortran90, 93
- fractions, 110
- function, 112, 156
 - abs_, 157
 - acos_, 174
 - acosh_, 174
 - asin_, 173
 - asinh_, 174
 - atan2_, 174
 - atan_, 174
 - atanh_, 174
 - bernoulli_, 157
 - binom_, 157
 - commuting, 8
 - conjg_, 157
 - content_, 157
 - cos_, 173
 - cosh_, 174
 - count_, 158
 - d_, 158
 - dd_, 158
 - delta_, 158
 - deltap_, 158
 - denom_, 158
 - distrib_, 136, 159
 - div_, 120, 159, 187
 - dum_, 159
 - dummy_, 160
 - dummyten_, 160
 - e_, 160
 - exp_, 160
 - exteuclidean_, 160
 - extrasymbol_, 161
 - fac_, 162
 - factorin_, 162
 - farg_, 162
 - firstbracket_, 162
 - firstterm_, 162
 - g5_, 162
 - g6_, 162
 - g7_, 163
 - g_, 163
 - gcd_, 120, 163, 187
 - gi_, 163
 - id_, 163
 - integer_, 164

- inverse_, 120, 164
- invfac_, 164
- li2_, 174
- lin_, 174
- ln_, 173
- makerational_, 164
- match_, 165
- max_, 165
- maxpowerof_, 165
- min_, 165
- minpowerof_, 165
- mod2_, 166
- mod_, 165
- moebius_, 166
- mul_, 120, 166
- nargs_, 166
- node_, 166
- non-commuting, 8
- nterms_, 166
- numfactors_, 167
- partitions_, 167
- pattern_, 168
- perm_, 168
- poly_, 168
- polyadd_, 168
- polydiv_, 159, 168
- polygcd_, 163, 168, 198
- polyintfac_, 168
- polymul_, 168
- polynorm_, 168
- polyrem_, 168, 170
- polysub_, 168
- prime_, 168
- putfirst_, 169
- random_, 43, 169, 170
- ranperm_, 170
- regular, 156
- rem_, 120, 170, 187
- replace_, 152, 170
- reverse_, 171
- root_, 171
- setfun_, 171
- sig_, 171
- sign_, 171
- sin_, 173
- sinh_, 174
- sizeof_, 171
- sqrt_, 173
- sum_, 171
- sump_, 172
- table_, 172
- tan_, 173
- tanh_, 174
- tbl_, 172
- term_, 172
- termsin_, 173
- termsinbracket_, 173
- theta_, 173
- thetap_, 173
- topologies_, 173
- function arguments, 87
- functionlevels, 229
- functions, 8, 94
 - commuting, 70, 73
 - noncommuting, 70, 98
- funpowers, 95
- g5_, 162, 213
- g6_, 162
- g7_, 163
- g_, 163, 213
- Gailly
 - Jean-loup, 200
- gamma algebra, 74
- gamma matrices, 8, 74, 148, 149, 162, 163, 213
- garbage collection, 224
- GCD, 162
- gcd_, 120, 163, 187
- gi_, 163, 214
- global, 96, 107
- GNU, 247
- goto, 96, 106
- Grace, 40
- greater than, 24
- greedy optimizations, 184
- group theory, 134
- guarantee, 111
- gzip, 120, 203
- harmonic polylogarithm, 136, 151
- harmonic polylogarithms, 69
- harmonic sum, 136, 139
- harmonic sums, 69, 151
- hide, 96, 104, 113, 116, 129, 153, 231, 232, 240
- hide file, 96
- hide files, 56
- hidesize, 229

- hierarchy, 101
- high energy physics, 213
- highfirst, 232
- Horner scheme, 183, 185
- i_, 6, 7, 214
- id, 57, 67, 81, 96, 97, 99, 108, 111, 123, 135, 137, 140, 170
- id statement, 81
- id_, 163
- identify, 97, 99
- idnew, 99
- idold, 99
- if, 83, 84, 100, 134, 142, 154
 - count, 142
- if statement, 83, 84
- ifmatch, 98, 102, 106
- ifnomatch, 102
- imaginary, 5, 8, 73, 94, 141, 219
- implode, 150
- improvement factor, 238
- inactive, 108
- incdir, 229, 231
- index, 6, 7, 70, 91, 102
 - bracket, 176
 - contracted, 133
 - dummy, 140, 156
 - main, 200
 - table, 156
 - trace line, 148
- index loop, 101, 133
- index tree, 176
- index_, 10
- indices, 7, 70, 80, 102
 - dummy, 13, 80, 131
 - fixed, 8
 - lower, 77, 218
 - numeric, 8
 - renumber, 13
 - summable, 8, 218
 - upper, 77, 218
- individual expressions, 126
- inexpression, 85, 103
- infinite loop, 132, 154
- input, 240
 - standard, 34
- inside, 85, 104
- insidefirst, 104, 229
- installation, 228
- instance, 29
- int_, 9
- integer_, 164
- intohide, 113
- inverse_, 120, 164
- invfac_, 164
- isdefined, 36
- isfactorized, 36
- islyndon, 151
- isnumerical, 36
- jumpratio, 229
- Karlsruhe, 237
- keep, 105
- keep brackets, 105, 176
- kill, 42, 246
- KILL signal, 41, 43, 246, 247
- killall, 42, 246
- Knuth, 225
- Kronecker, 7, 15, 77, 218, 220
- kronecker, 158
- label, 96, 98, 106
- large buffer, 224, 233
- largepatches, 233, 234
- largesize, 233, 234
- last, 149
- LCM, 162
- legal, 18
- less than, 24
- Levi-Civita, 77, 218
- Levi-Civita tensor, 160
- li2_, 174
- library, 40
 - making a, 40
- lin_, 174
- linebreaks, 126
- linefeed, 26, 48, 127
- LINUX, 239, 248
- listed loop, 30
- lists, 13
- ln_, 173
- load, 107, 134, 200, 202, 203
- load balancing, 239, 243
- loading dynamically, 91
- local, 96, 106, 107
- lock, 241

- log, 2, 126, 128
- logical, 35
- loop, 131
 - index, 133
 - infinite, 132
 - listed, 30
 - numerical, 31
 - while, 131
- loops
 - infinite, 154
- loopsize, 133
- lower, 77
- lowfirst, 232
- Lyndon word, 151
- macro, 30
- makeinteger, 107, 177
- makerational_, 164
- many, 97, 100, 108
- maple, 93
- master, 146, 222, 237, 238
- match, 100
- match_, 165
- mathematica, 93
- matrices
 - gamma, 8, 218
- matrix
 - unit, 153
- matrix like, 143
- max_, 165
- maxnumbersize, 230
- maxpowerof, 36
- maxpowerof_, 165
- maxtermsize, 75, 230, 231
- maxwildcards, 230
- MCTS, 183
- memory, 51
 - total amount of, 18
- merge, 136
- metric, 92, 108, 218, 219
- min_, 165
- mincer, 105
- minpowerof, 36
- minpowerof_, 165
- mixed statements, 51
- mod2_, 166
- mod_, 165
- mode
 - single term, 126
- module, 20, 51, 55
- module instruction, 51
- module number, 55
- ModuleOption, 104
- moduleoption, 52, 65, 109
 - inparallel, 109
 - local, 109, 242
 - maximum, 109, 242
 - minimum, 110, 242
 - noparallel, 109
 - notinparallel, 109
 - parallel, 109
 - polyfun, 109
 - processbucketsize, 109, 240
 - sum, 110, 242
- modulus, 110, 120
- moebius_, 166
- Monte Carlo tree search, 183
- mountains, 222
- MPI, 237, 240
- mpirun, 240
- MS-DOS, 41
- mul_, 120, 166
- mulargs, 151
- multi, 97, 100, 111
- Multiple Zeta Value, 151
- multiple zeta value, 136
- multiple zeta values, 69
- multipleof, 101
- multiplication, 23
- multiply, 111, 137
 - left, 111
 - right, 111
- MZV, 136, 151
- MZV data mine, 151
- NAME_, 21, 44
- names, 4
 - definition, 4
 - length, 5
- nargs_, 166
- ndrop, 112
- neg0_, 10
- neg_, 10
- nested, 46, 132, 146
- nesting, 20
- newline, 47, 245

- nfunction, 112
- nhide, 97, 113
- nintohide, 113
- non-commuting, 8
- noncommuting, 57, 70, 94, 98, 112, 115, 143, 145, 156
- nondeterministic, 65
- normal, 93
- normalization, 141
- normalize, 107, 113, 177
- nospaces, 92
- nospacesinnumbers, 230
- nosymmetrize, 214
- notation
 - ambiguous, 8
- nprint, 114
- nskip, 115, 137
- ntable, 115, 143
- ntensor, 115, 145
- nterms_, 166
- num_, 6
- number of FORM words, 171
- number of terms, 173
- number_, 10
- numerical loop, 31
- numerical output, 207
- numerical processing, 180
- numfactors_, 167
- numstorecaches, 230, 231
- nunhide, 116
- nwrite, 116
 - allnames, 117
 - allwarnings, 117
 - highfirst, 117
 - lowfirst, 117
 - names, 117
 - powerfirst, 117
 - setup, 117
 - shortstatistics, 116
 - shortstats, 116, 117
 - statistics, 116
 - stats, 116
 - warnings, 116
- nwritestatistics, 230
- nwritethreadstatistics, 230
- obsolete, 52, 95, 104, 116, 129, 155
- occurrence order, 183
- odd_, 10
- off, 55, 95, 116, 117, 129, 155, 203
 - allnames, 117
 - allwarnings, 117
 - backtrace, 117
 - checkpoint, 117
 - compress, 117
 - finalstats, 117
 - flint, 117
 - highfirst, 118
 - humanstats, 118
 - insidefirst, 118
 - lowfirst, 118
 - names, 118
 - nospacesinnumbers, 118
 - oldfactarg, 118
 - parallel, 118, 240
 - powerfirst, 118
 - processstats, 118
 - propercount, 118
 - properorder, 118
 - setup, 118
 - shortstatistics, 119
 - shortstats, 119
 - sortreallocate, 119
 - statistics, 119
 - stats, 118
 - threadloadbalancing, 119
 - threads, 119, 239
 - threadstats, 119
 - totalsize, 119
 - warnings, 119
 - wtimestats, 119
- old notation, 20
- OLDNUMEXTRASYNBOLS_, 21
- oldorder, 231
- on, 55, 95, 116, 129, 155, 203, 232
 - allnames, 119
 - allwarnings, 119
 - backtrace, 120
 - checkpoint, 120
 - compress, 120
 - fewerstatistics, 120
 - fewerstats, 120
 - finalstats, 120
 - flint, 120
 - highfirst, 120
 - humanstats, 121

- insidefirst, 121
- lowfirst, 121
- names, 121
- nospacesinnnumbers, 121
- oldfactarg, 121
- parallel, 121, 240
- powerfirst, 121
- processstats, 121, 241
- propercount, 121
- properorder, 121
- setup, 122
- shortstatistics, 122
- shortstats, 122
- sortreallocate, 122
- statistics, 122
- stats, 122
- threadloadbalancing, 122
- threads, 122, 239
- threadstats, 122, 239
- totalsize, 122
- warnings, 122
- wtimestats, 122
- once, 98, 100, 123
- only, 98, 100, 123
- open, 200, 202, 203
- opteron, 238
- optimization buffer, 181
- optimize, 92
- OPTIMMAXVAR_, 21
- optimmaxvar_, 180, 181
- OPTIMMINVAR_, 21
- optimminvar_, 180, 181
- OPTIMSCHEME_, 21
- optimscheme_, 183
- OPTIMVALUE_, 21
- optimvalue_, 183
- option
 - disorder, 81
 - select, 11
- or, 23
- order of terms, 241
- ordered set, 135
- output
 - standard, 34
- output channel, 47
- output specifications, 51
- overflow, 91
- overwrite, 128
- package
 - color, 134
- parallel, 109, 128
- parallel execution, 110
- parallel processing, 65
- parentheses, 175, 231
- ParFORM, 65, 103, 114, 128, 237, 240, 251
- partial fractioning, 129
- partitions_, 167
- patch, 224, 233
- PATH, 43, 246
- path, 25, 26, 37, 39, 229, 231, 233
 - search, 246
- pattern, 24, 57, 97
- pattern matcher, 100
- pattern matching, 11, 57
- pattern_, 168
- pentium, 238
- percentage, 75
- period, 5, 51
- perm_, 168
- permutation, 143
- permutations, 130, 131
- permute, 150
- pi_, 6, 263
- PID, 21, 249
- PID_, 21
- pipe, 38, 248
- PIPE1_, 248
- PIPE2_, 248
- pipelined job, 42
- PIPES_, 248
- poly_, 168
- polyadd_, 168
- polydiv_, 159, 168
- polyfun, 52, 109, 123, 177, 224
- polygcd_, 163, 168, 198
- polyintfac_, 168
- polylogarithms
 - harmonic, 69
- polymul_, 168
- polynorm_, 168
- polyratfun, 80, 109, 120, 124, 251
- polyrem_, 168, 170
- polysub_, 168
- pophide, 125, 129
- pos0_, 10
- pos_, 10

- POSIX, 237
- pow, 93
- power series, 53
- powerfirst, 232
- pre-opened external channels, 248
- precedence, 228
- prefactdollar, 120
- premature, 132
- preprocessor, 20
- preprocessor variable, 29, 40
 - numeric, 22
- preprocessor variables, 20, 130, 228
- prime_, 168
- print, 114, 126–128, 240, 241
 - %T, 126
 - %W, 126
 - %, 126
 - %%, 126
 - %t, 126
 - %w, 126
- print[], 127
- printf, 126
- printing, 92
- printtable, 128
- private, 240
- procedure, 26, 33, 39
- procedureextension, 231
- process
 - external, 244
- processbucketsize, 109, 128, 231, 240
- processor, 109
- prompt, 34, 40
- propagator, 218
- propercount, 129
- properorder, 98
- protocol
 - lowlevel, 248
- pushhide, 125, 129
- putfirst_, 169
- putinside, 129

- quadfortran, 93, 208
- quadruplefortran, 93
- question mark, 57
- quote, 20, 39

- random, 170
- random_, 43, 169
- range, 149, 206, 210

- ranperm_, 170
- ratio, 129
- rational, 92
- rational polynomials, 251
- rcyclesymmetrize, 130
- reading, 37
- real, 5, 8, 73, 94, 141, 210, 219
- recovery, 55
- recovery option, 55
- recursively, 89
- redefine, 31, 130
- REDUCE, 218
- reduce, 93
- rem_, 120, 170, 187
- remainder, 23, 138
- removespectator, 250
- renumber, 13, 131
- repeat, 85, 132
- repeat environment, 132
- replace, 150
- replace_, 59, 152, 163, 170
- replaceloop, 101, 133
- replacement, 170
- representation
 - textual, 62
- resetttimeonclear, 52, 231
- restriction
 - power, 5
- restrictions, 18
- reverse, 150
- reverse cycle symmetric, 73, 95
- reverse cycle symmetrize, 130
- reverse_, 171
- root of unity, 140, 141
- root_, 171
- rounding, 164
- run, 55
- run a script, 55
- running FORM, 1
- runtime error, 61

- save, 134
- save files, 56
- saved file, 107
- scalar objects, 58
- Schoonschip, 57, 130, 145, 162, 163, 172, 218
- scratch files, 56
- ScratchSize, 51

- scratchsize, 96, 231
- screen, 126
- script, 42
- select, 97, 100, 135
- selectargs, 151
- semicolon, 38, 48, 100
- set, 59, 97, 135, 138
 - ambiguous, 59
 - corresponding element, 60
 - dynamical, 23, 59
 - inside, 11
 - mixed, 135
 - ordered, 135
 - outside, 11
- set of indices, 59
- Set of symbols, 59
- setexitflag, 135
- setfun_, 171
- setrandom, 170
- sets, 9
 - array, 59
 - implicitly declared, 11
 - named, 12
- settings, 117
- setup, 226, 228
 - bracketindexsize, 228
 - commentchar, 228
 - compresssize, 229
 - constindex, 229
 - continuationlines, 229
 - define, 229
 - dotchar, 229
 - filepatches, 234
 - functionlevels, 229
 - hidesize, 229
 - incdir, 229, 231
 - insidefirst, 229
 - jumpratio, 229
 - largepatches, 234
 - largesize, 234
 - maxnumbersize, 230
 - maxtermsize, 230
 - maxwildcards, 230
 - nospacesinnnumbers, 230
 - numstorecaches, 230
 - nwritestatistics, 230
 - nwritethreadstatistics, 230
 - oldorder, 231
 - parentheses, 231
 - path, 229, 231
 - procedureEetension, 231
 - processbucketsize, 231
 - resetttimeonclear, 231
 - scratchsize, 231
 - sizestorecache, 231
 - smallestextension, 234
 - smallsize, 234
 - sortiosize, 234
 - sorttype, 232
 - tempdir, 232
 - tempsortdir, 232
 - termsinsmall, 234
 - threadbucketsize, 232
 - threadloadbalancing, 232
 - threads, 232
 - threadscratchoutsize, 232
 - threadscratchsize, 232
 - totalsize, 233
 - workspace, 233
 - wtimestats, 233
- setup file, 4
- shell, 42, 55, 244, 246
- short integer, 91
- shorthand, 102, 132
- SHOWINPUT_, 21, 44
- shuffle, 108, 140
- sig_, 171
- SIGKILL signal, 42, 246
- sign_, 171
- signal, 238
- sin_, 173
- single term mode, 126
- sinh_, 174
- sizeof_, 171
- sizestorecache, 231
- skip, 115, 137
- skipped expressions, 96
- slow, 176
- small buffer, 223, 233
- small extension, 224
- smallestextension, 224, 233, 234
- smallsize, 233, 234
- sort, 137, 145
 - final stage, 177
- sort file, 226
- sorting, 178, 223

- sortiosize, 226, 233, 234
- sorttype, 232
- spaces, 92
- sparse, 79, 143, 199, 201
- specifications, 51
- spin line, 149, 213
- spinline, 74
- splitarg, 87, 113, 137, 138
- splitfirstarg, 138
- splitlastarg, 138
- sqr_t_, 173
- square root, 23
- stack overflow, 89
- stages in the sorting, 226
- standard error, 43
- standard input, 34
- standard output, 34
- statements, 51
- stderr, 43, 246
- storage file, 79, 96
- store file, 134
- strict, 144
- string, 48
- strong typing, 12
- stub, 204
- stub function, 200
- student, 222
- stuffle, 136
- subexpressions, 234
- substaidentify, 163
- substitution
 - delayed, 29, 62
- substitutions, 57
- subtraction, 23
- SUBVERSION_, 21
- sum, 10, 140, 171
- sum_, 171
- summation
 - Einstein, 7, 9, 16
- sump_, 172
- sums
 - harmonic, 69
- superpatch, 233
- symbol, 141
- symbol_, 10
- symbols, 5, 70, 81, 180
 - built in, 6
- symmetric, 73, 94
 - anti, 134
 - cycle, 94, 133
 - reverse cycle, 95
- symmetrization, 142
- symmetrize, 130, 142, 214
 - reverse cycle, 130
- SYSTEMERROR_, 21, 46
- tab, 47
- table, 89, 91, 115, 128, 135, 143, 156, 199
 - check, 144
 - commuting, 78
 - matrix like, 143, 144
 - noncommuting, 115
 - onefill, 144
 - relax, 144
 - sparse, 79, 143, 144
 - strict, 144
 - zerofill, 144
- table elements, 199
- table index, 156
- table indices, 143
- table_, 172
- tablebase, 145, 146, 172, 199
- tables, 9
- tan_, 173
- tanh_, 174
- tbl_, 172, 199, 200, 204
- tempdir, 232
- temporary files, 226
- temporary variables, 180
- tempsortdir, 232
- tensor, 115, 135, 145, 156
 - commuting, 78
 - Levi-Civita, 77, 160, 218
 - metric, 218
 - noncommuting, 115
- tensors, 9
 - commuting, 71
 - noncommuting, 71, 98
- term, 86, 137, 145
- term_, 172
- terminate, 32
- termination, 55, 135
 - premature, 132
- terms, 4
- termsin, 36
- termsin_, 173

- termsinbracket_, 173
- termsinsmall, 233, 234
- testuse, 146, 200, 202, 204
- TFORM, 65, 103, 114, 146, 237, 238, 251
- theta_, 173
- thetap_, 173
- thread, 65, 109
- threadbucketsize, 146, 232, 238
- threadloadbalancing, 232, 239
- threads, 232
- threadscratchoutsize, 232
- threadscratchsize, 232
- time, 55
- time_, 21
- timer_, 21
- tointegralnotation, 150
- tolyndon, 151
- topologies_, 166, 173
- ToPolynomial, 181
- topolynomial, 87
- tospectator, 250
- tosumnotation, 150
- totalsize, 2, 122, 233
- totensor, 147, 148
 - !set, 147
 - !vector, 147
 - functions, 147
 - nosquare, 147
- tovector, 148
- trace, 213
- trace contractions, 81
- trace line, 148
- trace4, 148, 214
 - contract, 148
 - nocontract, 148
 - nosymmetrize, 148
 - notrick, 148
- tracen, 149, 214
- traces, 8
- traffic jam, 226
- Transform, 149
- transform
 - addargs, 151
 - cycle, 150
 - decode, 150
 - dedup, 150
 - dropargs, 151
 - encode, 150
 - explode, 150
 - implode, 150
 - islyndon, 151
 - mulargs, 151
 - permute, 150
 - replace, 150
 - reverse, 150
 - selectargs, 151
 - tointegralnotation, 150
 - tolyndon, 151
 - tosumnotation, 150
- translation, 205
- tree
 - balanced, 144
 - index, 176
- tree of losers, 225
- tree sort, 223
- tricks, 214
- tryreplace, 152
- twolog, 23
- underscore, 4, 5
- underscore character, 93
- unhide, 97, 116, 153
- unit matrix, 153, 214
- unittrace, 153
- UNIX, 1, 38, 41, 57, 233, 237, 247
- unpleasant, 104
- upper, 77
- use, 200, 202, 204
- variable
 - dollar, 90
 - preprocessor, 29, 40, 43, 47, 62
- variables
 - algebraic, 61
 - dollar, 4, 5, 15, 61
 - lists, 13
 - preprocessor, 4, 5, 33, 61, 130
 - total number of, 18
- vector, 154
- vectors, 6, 70
- VERSION_, 21, 44
- vertices, 133
- veto, 65
- virtual bracket, 159
- warning, 75, 89
- warnings, 206

- water, 222
- weight, 100
- while, 86, 154
- wildcard, 57, 63, 97, 156
 - \$-variable, 60
 - argument field, 58, 156
 - function, 57
 - index, 57
 - symbol, 58
 - vector, 58
- wildcarding, 10
- wildcards, 144
- Windows, 41
- word, 205
- worker, 238
- workers, 237
- workspace, 233
- write, 155
 - allnames, 155
 - allwarnings, 155
 - highfirst, 155
 - lowfirst, 155
 - names, 155
 - powerfirst, 155
 - setup, 155
 - shortstatistics, 155
 - shortstats, 155
 - statistics, 155
 - stats, 155
 - warnings, 155
- wtimestats, 233
- Zen, 222
- zero
 - dimension, 103
- zlib, 200
- zstd, 120